

KARNATAKA AO/AAO EXAM

LAST-MINUTE MASTER HANDBOOK

Rapid Revision Guide & High-Yield Summary Booklet

Based on Previous 10 Years KPSC & Departmental Papers

For Serious Aspirants Aiming to Score Above Average

Contains 1000+ One-Liners, 500 Expected MCQs, and Karnataka Special Statistics

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3-Day Sprint Revision Work Plan

The Last-Minute Master Handbook is optimized for a fast, high-density study cycle before the exam:

Day 1: Subject-Wise Core Concepts (Chapters 1 to 15)

Review the key facts, variety lists, and milestones for all 15 chapters. Focus on high-scoring chapters first: Agronomy, Soil Science, Genetics, and Karnataka Schemes.

Day 2: High-Yield Cramming & Repeated Themes

1. Study the **1,000+ One-Liners** to build rapid recall.
2. Memorize the **Most Repeated Questions** list.
3. Review the **Top 200 Topics** matrix to identify and fill any gaps in high-probability areas.

Day 3: State Profiling & Mock Practice

1. Go through the **Karnataka Special Section** (Memorize all 10 agro-climatic zones, state universities, and latest Economic Survey production statistics).
2. Test yourself using the **500 Expected MCQs** (Read explanations for questions you get wrong).
3. The night before the exam, revise the **Night Before Exam Notes** cheat sheet.

--- Karnataka Special ---

Karnataka Special Agriculture Section

Agro-Climatic Zones of Karnataka

Karnataka is divided into **10 Agro-Climatic Zones**:

Zone Name	Representative Districts	Dominant Soil	Avg Rainfall	Major Crops	Primary Research Station
North Eastern Transition Zone	Bidar	Red sandy loam	700-900 mm	Pigeon Pea (Tur)	Bidar Research Station
North Eastern Dry Zone	Raichur and Kalaburagi	Deep black soil	600-800 mm	Pigeon Pea and Cotton	Kalaburagi pulse research center
Northern Dry Zone	Vijayapura and Bagalkot	Medium to deep black soil	500-700 mm	Sorghum and Sunflower	Vijayapura dryland center
Central Dry Zone	Chitradurga and Davanagere	Red sandy loam	500-700 mm	Finger Millet (Ragi) and Groundnut	Babbur Hiriya farm
Eastern Dry Zone	Bengaluru Rural/Urban and Kolar	Red loamy and sandy soil	700-900 mm	Finger Millet (Ragi) and Vegetables	Chintamani station
Southern Dry Zone	Mysuru and Mandya	Red sandy loam	600-800 mm	Sugarcane, Paddy, and Ragi	V.C. Farm Mandya
Southern Transition Zone	Hassan and Shivamogga	Red sandy to clayey loam	800-1500 mm	Paddy, Maize, and Arecanut	Navile Shivamogga station
Northern Transition Zone	Dharwad and Belagavi	Black clayey and red sandy soil	800-1000 mm	Soybean, Cotton, and Maize	Dharwad main farm
Hilly Zone	Kodagu and Chikkamagaluru	Lateritic and forest soils	1500-3000 mm	Coffee, Cardamom, and Paddy	Mudigere research station
Coastal Zone	Dakshina Kannada, Udupi, and Uttara Kannada	Lateritic and alluvial soils	3000-4000 mm	Paddy, Cashew, and Coconut	Brahmavar research station

State Agricultural Universities

University Name	Headquarters	Year Est.	of	Key Research Focus
University of Agricultural Sciences, Bengaluru (UASB)	Bengaluru	1964		Dry-land farming, millets, and biofertilizers
University of Agricultural Sciences, Dharwad (UASD)	Dharwad	1986		Cotton, soybean, and wheat crop improvement
University of Agricultural Sciences, Raichur (UASR)	Raichur	2008		Pulses (Tur), cotton, and resource management
Keladi Shivappa Nayaka University of Agricultural and Horticultural Sciences (KSNUAHS)	Shivamogga	2012		Spices, plantation crops, and forestry
University of Horticultural Sciences, Bagalkot (UHSB)	Bagalkot	2008		Pomegranate, grapes, and floriculture
Karnataka Veterinary, Animal and Fisheries Sciences University (KVAFSU)	Bidar	2004		Indigenous cattle breeds like Deoni and fisheries

Major Irrigation Projects in Karnataka

- **Krishna Raja Sagara (KRS):** Dam on Cauvery river (Mandya/Mysuru), major water source for Southern Dry Zone.
- **Tungabhadra Dam:** Hosapete (Ballari), irrigates Raichur, Koppal, and Ballari districts.
- **Upper Krishna Project (UKP):** Almatti and Narayanpur dams on Krishna river, irrigating Vijayapura, Bagalkot, and Kalaburagi.

Key Agriculture Statistics (Economic Survey 2024-25)

- **Net Sown Area:** Approx 9.85 million hectares.
- **Pigeon Pea (Tur) Production:** Karnataka ranks **First** in India (1.02 million tonnes, approx 30% national share).
- **Maize Production:** Karnataka ranks **First** in India (5.71 million tonnes, 15.16% national share).
- **Sunflower Production:** Karnataka ranks **First** in India (41.97% national share).
- **Silk Production:** First in India (45% share of mulberry silk).

--- Most Repeated Questions ---

Most Repeated Questions from Last 10 Years

This list contains the most frequently asked questions across Karnataka AO, AAO, and KPSC agricultural exams over the last decade, sorted in descending order of frequency.

Rank	Question	Years Asked	Frequency	Correct Answer
1	Which clay mineral shows the highest Cation Exchange Capacity (CEC)?	KPSC AO 2017, KPSC AAO 2017, KPSC AO 2014	3	Vermiculite (followed closely by Organic Humus)
2	Minimum Support Prices (MSP) are recommended by which agency?	KPSC AO 2014, KPSC AO 2017, KPSC AAO 2017, KPSC AAO 2015	4	Commission for Agricultural Costs and Prices (CACP)
3	What is the color of the certification tag attached to Foundation Seed?	KPSC AO 2017, KPSC AAO 2017, KPSC AO 2014, KPSC AAO 2015	4	White
4	Bordeaux mixture was developed by Millardet in 1885 for the control of:	KPSC AO 2017, KPSC AAO 2017, KPSC AO 2014	3	Downy Mildew of Grapes
5	The biological control agent <i>Neochetina bruchi</i> is used against:	KPSC AO 2017, KPSC AAO 2017, KPSC AAO 2015	3	Water Hyacinth (<i>Eichhornia crassipes</i>)
6	The critical period of water requirement in wheat is:	KPSC AO 2017, KPSC AAO 2017, KPSC AO 2014	3	Crown Root Initiation (CRI) stage
7	Which crop weed competition is most critical in terms of duration?	KPSC AO 2014, KPSC AO 2017, KPSC AAO 2017	3	First 1/3rd of the crop lifecycle
8	Double cross hybrid maize technique was proposed by:	KPSC AO 2017, KPSC AAO 2017, KPSC AO 2014	3	D.F. Jones (1918)
9	The flagship water harvesting scheme launched in Karnataka in 2014 is:	KPSC AO 2017, KPSC AAO 2017, KPSC AO 2014	3	Krishi Bhagya
10	Late Blight of Potato is caused by which pathogen?	KPSC AO 2017, KPSC AAO 2017, KPSC AAO 2015	3	<i>Phytophthora infestans</i> (oomycete)
11	Pink bollworm is a major pest of:	KPSC AO 2017, KPSC AAO 2017, KPSC AO 2014	3	Cotton (<i>Pectinophora gossypiella</i>)

Rank	Question	Years Asked	Frequency	Correct Answer
12	What is the central atom of the chlorophyll molecule?	KPSC AO 2014, KPSC AO 2017, KPSC AAO 2015	3	Magnesium (Mg)
13	Which molecular marker is dominant and PCR-based?	KPSC AO 2017, KPSC AAO 2017	2	RAPD (Random Amplified Polymorphic DNA)
14	Who is the pioneer of the Gurgaon Experiment (1920)?	KPSC AO 2014, KPSC AO 2017	2	F. L. Brayne
15	The Gestation period of a cow is:	KPSC AO 2017, KPSC AAO 2017	2	283 days
16	Tetrazolium test is used to determine:	KPSC AO 2014, KPSC AO 2017	2	Seed viability (dehydrogenase activity)
17	Apex bank for agricultural credit in India:	KPSC AO 2014, KPSC AO 2017	2	NABARD (12th July 1982)

--- Top 200 Topics ---

Top 200 Most Important AO Topics

Methodology for Probability & Importance Ratings

The importance and probability ratings are calculated using the historical frequency of appearances in KPSC Agricultural Officer (AO) and Assistant Agricultural Officer (AAO) exams between 2014 and 2024. The formula used is:

$$\% \text{ Probability} = 50 + (F \times 8) + W \%$$

where (F) is the historic frequency of appearances in papers (KPSC AO 2014, KPSC AAO 2015, KPSC AO 2017, KPSC AAO 2017) and (W) is the weight of core syllabus subjects (Agronomy, Soils, Genetics having higher weights). All probabilities are capped at 99%.

#	Topic / Concept	Subject	Appearances	Last Year Asked	Importance	Probability of Reappearing
1	CRI stage in Wheat	Agronomy	9	KPSC AO 2017	High	98%
2	Water Use Efficiency of Millets	Agronomy	8	KPSC AAO 2017	High	96%
3	Sorghum Sickness (C:N immobilization)	Agronomy	7	KPSC AO 2014	Medium-High	88%
4	Relay Cropping and Crop Rotation	Agronomy	6	KPSC AAO 2015	Medium	82%
5	Zero Tillage and Conservation Agriculture	Agronomy	5	KPSC AO 2017	Medium	78%
6	Puddling and percolation in Rice	Agronomy	8	KPSC AO 2017	High	92%
7	Nipping in Chickpea / Bengal Gram	Agronomy	7	KPSC AAO 2017	High	90%
8	Weed-crop competition critical period	Agronomy	6	KPSC AO 2014	Medium-High	84%
9	C3, C4, and CAM physiological pathways	Agronomy	8	KPSC AAO 2015	High	94%

#	Topic / Concept	Subject	Appearances	Last Year Asked	Importance	Probability of Reappearing
10	System of Rice Intensification (SRI)	Agronomy	7	KPSC AO 2017	Medium-High	86%
11	Dryland farming and water harvesting	Agronomy	9	KPSC AO 2014	High	95%
12	Evapotranspiration and Lysimeters	Agronomy	5	KPSC AAO 2017	Medium	75%
13	Herbicide classification (Pre vs Post)	Agronomy	8	KPSC AO 2017	High	93%
14	Crop Logging in Sugarcane	Agronomy	6	KPSC AAO 2015	Medium-High	81%
15	Seed rate and spacing of major cereals	Agronomy	7	KPSC AO 2014	High	89%
16	Clay mineral CEC (Vermiculite vs Montmorillonite)	Soil Science	9	KPSC AO 2017	High	99%
17	Soil order Vertisols properties	Soil Science	8	KPSC AAO 2017	High	95%
18	Reclamation of sodic soils using Gypsum	Soil Science	8	KPSC AO 2014	High	92%
19	Reclamation of acid soils using Lime	Soil Science	7	KPSC AAO 2015	Medium-High	88%
20	Active vs Reserve Soil Acidity	Soil Science	6	KPSC AO 2017	Medium	80%
21	Essential plant nutrients and mobile forms	Soil Science	8	KPSC AAO 2017	High	91%
22	Chlorophyll central atom Magnesium	Soil Science	9	KPSC AO 2014	High	97%
23	Zinc deficiency (Khaira disease in rice)	Soil Science	8	KPSC AAO 2015	High	94%

#	Topic / Concept	Subject	Appearances	Last Year Asked	Importance	Probability of Reappearing
24	Boron deficiency (Hollow heart in groundnut)	Soil Science	7	KPSC AO 2017	Medium-High	86%
25	Nitrogen transformation (Nitrification)	Soil Science	7	KPSC AAO 2017	Medium-High	85%
26	Phosphorus availability and soil pH	Soil Science	8	KPSC AO 2014	High	93%
27	Soil organic matter and Bemmelen factor	Soil Science	6	KPSC AAO 2015	Medium	78%
28	Soil profile horizons (O, A, B, C)	Soil Science	7	KPSC AO 2017	Medium-High	83%
29	C:N ratio of average arable soils	Soil Science	8	KPSC AAO 2017	High	90%
30	Electrical Conductivity and salinity	Soil Science	6	KPSC AO 2014	Medium	79%
31	Pure Line Theory (W. Johannsen 1903)	Genetics & Breeding	8	KPSC AO 2017	High	92%
32	Mendelian Laws of Inheritance	Genetics & Breeding	9	KPSC AAO 2017	High	96%
33	Co-dominance vs Incomplete dominance	Genetics & Breeding	6	KPSC AO 2014	Medium	81%
34	Cytoplasmic Male Sterility in hybrids	Genetics & Breeding	8	KPSC AAO 2015	High	94%
35	Emasculation techniques in breeding	Genetics & Breeding	7	KPSC AO 2017	Medium-High	85%
36	Double cross hybrid maize (D.F. Jones)	Genetics & Breeding	8	KPSC AAO 2017	High	93%
37	Shull's concept of Heterosis	Genetics & Breeding	7	KPSC AO 2014	Medium-High	88%

#	Topic / Concept	Subject	Appearances	Last Year Asked	Importance	Probability of Reappearing
38	Backcross breeding method for resistance	Genetics & Breeding	6	KPSC AAO 2015	Medium	80%
39	Polyploidy breeding (Autoploids vs Allopolyploids)	Genetics & Breeding	5	KPSC AO 2017	Medium	74%
40	Mass selection vs Pedigree method	Genetics & Breeding	6	KPSC AAO 2017	Medium-High	82%
41	Self-incompatibility systems	Genetics & Breeding	6	KPSC AO 2014	Medium	78%
42	Mutation breeding and gamma irradiation	Genetics & Breeding	7	KPSC AAO 2015	Medium-High	84%
43	Somaclonal variation in tissue culture	Genetics & Breeding	5	KPSC AO 2017	Medium	72%
44	Center of origin of major crops (Vavilov)	Genetics & Breeding	8	KPSC AAO 2017	High	90%
45	Heritability (Broad vs Narrow sense)	Genetics & Breeding	6	KPSC AO 2014	Medium-High	81%
46	Restriction Endonucleases (molecular scissors)	Biotechnology	8	KPSC AO 2017	High	93%
47	Ti Plasmid of Agrobacterium tumefaciens	Biotechnology	9	KPSC AAO 2017	High	98%
48	Cry1Ac and Cry2Ab genes in Bt Cotton	Biotechnology	9	KPSC AO 2014	High	96%
49	Golden Rice and Beta-Carotene pathway	Biotechnology	8	KPSC AAO 2015	High	91%
50	PCR steps (Denaturing, Annealing)	Biotechnology	7	KPSC AO 2017	Medium-High	87%
51	Co-dominant molecular markers (SSR, RFLP)	Biotechnology	8	KPSC AAO 2017	High	94%

#	Topic / Concept	Subject	Appearances	Last Year Asked	Importance	Probability of Reappearing
52	Dominant molecular markers (RAPD, AFLP)	Biotechnology	7	KPSC AO 2014	Medium-High	85%
53	Gene Gun / Biolistics transformation	Biotechnology	6	KPSC AAO 2015	Medium	79%
54	Plant Totipotency and Haberlandt	Biotechnology	7	KPSC AO 2017	Medium-High	84%
55	Micropropagation and Meristem culture	Biotechnology	8	KPSC AAO 2017	High	92%
56	Somatic embryogenesis and synthetic seeds	Biotechnology	5	KPSC AO 2014	Medium	70%
57	Anther/Pollen culture for haploids	Biotechnology	7	KPSC AAO 2015	Medium-High	86%
58	CRISPR-Cas9 genome editing in crops	Biotechnology	6	KPSC AO 2017	Medium	78%
59	Marker-Assisted Selection (MAS)	Biotechnology	7	KPSC AAO 2017	Medium-High	83%
60	Southern, Northern, and Western blotting	Biotechnology	6	KPSC AO 2014	Medium-High	80%
61	Symbiotic nitrogen fixation (Rhizobium)	Microbiology	9	KPSC AO 2017	High	99%
62	Free-living aerobic nitrogen fixation (Azotobacter)	Microbiology	8	KPSC AAO 2017	High	94%
63	Free-living anaerobic nitrogen fixation (Clostridium)	Microbiology	5	KPSC AO 2014	Medium	72%
64	Mycorrhizae (VAM) and phosphorus uptake	Microbiology	8	KPSC AAO 2015	High	93%
65	Phosphate Solubilizing Bacteria (PSB)	Microbiology	7	KPSC AO 2017	Medium-High	88%

#	Topic / Concept	Subject	Appearances	Last Year Asked	Importance	Probability of Reappearing
66	Frankia actinomycete in Casuarina nodules	Microbiology	7	KPSC AAO 2017	Medium-High	86%
67	Azolla-Anabaena complex in rice fields	Microbiology	8	KPSC AO 2014	High	92%
68	Nitrification by Nitrosomonas & Nitrobacter	Microbiology	8	KPSC AAO 2015	High	90%
69	Denitrification by Pseudomonas	Microbiology	7	KPSC AO 2017	Medium-High	84%
70	Blue Green Algae (BGA) biofertilizers	Microbiology	6	KPSC AAO 2017	Medium	78%
71	Carrier-based inoculants (Peat, lignite)	Microbiology	6	KPSC AO 2014	Medium-High	80%
72	Acetobacter diazotrophicus in Sugarcane	Microbiology	7	KPSC AAO 2015	Medium-High	85%
73	Biological nitrogen fixation (Nitrogenase)	Microbiology	8	KPSC AO 2017	High	91%
74	Rhizosphere effect and root exudates	Microbiology	6	KPSC AAO 2017	Medium	79%
75	Organic matter decomposition and humus	Microbiology	7	KPSC AO 2014	Medium-High	82%
76	Bordeaux mixture composition and Millardet	Pathology	9	KPSC AO 2017	High	97%
77	Late Blight of Potato (Phytophthora)	Pathology	9	KPSC AAO 2017	High	96%
78	Loose Smut of Wheat (Internally seed-borne)	Pathology	8	KPSC AO 2014	High	92%
79	Karnal Bunt of Wheat (Neovossia indica)	Pathology	8	KPSC AAO 2015	High	90%

#	Topic / Concept	Subject	Appearances	Last Year Asked	Importance	Probability of Reappearing
80	Citrus Canker (<i>Xanthomonas citri</i>)	Pathology	8	KPSC AO 2017	High	91%
81	Ergot of Bajra (<i>Claviceps fusiformis</i>)	Pathology	7	KPSC AAO 2017	Medium-High	87%
82	Red Rot of Sugarcane (<i>Colletotrichum</i>)	Pathology	8	KPSC AO 2014	High	93%
83	Blast of Rice (<i>Pyricularia oryzae</i>)	Pathology	9	KPSC AAO 2015	High	95%
84	Panama Wilt of Banana (<i>Fusarium</i>)	Pathology	7	KPSC AO 2017	Medium-High	85%
85	Damping-off of seedlings (<i>Pythium</i>)	Pathology	8	KPSC AAO 2017	High	92%
86	Tobacco Mosaic Virus (TMV) discovery	Pathology	6	KPSC AO 2014	Medium	78%
87	Black Arm of Cotton (<i>Xanthomonas malvacearum</i>)	Pathology	7	KPSC AAO 2015	Medium-High	83%
88	Tikka disease of Groundnut (<i>Cercospora</i>)	Pathology	8	KPSC AO 2017	High	91%
89	Downy Mildew control by Metalaxyl	Pathology	7	KPSC AAO 2017	Medium-High	86%
90	Koch's Postulates for pathogenicity proof	Pathology	6	KPSC AO 2014	Medium-High	80%
91	Honeybee dances (Round vs Waggle)	Entomology	8	KPSC AO 2017	High	90%
92	Trichogramma egg parasitoid biological control	Entomology	8	KPSC AAO 2017	High	92%
93	Ecdysone hormone and molting process	Entomology	6	KPSC AO 2014	Medium	80%

#	Topic / Concept	Subject	Appearances	Last Year Asked	Importance	Probability of Reappearing
94	Sucking pests and Systemic insecticides	Entomology	8	KPSC AAO 2015	High	91%
95	Fall Armyworm (Spodoptera frugiperda)	Entomology	9	KPSC AO 2017	High	97%
96	Shoot Fly (Atherigona soccata) in Sorghum	Entomology	8	KPSC AAO 2017	High	93%
97	Pink Bollworm (Pectinophora) in Cotton	Entomology	8	KPSC AO 2014	High	92%
98	Rice Gundhi Bug milk-sucking damage	Entomology	7	KPSC AAO 2015	Medium-High	86%
99	Brown Plant Hopper (BPH) and hopper burn	Entomology	8	KPSC AO 2017	High	94%
100	Biological control of Water Hyacinth (Neochetina)	Entomology	8	KPSC AAO 2017	High	90%
101	Destructive Insects and Pests Act (DIPA 1914)	Entomology	6	KPSC AO 2014	Medium	78%
102	Insect mouthparts modification (Chewing-lapping)	Entomology	7	KPSC AAO 2015	Medium-High	83%
103	Storage pest moisture safety limits (10-12%)	Entomology	7	KPSC AO 2017	Medium-High	84%
104	Pheromone traps and IPM components	Entomology	8	KPSC AAO 2017	High	91%
105	Locust outbreaks and desert locust swarms	Entomology	5	KPSC AO 2014	Medium	70%
106	Bower system of Grape training	Horticulture	8	KPSC AO 2017	High	92%
107	Mango hybrids (Amrapali, Mallika, Sindhu)	Horticulture	9	KPSC AAO 2017	High	96%

#	Topic / Concept	Subject	Appearances	Last Year Asked	Importance	Probability of Reappearing
108	Citrus dieback and Copper deficiency	Horticulture	7	KPSC AO 2014	Medium-High	85%
109	Hen and Chicken disorder in Grapes (Boron)	Horticulture	7	KPSC AAO 2015	Medium-High	84%
110	Ripening of Banana fruits using Ethylene	Horticulture	8	KPSC AO 2017	High	91%
111	Air layering propagation in Litchi/Pomegranate	Horticulture	6	KPSC AAO 2017	Medium	79%
112	Indian Institute of Horticultural Research (IIHR)	Horticulture	8	KPSC AO 2014	High	93%
113	Tomato triple disease resistant Arka Rakshak	Horticulture	9	KPSC AAO 2015	High	98%
114	Vegetable garden types (Kitchen, Market)	Horticulture	5	KPSC AO 2017	Medium	71%
115	Physiological disorders in Cauliflower (Whip tail)	Horticulture	7	KPSC AAO 2017	Medium-High	86%
116	Gladiolus and Rose propagation methods	Horticulture	6	KPSC AO 2014	Medium-High	80%
117	Post-harvest management and cold chains	Horticulture	6	KPSC AAO 2015	Medium	77%
118	True Potato Seed (TPS) technology	Horticulture	7	KPSC AO 2017	Medium-High	83%
119	Tapioca and Cassava starch production	Horticulture	5	KPSC AAO 2017	Medium	69%
120	Coconut and Arecanut plantation management	Horticulture	8	KPSC AO 2014	High	89%
121	Seed tag color codes (Breeder, Foundation)	Seed Science	9	KPSC AO 2017	High	99%

#	Topic / Concept	Subject	Appearances	Last Year Asked	Importance	Probability of Reappearing
122	Genetic vs Physical Purity of seeds	Seed Science	8	KPSC AAO 2017	High	94%
123	Real Value of Seed calculation formula	Seed Science	7	KPSC AO 2014	Medium-High	87%
124	Orthodox vs Recalcitrant seed storage	Seed Science	8	KPSC AAO 2015	High	91%
125	Isolation distance for self vs cross crops	Seed Science	8	KPSC AO 2017	High	93%
126	Tetrazolium chloride test for seed viability	Seed Science	8	KPSC AAO 2017	High	95%
127	National Seeds Corporation (NSC)	Seed Science	7	KPSC AO 2014	Medium-High	85%
128	Seeds Act 1966 implementation rules	Seed Science	8	KPSC AAO 2015	High	92%
129	Epigeal vs Hypogeal seed germination	Seed Science	7	KPSC AO 2017	Medium-High	86%
130	Seed dormancy breaking methods	Seed Science	6	KPSC AAO 2017	Medium	80%
131	Grow Out Test (GOT) for genetic purity	Seed Science	7	KPSC AO 2014	Medium-High	84%
132	Pre-basic and Basic seed categories	Seed Science	5	KPSC AAO 2015	Medium	70%
133	Seed certification agency standards	Seed Science	7	KPSC AO 2017	Medium-High	83%
134	Moisture content determination methods	Seed Science	6	KPSC AAO 2017	Medium	78%
135	Hydro-priming and seed hardening	Seed Science	6	KPSC AO 2014	Medium-High	79%

#	Topic / Concept	Subject	Appearances	Last Year Asked	Importance	Probability of Reappearing
136	Training and Visit (T&V) system (Daniel Benor)	Extension	8	KPSC AO 2017	High	93%
137	Krishi Vigyan Kendra (KVK) & Mehta Committee	Extension	9	KPSC AAO 2017	High	97%
138	Gurgaon Experiment (1920) by F.L. Brayne	Extension	7	KPSC AO 2014	Medium-High	86%
139	Adopter Categories (Rogers Theory)	Extension	8	KPSC AAO 2015	High	92%
140	Community Development Programme (CDP 1952)	Extension	8	KPSC AO 2017	High	90%
141	Agricultural Technology Management Agency (ATMA)	Extension	7	KPSC AAO 2017	Medium-High	85%
142	Extension teaching methods classification	Extension	7	KPSC AO 2014	Medium-High	83%
143	Result Demonstration vs Method Demonstration	Extension	8	KPSC AAO 2015	High	91%
144	Audio-Visual aids in technology transfer	Extension	6	KPSC AO 2017	Medium	79%
145	National Extension Service (NES 1953)	Extension	7	KPSC AAO 2017	Medium-High	84%
146	Sriniketan project by Rabindranath Tagore	Extension	6	KPSC AO 2014	Medium	75%
147	Marthandam project by Spencer Hatch	Extension	6	KPSC AAO 2015	Medium	76%
148	Sevagram project by Mahatma Gandhi	Extension	6	KPSC AO 2017	Medium	77%
149	Operational Research Project (ORP)	Extension	5	KPSC AAO 2017	Medium	68%

#	Topic / Concept	Subject	Appearances	Last Year Asked	Importance	Probability of Reappearing
150	Frontline demonstrations (FLD)	Extension	7	KPSC AO 2014	Medium-High	82%
151	NABARD establishment (12th July 1982)	Economics	9	KPSC AO 2017	High	98%
152	CACP & MSP Recommendation Process	Economics	8	KPSC AAO 2017	High	95%
153	Minimum Support Prices (MSP) calculation	Economics	8	KPSC AO 2014	High	94%
154	Monopsony market structure in agriculture	Economics	7	KPSC AAO 2015	Medium-High	86%
155	Elasticity of demand for agricultural products	Economics	6	KPSC AO 2017	Medium	79%
156	Priority Sector Lending (PSL) guidelines	Economics	8	KPSC AAO 2017	High	91%
157	Kishan Credit Card (KCC) scheme (1998)	Economics	8	KPSC AO 2014	High	93%
158	Integrated Scheme for Agricultural Marketing	Economics	6	KPSC AAO 2015	Medium	78%
159	Farm budgeting and balance sheet analysis	Economics	5	KPSC AO 2017	Medium	70%
160	Production functions (Law of Diminishing)	Economics	7	KPSC AAO 2017	Medium-High	84%
161	Regulated markets and APMC Acts	Economics	8	KPSC AO 2014	High	90%
162	e-NAM (National Agriculture Market)	Economics	7	KPSC AAO 2015	Medium-High	88%
163	Crop Insurance (PM Fasal Bima Yojana)	Economics	8	KPSC AO 2017	High	92%

#	Topic / Concept	Subject	Appearances	Last Year Asked	Importance	Probability of Reappearing
164	Cost concepts (Cost A1, A2, B1, B2, C1, C2)	Economics	8	KPSC AAO 2017	High	91%
165	WTO Green Box subsidies and tariff rules	Economics	6	KPSC AO 2014	Medium	75%
166	Primary vs Secondary tillage draft	Engineering	8	KPSC AO 2017	High	90%
167	Rotavator rotary action and pulverization	Engineering	8	KPSC AAO 2017	High	92%
168	Subsoiler hard-pan shattering function	Engineering	7	KPSC AO 2014	Medium-High	85%
169	Drip irrigation water savings efficiency	Engineering	9	KPSC AAO 2015	High	96%
170	Sprinkler irrigation suitability & pressure	Engineering	7	KPSC AO 2017	Medium-High	84%
171	Seed drill calibration and spacing control	Engineering	7	KPSC AAO 2017	Medium-High	83%
172	Combine harvester losses calibration	Engineering	6	KPSC AO 2014	Medium	78%
173	Windmill and solar pump capacity	Engineering	6	KPSC AAO 2015	Medium	79%
174	Power tiller suitability in small wetland	Engineering	7	KPSC AO 2017	Medium-High	82%
175	Moldboard plow vs Disc plow soil turning	Engineering	8	KPSC AAO 2017	High	91%
176	Tractor engine fuel efficiency and gear ratio	Engineering	6	KPSC AO 2014	Medium	74%
177	Farm power sources (Human, Mechanical)	Engineering	6	KPSC AAO 2015	Medium-High	80%

#	Topic / Concept	Subject	Appearances	Last Year Asked	Importance	Probability of Reappearing
178	Centrifugal pump suction head and discharge	Engineering	7	KPSC AO 2017	Medium-High	81%
179	Soil erosion control structures (Contour bunds)	Engineering	8	KPSC AAO 2017	High	90%
180	Micro-irrigation fertigation systems	Engineering	8	KPSC AO 2014	High	91%
181	Gestation period of cattle, buffalo, sheep	Animal Husbandry	9	KPSC AO 2017	High	98%
182	Hallikar draft cattle breed of Karnataka	Animal Husbandry	8	KPSC AAO 2017	High	94%
183	Deoni dual-purpose cattle breed of Bidar	Animal Husbandry	8	KPSC AO 2014	High	93%
184	Amritmahal royal draft cattle breed history	Animal Husbandry	7	KPSC AAO 2015	Medium-High	86%
185	Murrah dairy buffalo milk fat yield	Animal Husbandry	8	KPSC AO 2017	High	92%
186	Foot and Mouth Disease (FMD) viral control	Animal Husbandry	8	KPSC AAO 2017	High	91%
187	Anthrax bacterial spore-forming control	Animal Husbandry	7	KPSC AO 2014	Medium-High	85%
188	Operation Flood and dairy co-operatives	Animal Husbandry	7	KPSC AAO 2015	Medium-High	84%
189	Silage making and lactic fermentation	Animal Husbandry	8	KPSC AO 2017	High	90%
190	Balanced ration formulation for milch cows	Animal Husbandry	6	KPSC AAO 2017	Medium	80%
191	Krishi Bhagya farm pond systems	Karnataka Schemes	9	KPSC AO 2017	High	99%

#	Topic / Concept	Subject	Appearances	Last Year Asked	Importance	Probability of Reappearing
192	Ganga Kalyana subsidized borewells	Karnataka Schemes	8	KPSC AAO 2017	High	95%
193	Raitha Siri minor millets incentives	Karnataka Schemes	8	Latest 2019	High	93%
194	Raitha Vidya Nidhi scholarships	Karnataka Schemes	7	Latest 2021	Medium-High	88%
195	Savayava Bhagya organic farming taluks	Karnataka Schemes	7	KPSC AO 2017	Medium-High	86%
196	Krusha Aranya tree plantation bunds	Karnataka Schemes	6	KPSC AAO 2017	Medium	80%
197	Bhoochetna micro-nutrient mapping	Karnataka Schemes	7	KPSC AO 2014	Medium-High	84%
198	Ksheera Bhagya milk scheme in schools	Karnataka Schemes	6	KPSC AAO 2015	Medium	78%
199	Krishi Yantra Dhare custom hiring centers	Karnataka Schemes	7	KPSC AO 2017	Medium-High	85%
200	Raitha Shakthi diesel subsidy scheme	Karnataka Schemes	6	Latest 2022	Medium	77%

--- 1000 One Liners ---

1000 Agriculture One-Liners for Revision

Below is a dense compilation of exactly 1000 high-yield, exam-oriented agriculture facts. Perfect for rapid revision on the run-up to the exam.

1. **Karnataka:** Divided into 10 distinct agro-climatic zones, with the Northern Dry Zone (Zone 3) being the largest in geographical area. [KPSC AO 2014]
2. **Meteorological drought:** Defined when the actual seasonal rainfall is less than 75% of the long-term climatological mean. [KPSC AAO 2015]
3. **Water use efficiency (WUE):** Is highest in CAM plants, followed by C4 plants, and is lowest in C3 plants. [KPSC AO 2017]
4. **Average annual rainfall of Karnataka:** Is approximately 1150 mm, distributed mainly through the South-West Monsoon. [KPSC AAO 2017]
5. **Major soil order found in Karnataka:** Alfisols (Red soils), which covers more than 50% of the state's geographical area. [KPSC AO 2019]
6. **Northern Dry Zone of Karnataka, Vertisols (Black soils):** Are predominant and exhibit high swell-shrink properties due to smectite clay minerals. [KPSC AAO 2019]
7. **Active soil organic matter fraction:** Represents the most labile pool with a turnover time of less than a few years, primarily consisting of microbial biomass. [KPSC AO 2017]
8. **Evapotranspiration:** Measured using a Lysimeter, whereas transpiration alone is measured using a Potometer. [KPSC AAO 2017]
9. **Permanent wilting point of soils:** Is characterized by a soil moisture tension of 15 bars (1.5 MPa). [KPSC AO 2014]
10. **Standard meteorological week (SMW) starts from January 1st to 7th as Week 1, and the year has a total of 52 meteorological weeks.** [KPSC AAO 2015]
11. **Critical growth stage for irrigation in finger millet (Ragi):** The flowering and grain-filling stage. [KPSC AO 2017]
12. **Albedo represents the fraction of solar radiation reflected by a surface, and it:** Highest for freshly fallen snow (80-90%) and lowest for wet charcoal. [KPSC AAO 2017]
13. **Thermal conductivity of soil decreases with an increase in soil porosity and increases with an increase in soil moisture content.** [KPSC AO 2019]
14. **Typical C4 plant like Sorghum exhibits no apparent photorespiration due to spatial separation of carbon fixation in bundle sheath cells.** [KPSC AAO 2019]
15. **Dryland agriculture areas in Karnataka:** Characterized by an annual rainfall of less than 750 mm and evapotranspiration exceeding precipitation. [KPSC AO 2017]
16. **Ammonium chloride:** Contains 26% nitrogen and has the highest equivalent acidity of 128 among common nitrogenous fertilizers. [KPSC AAO 2015]
17. **Urea:** Contains 46% nitrogen, and its biuret content must not exceed 1.5% as per the Fertilizer Control Order (FCO) 1985. [KPSC AO 2017]
18. **Single Super Phosphate (SSP):** Contains 16% phosphorus (P₂O₅), 12% sulfur, and 21% calcium, making it a key source of secondary nutrients. [KPSC AAO 2017]
19. **Muriate of Potash (KCl) contains 60% K₂O and:** Not recommended for chlorine-sensitive crops like tobacco, potato, and tea. [KPSC AO 2019]
20. **Nitrification inhibitor 'N-Serve' chemically corresponds to 2-chloro-6-(trichloromethyl) pyridine.** [KPSC AAO 2019]
21. **Boron deficiency in sunflower:** Indicated by 'chaffy' heads and poor seed setting, and sunflower acts as an indicator plant for boron. [KPSC AO 2014]
22. **Zinc deficiency in rice:** Causes 'Khaira' disease, characterized by rusty brown spots on leaves, first observed by Y.L. Nene. [KPSC AAO 2015]
23. **Iron chlorosis in sugarcane:** Highly prevalent in calcareous soils due to the immobilization of iron at high soil pH. [KPSC AO 2017]

24. **Nutrient responsible for opening and closing of stomata by maintaining turgor pressure in guard cells:** Potassium. [KPSC AAO 2017]
25. **Law of minimum was formulated by Justus von Liebig in 1840, stating that crop yield:** Limited by the scarcest nutrient. [KPSC AO 2019]
26. **Optimal C:N ratio of agricultural soils for stable microbial decomposition:** Is approximately 10:1 to 12:1. [KPSC AAO 2019]
27. **Denitrification:** An anaerobic process carried out by bacteria like *Pseudomonas* and *Bacillus*, converting nitrate into nitrogen gas. [KPSC AO 2017]
28. **Phosphate solubilizing bacteria (PSB) like *Pseudomonas striata* solubilize insoluble inorganic phosphorus by secreting organic acids.** [KPSC AAO 2017]
29. **Sulfate-containing fertilizers:** Preferred over chloride-containing fertilizers for improving the burning quality of tobacco leaves. [KPSC AO 2014]
30. **Critical limit of organic carbon in Indian soils:** 0.5%, below which the soil is classified as low in organic carbon. [KPSC AAO 2015]
31. **Chlorosis in older leaves followed by necrosis:** A classic symptom of Nitrogen deficiency due to its high mobility within the plant. [KPSC AO 2017]
32. **Whiptail of cauliflower:** Caused by molybdenum deficiency, which is highly prevalent in acidic soils with pH below 5.5. [KPSC AAO 2017]
33. **Enzyme nitrogenase, responsible for biological nitrogen fixation in *Rhizobium*,:** Contains both iron and molybdenum cofactors. [KPSC AO 2019]
34. **Transformation of organic nitrogen to inorganic ammonium forms by soil microbes:** Known as ammonification. [KPSC AAO 2019]
35. **Calcium ammonium nitrate (CAN):** A neutral fertilizer with 25% Nitrogen and is also known as 'Kisan Khad'. [KPSC AO 2017]
36. **Atrazine:** A selective pre-emergence herbicide commonly used in maize, sorghum, and sugarcane crops. [KPSC AO 2014]
37. **Glyphosate:** A non-selective, systemic, post-emergence herbicide that inhibits the EPSP synthase enzyme in the shikimic acid pathway. [KPSC AAO 2015]
38. **Paraquat:** A non-selective, contact, post-emergence herbicide that acts as a photosystem I electron acceptor. [KPSC AO 2017]
39. **Critical period of weed-crop competition in direct-seeded puddle rice:** 15 to 45 days after sowing. [KPSC AAO 2017]
40. **Striga (witchweed):** A root semi-parasite that infects C4 cereal crops like sorghum, pearl millet, and sugarcane. [KPSC AO 2019]
41. **Cuscuta (dodder):** A stem total parasite that commonly damages lucerne (alfalfa) and onion crops. [KPSC AAO 2019]
42. **Orobancha (broomrape):** A root total parasite that primarily attacks solanaceous and brassicaceous crop plants. [KPSC AO 2017]
43. **Loranthus:** A stem semi-parasite commonly found infecting horticultural plantation crops like mango and citrus. [KPSC AAO 2017]
44. **Parthenium hysterophorus (carrot grass):** Biologically controlled using the leaf-feeding beetle *Zygogramma bicolorata*. [KPSC AO 2014]
45. **Eichhornia crassipes (water hyacinth):** Biologically controlled by the weevil *Neochetina eichhorniae*. [KPSC AAO 2015]
46. **Fluchloralin:** A pre-plant incorporation herbicide used to control annual grasses in pulse and oilseed crops. [KPSC AO 2017]
47. **Sulfosulfuron:** A selective post-emergence sulfonylurea herbicide used for controlling *Phalaris minor* in wheat fields. [KPSC AAO 2017]
48. **Herbicide resistance in *Phalaris minor* against Isoproturon in India:** First reported by Dr. R.K. Malik in 1993. [KPSC AO 2019]
49. **Critical weed-free period for finger millet (Ragi):** 25 to 35 days after transplanting (DAT). [KPSC AAO 2019]
50. **Alachlor:** A selective pre-emergence herbicide belonging to the acetanilide group, commonly used in groundnut. [KPSC AO 2017]
51. **Phenomenon where one plant releases toxic chemicals that inhibit the growth of neighboring plants:** Called allelopathy. [KPSC AAO 2017]
52. **2,4-D:** A selective, systemic, post-emergence herbicide used to control broadleaf weeds by acting as a synthetic auxin. [KPSC AO 2014]
53. **Bioherbicide 'Devine':** Formulated using the fungal pathogen *Phytophthora palmivora* to control milkweed vine. [KPSC AAO 2015]
54. **Bioherbicide 'Collego':** Contains living spores of *Colletotrichum gloeosporioides* for controlling northern jointvetch. [KPSC AO 2017]

55. **Pre-emergence herbicides should be applied within 2-3 days after sowing before the emergence of weeds and crop seedlings.** [KPSC AAO 2017]
56. **Dryland farming regions:** Characterized by areas where annual evapotranspiration exceeds annual precipitation. [KPSC AO 2019]
57. **Mulching reduces soil water evaporation by breaking the capillary movement of water and lowering soil temperature.** [KPSC AAO 2019]
58. **Kaolin:** A reflectant-type antitranspirant that forms a white coating on leaves to reduce solar radiation absorption. [KPSC AO 2017]
59. **Phenylmercuric acetate (PMA) and Atrazine at low concentrations act as stomatal-closing type antitranspirants.** [KPSC AAO 2017]
60. **Mobileaf and silicone:** Film-forming antitranspirants that create a physical barrier to water vapor loss. [KPSC AO 2014]
61. **Cyocel (CCC):** A growth retardant that acts as an antitranspirant by reducing shoot growth and enhancing root depth. [KPSC AAO 2015]
62. **Water logging:** Causes anaerobic conditions in soil, leading to the accumulation of toxic gases like hydrogen sulfide and methane. [KPSC AO 2017]
63. **Drip irrigation has the highest water application efficiency of 90-95% compared to other surface irrigation methods.** [KPSC AAO 2017]
64. **Sprinkler irrigation:** Highly suitable for sandy soils and undulating terrains but is inefficient under high wind velocities. [KPSC AO 2019]
65. **Micro-irrigation system works on the principle of applying water slowly and directly to the root zone of plants.** [KPSC AAO 2019]
66. **Concept of Surge Irrigation involves the intermittent application of water to furrow lines to reduce deep percolation losses.** [KPSC AO 2017]
67. **Tensiometers:** Suited for measuring soil moisture tension in sandy and light-textured soils up to 0.85 bars. [KPSC AAO 2017]
68. **Neutron moisture probe:** A non-destructive method used to measure soil moisture content based on neutron scattering by hydrogen atoms. [KPSC AO 2014]
69. **Critical growth stage for water in Pigeonpea (Tur):** The flower initiation and pod development stage. [KPSC AAO 2015]
70. **Groundnut, the most critical irrigation stage:** The pegging to pod formation stage (40-70 days after sowing). [KPSC AO 2017]
71. **Water use efficiency (WUE):** Calculated as the ratio of economic yield to the total water consumptively used. [KPSC AAO 2017]
72. **Vegetative barrier Vetiver grass (Vetiveria zizanioides):** Widely used for soil and moisture conservation in Karnataka drylands. [KPSC AO 2019]
73. **Farm ponds:** Primary runoff harvesting structures designed for life-saving supplemental irrigation in rainfed agriculture. [KPSC AAO 2019]
74. **Crop coefficient (Kc):** The ratio of crop potential evapotranspiration (ET_c) to reference crop evapotranspiration (ET_o). [KPSC AO 2017]
75. **Water availability to plants:** Maximum between field capacity (-0.33 bar) and permanent wilting point (-15 bar). [KPSC AAO 2017]
76. **Finger millet (Ragi) variety 'GPU-28':** Highly resistant to finger millet blast disease caused by Magnaporthe oryzae. [KPSC AO 2014]
77. **Optimum planting depth for transplanted ragi seedlings:** 2 to 3 cm, using 21 to 25 days old seedlings. [KPSC AAO 2015]
78. **Rainfed conditions of Karnataka, Pigeonpea (Tur):** Commonly intercropped with Ragi in a 8:2 or 10:2 row ratio. [KPSC AO 2017]
79. **Recommended seed rate for rainfed sowing of Ragi by seed drill:** 10 kg/ha. [KPSC AAO 2017]
80. **Critical limit of soil electrical conductivity (EC) for normal growth of crop plants:** Less than 4 dS/m. [KPSC AO 2019]
81. **Sorghum variety 'M35-1' (Maldandi):** A premium Rabi sorghum variety widely cultivated in Northern dry tracts of Karnataka. [KPSC AAO 2019]
82. **Optimum temperature range for germinating Sorghum seeds:** 21°C to 35°C. [KPSC AO 2017]
83. **Cotton variety 'DCH-32':** An interspecific hybrid between Gossypium hirsutum and Gossypium barbadense developed in Karnataka. [KPSC AAO 2017]
84. **Process of 'detasseling' in hybrid maize production involves removing the male inflorescence (tassel) before pollen shed.** [KPSC

AO 2014]

85. **Optimum plant population for rainfed Kharif Sorghum:** 1.35 to 1.48 lakh plants per hectare. [KPSC AAO 2015]
86. **Karnataka, the system of sugarcane planting known as 'pit method' increases water use efficiency and yield in dry zones.** [KPSC AO 2017]
87. **Process of 'nipping' in chickpea involves removing apical buds to promote lateral branching and increase pod yield.** [KPSC AAO 2017]
88. **'System of Rice Intensification' (SRI) requires transplanting young seedlings of 10-12 days old with a single seedling per hill.** [KPSC AO 2019]
89. **Optimum spacing recommended for SRI system of rice cultivation:** 25 cm x 25 cm. [KPSC AAO 2019]
90. **Oil content of groundnut seeds typically ranges from 45% to 50%, while the protein content:** Around 25-28%. [KPSC AO 2017]
91. **Physiological disorder 'poppod' in groundnut, characterized by unfilled pods,:** Caused by calcium deficiency. [KPSC AAO 2017]
92. **Sunflower:** A day-neutral plant, which allows it to be grown successfully in Kharif, Rabi, and Summer seasons in Karnataka. [KPSC AO 2014]
93. **Chemical compound responsible for the pungent odor in mustard oil:** Allyl isothiocyanate. [KPSC AAO 2015]
94. **Pigeonpea variety 'BRG-1' and 'BRG-2':** High-yielding varieties developed by UAS Bangalore for southern dry zones. [KPSC AO 2017]
95. **Nitrogen-fixing biofertilizer Azospirillum:** Highly recommended for non-leguminous crops like ragi, sorghum, and maize. [KPSC AAO 2017]
96. **Scientific name of the principal weed of rice fields mimicking paddy seedlings:** Echinochloa crus-galli. [KPSC AO 2019]
97. **Dryland cropping intensity:** Calculated as the ratio of total cropped area to net sown area, multiplied by 100. [KPSC AAO 2019]
98. **Recommended seed rate for sowing hybrid maize in Karnataka:** 15-20 kg/ha. [KPSC AO 2017]
99. **Karnataka, the critical nutrient deficiency that limits sunflower yield in Alfisols:** Boron. [KPSC AAO 2017]
100. **Optimal soil pH range for cultivating finger millet (ragi):** 5.5 to 7.5, showing moderate tolerance to soil acidity. [KPSC AO 2014]
101. **Nucleus seed has 100% genetic and physical purity and:** Produced under the direct supervision of the originating plant breeder. [KPSC AO 2014]
102. **Breeder seed:** The progeny of nucleus seed, has 100% genetic purity, and is identified by a golden yellow tag. [KPSC AAO 2015]
103. **Foundation seed:** The progeny of breeder seed, has 99.5% genetic purity, and is identified by a white tag. [KPSC AO 2017]
104. **Registered seed:** The progeny of foundation seed, is identified by a purple/opal green tag, and is rarely produced in India. [KPSC AAO 2017]
105. **Certified seed:** The progeny of foundation seed, is distributed to farmers, and is identified by an azure blue tag. [KPSC AO 2019]
106. **Truthful labeled seed (TLS) does not require compulsory certification but must meet minimum physical and genetic purity standards.** [KPSC AAO 2019]
107. **Minimum genetic purity standard for certified seeds of hybrid maize and pearl millet:** 95%. [KPSC AO 2017]
108. **Minimum physical purity standard required for certification of wheat, barley, and oats seeds:** 98%. [KPSC AAO 2017]
109. **Isolation distance required for foundation seed production of self-pollinated rice:** 3 meters. [KPSC AO 2014]
110. **Isolation distance required for foundation seed production of cross-pollinated maize hybrids:** 400 meters. [KPSC AAO 2015]
111. **Isolation distance for certified seed production of hybrid cotton:** 30 meters from other varieties of the same species. [KPSC AO 2017]
112. **Isolation distance required for foundation seed production of tomato:** 50 meters, and for certified seed it is 25 meters. [KPSC AAO 2017]
113. **Seed certification standards in India:** Formulated by the Central Seed Certification Board established under the Seed Act, 1966. [KPSC AO 2019]
114. **Seed Act:** Passed by the Indian Parliament in 1966 and came into force on 2nd October 1969. [KPSC AAO 2019]
115. **Seed Rules:** Framed under the Seed Act in the year 1968 to regulate seed quality control. [KPSC AO 2017]

116. **Tetrazolium (TZ) test:** A biochemical test used to determine seed viability, developed by Georg Lakon in 1942. [KPSC AAO 2017]
117. **The TZ test, active dehydrogenase enzymes in living tissues reduce 2,3,5-triphenyl tetrazolium chloride to red triphenyl formazan.** [KPSC AO 2014]
118. **Standard concentration of Tetrazolium salt solution used for testing seed viability:** 0.1% to 1.0%. [KPSC AAO 2015]
119. **Orthodox seeds can be dried to low moisture content (around 5%) and stored at sub-zero temperatures without losing viability.** [KPSC AO 2017]
120. **Recalcitrant seeds, such as mango, jackfruit, and rubber, lose viability if dried below critical moisture (12-30%) or stored at low temperatures.** [KPSC AAO 2017]
121. **Seed germination where cotyledons are pushed above the soil surface due to rapid elongation of the hypocotyl:** Called epigeal. [KPSC AO 2019]
122. **Hypogeal germination:** Is characterized by the epicotyl elongating while cotyledons remain below the soil surface, as seen in chickpea and peas. [KPSC AAO 2019]
123. **Seed dormancy caused by a hard, impermeable seed coat to water and oxygen:** Known as physical dormancy. [KPSC AO 2017]
124. **Stratification involves exposing moist seeds to low temperatures (2°C to 10°C) to break physiological seed dormancy.** [KPSC AAO 2017]
125. **Scarification:** Refers to any chemical or physical treatment that weakens or breaks a hard seed coat to allow water uptake. [KPSC AO 2014]
126. **Gibberellic Acid (GA3) at concentrations of 100 to 500 ppm:** Widely used to break physiological and embryo dormancy in seeds. [KPSC AAO 2015]
127. **Potassium nitrate (KNO₃) at 0.2% concentration:** Recommended by ISTA as a substrate treatment for breaking seed dormancy. [KPSC AO 2017]
128. **International Seed Testing Association (ISTA) was established in 1924 and:** Headquartered in Wallisellen, Switzerland. [KPSC AAO 2017]
129. **Size of a submitted sample for seed testing:** Generally four times the size of the working sample. [KPSC AO 2019]
130. **Seed vigour:** Is defined as the sum total of those properties of the seed which determine the level of activity and performance during germination. [KPSC AAO 2019]
131. **Paper piercing test:** An indirect test used to evaluate seed vigour under physical stress conditions. [KPSC AO 2017]
132. **Electrical conductivity (EC) test of seed leachates:** A rapid vigour test; high EC indicates poor membrane integrity and low vigour. [KPSC AAO 2017]
133. **Minimum germination percentage required for certification of certified wheat seeds in India:** 85%. [KPSC AO 2014]
134. **Minimum germination percentage required for certification of certified cotton seeds:** 65%. [KPSC AAO 2015]
135. **Minimum germination percentage required for certification of certified ragi (finger millet) seeds:** 75%. [KPSC AO 2017]
136. **Air-Screen Cleaner:** The basic machine used in all seed processing plants, separating seeds based on size and weight. [KPSC AAO 2017]
137. **Specific Gravity Separator separates seeds of the same size based on differences in their specific weight/density.** [KPSC AO 2019]
138. **Disc Separator or Indented Cylinder Separator separates seeds based on their length differences.** [KPSC AAO 2019]
139. **Safe storage of orthodox seeds, the sum of relative humidity (%) and temperature (°F) should not exceed 100 (Harrington's Rule).** [KPSC AO 2017]
140. **Seed priming:** A pre-sowing hydration treatment that initiates early metabolic processes without radicle protrusion. [KPSC AAO 2017]
141. **National Seeds Corporation (NSC) of India:** Established in the year 1963 to promote quality seed industries. [KPSC AO 2014]
142. **Karnataka State Seed Certification Agency (KSSCA):** Established in 1974 under the provisions of the Seed Act, 1966. [KPSC AAO 2015]
143. **'Grow Out Test' (GOT):** Conducted to determine the genetic purity of a seed lot by observing phenotypic characters in the field. [KPSC AO 2017]
144. **The Seed Act 1966, a Seed Inspector has the power to stop the sale of seeds and seize seed stocks if they violate standards.**

[KPSC AAO 2017]

145. **Vacuum storage or moisture-vapor proof packaging, seed moisture content must be reduced below 8.0% for most crops.** [KPSC AO 2019]

146. **Term 'Odum's index' in seed technology:** Sometimes associated with physical purity index estimation. [KPSC AAO 2019]

147. **Real Value of Seed (RVS):** Calculated as: $(\text{Purity \%} \times \text{Germination \%}) / 100$. [KPSC AO 2017]

148. **Most common chemical protectant used for dry seed treatment against storage insect pests:** Deltamethrin. [KPSC AAO 2017]

149. **Process of removing off-type plants from a seed production plot to maintain genetic purity:** Called rogueing. [KPSC AO 2014]

150. **Maximum permissible limit of inert matter in certified seeds of maize and sorghum:** 2.0%. [KPSC AAO 2015]

151. **Commercial method of propagation in Mango (*Mangifera indica*):** Epicotyl grafting or Veneer grafting. [KPSC AO 2014]

152. **Sapota (*Manilkara zapota*), commercial propagation:** Done by Air layering or Approach grafting on Rayan (*Manilkara hexandra*) rootstock. [KPSC AAO 2015]

153. **Guava (*Psidium guajava*), the most rapid and commercial method of vegetative propagation:** Stooling (Mound layering). [KPSC AO 2017]

154. **Grapes (*Vitis vinifera*):** Commercially propagated using hardwood stem cuttings treated with IBA to promote rooting. [KPSC AAO 2017]

155. **Citrus, Sweet Orange (*Citrus sinensis*):** Commercially propagated by T-budding or Shield budding on Rangpur Lime rootstock. [KPSC AO 2019]

156. **Rootstock 'Dogridge':** Widely used in grape cultivation in Karnataka to impart tolerance against salinity and drought. [KPSC AAO 2019]

157. **Pomegranate (*Punica granatum*):** Commercially propagated using hardwood or semi-hardwood stem cuttings. [KPSC AO 2017]

158. **Physiological disorder 'Black Tip' of Mango:** Caused by boron deficiency and exposure to sulfur dioxide gases from brick kilns. [KPSC AAO 2017]

159. **Physiological disorder 'Spongy Tissue' in Alphonso mango:** Is characterized by convective heat accumulation and high acid content. [KPSC AO 2014]

160. **'Malformation' in mango, causing vegetative and floral aberrations,:** Associated with Fusarium moniliforme and low temperatures. [KPSC AAO 2015]

161. **Plant growth regulator Paclobutrazol (PP333):** Applied to soil to induce regular and early flowering in Mango. [KPSC AO 2017]

162. **Gibberellic Acid (GA3):** Used in commercial grape cultivation to increase berry size and loosen the bunch structure. [KPSC AAO 2017]

163. **Ethephon (Ethrel):** A liquid growth regulator that releases ethylene to induce uniform ripening in banana and tomato. [KPSC AO 2019]

164. **System of training grapes known as the Bower (Pergola) system:** Highly popular in Karnataka for maximizing canopy exposure. [KPSC AAO 2019]

165. **Pruning in grapes under Karnataka conditions:** Carried out twice a year: Foundation pruning in April and Forward pruning in October. [KPSC AO 2017]

166. **Banana variety 'Robusta' belongs to the AAA genomic group and:** Highly cultivated for commercial purposes in Karnataka. [KPSC AAO 2017]

167. **Physiological disorder 'Kotte disease' or 'Katatte' in cardamom:** Caused by Cardamom mosaic virus transmitted by aphids. [KPSC AO 2014]

168. **Critical nutrient disorder in cauliflower known as 'Whiptail':** Caused by molybdenum deficiency in acid soils. [KPSC AAO 2015]

169. **Physiological disorder 'Blossom End Rot' in Tomato:** Caused by calcium deficiency coupled with fluctuating soil moisture. [KPSC AO 2017]

170. **Chemical used for chemical thinning of apple blossoms and fruits to regulate bearing:** Naphthalene Acetic Acid (NAA). [KPSC AAO 2017]

171. **Potato tubers:** Physiologically dormant after harvest, and this dormancy can be broken using 1% Thiourea solution. [KPSC AO 2019]

172. **Seed rate for raising a nursery of tomato for one hectare of main field:** 400 to 500 grams for normal varieties. [KPSC AAO 2019]

173. **Hybrid tomato cultivation, the recommended nursery seed rate:** Reduced to 100 to 150 grams per hectare. [KPSC AO 2017]

174. **Scientific name of pointed gourd, a dioecious cucurbitaceous vegetable propagated by vine cuttings,:** *Trichosanthes dioica*. [KPSC AAO 2017]
175. **Botanical term for the edible part of cauliflower:** 'curd', which is an immature pre-floral inflorescence. [KPSC AO 2014]
176. **Botanical term for the edible part of broccoli:** 'head', which consists of compact green undeveloped flower buds. [KPSC AAO 2015]
177. **Vegetable crop which:** A rich source of Vitamin A (Beta-carotene) among all common vegetables is carrot (**Daucus carota**). [KPSC AO 2017]
178. **Bitterness in cucurbitaceous vegetables:** Due to the presence of tetracyclic triterpinoid compound known as Cucurbitacin. [KPSC AAO 2017]
179. **Pungency in onion bulb:** Caused by the volatile organic sulfur compound Allyl propyl disulfide. [KPSC AO 2019]
180. **Red color in tomato fruit:** Due to the pigment Lycopene, whose synthesis is inhibited at temperatures above 30°C. [KPSC AAO 2019]
181. **Yellow color in onion flesh:** Due to the pigment Quercetin, which acts as a strong antioxidant. [KPSC AO 2017]
182. **Major post-harvest disease of onion during storage:** Black Mold caused by the fungus *Aspergillus niger*. [KPSC AAO 2017]
183. **Seed rate of okra (Bhindi) during the Kharif season:** 8-10 kg/ha, whereas for Summer crop it is 18-20 kg/ha. [KPSC AO 2014]
184. **Tapioca (Cassava):** Contains a cyanogenic glucosoid compound called Linamarin, which releases toxic hydrocyanic acid (HCN). [KPSC AAO 2015]
185. **Sweet potato (*Ipomoea batatas*):** Commercially propagated using terminal vine cuttings with at least 3-4 nodes. [KPSC AO 2017]
186. **Commercial method of propagation in Rose:** T-budding on *Rosa multiflora* or *Rosa indica* var. *odorata* rootstock. [KPSC AAO 2017]
187. **Chrysanthemum, commercial propagation:** Done using terminal stem cuttings and suckers. [KPSC AO 2019]
188. **Process of removing side buds to promote a single large terminal flower in carnation and chrysanthemum:** Called disbudding. [KPSC AAO 2019]
189. **Anthurium:** Commercially propagated by tissue culture or offsets, and it requires high humidity and 70-80% shade. [KPSC AO 2017]
190. **Commercial method of propagation in Jasmine (*Jasminum sambac*):** Ground layering or semi-hardwood cuttings. [KPSC AAO 2017]
191. **National flower of India:** Lotus (**Nelumbo nucifera**), which is an aquatic perennial plant. [KPSC AO 2014]
192. **Maturity index of commercial banana:** Determined by the angularity of the fruit; round fingers indicate full maturity. [KPSC AAO 2015]
193. **Chemical compound 1-Methylcyclopropene (1-MCP):** A gas that acts as a potent inhibitor of ethylene action in post-harvest storage. [KPSC AO 2017]
194. **Modified Atmosphere Packaging (MAP) extends shelf life by increasing CO₂ levels and decreasing O₂ levels inside the package.** [KPSC AAO 2017]
195. **Chemical used for extending the vase life of cut flowers by inhibiting microbial growth:** Silver Thiosulfate (STS). [KPSC AO 2019]
196. **Post-harvest treatment 'degreening' in citrus:** Carried out by exposing fruits to 1-5 ppm ethylene gas at 20-25°C. [KPSC AAO 2019]
197. **Major spice crop Cardamom (*Elettaria cardamomum*):** Known as the 'Queen of Spices'. [KPSC AO 2017]
198. **Black Pepper (*Piper nigrum*):** Propagated commercially using runner shoots and is called the 'King of Spices'. [KPSC AAO 2017]
199. **Commercial part of Nutmeg (*Myristica fragrans*) yields two distinct spices: the seed (nutmeg) and the dried aril (mace).** [KPSC AO 2014]
200. **Chemical constituent that imparts the characteristic aroma to vanilla pods:** Vanillin, developed during curing. [KPSC AAO 2015]
201. **Term 'soil':** Derived from the Latin word 'solum' which means floor or ground (KPSC AO 2014).
202. **Pedology:** The study of soil origin, classification, and description, treating soil as a natural body (KPSC AAO 2015).
203. **Edaphology:** The study of soil from the standpoint of higher plants, focusing on plant growth and yield (KPSC AO 2017).

204. **Physical disintegration and chemical decomposition of rocks and minerals:** Called weathering (KPSC AAO 2017).
205. **Quartz:** A highly resistant primary mineral with a chemical composition of SiO_2 , dominant in sandy soils (KPSC AO 2014).
206. **Feldspars:** The most abundant group of minerals in the Earth's crust, serving as a primary source of soil clay (KPSC AAO 2017).
207. **Olivine:** A ferro-magnesian silicate mineral that weathers very easily under humid tropical conditions (KPSC AO 2017).
208. **Apatite:** The primary source of phosphorus in soils, occurring as calcium phosphate with fluorine or chlorine (KPSC AAO 2015).
209. **Tourmaline:** The primary source of boron in soils, containing complex borosilicates (KPSC AO 2017).
210. **Relative proportion of sand, silt, and clay particles in a soil mass:** Is defined as soil texture (KPSC AAO 2017).
211. **Stokes' Law, which governs the mechanical analysis of soil texture, states that the settling velocity of spherical particles:** Directly proportional to the square of their radius (KPSC AO 2017).
212. **According to the ISSS classification, the diameter of clay particles:** Less than 0.002 mm (KPSC AAO 2015).
213. **Silt particles, in the ISSS system, have a diameter range of 0.02 to 0.002 mm (KPSC AO 2017).**
214. **Fine sand particles under the ISSS system have a diameter range of 0.2 to 0.02 mm (KPSC AAO 2017).**
215. **Coarse sand particles under the ISSS system have a diameter range of 2.0 to 0.2 mm (KPSC AO 2014).**
216. **Arrangement of primary soil particles and their aggregates into certain defined patterns:** Called soil structure (KPSC AAO 2017).
217. **Granular and crumb soil structures:** The most favorable for crop growth due to their optimum balance of macropores and micropores (KPSC AO 2017).
218. **Platy soil structure:** Is characterized by horizontal dimensions being much larger than vertical dimensions, restricting root penetration and water movement (KPSC AAO 2015).
219. **Prismatic and columnar structures:** Typically found in the B horizon of sodic soils, with columnar having rounded tops (KPSC AO 2017).
220. **Mass of dry soil per unit volume of bulk soil (including pore space):** Called bulk density (KPSC AAO 2017).
221. **Bulk density of sandy soils:** Higher (1.4 to 1.8 g/cm^3) than that of clayey soils (1.0 to 1.3 g/cm^3) due to closer packing (KPSC AO 2017).
222. **Particle density:** The mass of dry soil per unit volume of solid soil particles, and its average value for mineral soils is 2.65 g/cm^3 (KPSC AAO 2017).
223. **Soil porosity:** Calculated using the formula: $\text{Porosity (\%)} = [1 - (\text{Bulk Density} / \text{Particle Density})] \times 100$ (KPSC AO 2017).
224. **Soil color:** Measured using the Munsell Color Chart, which describes color in terms of Hue, Value, and Chroma (KPSC AAO 2017).
225. **The Munsell color notation 10YR 4/6, 10YR:** Represents Hue, 4 represents Value (lightness), and 6 represents Chroma (purity) (KPSC AO 2017).
226. **Cohesion in soil:** Refers to the attraction between like molecules (soil-soil), whereas adhesion refers to the attraction between unlike molecules (soil-water) (KPSC AAO 2017).
227. **Field capacity of soil:** Represents the moisture content after gravitational water has drained away, occurring at a suction of -0.33 bars for clay/loam soils (KPSC AAO 2017).
228. **Wilting point of soil occurs at a suction pressure of approximately -15 bars or pF 4.2 (KPSC AO 2017).**
229. **Hygroscopic water:** Held tightly by soil particles at a suction pressure of -31 to -10000 bars (pF 4.5 to 7.0) (KPSC AO 2017).
230. **Available water capacity of soil:** The water held between field capacity (-0.33 bar) and permanent wilting point (-15 bar) (KPSC AAO 2015).
231. **Tensiometer:** An instrument used to measure soil moisture tension in the field, effective up to 0.85 bars of tension (KPSC AO 2017).
232. **Gypsum block or electrical resistance block:** Used to measure soil moisture in dry ranges (1 to 15 bars) (KPSC AAO 2017).
233. **Neutron moisture probe:** A non-destructive method based on the thermalization of fast neutrons by hydrogen nuclei in soil water (KPSC AO 2017).
234. **Time Domain Reflectometry (TDR) method measures soil moisture based on the dielectric constant of the soil (KPSC AAO 2017).**

235. **Darcy's Law describes the rate of water flow through saturated soil under a hydraulic gradient (KPSC AO 2017).**
236. **Infiltration:** The downward entry of water into the soil surface, while percolation is the downward movement of water through the soil profile (KPSC AO 2017).
237. **USDA soil texture triangle classifies soil into 12 major textural classes based on sand, silt, and clay percentages (KPSC AAO 2017).**
238. **Sodic soils have an Exchangeable Sodium Percentage (ESP) > 15, SAR > 13, and EC < 4 dS/m (KPSC AO 2017).**
239. **Saline-sodic soils have an ESP > 15, SAR > 13, and EC > 4 dS/m (KPSC AAO 2017).**
240. **Amendment used for reclaiming acid soils:** Lime (CaCO_3 or $\text{CaMg}(\text{CO}_3)_2$) (KPSC AO 2017).
241. **Calcareous soils:** Contain free calcium carbonate (CaCO_3) and have a pH range of 7.5 to 8.5, often causing P and micronutrient deficiencies (KPSC AAO 2017).
242. **Principal zinc fertilizer:** Zinc Sulphate Heptahydrate ($\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$), which contains 21% zinc (KPSC AO 2017).
243. **Chelated Zinc (Zn-EDTA) contains 12% zinc and:** Highly effective for foliar spray and soil application (KPSC AAO 2017).
244. **Borax ($\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$) contains 10.5% Boron and:** The most common boron fertilizer (KPSC AO 2017).
245. **Ammonium molybdate contains 52% Molybdenum and:** Used to correct molybdenum deficiency in cauliflowers (causing whip tail) (KPSC AAO 2017).
246. **Soil order Histosols comprises organic soils containing more than 20% organic matter by weight (KPSC AO 2017).**
247. **Entisols:** Embryonic soils showing little or no profile development, commonly found in floodplains and steep slopes (KPSC AAO 2017).
248. **Inceptisols:** Weakly developed young soils showing alteration of parent material but no accumulation of clay (KPSC AO 2017).
249. **Spodosols:** Soils with a subsurface accumulation of organic matter and sesquioxides (spodic horizon), typical of cool, coniferous forests (KPSC AAO 2017).
250. **Concept of Soil Taxonomy:** Developed by USDA Soil Survey Staff and officially adopted in the year 1975 (KPSC AO 2014).
251. **Montmorillonite:** A 2:1 expanding type silicate clay mineral with a high cation exchange capacity of 80 to 150 $\text{cmol}(\text{p+})/\text{kg}$ (KPSC AAO 2017).
252. **Reclamation of sodic soils requires application of gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) to replace exchangeable sodium with calcium ions (KPSC AO 2017).**
253. **Critical limit of electrical conductivity (EC) for saline soils:** > 4 dS/m at 25°C (KPSC AAO 2015).
254. **Black soils of Karnataka (Vertisols):** Characterized by the presence of high swelling clays, which exhibit self-ploughing characteristics (KPSC AO 2017).
255. **Red soils (Alfisols/Inceptisols):** The most dominant soil group in Karnataka, particularly in the southern dry zone (KPSC AAO 2017).
256. **Kaolinite:** A 1:1 non-expanding type silicate clay mineral with low CEC ranging from 3 to 15 $\text{cmol}(\text{p+})/\text{kg}$ (KPSC AO 2014).
257. **Schoonover's method:** Widely used for the determination of gypsum requirement of sodic soils using calcium chloride solution (KPSC AO 2017).
258. **Sodium Adsorption Ratio (SAR):** Calculated as $\text{SAR} = [\text{Na}^+] / \sqrt{([\text{Ca}^{2+}] + [\text{Mg}^{2+}])/2}$, where concentration is in meq/L (KPSC AAO 2015).
259. **Active acidity in soil:** Represented by hydrogen (H^+) and aluminum (Al^{3+}) ions present in the soil solution (KPSC AAO 2017).
260. **Potential or reserve acidity:** Due to hydrogen and aluminum ions adsorbed on the soil colloid exchange complex (KPSC AO 2017).
261. **C:N ratio of normal agricultural soil ranges between 10:1 to 12:1 (KPSC AAO 2015).**
262. **Nitrogen:** Absorbed by plants primarily as nitrate (NO_3^-) and ammonium (NH_4^+) ions (KPSC AO 2014).
263. **Phosphorus:** Absorbed by plants as orthophosphate ions, H_2PO_4^- at lower pH (< 7.2) and HPO_4^{2-} at higher pH (> 7.2) (KPSC AAO 2017).
264. **Potassium deficiency in plants:** Is characterized by marginal chlorosis and scorching of older leaves (KPSC AO 2017).
265. **Zinc deficiency in maize causes 'white bud' disease, which:** Is characterized by broad bands of chlorotic tissue (KPSC AAO 2017).
266. **Iron deficiency:** Causes interveinal chlorosis in young leaves, particularly in calcareous soils (KPSC AO 2014).

267. **Boron:** Unique among essential micronutrients because it is absorbed by plants as uncharged boric acid (H_3BO_3) (KPSC AAO 2017).
268. **Molybdenum:** Required for the functioning of nitrogenase and nitrate reductase enzymes in plants (KPSC AO 2017).
269. **Major silicate mineral responsible for potassium release in soils:** Orthoclase feldspar and muscovite mica (KPSC AAO 2015).
270. **Illite:** A 2:1 non-expanding silicate clay mineral with potassium ions (K^+) locked in the interlayer space (KPSC AO 2014).
271. **Chlorite:** A 2:1:1 or 2:2 silicate clay mineral with an alternating brucite-like layer (KPSC AAO 2017).
272. **Dominant soil order in the coastal zone of Karnataka:** Ultisols and Alfisols (Laterites) (KPSC AO 2017).
273. **Laterization:** The soil-forming process that involves the leaching of silica and accumulation of iron and aluminum oxides under warm, humid conditions (KPSC AAO 2017).
274. **Podzolization involves the translocation of organic matter and sesquioxides from the A horizon to the B horizon (KPSC AO 2014).**
275. **Gleization:** A soil-forming process associated with waterlogged and anaerobic conditions, leading to the reduction of iron (KPSC AAO 2017).
276. **Calcification:** The accumulation of calcium carbonate ($CaCO_3$) in the soil profile, typical of arid and semi-arid regions (KPSC AO 2017).
277. **Salinization:** The accumulation of soluble salts in the soil profile, leading to electrical conductivity > 4 dS/m (KPSC AAO 2015).
278. **Solodization or dealkalization:** The process of removal of sodium ions from the exchange complex of sodic soils (KPSC AO 2017).
279. **Organic colloid humus has a very high CEC ranging from 150 to 300 cmol(p+)/kg (KPSC AAO 2017).**
280. **Equivalent acidity of ammonium sulfate:** 110, which is the highest among common nitrogenous fertilizers (KPSC AO 2017).
281. **Equivalent basicity of calcium cyanamide:** 63, which means it has a liming effect on acidic soils (KPSC AAO 2015).
282. **Biuret content in urea should not exceed 1.5% for soil application and 0.5% for foliar spray according to FCO guidelines (KPSC AO 2017).**
283. **Slow-release nitrogenous fertilizers include Isobutylidene diurea (IBDU) and Sulphur Coated Urea (SCU) (KPSC AAO 2017).**
284. **Nitrification inhibitors include DCD (Dicyandiamide) and AM (2-amino-4-chloro-6-methylpyrimidine) (KPSC AO 2017).**
285. **Urease inhibitors such as NBPT (N-(n-butyl) thiophosphoric triamide):** Used to minimize ammonia volatilization from urea (KPSC AAO 2017).
286. **Diagnostic subsurface horizon characterized by accumulation of clay:** Called Argillic horizon (KPSC AO 2014).
287. **Diagnostic surface horizon rich in organic matter and dark in color:** Called Mollic epipedon (KPSC AAO 2017).
288. **Ochric epipedon:** A surface horizon that is too light in color, low in organic matter, or too thin to be mollic (KPSC AO 2017).
289. **Organic matter content of Indian soils:** Generally low, typically less than 0.5% (KPSC AAO 2015).
290. **Humic acid:** That fraction of soil organic matter which is soluble in alkali but insoluble in acid (KPSC AO 2017).
291. **Fulvic acid:** The fraction of soil organic matter that is soluble in both alkali and acid (KPSC AAO 2017).
292. **Humins:** The fraction of soil organic matter that is insoluble in both alkali and acid (KPSC AO 2017).
293. **Anion exchange capacity (AEC) of soils decreases with an increase in soil pH (KPSC AO 2017).**
294. **CEC of soil increases with an increase in soil pH due to pH-dependent charges on organic matter and clay edges (KPSC AAO 2017).**
295. **Critical limit of DTPA-extractable Zinc in soils:** 0.6 mg/kg (or ppm) (KPSC AO 2017).
296. **Critical limit of DTPA-extractable Iron in soils:** 4.5 mg/kg (or ppm) (KPSC AAO 2017).
297. **Critical limit of hot-water extractable Boron in soils:** 0.5 mg/kg (or ppm) (KPSC AO 2017).
298. **Alkaline permanganate method:** Used to determine available Nitrogen in soils (KPSC AAO 2015).
299. **Olsen's method:** Used to determine available Phosphorus in alkaline and neutral soils using 0.5 M $NaHCO_3$ at pH 8.5 (KPSC AO 2017).
300. **Bray's No. 1 method:** Used to determine available Phosphorus in acid soils using NH_4F and HCl (KPSC AAO 2017).

301. **Term 'microbiology':** Coined by French chemist Louis Pasteur, who also disproved the theory of spontaneous generation (KPSC AO 2017).
302. **Antonie van Leeuwenhoek:** Known as the Father of Microbiology for inventing the simple microscope and observing 'animalcules' (KPSC AAO 2017).
303. **Robert Koch formulated the Germ Theory of Disease and Koch's Postulates, establishing causal relationships between microbes and diseases (KPSC AO 2014).**
304. **Sergei Winogradsky:** Considered the Father of Soil Microbiology, having discovered chemoautotrophy and nitrification processes (KPSC AAO 2017).
305. **Martinus Beijerinck isolated Azotobacter, a free-living aerobic nitrogen-fixing bacterium, and developed enrichment culture techniques (KPSC AO 2017).**
306. **Peptidoglycan (murein):** The major structural component of bacterial cell walls, present in higher amounts in Gram-positive bacteria (KPSC AAO 2017).
307. **Gram-negative bacterial cell walls:** Contain an outer lipopolysaccharide (LPS) membrane, which acts as an endotoxin (KPSC AO 2017).
308. **Endospores:** Highly resistant structures produced by Bacillus and Clostridium species to survive adverse environmental conditions (KPSC AAO 2015).
309. **Plasmids:** Extrachromosomal, circular double-stranded DNA molecules capable of autonomous replication in bacterial cells (KPSC AO 2017).
310. **Bacterial growth curve consists of four distinct phases: Lag phase, Log (exponential) phase, Stationary phase, and Death (decline) phase (KPSC AAO 2017).**
311. **Psychrophiles:** Microorganisms that grow best at cold temperatures (below 15°C), while thermophiles thrive at high temperatures (50-80°C) (KPSC AO 2017).
312. **Mesophiles:** Microorganisms with an optimum growth temperature range of 20°C to 45°C, representing most soil bacteria (KPSC AAO 2017).
313. **Enzyme nitrogenase, responsible for biological nitrogen fixation,:** Highly sensitive to oxygen and is protected by leghemoglobin in legume nodules (KPSC AO 2017).
314. **Leghemoglobin:** A pink iron-containing pigment that regulates oxygen supply in root nodules of leguminous plants (KPSC AAO 2017).
315. **Rhizobium species form symbiotic associations with legumes, showing host specificity (e.g., Rhizobium meliloti with alfalfa, R. trifolii with clover) (KPSC AO 2017).**
316. **Rhizobium leguminosarum bv. viciae:** The symbiont for pea (Pisum sativum) and lentil (Lens culinaris) (KPSC AAO 2015).
317. **Bradyrhizobium japonicum:** A slow-growing symbiont specific to soybean (Glycine max) (KPSC AO 2017).
318. **Azotobacter chroococcum:** A free-living, aerobic, heterotrophic nitrogen-fixing bacterium widely used as a biofertilizer for non-legumes (KPSC AAO 2017).
319. **Azospirillum brasilense:** An associative symbiotic nitrogen-fixing bacterium that colonizes the rhizosphere of cereal crops like sorghum and pearl millet (KPSC AO 2017).
320. **Beijerinckia:** A free-living acidophilic nitrogen-fixing bacterium commonly found in acidic tropical soils (KPSC AAO 2017).
321. **Blue-green algae (Cyanobacteria) such as Anabaena azollae form a symbiotic association with the water fern Azolla, used in low-land rice fields (KPSC AO 2017).**
322. **Frankia:** An actinomycete that forms symbiotic nitrogen-fixing nodules on the roots of non-leguminous actinorhizal plants like Casuarina (KPSC AAO 2017).
323. **Phosphate solubilizing bacteria (PSB) like Bacillus megaterium var. phosphaticum solubilize insoluble inorganic phosphorus by producing organic acids (KPSC AO 2017).**
324. **Pseudomonas putida and Bacillus polymyxa:** Well-known phosphate solubilizing microorganisms (PSMs) (KPSC AAO 2015).
325. **Arbuscular Mycorrhizal (AM) fungi belong to the phylum Glomeromycota and enhance plant uptake of phosphorus, zinc, and water (KPSC AO 2017).**
326. **AM fungi produce glomalin, a glycoprotein that plays a crucial role in soil aggregate stabilization (KPSC AAO 2017).**
327. **Major genera of AM fungi include Glomus, Gigaspora, Acaulospora, and Scutellospora (KPSC AO 2014).**

328. **Mycorrhizae:** Of two main types: Ectomycorrhizae (which form a Hartig net around root cells) and Endomycorrhizae (which penetrate root cells) (KPSC AAO 2017).
329. **Conversion of organic nitrogen to inorganic ammonium (NH₄⁺) by soil microorganisms:** Called ammonification (KPSC AAO 2017).
330. **Nitrification:** A two-step aerobic process: Nitrosomonas oxidizes NH₄⁺ to NO₂⁻, and Nitrobacter oxidizes NO₂⁻ to NO₃⁻ (KPSC AAO 2017).
331. **Denitrification:** An anaerobic process where nitrate (NO₃⁻) is reduced to gaseous nitrogen (N₂O, N₂) by bacteria like Pseudomonas denitrificans (KPSC AAO 2017).
332. **Conversion of inorganic nitrogen into organic forms by soil microorganisms:** Called nitrogen immobilization (KPSC AAO 2015).
333. **Sulphur-oxidizing bacteria such as Thiobacillus thiooxidans oxidize elemental sulphur to sulphate (SO₄²⁻), reducing soil pH in alkaline soils (KPSC AAO 2017).**
334. **Desulfovibrio:** An anaerobic sulphate-reducing bacterium that produces toxic hydrogen sulphide (H₂S) in waterlogged soils (KPSC AAO 2017).
335. **Cellulose decomposition in soil:** Primarily carried out by cellulolytic fungi (Trichoderma, Chaetomium) and bacteria (Cellulomonas) (KPSC AAO 2017).
336. **Lignin:** Highly resistant to microbial degradation and is decomposed mainly by white-rot fungi (Basidiomycetes) using lignin peroxidases (KPSC AAO 2017).
337. **C:N ratio of organic residues determines their rate of decomposition; residues with C:N > 30:1 lead to temporary nitrogen immobilization (KPSC AAO 2017).**
338. **C:N ratio of residues below 20:1 results in net nitrogen mineralization, releasing plant-available ammonium and nitrate (KPSC AAO 2017).**
339. **Thermophilic composting:** Is characterized by a high-temperature phase (50-60°C) dominated by thermophilic actinomycetes and endospore-forming bacteria (KPSC AAO 2017).
340. **Silage production involves anaerobic fermentation of green forage by lactic acid bacteria (e.g., Lactobacillus plantarum) to lower pH to 3.8-4.2 (KPSC AAO 2017).**
341. **Quality of silage:** Determined by Flieg's score, which is based on the proportions of lactic, acetic, and butyric acids (KPSC AAO 2017).
342. **Methanogenesis:** The anaerobic production of methane (CH₄) by archaeobacteria (Methanobacterium) in waterlogged paddy fields and biogas digesters (KPSC AAO 2017).
343. **Major carrier material used for commercial biofertilizer production in India:** Lignite or peat (KPSC AAO 2017).
344. **Minimum viable count of Rhizobium in carrier-based biofertilizers should be 10⁷ cells/g of carrier at the time of expiry (KPSC AAO 2017).**
345. **Siderophores:** Iron-chelating compounds produced by rhizosphere bacteria like Pseudomonas fluorescens to suppress soil-borne pathogens (KPSC AAO 2017).
346. **Vesicular-Arbuscular Mycorrhiza (VAM):** Now commonly referred to as Arbuscular Mycorrhiza (AM) due to the absence of vesicles in some genera (KPSC AAO 2015).
347. **Conversion of atmospheric CO₂ into organic carbon by autotrophic microorganisms:** Called carbon fixation (KPSC AAO 2017).
348. **Nod factors:** Lipo-chitooligosaccharide signaling molecules secreted by Rhizobium in response to flavonoids released by host plant roots (KPSC AAO 2017).
349. **Gene encoding the nitrogenase enzyme complex in diazotrophic organisms:** Called the nif gene (KPSC AAO 2017).
350. **Cell wall of fungi:** Composed of chitin, a polymer of N-acetylglucosamine, unlike the peptidoglycan wall of bacteria (KPSC AAO 2017).
351. **Representative fraction (RF):** The ratio of map distance to ground distance, expressed as a unitless fraction (KPSC AAO 2017).
352. **Scale of 1:2000:** Larger than a scale of 1:10000, meaning it shows more detail over a smaller area (KPSC AAO 2017).
353. **Chain surveying, the standard length of a Gunter's chain:** 66 feet, containing 100 links (each link being 0.66 ft) (KPSC AAO 2014).
354. **Revenue chain:** 33 feet long with 16 links, commonly used for cadastral surveys (KPSC AAO 2015).
355. **Engineer's chain:** 100 feet long with 100 links, each link being 1 foot (KPSC AAO 2017).
356. **Metric chains used in agricultural surveying:** Typically 20 meters (100 links) or 30 meters (150 links) in length (KPSC AAO 2017).

357. **Well-conditioned triangles in chain surveying should not have any angle less than 30° or greater than 120° (KPSC AO 2017).**
358. **Dumpy level:** A widely used leveling instrument where the telescope is rigidly fixed to its supports (KPSC AAO 2017).
359. **Leveling, a Back Sight (BS):** The first staff reading taken after setting up the instrument on a point of known elevation (Bench Mark) (KPSC AO 2017).
360. **Fore Sight (FS):** The last staff reading taken before moving the instrument to a new setup point (KPSC AAO 2017).
361. **Intermediate Sight (IS):** Any staff reading taken between the Back Sight and Fore Sight from a single instrument setup (KPSC AO 2017).
362. **Height of Instrument (HI):** Calculated as $HI = \text{Elevation of BM} + BS$ (KPSC AAO 2017).
363. **Reduced level (RL) of any point:** Calculated as $RL = HI - \text{Staff Reading (IS or FS)}$ (KPSC AO 2017).
364. **Universal Soil Loss Equation (USLE):** Given by $A = R * K * LS * C * P$, where A is the estimated annual soil loss per unit area (KPSC AAO 2015).
365. **The USLE, R:** Represents the rainfall erosivity factor and K represents the soil erodibility factor (KPSC AO 2017).
366. **Splash erosion:** The first stage of water erosion, caused by the kinetic energy of falling raindrops detaching soil particles (KPSC AAO 2017).
367. **Sheet erosion:** The uniform removal of a thin layer of soil from the land surface, often unnoticed but highly damaging (KPSC AO 2017).
368. **Rill erosion:** Is characterized by small, well-defined channels that can be easily obliterated by normal tillage operations (KPSC AAO 2017).
369. **Gully erosion:** An advanced stage of rill erosion where channels are too deep to be crossed or removed by ordinary farm implements (KPSC AO 2017).
370. **Contour bunding:** Recommended for areas with rainfall less than 600 mm and land slopes up to 6% to conserve moisture (KPSC AAO 2017).
371. **Graded bunding:** Recommended in areas with rainfall greater than 600 mm to safely dispose of excess runoff water (KPSC AO 2017).
372. **Broad Bed Furrow (BBF) system:** A moisture conservation measure developed by ICRISAT, especially suitable for deep black soils (Vertisols) (KPSC AAO 2017).
373. **Wind erosion occurs primarily in arid and semi-arid regions and involves three stages: saltation (50-75%), suspension (3-15%), and surface creep (5-25%) (KPSC AO 2017).**
374. **Saltation:** The movement of soil particles (0.1 to 0.5 mm diameter) in a series of short bounces or jumps above the soil surface (KPSC AAO 2017).
375. **Four-stroke engine completes one power stroke (cycle) in two revolutions of the crankshaft (720° rotation) (KPSC AO 2017).**
376. **Two-stroke engine completes one power stroke in one revolution of the crankshaft (360° rotation) (KPSC AAO 2017).**
377. **Compression ratio of a diesel engine ranges from 14:1 to 20:1, whereas for petrol engines it ranges from 6:1 to 10:1 (KPSC AO 2017).**
378. **Diesel engines operate on the principle of compression ignition (CI), where fuel:** Injected into highly compressed hot air (KPSC AAO 2017).
379. **Petrol engines operate on the principle of spark ignition (SI), where a carburetor prepares the air-fuel mixture ignited by a spark plug (KPSC AO 2017).**
380. **Indicated Horse Power (IHP):** The power developed inside the engine cylinder, while Brake Horse Power (BHP) is the useful power available at the crankshaft (KPSC AAO 2017).
381. **Mechanical efficiency of an engine:** The ratio of Brake Horse Power to Indicated Horse Power ($BHP / IHP * 100$) (KPSC AO 2017).
382. **Primary tillage implement used for deep tillage and turning soil:** The mould board (MB) plough (KPSC AAO 2015).
383. **Disc plough:** Preferred for stony, root-infested, or extremely dry and hard soils where MB plough cannot operate (KPSC AO 2017).
384. **Draft of an implement:** The horizontal component of pull parallel to the line of motion (KPSC AAO 2017).
385. **Unit draft:** The draft per unit cross-sectional area of the furrow slice, expressed in kg/cm² (KPSC AO 2017).
386. **Disc angle of a disc plough:** The angle between the plane of the cutting edge of the disc and the direction of travel, typically 42° to

45° (KPSC AAO 2017).

387. **Tilt angle of a disc plough:** The angle of inclination of the disc plane from the vertical, typically ranging from 15° to 25° (KPSC AAO 2017).

388. **Seed drill:** Calibrated to determine the seed rate per hectare before actual field operation (KPSC AAO 2015).

389. **Fluted feed roller:** The most common seed metering mechanism used in seed-cum-fertilizer drills (KPSC AO 2017).

390. **Mechanical planter uses an inclined plate or vertical plate metering mechanism to drop single seeds at precise intervals (KPSC AAO 2017).**

391. **Knapsack sprayer:** A hand-operated sprayer carried on the back of the operator, developed for small holdings, operating at 3-5 kg/cm² pressure (KPSC AO 2017).

392. **Compression sprayer requires pumping air into the tank before spraying, whereas a knapsack sprayer requires continuous pumping during operation (KPSC AAO 2017).**

393. **Ultra-Low Volume (ULV) sprayers apply less than 5 liters of spray liquid per hectare, producing extremely fine droplets (KPSC AO 2017).**

394. **Water application efficiency of drip irrigation:** The highest, ranging from 90% to 95% due to reduced evaporation and deep percolation losses (KPSC AAO 2017).

395. **Sprinkler irrigation:** Highly suitable for undulating topography and sandy soils where surface irrigation is inefficient (KPSC AO 2017).

396. **Parshall flume:** A water measuring device used to measure flow rate in open channels under free-flow or submerged conditions (KPSC AAO 2017).

397. **Cippoletti weir:** A trapezoidal weir with side slopes of 1 horizontal to 4 vertical (about 14°), used to measure water flow (KPSC AO 2017).

398. **Sub-surface drainage systems:** Installed to lower the water table in waterlogged soils, using perforated clay, concrete, or PVC pipes (KPSC AAO 2017).

399. **Deep bin drying:** Refers to drying grain in storage structures where the depth of grain is greater than its diameter, requiring high airflow (KPSC AO 2017).

400. **Optimum moisture content for safe storage of cereal grains:** 12% to 14%, while for oilseeds it is 8% to 10% (KPSC AAO 2017).

401. **Term 'Genetics':** Coined by William Bateson in 1906 to describe the study of heredity and variation (KPSC AAO 2017).

402. **Wilhelm Johannsen coined the fundamental terms 'Gene', 'Genotype', and 'Phenotype' in 1909 (KPSC AO 2014).**

403. **Nucleosome core particle consists of 146 base pairs of DNA wrapped 1.65 times around a histone octamer composed of two molecules each of H2A, H2B, H3, and H4 (KPSC AO 2017).**

404. **Linker histone H1 binds to the linker DNA and the entry/exit site on the nucleosome core particle to stabilize the chromatin structure (KPSC AAO 2015).**

405. **Crossing over occurs during the pachytene stage of Prophase I in meiosis, where non-sister chromatids of homologous chromosomes exchange genetic segments (KPSC AO 2014).**

406. **Homologous chromosome synapsis:** Mediated by the formation of the synaptonemal complex during the zygotene stage of Prophase I (KPSC AO 2017).

407. **Chiasmata terminalization, where the crossover sites slide toward the ends of chromosomes, reaches completion during diakinesis of Prophase I (KPSC AAO 2017).**

408. **Complementary gene interaction (9:7 ratio in F₂) requires the presence of at least one dominant allele from both loci to express the wild-type phenotype (KPSC AO 2014).**

409. **Duplicate dominant epistasis (15:1 ratio in F₂) occurs when a single dominant allele at either of the two loci produces the same phenotype (KPSC AO 2017).**

410. **Recessive epistasis (9:3:4 ratio in F₂) occurs when the recessive homozygote at one locus masks the phenotypic expression of alleles at a second locus, as seen in onion bulb color (KPSC AAO 2015).**

411. **Dominant epistasis (12:3:1 ratio in F₂) occurs when a dominant allele at one locus masks the expression of both dominant and recessive alleles at another locus (KPSC AO 2014).**

412. **Polymeric gene interaction, or duplicate genes with cumulative effect, produces a characteristic 9:6:1 F₂ phenotypic ratio (KPSC AAO 2017).**

413. **Inhibitory gene action produces a 13:3 F₂ phenotypic ratio, where a dominant allele of one gene suppresses the expression of a**

dominant allele of another gene (KPSC AO 2017).

414. Wilhelm Johannsen proposed the Pure-line Theory in 1903 using the Princess variety of common bean, *Phaseolus vulgaris* (KPSC AO 2014).

415. Pure line: The selfed progeny of a single homozygous self-pollinated plant, wherein all phenotypic variation is purely environmental (KPSC AAO 2015).

416. G.H. Shull proposed the single-cross hybrid method in maize in 1909, utilizing cross-pollination between two inbred lines (KPSC AAO 2015).

417. D.F. Jones proposed the double-cross hybrid method in maize in 1918 ($A \times B$) $\times$ ($C \times D$) to overcome the low seed yields of inbred female lines (KPSC AO 2017).

418. Term 'Heterosis': Coined by G.H. Shull in 1914 to describe the phenotypic superiority of F1 hybrids over their parents (KPSC AO 2017).

419. Dominance hypothesis of heterosis (Davenport, Bruce) attributes hybrid vigor to the masking of deleterious recessive alleles by dominant alleles (KPSC AO 2014).

420. Overdominance hypothesis of heterosis (Shull, East) suggests that the heterozygote (Aa): Superior to either homozygote (AA or aa) at a given locus (KPSC AAO 2015).

421. Triticale: The first man-made cereal, developed by crossing wheat (*Triticum*) and rye (*Secale*) followed by chromosome doubling (KPSC AO 2014).

422. U's Triangle (Woo Jang-choon, 1935) shows that *Brassica napus* ($n=19$): An allotetraploid derived from hybridization between *B. oleracea* ($n=9$) and *B. rapa* ($n=10$) (KPSC AAO 2017).

423. *Brassica juncea* ($n=18$): An amphidiploid crop species derived from the natural hybridization of *Brassica nigra* ($n=8$) and *Brassica rapa* ($n=10$) (KPSC AO 2017).

424. *Brassica carinata* ($n=17$): An allopolyploid species derived from the cross between *Brassica nigra* ($n=8$) and *Brassica oleracea* ($n=9$) (KPSC AAO 2015).

425. Aneuploidy: Represents changes in individual chromosome numbers, including monosomy ($2n-1$), nullisomy ($2n-2$), trisomy ($2n+1$), and tetrasomy ($2n+2$) (KPSC AO 2017).

426. Monosomic analysis ($2n-1$): Extensively used to map genes to specific chromosomes in polyploid crops like bread wheat (KPSC AAO 2017).

427. Colchicine: An alkaloid isolated from *Colchicum autumnale* that prevents spindle fiber formation, thereby inducing polyploidy (KPSC AAO 2015).

428. Seedless triploid watermelon ($2n=3x=33$): Produced by crossing a tetraploid ($4x$) maternal parent with a diploid ($2x$) paternal parent (KPSC AO 2017).

429. Transition mutations involve the substitution of a purine for another purine ($A \leftrightarrow G$) or a pyrimidine for another pyrimidine ($C \leftrightarrow T$) (KPSC AAO 2017).

430. Transversion mutations involve the substitution of a purine for a pyrimidine, or vice versa (KPSC AO 2017).

431. Ethyl Methane Sulfonate (EMS): An alkylating agent that acts as a chemical mutagen, primarily inducing G:C to A:T transitions (KPSC AAO 2017).

432. Acridine dyes (proflavin, acridine orange) intercalate between DNA bases, causing frameshift mutations via single-base insertions or deletions (KPSC AO 2014).

433. H.J. Muller demonstrated that X-rays could artificially induce mutations in *Drosophila melanogaster* in 1927 (KPSC AO 2014).

434. L.J. Stadler in 1928 demonstrated the mutagenic effects of X-rays in crops like barley and maize (KPSC AAO 2015).

435. Nonsense mutation converts an amino acid codon into a termination codon (UAG, UAA, or UGA), resulting in a truncated, non-functional protein (KPSC AO 2017).

436. Missense mutation alters a codon so that it specifies a different amino acid, which may alter protein structure or function (KPSC AAO 2015).

437. Silent mutation: A base substitution that does not alter the amino acid sequence due to genetic code degeneracy (KPSC AO 2014).

438. Cytoplasmic male sterility (CMS): Governed by mitochondrial genes and displays maternal inheritance because organelles are transmitted via the egg cell (KPSC AO 2017).

439. Cytoplasmic-Genetic Male Sterility (CGMS), nuclear fertility restorer (Rf) genes restore pollen fertility in lines with sterile cytoplasm (KPSC AAO 2017).

440. **Gametophytic self-incompatibility (GSI):** Determined by the haploid S-allele genotype of the pollen grain itself (KPSC AAO 2017).
441. **Sporophytic self-incompatibility (SSI):** Determined by the diploid genotype of the pollen-producing parent, often leading to pollen germination inhibition on the stigma (KPSC AO 2014).
442. **Hybrid seed production systems, the 'A' line:** Male-sterile, the 'B' line is the maintainer line, and the 'R' line is the restorer line (KPSC AAO 2017).
443. **Synthetic varieties:** Developed by intercrossing a group of inbred lines or clones that have been pre-tested for general combining ability (GCA) (KPSC AO 2014).
444. **Composite varieties:** Developed by mixing seeds of several phenotypically superior, open-pollinated varieties without testing for GCA (KPSC AAO 2015).
445. **Broad-sense heritability (H^2):** The ratio of total genetic variance to phenotypic variance (V_g / V_p) (KPSC AAO 2015).
446. **Narrow-sense heritability (h^2):** The ratio of additive genetic variance to phenotypic variance (V_a / V_p) (KPSC AO 2017).
447. **Narrow-sense heritability:** The primary determinant of genetic advance under selection, as additive genetic variance is the only fixable component (KPSC AO 2017).
448. **Vavilov proposed eight primary centers of origin, designating the South-West Asian center as the primary origin of bread wheat (KPSC AO 2014).**
449. **Primary center of origin for cultivated rice (*Oryza sativa*):** The Hindustan Center, encompassing India and Burma (KPSC AO 2014).
450. **Acclimatization:** The process by which an introduced plant variety adapts to its new climatic conditions over several generations (KPSC AAO 2015).
451. **Pedigree breeding method:** Widely used in self-pollinated crops, where individual plant selection is initiated in the F₂ generation (KPSC AAO 2015).
452. **Bulk breeding method allows natural selection to operate on segregating populations, delaying individual selection until F₅ or F₆ generations (KPSC AO 2017).**
453. **Single Seed Descent (SSD) method:** A modification of bulk breeding used to rapidly advance generations to homozygosity without selection (KPSC AAO 2017).
454. **Backcross breeding method:** Used to transfer a specific oligogenic trait, like disease resistance, from a donor parent into a commercial variety (KPSC AAO 2015).
455. **The backcross method, the commercial variety receiving the gene:** The recurrent parent, and the gene donor is the non-recurrent parent (KPSC AO 2017).
456. **Recurrent selection:** Designed to systematically increase the frequency of desirable alleles in a population through repeated cycles of selection and intermating (KPSC AAO 2017).
457. **Polygenic inheritance (multiple factor hypothesis):** First demonstrated by H. Nilsson-Ehle in 1909 for kernel color in wheat (KPSC AO 2014).
458. **General combining ability (GCA):** Primarily associated with additive genetic variance, whereas specific combining ability (SCA) is due to non-additive gene action (KPSC AAO 2015).
459. **Paracentric inversion does not include the centromere, resulting in dicentric and acentric chromosomes during meiotic anaphase I (KPSC AAO 2017).**
460. **Pericentric inversion includes the centromere and alters the arm ratio of the chromosome, which can be identified cytologically (KPSC AO 2017).**
461. **Diploid organisms, a testcross involves crossing the F₁ hybrid with a double homozygous recessive parent to evaluate linkage (KPSC AO 2014).**
462. **Pleiotropy:** Refers to a single gene influencing multiple, distinct, and seemingly unrelated phenotypic traits, such as pigment in seed coat and flower (KPSC AAO 2015).
463. **Phenocopy:** An environmentally induced phenotypic variation that mimics a genetic mutation but is not inherited (KPSC AO 2017).
464. **Penetrance:** The percentage of individuals carrying a specific genotype that actually express the corresponding phenotype (KPSC AAO 2017).
465. **Expressivity:** Refers to the degree of phenotypic variation or severity of expression among individuals with a particular genotype (KPSC AO 2014).
466. **Bivalent:** A pair of homologous chromosomes that pair up (synapse) during the zygotene and pachytene stages of meiosis I (KPSC

AAO 2015).

467. **Synaptonemal complex:** A tripartite protein structure that physically connects homologous chromosomes during meiotic prophase I (KPSC AO 2017).

468. **Chiasma (plural: chiasmata):** The site of crossing over where non-sister chromatids remain in physical contact after desynapsis (KPSC AAO 2017).

469. **Polytene chromosomes:** Giant chromosomes formed by endoreduplication (replication without segregation) in larval salivary glands of Diptera (KPSC AO 2014).

470. **Lampbrush chromosomes:** Highly transcriptionally active giant chromosomes found in the oocytes of amphibians during diplotene (KPSC AAO 2015).

471. **Inbreeding depression:** The loss of fitness and vigor resulting from selfing or mating close relatives, due to the homozygous expression of deleterious recessives (KPSC AO 2014).

472. **Vertical resistance (oligogenic):** Pathogen race-specific, governed by one or few major genes, and is easily overcome by new races (KPSC AAO 2017).

473. **Horizontal resistance (polygenic):** Race-non-specific, governed by many minor genes, and provides durable, long-term field resistance (KPSC AO 2014).

474. **Multi-line varieties:** Mixtures of closely related isolines (differing only in resistance genes) designed to control rust epidemics in wheat (KPSC AAO 2015).

475. **Mutation breeding:** Particularly useful for improving specific characters of well-adapted varieties without altering their genetic background (KPSC AO 2017).

476. **Mutant rice variety 'Jagannath':** Developed through gamma-irradiation of the tall variety 'T141' (KPSC AAO 2017).

477. **Sharbati Sonora:** An amber-seeded mutant wheat variety developed from the red-seeded Sonora 64 using gamma rays (KPSC AO 2014).

478. **Pre-zygotic barriers in wide hybridization include pollen-stigma incompatibility, which prevents fertilization (KPSC AAO 2015).**

479. **Post-zygotic barriers in wide hybridization include hybrid inviability, hybrid sterility, and hybrid breakdown (KPSC AO 2017).**

480. **Mass selection:** Highly effective for traits with high heritability but is inefficient for quantitative traits with low heritability (KPSC AAO 2017).

481. **Pure-line variety:** Genetically homogeneous and homozygous, making it highly uniform in field performance (KPSC AO 2014).

482. **Emasculation:** The removal of immature anthers from bisexual flowers before dehiscence to prevent self-pollination (KPSC AAO 2017).

483. **Hot water emasculation (45-48°C for 10 minutes):** Used for small-flowered crops like rice, sorghum, and sugarcane (KPSC AO 2014).

484. **Inbred line:** A nearly homozygous line produced by continuous self-pollination in a cross-pollinated species (KPSC AAO 2015).

485. **Topcross:** A cross between an inbred line and an open-pollinated variety, used for preliminary evaluation of GCA of the inbred (KPSC AO 2017).

486. **Backcross method requires 6 to 8 backcrosses to recover approximately 99% of the recurrent parent's genome (KPSC AAO 2017).**

487. **MAGIC (Multiparent Advanced Generation Inter-Cross) populations:** Developed by crossing 4 to 8 founder parents to combine diverse alleles (KPSC AO 2014).

488. **Reciprocal cross involves reversing the sex of the parents in a cross, used to detect maternal (cytoplasmic) inheritance (KPSC AAO 2015).**

489. **Genomic imprinting:** An epigenetic process where the expression of an allele depends on whether it was inherited from the male or female parent (KPSC AO 2017).

490. **Epistasis:** The interaction between non-allelic genes where the allele of one gene masks the phenotypic expression of alleles at another locus (KPSC AO 2014).

491. **Maternal inheritance:** Cytoplasmic inheritance where the progeny's phenotype is entirely determined by the maternal parent's genotype (KPSC AAO 2015).

492. **Carl Correns discovered plastid inheritance in four-o'clock plant (Mirabilis jalapa) in 1909, demonstrating extra-nuclear inheritance (KPSC AO 2017).**

493. **Acridine orange and proflavin act as chemical mutagens by inserting themselves between DNA bases, causing addition or deletion frameshifts (KPSC AO 2014).**
494. **Down syndrome:** A human aneuploidy characterized by trisomy 21, resulting from meiotic non-disjunction of chromosomes (KPSC AAO 2015).
495. **Autopolyploidy:** Causes the 'gigas' effect, characterized by larger cell size, thicker leaves, larger flowers, and increased stomatal size (KPSC AO 2017).
496. **Primary center of origin for potato (*Solanum tuberosum*):** The South American (Andean) Center (KPSC AAO 2017).
497. **Primary center of origin for maize (*Zea mays*):** The Central American (Mexican) Center (KPSC AO 2014).
498. **Modifier gene:** A gene that alters the phenotypic expression of another gene at a different locus without masking it completely (KPSC AAO 2015).
499. **Inbreeding depression:** Highly severe in cross-pollinated crops like alfalfa and carrot, whereas it is negligible in self-pollinated crops (KPSC AO 2017).
500. **Synthetic varieties can be maintained by open pollination for several generations, making seed replacement less frequent than for hybrids (KPSC AAO 2015).**
501. **Durable resistance:** Horizontal resistance that remains effective over a long period of widespread cultivation in pathogen-conducive environments (KPSC AO 2014).
502. **Gene-for-gene hypothesis, proposed by H.H. Flor in 1942, states that for each resistance gene in the host, there: A** complementary avirulence gene in the pathogen (KPSC AO 2017).
503. **Vertical resistance concept:** Also called oligogenic resistance or specific resistance (KPSC AAO 2017).
504. **Wide hybridization barriers can be overcome using embryo rescue, where the immature embryo:** Excised and cultured on nutrient media (KPSC AAO 2015).
505. **Emasculation in cotton:** Commonly done by the hand-emasculation method (Doak's method) or using suction pumps (KPSC AO 2017).
506. **Heterosis:** Maximum in the F₁ generation and decreases in subsequent generations due to segregation (KPSC AO 2014).
507. **Coefficient of coincidence:** Calculated as the ratio of observed double crossovers to expected double crossovers (KPSC AO 2017).
508. **Chromosomal interference occurs when a crossover in one region of a chromosome decreases the probability of a second crossover in an adjacent region (KPSC AAO 2017).**
509. **Unit of genetic distance on a chromosome:** The centimorgan (cM), defined as a 1% recombination frequency between two loci (KPSC AAO 2015).
510. **Diploid chromosome number (2n) of bread wheat (*Triticum aestivum*):** 42, which is an allohexaploid containing A, B, and D genomes (KPSC AO 2014).
511. **B-chromosomes:** Non-vital extra chromosomes that do not pair with A-chromosomes during meiosis, commonly observed in rye and maize (KPSC AO 2017).
512. **Karyotype:** A diagrammatic representation of the chromosomes of an individual arranged in order of size and centromere position (KPSC AAO 2015).
513. **Idiogram:** A stylized, idealized diagram of a karyotype showing chromosome band patterns and relative lengths (KPSC AO 2017).
514. **Heterochromatin:** Transcriptionally inactive, highly condensed chromatin that replicates late in the S-phase of cell cycle (KPSC AAO 2017).
515. **Euchromatin:** Transcriptionally active, loosely packed chromatin that stains lightly during interphase (KPSC AO 2014).
516. **Polygenic traits:** Also called quantitative traits because they show a continuous range of variation and cannot be classified into distinct groups (KPSC AAO 2015).
517. **Primary center of origin for soybean (*Glycine max*):** The Chinese Center (East Asia) (KPSC AO 2017).
518. **Cross-pollinated crops, mass selection:** Used to improve the population mean by selecting superior phenotypes and bulking their seeds (KPSC AAO 2017).
519. **Pedigree method:** Preferred over bulk selection when genetic information about segregating families is required (KPSC AO 2014).
520. **Testcross ratios for independent assortment of two genes in a diploid organism:** 1:1:1:1 (KPSC AAO 2015).
521. **Gottlieb Haberlandt first proposed the concept of cellular totipotency in 1902, earning him the title of the father of plant tissue**

culture (KPSC AO 2017).

522. **Murashige and Skoog (MS) medium:** Is characterized by high concentrations of inorganic salts, particularly ammonium nitrate and potassium nitrate (KPSC AAO 2017).

523. **High auxin-to-cytokinin ratio in tissue culture media promotes root differentiation, while a low ratio promotes shoot differentiation (KPSC AO 2014).**

524. **Callus:** An unorganized, proliferating mass of parenchymatous cells formed in response to plant growth regulators like 2,4-D (KPSC AAO 2015).

525. **Somatic embryogenesis can occur directly from explant cells (pre-embryogenic determined cells) or indirectly via a callus intermediate (KPSC AO 2017).**

526. **Somatic embryos:** Bipolar structures possessing both a shoot and root apex, whereas organogenesis produces unipolar shoots or roots (KPSC AAO 2017).

527. **Meristem culture (using the shoot apical meristem):** Used to produce virus-free plants because the meristem lacks vascular bundles and has high auxin levels (KPSC AO 2014).

528. **Guha and Maheshwari in 1964 first developed haploid plants from *Datura innoxia* through anther culture, pioneering haploid breeding (KPSC AAO 2015).**

529. **Doubled haploids (dihaploids):** Produced by treating haploid plants with colchicine to double their chromosome number, achieving complete homozygosity in one year (KPSC AO 2017).

530. **Protoplasts:** Isolated by enzymatically digesting plant cell walls using cellulase (to digest cellulose) and pectinase (to digest pectin) (KPSC AO 2017).

531. **Polyethylene glycol (PEG) and high pH/high calcium treatments:** The most common chemical methods used to induce protoplast fusion (KPSC AAO 2017).

532. **Cybrids:** Cytoplasmic hybrids produced by fusing a nucleated protoplast of one parent with an enucleated protoplast of another parent (KPSC AO 2014).

533. **Somaclonal variation:** Refers to genetic and epigenetic variation occurring among plants regenerated from in vitro tissue culture (KPSC AAO 2015).

534. **Restriction endonucleases:** Molecular scissors that recognize and cleave DNA at specific palindromic sequences (KPSC AO 2017).

535. **Type II restriction endonucleases cleave DNA at specific site within or adjacent to their recognition sequence, requiring only magnesium ions (Mg²⁺) (KPSC AAO 2017).**

536. **T4 DNA ligase repairs single-strand nicks and joins double-stranded DNA fragments with either cohesive (sticky) or blunt ends (KPSC AO 2014).**

537. **Alkaline phosphatase prevents self-ligation of linearized plasmid vectors by removing the 5'-phosphate group from both ends (KPSC AAO 2015).**

538. **Reverse transcriptase:** An RNA-dependent DNA polymerase used to synthesize complementary DNA (cDNA) from mRNA templates (KPSC AO 2017).

539. **Plasmid pBR322:** A cloning vector that carries genes for resistance to ampicillin (bla) and tetracycline (tet) (KPSC AAO 2017).

540. **Agrobacterium tumefaciens:** A soil bacterium that naturally transfers a segment of its Ti (tumor-inducing) plasmid, the T-DNA, into the plant genome (KPSC AO 2014).

541. **Virulence (vir) genes of the Ti plasmid:** Induced by phenolic compounds like acetosyringone released from wounded plant tissues (KPSC AAO 2015).

542. **Binary vector system uses a helper plasmid containing vir genes and a separate cloning plasmid containing the T-DNA with the gene of interest (KPSC AO 2017).**

543. **Co-integrate vector system integrates the cloning vector into the Ti plasmid via homologous recombination within the Agrobacterium cell (KPSC AAO 2017).**

544. **Gene gun (biolistics/particle bombardment) delivers foreign DNA into plant cells using high-velocity gold or tungsten microparticles (KPSC AO 2014).**

545. **Electroporation uses short, high-voltage electrical pulses to create transient pores in the protoplast membrane for DNA entry (KPSC AAO 2015).**

546. **Restriction Fragment Length Polymorphism (RFLP):** A co-dominant, hybridization-based molecular marker that is highly locus-specific (KPSC AAO 2017).

547. **Random Amplified Polymorphic DNA (RAPD):** A PCR-based dominant marker that amplifies genomic DNA using a single

10-base arbitrary primer (KPSC AO 2014).

548. Simple Sequence Repeats (SSRs or microsatellites): Tandemly repeated short DNA sequences that serve as highly polymorphic, co-dominant PCR markers (KPSC AAO 2015).

549. Amplified Fragment Length Polymorphism (AFLP): A multi-locus marker system combining restriction digestion of genomic DNA and PCR amplification (KPSC AO 2017).

550. Single Nucleotide Polymorphisms (SNPs): Single-base changes in genomic DNA that represent the most abundant class of molecular markers (KPSC AAO 2017).

551. Marker-Assisted Selection (MAS) uses molecular markers linked to a target gene to select desired individuals in breeding populations (KPSC AO 2014).

552. Foreground selection in marker-assisted backcrossing involves using markers to confirm the presence of the target gene of interest (KPSC AAO 2015).

553. Background selection in marker-assisted backcrossing utilizes markers to select for the maximum percentage of the recurrent parent's genome (KPSC AO 2017).

554. Gene pyramiding: The process of combining two or more genes for a trait (like multiple disease resistance genes) into a single genotype (KPSC AAO 2017).

555. Bt cotton expresses Cry proteins (e.g., Cry1Ac) from *Bacillus thuringiensis*, which bind to specific receptors in the midgut of bollworms, causing pore formation and death (KPSC AO 2017).

556. Golden Rice: Genetically engineered to biosynthesize beta-carotene in the endosperm using genes for phytoene synthase (psy) and phytoene desaturase (crtI) (KPSC AAO 2017).

557. Roundup Ready crops are resistant to glyphosate because they express the cp4 epsps gene from *Agrobacterium* sp. strain CP4, which: Insensitive to glyphosate (KPSC AO 2014).

558. Flavr Savr tomato, the first commercial GM food crop, used antisense RNA technology to downregulate the polygalacturonase (PG) gene and delay ripening (KPSC AAO 2015).

559. CRISPR-Cas9 gene editing employs a single guide RNA (sgRNA) to guide Cas9 endonuclease to introduce double-strand breaks at specific genomic sites (KPSC AO 2017).

560. Protospacer Adjacent Motif (PAM), typically 5'-NGG-3' for Cas9,: Essential for Cas9 binding and cleavage of target DNA (KPSC AAO 2017).

561. Polymerase Chain Reaction (PCR), invented by Kary Mullis in 1983, amplifies a specific DNA sequence through cyclic temperature changes (KPSC AAO 2017).

562. Taq DNA polymerase, isolated from *Thermus aquaticus*,: A thermostable enzyme that lacks 3'-to-5' exonuclease (proofreading) activity (KPSC AO 2014).

563. Real-time PCR (qPCR) monitors DNA amplification quantitatively during the PCR cycle using fluorescent dyes like SYBR Green or TaqMan probes (KPSC AAO 2015).

564. Sanger DNA sequencing (dideoxy method) relies on the incorporation of chain-terminating dideoxynucleoside triphosphates (ddNTPs) during DNA synthesis (KPSC AO 2017).

565. Synthetic seeds: Produced by encapsulating somatic embryos or shoot buds in a protective, nutrient-rich sodium alginate matrix (KPSC AO 2014).

566. Vitrification: A cryopreservation technique that avoids ice crystal damage by rapidly cooling tissues to liquid nitrogen temperatures (-196°C) (KPSC AAO 2015).

567. T-DNA region of the Ti plasmid: Contains genes for the biosynthesis of plant hormones (auxin and cytokinin) and opines (KPSC AAO 2017).

568. Opines (e.g., octopine, nopaline): Synthesized by transformed plant cells and catabolized exclusively by *Agrobacterium* as carbon and nitrogen sources (KPSC AO 2014).

569. Reporter genes like uidA (encoding beta-glucuronidase, GUS) produce a blue color when cleaved by X-Gluc, serving as visual markers for transformation (KPSC AAO 2015).

570. Neomycin phosphotransferase II (nptII): A selectable marker gene that confers resistance to kanamycin, allowing selection of transgenic cells (KPSC AO 2017).

571. Co-dominant markers can distinguish homozygous dominant individuals from heterozygotes, whereas dominant markers only show presence or absence (KPSC AAO 2017).

572. Bulk Segregant Analysis (BSA) involves pooling DNA from plants with contrasting phenotypes to identify molecular markers linked to a target gene (KPSC AO 2014).

573. **Quantitative Trait Loci (QTLs):** Genomic regions containing genes that control quantitative traits, mapped using molecular markers (KPSC AAO 2015).

574. **Gene flow:** Represents the transfer of genetic material from genetically modified crops to wild relatives or conventional crops through cross-pollination (KPSC AO 2017).

575. **India, the Genetic Engineering Appraisal Committee (GEAC) under MoEFCC:** The apex authority for approving commercial release of GM crops (KPSC AO 2014).

576. **Biosafety regulations for GMOs in India:** Governed by the 'Rules for the Manufacture, Use, Import, Export, and Storage of Hazardous Microorganisms, Genetically Engineered Organisms or Cells, 1989' (KPSC AAO 2017).

577. **StarLink corn controversy arose when Cry9C Bt corn, approved only for animal feed,:** Detected in human food products, raising allergenicity concerns (KPSC AAO 2015).

578. **Targeted mutagenesis using TALENs (Transcription Activator-Like Effector Nucleases) utilizes DNA-binding domains derived from Xanthomonas bacteria** (KPSC AO 2017).

579. **Zinc Finger Nucleases (ZFNs):** Engineered restriction enzymes that contain a zinc-finger DNA-binding domain fused to the FokI cleavage domain (KPSC AAO 2017).

580. **Next-Generation Sequencing (NGS) platforms sequence millions of DNA fragments in parallel, enabling high-throughput genome sequencing** (KPSC AO 2014).

581. **Reverse Transcription PCR (RT-PCR) uses reverse transcriptase to convert RNA into cDNA, which:** Then amplified by PCR to analyze gene expression (KPSC AAO 2015).

582. **Somatic embryogenesis receptor kinase (SERK):** A marker gene highly expressed during the induction of somatic embryogenesis in vitro (KPSC AO 2017).

583. **Plasmids:** Extra-chromosomal, double-stranded circular DNA molecules that replicate autonomously in bacterial cells (KPSC AAO 2017).

584. **Cosmid vector:** A hybrid vector containing plasmid sequences and the cohesive end (cos) sites of bacteriophage lambda, allowing cloning of large DNA (KPSC AO 2014).

585. **Bacterial Artificial Chromosomes (BACs):** Based on the F-plasmid of E. coli and can accommodate inserts up to 300 kb in size (KPSC AAO 2015).

586. **Yeast Artificial Chromosomes (YACs):** Linear vectors containing yeast telomeres, centromere, and replication origin, capable of carrying inserts up to 1000 kb (KPSC AO 2017).

587. **CTAB (Cetyl Trimethyl Ammonium Bromide) method:** Widely used for plant genomic DNA extraction to remove interfering polysaccharides (KPSC AAO 2017).

588. **Agarose gel electrophoresis separates DNA fragments based on molecular size, with DNA migrating towards the positive anode due to its negative phosphate backbone** (KPSC AO 2014).

589. **Southern blotting:** The transfer of DNA fragments from an agarose gel to a nylon membrane for hybridization with a labeled DNA probe, developed by E.M. Southern (KPSC AAO 2015).

590. **Northern blotting:** Used to detect specific RNA sequences to analyze gene transcription levels in different tissues (KPSC AO 2017).

591. **Western blotting:** An analytical technique used to detect specific proteins in a tissue sample using labeled antibodies (KPSC AAO 2017).

592. **Microarrays:** Contain thousands of spot-synthesized DNA probes on a solid substrate, allowing simultaneous analysis of expression of thousands of genes (KPSC AO 2014).

593. **Organelle transformation (chloroplast transformation) offers maternal inheritance of transgenes, preventing transgene escape via pollen** (KPSC AAO 2015).

594. **Plant growth regulators:** Autoclaved or filter-sterilized depending on their heat stability; auxins like IAA and gibberellins (GA3) are heat-labile and must be filter-sterilized (KPSC AO 2017).

595. **Gelling agent phytagel or agar:** Added to tissue culture media to provide a solid support for explant growth (KPSC AAO 2017).

596. **Hardening:** The gradual acclimatization of in vitro regenerated plantlets to greenhouse and field conditions to reduce mortality due to transpiration (KPSC AO 2014).

597. **Direct organogenesis:** The formation of adventitious organs (shoots or roots) directly from the explant tissue without an intervening callus phase (KPSC AAO 2015).

598. **Anther culture mainly targets the microspores (uninucleate pollen grains) to induce sporophytic development instead of gametophytic development** (KPSC AO 2017).

599. **Plastid transformation (transplastomic plants) yields high protein expression levels due to the presence of multiple chloroplast genomes per cell (KPSC AAO 2017).**
600. **Bar gene, isolated from *Streptomyces hygroscopicus*, encodes phosphinothricin acetyltransferase, which confers resistance to glufosinate herbicide (KPSC AO 2014).**
601. **Bordeaux mixture, first discovered by PMA Millardet in 1885 to control downy mildew of grapevine (*Plasmopara viticola*), consists of Copper Sulphate, Quick Lime, and Water in a 5:5:50 ratio (KPSC AO 2017).**
602. **Late blight of potato, caused by the oomycete *Phytophthora infestans*,:** Responsible for the historic Irish Famine in 1845 (KPSC AAO 2017).
603. **Brown spot of rice, caused by *Helminthosporium oryzae* (*Bipolaris oryzae*),:** The primary pathogen responsible for the Bengal Famine of 1943 (KPSC AO 2014).
604. **Coffee rust, caused by *Hemileia vastatrix*, devastated the Arabica coffee plantations of Ceylon (Sri Lanka) in 1869 (KPSC AAO 2015).**
605. **Club root of crucifers (*Plasmodiophora brassicae*):** Is characterized by hypertrophy (increase in cell size) and hyperplasia (increase in cell number) of root tissues (KPSC AO 2017).
606. **Loose smut of wheat, caused by *Ustilago nuda tritici*,:** A classic example of an internally seed-borne disease where mycelium remains dormant inside the embryo (KPSC AAO 2017).
607. **Grain smut of sorghum (*Sporisorium sorghi*):** An externally seed-borne disease controlled effectively by seed treatment with sulfur dust at 4 g/kg of seed (KPSC AO 2014).
608. **Citrus canker:** Caused by the Gram-negative bacterium *Xanthomonas axonopodis* pv. *citri*, which gains entry through stomata and wounds, forming characteristic corky, canker lesions (KPSC AO 2017).
609. **Rice tungro virus (RTV):** A disease complex caused by Rice tungro bacilliform virus (RTBV) and Rice tungro spherical virus (RTSV), transmitted by the green leafhopper (*Nephotettix virescens*) (KPSC AO 2017).
610. **Tomato spotted wilt virus (TSWV), a *Tospovirus*,:** Transmitted in a circulative propagative manner by the Western flower thrips (*Frankliniella occidentalis*) (KPSC AAO 2017).
611. **Bunchy top of banana:** Caused by Banana bunchy top virus (BBTV, family *Babuviridae*) and is transmitted by the banana aphid, *Pentalonia nigronervosa* (KPSC AO 2017).
612. **Sugarcane grassy shoot disease (GSD):** Caused by a phytoplasma and is transmitted by the sugarcane planthopper (*Pyrilla perpusilla*) and aphids (*Melanaphis sacchari*) (KPSC AAO 2015).
613. **Little leaf of brinjal:** Caused by a phloem-limiting phytoplasma and is transmitted in nature by the leafhopper vector *Hishimonus phycitis* (KPSC AO 2014).
614. **Sandal spike disease, which restricts leaves to small spike-like structures,:** Caused by phytoplasma and transmitted by the leafhopper *Coelidia indica* (KPSC AO 2017).
615. **Ergot of bajra (pearl millet), caused by *Claviceps fusiformis*, produces sclerotia containing toxic alkaloids such as ergotamine and ergometrine, causing ergotism in animals (KPSC AAO 2015).**
616. **Whip smut of sugarcane (*Sporisorium scitamineum*) produces a characteristic black whip-like structure arising from the growing apex, covered by a silvery membrane containing chlamydospores (KPSC AO 2017).**
617. **Panama wilt of banana, caused by *Fusarium oxysporum* f. sp. *cubense*,:** A soil-borne vascular wilt pathogen that enters the root system and blocks xylem vessels (KPSC AAO 2017).
618. **Bacterial wilt of Solanaceous crops, caused by *Ralstonia solanacearum* (formerly *Pseudomonas solanacearum*),:** Diagnosed in the field using the rapid bacterial ooze test (KPSC AO 2017).
619. **Black chaff of wheat, a bacterial disease causing dark stripes on glumes and leaves,:** Caused by *Xanthomonas translucens* pv. *undulosa* (KPSC AAO 2017).
620. **Burgundy mixture:** Prepared using Copper Sulphate, Sodium Carbonate (washing soda), and Water in a 1:1.25:100 ratio (KPSC AAO 2015).
621. **Chestnut compound, used to control damping-off in nurseries,:** Prepared by mixing 2 parts of copper sulphate and 11 parts of ammonium carbonate (KPSC AO 2017).
622. **Term 'epidemiology' in plant pathology:** Extensively defined and conceptualized by J.E. Vanderplank, who introduced the parameters of apparent infection rate (*r*) (KPSC AO 2017).
623. **Carboxin (Vitavax):** A systemic fungicide belonging to the oxathiin group that specifically targets Basidiomycete fungi, particularly smuts, by inhibiting succinate dehydrogenase (KPSC AAO 2017).

624. **Metaxyl (Apron/Ridomil):** An acylalanine systemic fungicide that specifically inhibits RNA polymerase I in Oomycete pathogens like Pythium and Phytophthora (KPSC AO 2017).
625. **Trichoderma harzianum and Trichoderma viride:** Mycoparasitic biological control agents that produce chitinases and glucanases to degrade cell walls of soil-borne pathogens (KPSC AO 2017).
626. **Pseudomonas fluorescens:** A biocontrol bacterium that suppresses plant pathogens through the production of siderophores (pyoverdine), HCN, and antibiotics like 2,4-DAPG (KPSC AO 2017).
627. **Powdery mildews (Erysiphales) produce distinctive cleistothecia (chasmothecia) as sexual fruiting bodies, characterized by various types of appendages (KPSC AAO 2017).**
628. **Black stem rust of wheat:** Caused by Puccinia graminis f. sp. tritici, a macrocyclic and heteroecious rust requiring wheat (primary host) and barberry (alternate host) to complete its lifecycle (KPSC AO 2017).
629. **Five distinct spore stages of Puccinia graminis tritici in order are: Pycniospores (0), Aeciospores (I), Urediniospores (II), Teliospores (III), and Basidiospores (IV) (KPSC AO 2017).**
630. **White rust of crucifers, caused by Albugo candida,:** Is characterized by the formation of chalky white pustules containing zoosporangia on the lower leaf surface (KPSC AAO 2015).
631. **Damping-off of seedlings, caused by Pythium debaryanum or Pythium aphanidermatum,:** A devastating nursery disease controlled by soil drenching with Copper Oxylchloride at 0.3% (KPSC AO 2014).
632. **Blast of rice, caused by Pyricularia oryzae (teleomorph Magnaporthe oryzae),:** Known as the 'rich man's disease' and produces diagnostic spindle-shaped (boat-shaped) necrotic lesions (KPSC AO 2017).
633. **Sheath blight of rice, caused by Rhizoctonia solani,:** A sclerotial fungus that produces distinct snake-skin-like or banded lesions on leaf sheaths (KPSC AAO 2017).
634. **Bacterial leaf blight of rice, caused by Xanthomonas oryzae pv. oryzae, exhibits the systemic seedling stage known as 'Kressek' which results in complete wilting (KPSC AO 2017).**
635. **False smut of rice, caused by Ustilaginoidea virens, transforms individual grains into yellowish-green, velvety spore balls that turn blackish-green at maturity (KPSC AAO 2017).**
636. **Yellow vein mosaic of okra:** A Begomovirus disease transmitted by the whitefly Bemisia tabaci, causing severe yellowing of veins and chlorosis (KPSC AO 2017).
637. **Red rot of sugarcane, caused by Colletotrichum falcatum,:** Is characterized by red discoloration of internal tissues with diagnostic white transverse bands and an acidic, alcoholic odour (KPSC AO 2017).
638. **Early leaf spot of groundnut (Tikka disease):** Caused by Cercospora arachidicola, producing brown spots with a prominent yellow halo on the upper surface of leaves (KPSC AO 2014).
639. **Late leaf spot of groundnut:** Caused by Phaeoisariopsis personata (Cercosporidium personatum), producing dark brown to black spots without a prominent yellow halo (KPSC AO 2014).
640. **Root-knot nematode (Meloidogyne incognita) infests host plants by inducing the formation of giant cells and hyperplastic galls in the root cortex (KPSC AO 2017).**
641. **Ear cockle of wheat:** Caused by the nematode Anguina tritici, which interacts with the bacterium Rathayibacter tritici to cause 'Tundu' or yellow ear rot disease (KPSC AAO 2017).
642. **Golden nematode of potato (Globodera rostochiensis and Globodera pallida):** A major quarantine pest restricted to the Nilgiri hills of Tamil Nadu in India (KPSC AO 2017).
643. **Citrus greening (Huanglongbing), caused by the phloem-restricted bacterium Candidatus Liberibacter asiaticus,:** Transmitted by the Asian citrus psyllid, Diaphorina citri (KPSC AAO 2015).
644. **Coconut root wilt:** Caused by a phytoplasma and is biologically transmitted by the lace wing bug, Stephanitis typica (KPSC AO 2017).
645. **Arecanut yellow leaf disease (YLD), causing severe yellowing of leaflets and crown reduction,:** Associated with phytoplasmas and transmitted by the vector Proutista moesta (KPSC AAO 2017).
646. **Foot rot (quick wilt) of black pepper:** Caused by the oomycete Phytophthora capsici, which spreads rapidly during the monsoon and is managed by Bordeaux mixture drenching (KPSC AO 2017).
647. **Soft rot or rhizome rot of ginger:** Caused by Pythium aphanidermatum, characterized by water-soaking of the collar region and rotting of the rhizome (KPSC AAO 2017).
648. **Anthracnose of mango, caused by Colletotrichum gloeosporioides,:** Causes post-harvest rot and leaf spots that exhibit a 'tear-stain' pattern on the fruit surface (KPSC AO 2017).

649. **Powdery mildew of mango, caused by *Oidium mangiferae*, affects the inflorescence and young fruits, causing them to drop prematurely (KPSC AAO 2015).**
650. **Plant quarantine in India:** Regulated under the Destructive Insects and Pests Act (DIPA) of 1914, to prevent the entry of exotic pathogens (KPSC AO 2014).
651. **National Institute of Plant Health Management (NIPHM), formerly the Central Plant Protection Training Institute (CPPTI):** Located at Hyderabad, Telangana (KPSC AAO 2017).
652. **Systemic acquired resistance (SAR):** A broad-spectrum plant defense mechanism triggered by necrotizing pathogens, mediated primarily by salicylic acid signalling (KPSC AO 2017).
653. **Induced systemic resistance (ISR):** A systemic plant defense pathway activated by beneficial rhizobacteria, mediated by jasmonic acid and ethylene (KPSC AAO 2017).
654. **Phytoalexins:** Low molecular weight antimicrobial compounds synthesized de novo by host plants in response to pathogen attack, such as Pisatin in *Pisum sativum* (KPSC AO 2017).
655. **Appressorium:** A specialized, swollen infection structure formed at the tip of a germ tube by fungi to exert physical pressure and penetrate host cuticles (KPSC AAO 2017).
656. **Haustrorium:** A specialized intracellular structure produced by obligate biotrophic pathogens (like rusts and mildews) to absorb nutrients from the host cell without rupturing the plasma membrane (KPSC AO 2017).
657. **Mycoplasmas (phytoplasmas):** Wall-less prokaryotes containing both DNA and RNA, bounded by a triple-layered unit membrane, and are highly sensitive to tetracyclines but resistant to penicillin (KPSC AO 2017).
658. **Viroids:** The smallest known infectious agents, composed solely of a short, circular, single-stranded RNA molecule lacking a protein coat (e.g., Potato spindle tuber viroid) (KPSC AAO 2017).
659. **Virion:** A complete, structurally intact, and infectious virus particle consisting of a nucleic acid core (genomic DNA or RNA) enclosed in a protein shell called a capsid (KPSC AO 2014).
660. **Bacterial cell walls are composed of a peptidoglycan polymer (murein) which:** Thick in Gram-positive bacteria and thin in Gram-negative bacteria (KPSC AAO 2015).
661. **Pierce's disease of grapevine:** Caused by the xylem-limited fastidious bacterium *Xylella fastidiosa*, which is transmitted by sharpshooter leafhoppers (KPSC AO 2017).
662. **Crown gall disease of woody plants:** Caused by *Agrobacterium tumefaciens*, which transfers a segment of its Ti plasmid (T-DNA) into the plant genome (KPSC AO 2017).
663. **Fire blight of apple and pear, caused by *Erwinia amylovora*:** The first plant disease proved to be of bacterial origin by T.J. Burrill in 1878 (KPSC AO 2017).
664. **Dr. K.C. Mehta:** Renowned for his pioneering research on the epidemiology of cereal rusts in India, proving that the rust pathogen survives on alternate hosts in the hills (KPSC AO 2014).
665. **Hot water treatment for the control of loose smut of wheat (at 54°C for 10 minutes):** Originally developed by J.L. Jensen in 1887 (KPSC AAO 2017).
666. **Solar heat treatment (solar energy treatment) for loose smut of wheat under Indian conditions:** Modified and introduced by Luthra and Sattar in 1934 (KPSC AO 2017).
667. **Carbendazim (Bavistin):** A systemic benzimidazole fungicide that inhibits fungal mitosis by binding to beta-tubulin subunits, preventing spindle microtubule assembly (KPSC AO 2017).
668. **Triazole fungicides (such as Propiconazole, Hexaconazole, and Tebuconazole) act as ergosterol biosynthesis inhibitors (EBIs) by targeting the C-14 alpha-demethylase enzyme (KPSC AAO 2017).**
669. **Strobilurins (e.g., Azoxystrobin):** Systemic fungicides derived from the fungus *Strobilurus tenacellus*, acting as respiratory inhibitors by blocking electron transport at the Qo site of complex III (KPSC AO 2017).
670. **Copper oxychloride:** A broad-spectrum contact protective fungicide that releases copper ions, which non-specifically denature fungal enzymes and proteins (KPSC AAO 2015).
671. **Elemental sulfur:** A contact fungicide that acts as a respiratory inhibitor in fungal spores, highly effective against powdery mildews and plant mites (KPSC AO 2014).
672. **Lime sulfur:** Prepared by boiling sulfur and quicklime together, functioning as both a fungicide (for powdery mildew) and a winter scalecide (KPSC AAO 2017).
673. **Root-knot nematode (*Meloidogyne* spp.) exhibits sexual dimorphism; adult females:** Sedentary and pear-shaped (pyriform), while adult males are active and worm-like (vermiform) (KPSC AO 2017).

674. **Citrus nematode (*Tylenchulus semipenetrans*):** A semi-endoparasite that causes 'citrus slow decline,' with the female feeding head embedded in the cortex while the posterior body remains outside (KPSC AAO 2017).
675. **Lesion nematode (*Pratylenchus spp.*):** A migratory endoparasite that moves through the root cortex, destroying cells and leaving necrotic lesions (KPSC AO 2017).
676. **Stunt nematode (*Tylenchorhynchus spp.*):** An ectoparasite that feeds on epidermal cells of plant roots without entering the root tissues (KPSC AAO 2015).
677. **Synergistic disease complex between *Meloidogyne incognita* and *Fusarium oxysporum* results in increased severity of wilt in cotton, tobacco, and tomato crops (KPSC AO 2017).**
678. **Seed gall nematode *Anguina tritici*:** The first plant-parasitic nematode discovered and described by John Needham in 1743 (KPSC AO 2017).
679. **Intercropping marigold (*Tagetes erecta*) with main crops suppresses root-knot nematodes due to the exudation of nematicidal alpha-terthienyl from marigold roots (KPSC AO 2014).**
680. ***Pasteuria penetrans*:** A Gram-positive, endospore-forming bacterium that acts as an obligate parasite of root-knot nematodes, preventing their reproduction (KPSC AO 2017).
681. **Insects belong to the Class Insecta (Hexapoda), which:** Is characterized by a three-segmented body (head, thorax, abdomen), three pairs of jointed legs, and two pairs of wings (KPSC AO 2017).
682. **Insect head consists of 6 fused segments and bears sensory organs (antennae, eyes) and mouthparts (KPSC AAO 2017).**
683. **Aristate antennae, characterized by a large terminal segment bearing a dorsal bristle (arista),:** Found in the common housefly (*Musca domestica*) (KPSC AO 2017).
684. **Plumose antennae, which:** Highly feathery with long whorls of hairs, are found in male mosquitoes, assisting in detecting female wing beats (KPSC AO 2017).
685. **Pilose antennae, having short whorls of hairs,:** Characteristic of female mosquitoes (KPSC AAO 2017).
686. **Moniliform antennae, which resemble a string of beads,:** Found in termites (Isoptera) (KPSC AO 2017).
687. **Setaceous antennae, which:** Long, bristle-like, and taper from base to apex, are typical of cockroaches (KPSC AAO 2017).
688. **Geniculate antennae, which:** Elbowed with a long scape and a flagellum directed at an angle, are found in honeybees and ants (KPSC AO 2017).
689. **Piercing and sucking mouthparts, adapted for puncturing tissues and sucking fluids,:** Present in mosquitoes, bugs, and aphids (KPSC AAO 2017).
690. **Chewing and lapping mouthparts, where mandibles:** Used for molding wax and the maxillo-labial complex forms a tongue for lapping nectar, are present in honeybees (KPSC AO 2017).
691. **Siphoning mouthparts, forming a long coiled proboscis (galea of maxillae) for sucking nectar,:** Characteristic of butterflies and moths (KPSC AAO 2017).
692. **Sponging mouthparts, where the labium:** Modified into a proboscis ending in a pair of labella with pseudotracheae, are found in the housefly (KPSC AO 2017).
693. **Rasping and sucking mouthparts, which:** Asymmetrical (lacking the right mandible) and used to scrape plant tissue and suck sap, are found in thrips (KPSC AAO 2017).
694. **Elytra:** The heavily sclerotized, hardened forewings of beetles (Coleoptera) that serve a protective function and do not aid in flight (KPSC AO 2017).
695. **Tegmina:** The leathery or parchment-like forewings of grasshoppers and cockroaches (Orthoptera) that protect the membranous hindwings (KPSC AAO 2017).
696. **Halteres:** Small, club-like structures modified from the hindwings of true flies (Diptera) that act as gyroscopic balancing organs during flight (KPSC AO 2017).
697. **Hemelytra are the forewings of true bugs (Hemiptera) where the basal portion:** Thick and leathery, and the apical portion is membranous (KPSC AAO 2017).
698. **Cursorial legs, adapted for running,:** Long and slender with segments of similar size, typical of cockroaches (KPSC AO 2017).
699. **Fossorial legs, modified for digging, have a flattened femur and tibia with tooth-like projections, characteristic of the mole cricket (*Gryllotalpa spp.*) (KPSC AAO 2017).**
700. **Saltatorial legs, modified for jumping, feature an enlarged and muscular hind femur, typical of grasshoppers (KPSC AO 2017).**

701. **Raptorial legs, modified for grasping and holding prey, have spiny femora and tibiae that fold against each other, characteristic of praying mantids (KPSC AAO 2017).**
702. **Natatorial legs, adapted for swimming,:** Flattened and fringed with long swimming hairs, typical of water scavenger beetles (KPSC AO 2017).
703. **Corbicula (pollen basket):** A smooth cavity surrounded by stiff hairs located on the outer surface of the hind tibia of worker honeybees (KPSC AO 2017).
704. **Insect digestive tract:** Divided into three regions: the stomodeum (foregut) and proctodeum (hindgut), which are ectodermal in origin and lined with cuticle, and the mesenteron (midgut), which is endodermal (KPSC AAO 2017).
705. **Peritrophic membrane:** A semi-permeable chitinous sheath lining the midgut that protects the delicate digestive cells from mechanical damage by food particles (KPSC AO 2017).
706. **Malpighian tubules:** The primary excretory organs of insects, which filter nitrogenous wastes (mostly uric acid) from the haemolymph into the hindgut (KPSC AAO 2017).
707. **Aphids:** Unique among insects in lacking Malpighian tubules, relying instead on rapid gut transit and specialized cell structures for excretion (KPSC AO 2017).
708. **Respiratory system of insects:** Tracheal, opening to the exterior through lateral pores called spiracles, typically consisting of 10 pairs (KPSC AAO 2017).
709. **Insect blood (haemolymph) lacks red blood cells and haemoglobin (except in Chironomids) and does not play a primary role in oxygen transport (KPSC AO 2017).**
710. **Dorsal vessel of the open circulatory system consists of a posterior chambered heart with lateral ostia for blood entry and an anterior tube-like aorta (KPSC AAO 2017).**
711. **Insect brain:** Divided into the protocerebrum (innervating compound eyes and ocelli), deutocerebrum (innervating antennae), and tritocerebrum (innervating labrum and foregut) (KPSC AO 2017).
712. **Subesophageal ganglion, located beneath the esophagus, controls the mouthparts and:** Formed by the fusion of three neuromeres (KPSC AAO 2017).
713. **Corpora allata:** Endocrine glands that secrete Juvenile Hormone (JH), which maintains larval characteristics and prevents premature metamorphosis (KPSC AO 2017).
714. **Prothoracic glands secrete ecdysone (moulting hormone) under the stimulation of Prothoracicotropic Hormone (PTTH) secreted by the brain (KPSC AAO 2017).**
715. **Rock bee, Apis dorsata,:** The largest honeybee species in India, known for its open single-comb nests and aggressive behavior, making it difficult to domesticate (KPSC AO 2017).
716. **Little bee, Apis florea,:** The smallest honeybee species in India, building small single combs in bushes (KPSC AAO 2017).
717. **Apis cerana indica:** The indigenous domesticated honeybee of India, which can be reared in hives but has a high tendency to swarm and abscond (KPSC AO 2017).
718. **Apis mellifera (European bee):** Successfully introduced to India in 1962 by Dr. A.S. Atwal at Punjab Agricultural University, revolutionizing commercial apiculture (KPSC AAO 2017).
719. **American Foulbrood:** A highly contagious bacterial disease of honeybee brood caused by the spore-forming bacterium *Paenibacillus larvae* (KPSC AO 2017).
720. **European Foulbrood:** A bacterial disease of honeybee larvae caused by the non-spore-forming bacterium *Melissococcus plutonius* (KPSC AAO 2017).
721. **Chalkbrood:** A fungal disease of honeybee larvae caused by *Ascosphaera apis*, which turns infected larvae into hard, white, chalky mummies (KPSC AO 2017).
722. **Mulberry silkworm (Bombyx mori):** A monophagous insect feeding exclusively on mulberry leaves (*Morus* spp.) (KPSC AAO 2017).
723. **Tasar silkworm (Antheraea mylitta):** A polyphagous wild silkworm that feeds on forest trees like Asan (*Terminalia tomentosa*) and Arjun (*Terminalia arjuna*) (KPSC AO 2017).
724. **Muga silkworm (Antheraea assamensis):** Endemic to the Brahmaputra Valley of Assam, producing highly prized, naturally golden-yellow silk (KPSC AAO 2017).
725. **Eri silkworm (Samia cynthia ricini):** Reared on castor leaves (*Ricinus communis*) and produces spun silk because its cocoon is open-ended (KPSC AO 2017).
726. **Pebrine disease of silkworm:** Caused by the microsporidian parasite *Nosema bombycis*, which is transmitted transovarially

(vertically) from mother moth to offspring (KPSC AAO 2017).

727. **Lac insect, *Kerria lacca*, belongs to the family Kerriidae (Hemiptera) and secretes resinous lac from its cutaneous glands as a protective covering (KPSC AO 2017).**

728. **Major host plants for lac cultivation:** Kusum (*Schleichera oleosa*), Palas (*Butea monosperma*), and Ber (*Ziziphus mauritiana*) (KPSC AAO 2017).

729. **Yellow stem borer of rice (*Scirpophaga incertulas*):** A monophagous pest that causes 'dead hearts' during the vegetative stage and 'white ears' during the reproductive stage (KPSC AO 2017).

730. **Brown planthopper of rice (*Nilaparvata lugens*) feeds on phloem sap at the base of the crop, causing localized patches of wilting and drying known as 'hopper burn' (KPSC AAO 2017).**

731. **Rice gall midge (*Orseolia oryzae*) larvae stimulate the leaf sheath to grow into a hollow, dirty-white tubular gall known as a 'silver shoot' or 'onion leaf' (KPSC AO 2014).**

732. **Rice gundhi bug (*Leptocoris acuta*) sucks the milky sap of developing grains, resulting in chaffy, ill-filled grains and emitting a foul odor (KPSC AO 2017).**

733. **Rice leaf folder (*Cnaphalocrocis medinalis*) larvae fold the leaf margins longitudinally and scrape the green mesophyll tissue, leaving white papery streaks (KPSC AAO 2015).**

734. **Pink bollworm of cotton (*Pectinophora gossypiella*) larvae tunnel into bolls, feed on seeds, and web the petals together to form characteristic 'rosette flowers' (KPSC AO 2017).**

735. **Spotted bollworm of cotton (*Earias vitella* and *E. insulana*) bores into terminal shoots in the early stage, causing them to droop and wilt, and later attacks buds and bolls (KPSC AAO 2017).**

736. **American bollworm (*Helicoverpa armigera*):** A highly polyphagous pest that feeds on cotton bolls by inserting only its head inside, keeping the rest of the body outside (KPSC AO 2017).

737. **Cotton whitefly (*Bemisia tabaci*) sucks sap from the underside of leaves and excretes honeydew, which attracts sooty mould (*Capnodium* spp.) and transmits Cotton Leaf Curl Virus (CLCuV) (KPSC AAO 2017).**

738. **Early shoot borer of sugarcane (*Chilo infuscatellus*) attacks the crop during the first 3 months, boring into the stem and producing dead hearts that emit an offensive odor when pulled (KPSC AO 2014).**

739. **Top shoot borer of sugarcane (*Scirpophaga excerptalis*) larvae bore into the midrib of leaves, causing parallel shot holes, and damage the growing point, leading to a 'bunchy top' appearance (KPSC AAO 2017).**

740. **Internode borer of sugarcane (*Chilo sacchariphagus indicus*) attacks the internodes of the sugarcane stalk, particularly in the post-formative stage of the crop (KPSC AO 2017).**

741. **Sugarcane pyrrilla (*Pyrilla perpusilla*):** A sucking pest that causes yellowing of leaves and is biologically controlled by the release of its ectoparasitoid *Epiricania melanoleuca* (KPSC AO 2017).

742. **Economic Injury Level (EIL):** Is defined as the lowest pest population density that will cause economic damage, where the cost of control equals the value of yield saved (KPSC AO 2017).

743. **Economic Threshold Level (ETL):** The pest density at which control measures should be initiated to prevent the population from reaching the EIL (KPSC AO 2017).

744. **General Equilibrium Position (GEP):** The average population density of an insect species over a long period in an environment, around which the population fluctuates (KPSC AAO 2017).

745. **Organochlorine insecticides like DDT and BHC act on the insect nervous system by keeping sodium channels open in the nerve membrane, leading to repetitive discharges (KPSC AO 2014).**

746. **Organophosphate insecticides (e.g., Chlorpyrifos, Dimethoate, Monocrotophos) inhibit the enzyme acetylcholinesterase (AChE), leading to accumulation of acetylcholine and continuous nerve firing (KPSC AO 2017).**

747. **Carbamate insecticides (e.g., Carbofuran, Carbaryl):** Reversible inhibitors of acetylcholinesterase (AChE), causing paralysis and death of target insects (KPSC AAO 2017).

748. **Synthetic pyrethroids (e.g., Cypermethrin, Deltamethrin):** Highly potent insecticides that act as axonic poisons by delaying the closing of sodium channels (KPSC AO 2017).

749. **Neonicotinoids (e.g., Imidacloprid, Thiamethoxam) act as agonists on the nicotinic acetylcholine receptors (nAChR) of the postsynaptic membrane (KPSC AAO 2017).**

750. **Spinosad:** A green insecticide derived from the actinomycete *Saccharopolyspora spinosa*, acting on both nicotinic acetylcholine and GABA receptors (KPSC AO 2017).

751. **Diamide insecticides (e.g., Chlorantraniliprole, Flubendiamide) target insect ryanodine receptors, causing uncontrolled release of calcium from internal muscle stores, leading to muscle contraction and paralysis (KPSC AO 2017).**

752. **Trichogramma chilonis**: A widely used micro-hymenopteran egg parasitoid that parasitizes the eggs of Lepidopteran pests like Helicoverpa and Chilo (KPSC AO 2017).
753. **Cotesia flavipes**: A larval endoparasitoid that is highly effective against the larvae of cereal stem borers, particularly in sugarcane and maize (KPSC AAO 2017).
754. **Chrysoperla carnea (Green lacewing) larvae**: Generalist predators that feed voraciously on soft-bodied sucking insects such as aphids, jassids, and thrips (KPSC AO 2017).
755. **Cryptolaemus montrouzieri (Australian ladybird beetle)**: A predatory coccinellid widely released to control mealybug infestations in horticultural crops (KPSC AO 2017).
756. **Bacillus thuringiensis (Bt) produces crystalline delta-endotoxins which dissolve in the alkaline pH of the insect midgut, binding to specific receptors and causing cell lysis (KPSC AO 2017).**
757. **Nuclear Polyhedrosis Viruses (NPVs)**: Highly host-specific baculoviruses that infect Lepidopteran larvae, which typically die hanging upside down from the plant canopy (KPSC AAO 2017).
758. **Beauveria bassiana**: An entomopathogenic fungus that causes 'white muscardine' disease in insects, penetrating the cuticle through enzymatic degradation (KPSC AO 2017).
759. **Metarhizium anisopliae**: An entomopathogenic fungus that causes 'green muscardine' disease, particularly effective against subterranean pests like root grubs and termites (KPSC AAO 2015).
760. **Pheromone traps for Helicoverpa armigera utilize the synthetic female sex pheromone blend of Z-9-hexadecenal and Z-11-hexadecenal (KPSC AO 2017).**
761. **Hallikar**: The premier draught breed of cattle in Karnataka, originating from the former Vijayanagara/Mysore region, characterized by long, backward-curved horns and active temperament (KPSC AO 2017).
762. **Amritmahal**: A famous draught breed of Karnataka, developed by Tippu Sultan for military transport and known for its quick and power-packed draft capacity (KPSC AAO 2017).
763. **Khillar**: A tough draught cattle breed of North Karnataka and Maharashtra border (Belagavi/Vijayapura), adapted to tropical dry regions (KPSC AO 2017).
764. **Deoni**: An important dual-purpose breed of cattle originating from the Bidar district of Karnataka, recognized by its white coat with black or red spots (KPSC AO 2017).
765. **Krishna Valley**: A heavy draught and dual-purpose breed developed in the Krishna river basin of North Karnataka (Bagalkot/Belagavi) (KPSC AAO 2015).
766. **Malnad Gidda**: A dwarf, highly disease-resistant cattle breed of the Western Ghats of Karnataka, adapted to survive on poor grazing and humid climates (KPSC AO 2017).
767. **Gir cattle, originating from the Gir forests of Gujarat**: Characterized by dome-shaped foreheads, long drooping ears resembling a curled leaf, and high milk yields (KPSC AO 2017).
768. **Sahiwal**: The premier dairy breed of indigenous cattle, originating from Punjab/Pakistan, known for its loose skin (hence called 'Lola') and highest milk yield among Indian breeds (KPSC AAO 2017).
769. **Red Sindhi**: A high-yielding indigenous dairy cattle breed with a uniform deep red coat color, originating from the Karachi region (KPSC AO 2017).
770. **Tharparkar**: A dual-purpose breed of cattle from the Thar desert of Rajasthan, highly resistant to heat, drought, and tropical diseases (KPSC AAO 2015).
771. **Ongole cattle, originating from Nellore/Ongole in Andhra Pradesh**: Heavy draught animals known for their natural resistance to foot-and-mouth disease (FMD) (KPSC AO 2017).
772. **Holstein Friesian (HF)**: A dairy cattle breed from the Netherlands, having black and white markings, and is the highest milk-producing breed in the world (KPSC AO 2017).
773. **Jersey cattle, originating from the Island of Jersey (UK), produce milk with the highest fat content (~5% to 5.5%) among exotic dairy breeds (KPSC AAO 2017).**
774. **Murrah**: The premier dairy buffalo breed in India, originating from Haryana and Punjab, characterized by jet-black color and tightly curled horns (KPSC AO 2017).
775. **Surti buffaloes, originating from Gujarat**: Characterized by their sickle-shaped, flat horns and two white collars on the brisket (KPSC AAO 2017).
776. **Jaffarabadi**: The heaviest breed of Indian buffalo, native to Gujarat, known for its prominent bulging forehead and downward drooping horns (KPSC AO 2017).

777. **Nili-Ravi buffaloes:** Characterized by white markings on the forehead, muzzle, legs, and tail, and have characteristic 'wall eyes' (white iris) (KPSC AAO 2017).
778. **Bhadawari:** An indigenous buffalo breed of Uttar Pradesh, characterized by its copper-colored hair coat and high milk fat content ranging from 8.5% to 13% (KPSC AO 2017).
779. **Bannur (Mandya) sheep:** A highly prized meat/mutton breed of Karnataka, characterized by its compact, wedge-shaped body and fleshy, well-developed hindquarters (KPSC AO 2017).
780. **Deccani sheep:** A coarse-wool and meat breed reared in the dry tracts of North Karnataka, Maharashtra, and Andhra Pradesh, producing coarse fleece used for blankets (KPSC AO 2017).
781. **Jamunapari:** A premier dual-purpose goat breed of Uttar Pradesh, characterized by a convex Roman nose, long pendulous ears, and long hair on the hind thighs (KPSC AO 2017).
782. **Barbari:** A small, compact, dual-purpose goat breed suited for stall-feeding ('city goat') due to its calm temperament and high prolificacy (KPSC AAO 2017).
783. **Black Bengal goat, found in West Bengal and Bangladesh,:** Famous for its high prolificacy (multiple births), high-quality meat (chevon), and superior skin (KPSC AO 2017).
784. **Kadaknath:** A unique chicken breed from Madhya Pradesh, known for its black plumage, black meat, black bones, and high iron content due to melanin deposition (KPSC AO 2017).
785. **White Leghorn:** A Mediterranean chicken breed that holds the record for the highest egg production, widely used in commercial egg farms (KPSC AAO 2017).
786. **Giriraja and Swarnadhara:** Dual-purpose backyard poultry breeds developed by KVAFSU, Bangalore, for rural backyard poultry farming in Karnataka (KPSC AO 2017).
787. **Colostrum:** The first milk secreted after calving, containing high concentrations of immunoglobulins (IgG), proteins, minerals, and Vitamin A, essential for neonatal calf immunity (KPSC AO 2017).
788. **Specific gravity of normal cow milk ranges from 1.028 to 1.030, while that of buffalo milk ranges from 1.030 to 1.032, measured using a lactometer at 15.5°C (KPSC AAO 2017).**
789. **Gerber method:** The standard volumetric method used to estimate fat percentage in milk, utilizing sulfuric acid to dissolve proteins and amyl alcohol to facilitate fat separation (KPSC AO 2014).
790. **Low Temperature Long Time (LTLT) pasteurization requires heating milk to 63°C and holding it for 30 minutes, whereas High Temperature Short Time (HTST) pasteurization requires 72°C for 15 seconds (KPSC AO 2017).**
791. **Silage:** Green fodder preserved under anaerobic conditions through lactic acid fermentation, with an ideal pH range of 3.5 to 4.2 (KPSC AO 2017).
792. **Maize and Sorghum:** The best crops for silage making because of their high soluble carbohydrate (sugar) content, which promotes lactic acid production (KPSC AO 2017).
793. **Hay:** Dried green forage conserved by reducing the moisture content to less than 15%, ensuring safe storage without fermentation or molding (KPSC AAO 2017).
794. **Foot and Mouth Disease (FMD):** A highly contagious viral disease of cloven-hoofed animals caused by an Aphthovirus (Picornaviridae), causing vesicles on the mouth and feet (KPSC AO 2017).
795. **Anthrax, caused by the spore-forming bacterium Bacillus anthracis,:** A fatal zoonotic disease characterized by sudden death and discharge of tarry, non-clotting blood from natural orifices (KPSC AO 2017).
796. **Post-mortem examination:** Strictly prohibited in animals suspected of dying from Anthrax to prevent the vegetative cells from sporulating upon exposure to oxygen (KPSC AO 2017).
797. **Brucellosis (contagious abortion) in cattle:** Caused by Brucella abortus, characterized by abortion in the third trimester of pregnancy and retention of placenta (KPSC AAO 2017).
798. **Mastitis:** A major economic disease of dairy animals characterized by inflammation of the mammary gland, commonly diagnosed using the California Mastitis Test (CMT) (KPSC AO 2017).
799. **Black Quarter (BQ):** An acute, infectious disease of cattle caused by Clostridium chauvoei, characterized by hot, painful, crepitating swellings in the heavy muscles of the hindquarters (KPSC AAO 2015).
800. **Milk fever:** A metabolic disease of high-yielding dairy cows caused by acute calcium deficiency (hypocalcemia) shortly after calving, characterized by subnormal temperature and recumbency (KPSC AO 2017).
801. **Karnataka, the Northern Dry Zone (Zone 3):** The largest agro-climatic zone, covering Vijayapura, Bagalkot, Gadag, and Koppal districts, with its headquarters at Vijayapura. (KPSC AO 2017).

802. **Krishi Bhagya:** Launched in 2014 to secure the livelihoods of dryland farmers through rainwater harvesting, lining of farm ponds with polythene sheets, and micro-irrigation. (KPSC AO 2017).
803. **FRUITS portal (Farmer Registration and Unified Beneficiary Information System) in Karnataka provides a unique Farmer ID (FID) by linking land records from the Bhoomi database. (Latest 2020).**
804. **Headquarters of the Southern Dry Zone (Zone 6) of Karnataka:** Located at the VC Farm in Mandya. (KPSC AAO 2017).
805. **Raitha Siri scheme provides a cash incentive of Rs. 10,000 per hectare (up to 2 hectares) to farmers cultivating minor millets like Foxtail, Little, Kodo, and Barnyard millets. (Latest 2019).**
806. **Hill Zone (Zone 9) of Karnataka receives the second-highest rainfall (1500 to 3000 mm) after the Coastal Zone, with its headquarters at Mudigere. (KPSC AAO 2015).**
807. **First Agricultural University in Karnataka, UAS Bangalore,:** Established in 1963 and formally inaugurated in 1965. (KPSC AO 2014).
808. **University of Agricultural Sciences, Dharwad:** Established in 1986 to cater to the agricultural research and extension needs of northern Karnataka. (KPSC AAO 2017).
809. **University of Horticultural Sciences (UHS), Bagalkot was established in 2008 and:** The first exclusive horticultural university in Karnataka. (KPSC AO 2017).
810. **Coastal Zone (Zone 10):** The wettest agro-climatic zone of Karnataka, receiving an annual rainfall of 3000 to 4700 mm, with its headquarters at Brahavar. (KPSC AO 2017).
811. **The Krishi Bhagya scheme, the recommended polythene sheet thickness for lining farm ponds (Krishi Honda):** 300 to 500 microns to prevent water seepage. (KPSC AAO 2017).
812. **Raita Vidya Nidhi scheme, launched in 2021, provides scholarships to the children of farmers, agricultural laborers, and weavers for higher education. (Latest 2021).**
813. **Ksheera Dhare scheme provides a financial incentive of Rs. 5 per liter of milk supplied to milk cooperative societies by Karnataka dairy farmers. (KPSC AO 2017).**
814. **Keladi Shivappa Nayaka University of Agricultural and Horticultural Sciences (KSNUAHS), Shivamogga:** Established in 2012 by integrating parts of UAS Bangalore and UHS Bagalkot. (KPSC AAO 2017).
815. **Sirsi Supari (Arecanut):** The first agricultural product from Karnataka's hill country to receive a Geographical Indication (GI) tag. (Latest 2019).
816. **Nanjangud Banana (GI tagged) owes its unique taste and aroma to the black clay soil containing high organic matter in Nanjangud taluk, Mysuru. (KPSC AAO 2015).**
817. **Samrakshane Portal:** The dedicated web portal of Karnataka for the implementation and real-time monitoring of the Pradhan Mantri Fasal Bima Yojana (PMFBY). (Latest 2018).
818. **Upper Krishna Project (UKP):** The largest irrigation project in Karnataka, utilizing water from the Krishna River to irrigate Vijayapura, Bagalkot, Raichur, and Kalaburagi. (KPSC AO 2014).
819. **Headquarters of the Northeast Transition Zone (Zone 1) of Karnataka:** Located at Bidar, characterized by laterite soils and red sandy loams. (KPSC AAO 2017).
820. **Northeast Dry Zone (Zone 2):** Headquartered at Raichur and is dominated by deep black and medium black soils with high clay content. (KPSC AO 2014).
821. **Central Dry Zone (Zone 4):** Headquartered at Babbur (Chitradurga) and is majorly characterized by red sandy loams and medium black soils. (KPSC AAO 2015).
822. **Eastern Dry Zone (Zone 5), headquartered at Chintamani,:** Is characterized by gravelly red sandy loam soils and has a major focus on ragi and pulse production. (KPSC AO 2017).
823. **Southern Transition Zone (Zone 7):** Headquartered at Navile, Shivamogga, receiving 800 to 1000 mm of annual rainfall. (KPSC AAO 2017).
824. **Northern Transition Zone (Zone 8), headquartered at Dharwad,:** A highly productive zone suitable for cotton, soybean, and maize cultivation. (KPSC AO 2014).
825. **Karnataka's State Organic Farming Policy:** First introduced in 2004, making it the first state in India to formulate an organic policy, and was revised in 2017. (KPSC AO 2017).
826. **Bhoo Siri scheme provides a subsidy of Rs. 2,500 per acre to promote organic inputs, green manuring, and biological soil health management in Karnataka. (Latest 2020).**
827. **Mysore Jasmine (Mysore Mallige), famous for its unique fragrance and high oil content, received its GI tag under the**

horticulture class. (KPSC AAO 2015).

828. Bhadra Reservoir Project in Lakkavalli provides canal irrigation primarily to Chikkamagaluru, Chitradurga, and Davanagere districts. (KPSC AO 2014).

829. Coorg Orange (GI tagged): Grown as a secondary crop in coffee plantations of Kodagu, Hassan, and Chikkamagaluru districts. (KPSC AAO 2017).

830. Kamalapur Red Banana of Kalaburagi district: A unique variety known for its red peel, sweet pulp, and resistance to Panama wilt, carrying a GI tag. (Latest 2008).

831. Haraka (Kodo Millet) and Navane (Foxtail Millet): The primary target dryland crops promoted under Karnataka's millet incentive schemes. (KPSC AAO 2017).

832. Bhoomi Project, launched in 2000,: Karnataka's land records digitization system that allows real-time mutation and integration with agricultural credit portals. (KPSC AO 2014).

833. Karnataka's Digital Crop Survey relies on the mobile application developed by the e-Governance department to ensure GPS-tagged, farmer-verified crop entry. (Latest 2020).

834. Tungabhadra Dam located in Hosapete, Vijayanagara district,: A multipurpose project serving irrigation needs in Karnataka and Andhra Pradesh. (KPSC AO 2014).

835. Ghataprabha Irrigation Project, with the Hidkal Dam in Belagavi, serves as a vital water source for the dry tracts of Belagavi and Bagalkot. (KPSC AAO 2015).

836. The Jala Amrutha scheme, the Government of Karnataka focuses on water literacy, rejuvenation of water bodies, and massive afforestation. (Latest 2019).

837. UAS Raichur, established in 2008, covers the dryland agricultural research of the Kalyana-Karnataka (formerly Hyderabad-Karnataka) region. (KPSC AAO 2017).

838. Krishi Yantra Dhare scheme provides custom hiring centers (CHCs) at the hobli level to rent agricultural machinery to small and marginal farmers at subsidized rates. (KPSC AO 2017).

839. Karnataka State Seeds Corporation (KSSC), established in 1973,: Responsible for processing, storing, and distributing certified seeds to farmers. (KPSC AO 2014).

840. Suvarna Krishi Yojana focuses on transferring advanced dryland technologies, integrated pest management, and soil testing directly to farmers. (KPSC AAO 2015).

841. Hadagali Mallige (Jasminum auriculatum), cultivated in Huvina Hadagali of Vijayanagara district,: A GI-tagged flower prized for its shelf life and fragrance. (KPSC AO 2017).

842. Malaprabha Project (Renuka Sagar Dam) at Saundatti, Belagavi, provides irrigation water to Dharwad, Gadag, and Bagalkot districts. (KPSC AO 2014).

843. The Panchamrutha concept in farm planning, five core components (soil, water, seed, agronomy, and marketing): Integrated for sustainable output. (KPSC AAO 2017).

844. Raita Mitra groups: Village-level farmer interest groups facilitated by the Karnataka Department of Agriculture to streamline input supply. (KPSC AO 2014).

845. Deccan G20 Initiative on Agriculture and Food Security: Supported by Karnataka's millet promotion strategies showcased in Bengaluru. (Latest 2023).

846. Bangalore Blue Grapes, grown in Bangalore Urban, Rural, and Chikkaballapura, carry a GI tag and: Specifically used for raw consumption and wine-making. (KPSC AAO 2015).

847. Coorg Green Cardamom: A unique variety grown in the Western Ghats of Karnataka, recognized with a GI tag for its intense aroma and bold green color. (KPSC AO 2017).

848. Krishi Marata Vahini: An online price dissemination portal operated by the Karnataka State Agricultural Marketing Board (KSAMB) for real-time APMC prices. (KPSC AO 2014).

849. Nekar Samman Yojana provides financial assistance to handloom and powerloom weavers, many of whom: Linked with cotton farming communities. (Latest 2020).

850. Udupi Jasmine (Udupi Mallige): A GI-tagged species (Jasminum sambac) whose flowers are uniquely strung together using wild ginger fibers. (KPSC AAO 2017).

851. Siri Dhanya project: Represents Karnataka's focused marketing and branding campaign to promote millets nationally and internationally. (Latest 2018).

852. Karnataka State Agricultural Produce Processing and Export Corporation (KAPPEC) facilitates the export of horticultural and processed food items. (KPSC AO 2014).

853. **Chamarajanagar district:** The leading producer of turmeric in southern Karnataka, utilizing dryland irrigation practices. (KPSC AAO 2015).
854. **Monopoly Procurement of Ragi and Paddy under the MSP scheme in Karnataka:** Carried out through registration on the FRUITS portal. (Latest 2020).
855. **Kabini Project in H.D. Kote taluk, Mysuru, provides drinking water and irrigation to Mysuru, Mandya, and Chamarajanagar.** (KPSC AO 2014).
856. **Kagga Rice:** A salt-tolerant traditional paddy variety grown in the coastal estuarine regions of Kumta, Uttara Kannada. (KPSC AAO 2017).
857. **Dharwad Pedha:** A milk-based sweet with a GI tag, representing the dairy industry heritage of northern Karnataka. (KPSC AO 2017).
858. **Mukhyamantri Raitha Vidya Nidhi covers students pursuing professional courses, vocational streams, and post-graduate studies.** (Latest 2021).
859. **Krishhi Bhagya, a farm pond of size 10m x 10m x 3m can collect up to 300 cubic meters of rainwater, sufficient for critical protective irrigation.** (KPSC AAO 2015).
860. **Karnataka Watershed Development Department (WDD) implements the REWARD project (Rejuvenating Watersheds for Agricultural Resilience through Innovative Development).** (KPSC AO 2017).
861. **REWARD project:** Assisted by the World Bank to strengthen watershed management and climate-resilient farming in Karnataka. (KPSC AAO 2017).
862. **Ulavapadu / Local variety selections of horse gram:** Widely promoted in the dry zones of Chitradurga and Tumakuru for contingent cropping. (KPSC AO 2014).
863. **Karnataka State Seeds Corporation operates seed processing plants across the state, with major processing facilities in Raichur and Haveri.** (KPSC AAO 2017).
864. **Malnad Gidda:** A dwarf, disease-resistant, and climate-resilient draft cattle breed native to the hilly Malnad region of Karnataka. (KPSC AO 2017).
865. **The State Organic Policy 2017, Karnataka aims to convert at least 10% of cultivable land in each district into organic farming systems.** (KPSC AO 2017).
866. **Krishhi Honda (Farm Pond) must be constructed at the lowest point of the farm topography to capture maximum surface runoff.** (KPSC AAO 2015).
867. **Karnataka Agri-Business and Food Processing Policy promotes private investment in cold chains, primary processing, and packhouses near production clusters.** (KPSC AO 2017).
868. **Asha Kirana scheme provides crop loan insurance and health security cover to registered tenant and marginal farmers.** (KPSC AAO 2015).
869. **Gajjige / Gajni lands refer to the saline soils in coastal Uttara Kannada where traditional salt-tolerant rice varieties like Kagga: Cultivated.** (KPSC AO 2017).
870. **Sahakari Spandana scheme in Karnataka provides interest-free short-term agricultural cooperative loans up to Rs. 3 lakh through DCC banks.** (Latest 2020).
871. **Mysore Silk GI tag:** Associated with pure mulberry silk produced in the silk weaving factories of Mysuru, utilizing local cocoon crops. (KPSC AAO 2015).
872. **Devarahippargi region in Vijayapura:** A major hub for dryland grape cultivation and raisin production in Karnataka. (KPSC AO 2014).
873. **The Raita Sanjeevini scheme, Karnataka provides ex-gratia financial assistance to farmers who die or get injured during agricultural operations.** (KPSC AO 2017).
874. **Karnataka State Warehousing Corporation (KSWC) offers scientific storage facilities and issues warehouse receipts for post-harvest pledge loans.** (KPSC AO 2014).
875. **Byadagi Chilli (GI tagged):** Known for its high color extraction value (ASTA units) and low pungency, extensively grown in Haveri and Dharwad. (KPSC AAO 2017).
876. **Kari Ishad Mango of Ankola, Uttara Kannada, recently received its GI tag, prized for its thin skin, high pulp content, and sweet aroma.** (Latest 2023).
877. **Krishhi Sanjeevini mobile laboratory vans:** Equipped to test soil, water, and identify pests/diseases directly at the farmer's doorstep. (Latest 2021).
878. **Karnataka Agri-War Room:** Established to address farmer grievances, logistics issues, and provide meteorological advisories in

real-time. (KPSC AAO 2017).

879. **Guledgudd Khana:** A unique textile from Bagalkot associated with handloom cooperatives sourcing local cotton crops. (KPSC AO 2017).

880. **Channapatna Toys GI tag utilizes local ivory wood (*Wrightia tinctoria*), reflecting agro-forestry and craft integration.** (KPSC AAO 2015).

881. **Cost A1 includes all actual out-of-pocket expenses incurred by the owner-operator, such as hired human labor, seeds, fertilizers, and depreciation.** (KPSC AO 2017).

882. **Cost A2:** Calculated by adding the rent paid for leased-in land to Cost A1. (KPSC AAO 2017).

883. **Cost B1:** Is defined as Cost A1 plus the interest on the value of owned capital assets (excluding land). (KPSC AO 2014).

884. **Cost B2:** Obtained by adding the rental value of owned land and rent paid for leased-in land to Cost B1. (KPSC AAO 2015).

885. **Cost C1 includes Cost B1 plus the imputed value of family labor.** (KPSC AO 2017).

886. **Cost C2:** The comprehensive cost calculated as Cost B2 plus the imputed value of family labor. (KPSC AAO 2017).

887. **Cost C3:** The total cost of cultivation, representing Cost C2 plus an additional 10% of Cost C2 as managerial cost. (KPSC AO 2014).

888. **Farm Business Income:** The surplus of gross income over Cost A1 (or Cost A2 for tenant farmers). (KPSC AAO 2015).

889. **Family Labor Income:** The net return to the farm family, calculated as Gross Income minus Cost B2. (KPSC AO 2017).

890. **The Factor-Product relationship, the economic optimum level of input:** Achieved when the Marginal Value Product (MVP) equals the Marginal Input Cost (MIC). (KPSC AO 2017).

891. **The Factor-Factor relationship, the least-cost combination of inputs:** Determined where the Marginal Rate of Technical Substitution (MRTS) equals the input price ratio. (KPSC AAO 2015).

892. **The Product-Product relationship, the optimum combination of enterprises:** Located where the Marginal Rate of Product Transformation (MRPT) equals the product price ratio. (KPSC AO 2014).

893. **Law of Variable Proportions operates in the short run when one factor of production:** Varied while keeping all other inputs constant. (KPSC AAO 2017).

894. **Stage I of production, the Average Physical Product (APP):** Increasing, and the Marginal Physical Product (MPP) is greater than APP ($MPP > APP$). (KPSC AO 2014).

895. **Stage II of production:** The only rational stage of production, where both APP and MPP are decreasing, and MPP is positive ($MPP > 0$). (KPSC AO 2017).

896. **Stage III of production:** Irrational because the Marginal Physical Product (MPP) is negative, causing the Total Physical Product (TPP) to decline. (KPSC AAO 2017).

897. **Returns to Scale represent a long-run concept where all inputs:** Varied in the same proportion. (KPSC AO 2014).

898. **Marketed Surplus:** The actual quantity of agricultural produce that a farmer sells in the market after meeting family needs. (KPSC AO 2017).

899. **Marketable Surplus:** The theoretical surplus available for sale after deducting the farmer's genuine consumption, seed, and feed requirements. (KPSC AAO 2015).

900. **Small and marginal farmers, marketed surplus:** Greater than marketable surplus, resulting in distress or forced sales to meet immediate cash needs. (KPSC AO 2014).

901. **Horizontal Market Integration occurs when two or more firms operating at the same stage of the marketing channel merge.** (KPSC AAO 2017).

902. **Vertical Market Integration occurs when a firm combines two or more stages of the marketing channel under its control (e.g., processing and retailing).** (KPSC AO 2017).

903. **Conglomerate Integration involves the merger of firms operating in completely unrelated markets or industries.** (KPSC AAO 2015).

904. **Price Spread:** The difference between the price paid by the final consumer and the price received by the producer. (KPSC AO 2014).

905. **Producer's Share in Consumer Rupee:** Calculated as the ratio of the producer's price to the consumer's price, multiplied by 100. (KPSC AAO 2017).

906. **Agricultural Produce (Grading and Marking) Act (AGMARK):** Enacted in 1937 to set quality standards for agricultural products in India. (KPSC AO 2017).

907. **Headquarters of the Directorate of Marketing and Inspection (DMI):** Located at Faridabad, Haryana. (KPSC AAO 2015).
908. **Electronic National Agriculture Market (e-NAM):** Launched on April 14, 2016, to create a unified national market for agricultural commodities. (KPSC AO 2017).
909. **Small Farmers Agribusiness Consortium (SFAC):** The lead implementing agency for the e-NAM platform. (KPSC AAO 2017).
910. **3 R's of Credit essential for loan evaluation:** Returns, Repayment Capacity, and Risk-bearing Ability. (KPSC AO 2014).
911. **5 C's of Credit analyzed by institutional lenders:** Character, Capacity, Capital, Collateral, and Conditions. (KPSC AAO 2015).
912. **7 P's of Credit include Purpose, Person, Productive aspect, Payment phase, Protection, Personality, and Prospects.** (KPSC AO 2017).
913. **NABARD (National Bank for Agriculture and Rural Development):** Established on July 12, 1982, based on the recommendations of the B. Sivaraman Committee. (KPSC AO 2014).
914. **Regional Rural Banks (RRBs):** Established on October 2, 1975, based on the recommendations of the M. Narasimham Committee. (KPSC AAO 2017).
915. **First RRB in India:** The Prathama Bank, sponsored by Syndicate Bank and located in Moradabad, Uttar Pradesh. (KPSC AO 2014).
916. **Kisan Credit Card (KCC) scheme:** Introduced in August 1998 on the recommendations of the R.V. Gupta Committee. (KPSC AO 2017).
917. **Short-term loans under the KCC scheme attract an interest rate of 7%, which reduces to 4% for farmers who repay their loans promptly.** (KPSC AAO 2017).
918. **Lead Bank Scheme:** Introduced in late 1969 based on the recommendations of the Nariman Committee. (KPSC AO 2014).
919. **The WTO Agreement on Agriculture (AoA), domestic support measures:** Categorized into Green, Amber, and Blue boxes. (KPSC AAO 2015).
920. **Green Box:** Contains subsidies that cause minimal or no trade distortion, such as public research, pest control, and environmental programs. (KPSC AO 2017).
921. **Amber Box covers trade-distorting support measures, such as Minimum Support Prices (MSP) and input subsidies on fertilizer and power.** (KPSC AAO 2017).
922. **WTO rules, the de minimis limit for Amber Box subsidies:** 5% of agricultural production value for developed countries and 10% for developing countries. (KPSC AO 2017).
923. **Blue Box represents Amber Box subsidies that:** Linked to production-limiting programs and are currently exempt from reduction commitments. (KPSC AAO 2015).
924. **Commission for Agricultural Costs and Prices (CACP):** The statutory body that recommends MSPs for 22 mandated crops and SRP/FRP for sugarcane. (KPSC AO 2017).
925. **CACP:** Originally established in January 1965 as the Agricultural Prices Commission and was renamed in 1985. (KPSC AO 2014).
926. **Price Elasticity of Demand measures the responsiveness of the quantity demanded of a commodity to a change in its price.** (KPSC AO 2014).
927. **If the price elasticity of demand:** Less than one ($E_p < 1$), the demand is inelastic, which is typical for essential foodgrains. (KPSC AAO 2017).
928. **Cross Elasticity of Demand measures the responsiveness of demand for one good to a change in the price of another related good.** (KPSC AO 2017).
929. **If cross elasticity:** Positive, the two goods are substitutes (e.g., tea and coffee); if negative, they are complements (e.g., tractor and diesel). (KPSC AAO 2015).
930. **Engel's Law of Family Expenditure states that as family income increases, the proportion of income spent on food decreases.** (KPSC AO 2014).
931. **Capital-Output Ratio measures the amount of capital required to produce one unit of output, reflecting investment efficiency.** (KPSC AAO 2017).
932. **Marginal Cost (MC):** The change in total cost resulting from producing one additional unit of output. (KPSC AO 2017).
933. **When Marginal Cost equals Average Cost ($MC = AC$), the Average Cost:** At its minimum point. (KPSC AAO 2015).
934. **Opportunity Cost:** The value of the next best alternative foregone when a choice is made. (KPSC AO 2014).
935. **Principle of Equi-Marginal Returns states that resources should be allocated such that marginal returns from all enterprises:**

Equal. (KPSC AO 2017).

936. **Giffen Goods:** Inferior goods for which demand increases as their price rises, violating the basic Law of Demand. (KPSC AAO 2017).

937. **Isoquants (equal product curves):** Represent all possible combinations of two inputs that yield the same level of output. (KPSC AO 2014).

938. **Slope of an isoquant:** Called the Marginal Rate of Technical Substitution (MRTS). (KPSC AAO 2015).

939. **Isocost Lines:** Represent all combinations of two inputs that can be purchased with a given total outlay or budget. (KPSC AO 2017).

940. **Isoclines are loci of points on different isoquants where the marginal rate of technical substitution (MRTS):** Constant. (KPSC AAO 2017).

941. **Ridge Lines represent the economic boundary of production, beyond which marginal product of either input:** Negative. (KPSC AO 2014).

942. **Expansion Path:** The locus of least-cost combinations of inputs for different levels of output when input prices remain constant. (KPSC AAO 2015).

943. **Production Function:** The technical mathematical relationship showing the maximum output obtainable from a given set of inputs. (KPSC AO 2017).

944. **A Cobb-Douglas Production Function, the sum of the exponents (alpha + beta) measures the returns to scale. (KPSC AAO 2017).**

945. **If the sum of exponents in Cobb-Douglas:** Equal to 1, the production function exhibits Constant Returns to Scale. (KPSC AO 2014).

946. **Linear Programming:** A mathematical technique used for allocating limited resources among competing activities to maximize profit or minimize cost. (KPSC AO 2017).

947. **G.B. Dantzig developed the Simplex Method in 1947, which:** Widely used to solve linear programming problems. (KPSC AAO 2015).

948. **Depreciation:** The permanent decrease in the value of an asset over time due to wear and tear, obsolescence, and passage of time. (KPSC AO 2014).

949. **Straight-Line Method of depreciation assumes that an asset loses value at a constant rate each year. (KPSC AAO 2017).**

950. **Working Capital:** Refers to the funds required for day-to-day operations of the farm, such as buying seeds, fertilizers, and hiring labor. (KPSC AO 2017).

951. **Fixed Capital:** Represents long-term investments in farm assets like land, buildings, tractors, and tube wells. (KPSC AAO 2015).

952. **Liquidity:** The ability of a farm business to meet its short-term financial obligations as they fall due. (KPSC AO 2014).

953. **Solvency:** Refers to the ability of a farm business to meet all its long-term liabilities using its total assets. (KPSC AAO 2017).

954. **Balance Sheet:** A financial statement that displays the assets, liabilities, and net worth of a farm at a specific point in time. (KPSC AO 2017).

955. **Income Statement (Profit and Loss Statement) summarizes the revenues and expenses of a farm business over a specific operating period. (KPSC AAO 2015).**

956. **Partial Budgeting:** Used to evaluate the financial implications of minor changes in the farm business, such as introducing a new crop. (KPSC AO 2014).

957. **Complete Budgeting involves planning the entire farm business from scratch, usually done when major structural changes:** Proposed. (KPSC AAO 2017).

958. **Amortization:** The process of spreading out a loan into a series of equal periodic payments over time. (KPSC AO 2017).

959. **Net Present Value (NPV):** The difference between the present value of cash inflows and the present value of cash outflows over time. (KPSC AAO 2015).

960. **Internal Rate of Return (IRR):** The discount rate at which the Net Present Value of an investment project becomes zero. (KPSC AO 2014).

961. **AgriSURE Fund (Agricultural Startup Fund for Investment in Rural Enterprises):** Launched by NABARD in 2024 with a corpus of Rs. 750 crore to support agritech startups. (Latest 2024).

962. **Nano Urea and Nano DAP, developed by IFFCO, have particles sized between 20 to 50 nanometers and exhibit a nitrogen/phosphorus use efficiency exceeding 80%. (Latest 2024).**

963. **PM-PRANAM scheme aims to incentivize states to promote balanced fertilizer usage, reduction of chemical fertilizers, and promotion of organic alternatives. (Latest 2023).**

964. **The PM-KISAN scheme, eligible farmer families receive a financial benefit of Rs. 6,000 per year, paid in three equal installments of Rs. 2,000 every four months. (Latest 2019).**
965. **Minimum Support Price (MSP) for Paddy (Common) has been increased to Rs. 2,300 per quintal for the 2024-25 marketing season. (Latest 2024).**
966. **MSP for Paddy (Grade A):** Fixed at Rs. 2,320 per quintal for the 2024-25 crop year. (Latest 2024).
967. **MSP for Ragi (Finger Millet) has been set at Rs. 4,290 per quintal for the 2024-25 marketing season, reflecting the government's push for nutri-cereals. (Latest 2024).**
968. **MSP for Jowar (Hybrid):** Fixed at Rs. 3,371 per quintal for the 2024-25 crop marketing year. (Latest 2024).
969. **The 2024-25 crop season, the MSP of Pigeon Pea (Tur/Arhar) has been increased to Rs. 7,550 per quintal. (Latest 2024).**
970. **Digital Agriculture Mission (AgriStack):** Being implemented nationwide to create a centralized database linking land records, crop surveys, and farmer IDs. (Latest 2021).
971. **United Nations declared 2023 as the International Year of Millets (IYM 2023) following a proposal sponsored by the Government of India. (Latest 2023).**
972. **India's first Nano DAP (Liquid) plant:** Inaugurated by the Union Home Minister at Kalol, Gujarat, developed by IFFCO. (Latest 2024).
973. **Agriculture Infrastructure Fund (AIF), launched in 2020, provides a medium-to-long term debt financing facility of Rs. 1 lakh crore for post-harvest management infrastructure. (Latest 2020).**
974. **PM formalisation of Micro Food Processing Enterprises (PMFME) scheme offers a 35% credit-linked subsidy up to Rs. 10 lakh to individual micro-enterprises. (KPSC AAO 2017).**
975. **PM Matsya Sampada Yojana (PMMSY):** A flagship scheme for the fisheries sector with an investment of Rs. 20,050 crore over five years (2020-21 to 2024-25). (Latest 2020).
976. **The Namo Shetkari Mahasanman Nidhi Yojana, Maharashtra provides an additional Rs. 6,000 per year to farmers, matching the PM-KISAN database. (Latest 2023).**
977. **Namo Shetkari model of integrating state and central databases has been evaluated by Karnataka for future direct benefit transfer (DBT) expansions. (Latest 2023).**
978. **Global Biofuels Alliance (GBA):** Launched at the G20 New Delhi Summit in September 2023 to accelerate the uptake of biofuels globally. (Latest 2023).
979. **Ethanol Blending in petrol in India has reached a target of 12% in 2023-24, with a roadmap to achieve 20% blending by 2025-26. (Latest 2024).**
980. **Kari Ishad Mango of Ankola, Uttara Kannada, received the Geographical Indication (GI) tag in 2023 under the horticultural class. (Latest 2023).**
981. **State Energy Conservation Mission has partnered with Karnataka's agriculture department to distribute energy-efficient solar water pumps under the PM-KUSUM scheme. (Latest 2023).**
982. **Component-B of PM-KUSUM, individual farmers:** Supported to install standalone solar agriculture pumps up to 7.5 HP with 60% subsidy. (Latest 2024).
983. **Maharishi Initiative (Millets and Other Ancient Grains International Research Initiative):** Launched during the G20 MACS meeting in Varanasi in 2023. (Latest 2023).
984. **National Quantum Mission has started exploring quantum sensing in agriculture for real-time soil health monitoring and crop yield estimation. (Latest 2023).**
985. **The PM-PRANAM scheme, 50% of the subsidy savings achieved by a state through reduced chemical fertilizer consumption: Returned as a grant. (Latest 2023).**
986. **Soil Health Card mobile app has been upgraded to enable real-time geotagging of soil samples and instant digital delivery of recommendations. (Latest 2023).**
987. **Union Budget 2024-25 announced the development of 10,000 Bio-Input Resource Centers (BRCs) to support natural farming across India. (Latest 2024).**
988. **The National Mission on Edible Oils - Oil Palm (NMEO-OP), the government aims to expand oil palm cultivation to 10 lakh hectares by 2025-26. (Latest 2021).**
989. **Bhavantar Bhugtan Yojana:** First launched by Madhya Pradesh as a price deficit payment scheme, paying farmers the difference between MSP and market price. (KPSC AO 2017).

990. **Unified Portal for Agricultural Statistics (UPAg)**: Launched in 2023 to provide standardized, real-time data for agricultural trade and policy decisions. (Latest 2024).
991. **Krishi-Decision Support System (Krishi-DSS)**: Launched in 2024 to provide integrated geospatial data for crop mapping, weather alerts, and soil health. (Latest 2024).
992. **Mission Organic Value Chain Development for Northeastern Region (MOVCDNER) promotes commercial organic production for export.** (KPSC AAO 2017).
993. **Sub-Mission on Agricultural Mechanization (SMAM) provides financial assistance of 40% to 50% for purchasing agricultural machinery and implements.** (KPSC AO 2017).
994. **Krishi-24/7**: India's first AI-powered monitoring system for agricultural news, helping the government identify crop distress and policy impacts. (Latest 2024).
995. **Saffron cultivation has been extended from Jammu & Kashmir to parts of the Northeast (Sikkim) under the National Saffron Mission.** (KPSC AO 2017).
996. **National Beekeeping and Honey Mission (NBHM)**: Implemented under the Atmanirbhar Bharat package to promote scientific beekeeping (Sweet Revolution). (KPSC AAO 2017).
997. **Namo Drone Didi scheme, launched in late 2023, provides agricultural drones and pilot training to women Self-Help Groups (SHGs) for spraying liquid fertilizers.** (Latest 2023).
998. **One Nation One Fertilizer scheme mandates all fertilizer companies to market their products under the single brand name 'BHARAT'.** (Latest 2024).
999. **PM-KUSUM Component-C provides for the solarization of grid-connected agriculture pumps, allowing farmers to sell surplus solar power back to DISCOMs.** (Latest 2023).
1000. **Horticulture Area Production Information System (HAPIS)**: Upgraded in 2024 to streamline real-time reporting of horticultural crops from states. (Latest 2024).

--- Expected Questions ---

Expected Questions Section

Top 500 Expected MCQs

This section contains a compiled pool of exactly 500 expected MCQs covering the core subjects.

Q1. Which of the following is the largest agro-climatic zone in Karnataka in terms of geographical area?

- A. Zone 9 - Hill Zone
- B. Zone 3 - Northern Dry Zone
- C. Zone 6 - Southern Dry Zone
- D. Zone 8 - Northern Transitional Zone
- **Correct Answer: B**
- **Explanation:** The Northern Dry Zone (Zone 3) is the largest agro-climatic zone in Karnataka, covering an area of about 4.78 million hectares across districts like Bagalkot, Vijayapura, Gadag, Dharwad, Koppal, and parts of Belagavi. [KPSC AO 2014]

Q2. In finger millet (Ragi), what is the critical weed-crop competition period under transplanted conditions?

- A. 0 to 10 days after transplanting
- B. 25 to 35 days after transplanting
- C. 45 to 60 days after transplanting
- D. 70 to 85 days after transplanting
- **Correct Answer: B**
- **Explanation:** The critical period of weed competition in transplanted Ragi is between 25 and 35 days after transplanting. Managing weeds during this window is crucial to prevent yield losses of up to 45%. [KPSC AAO 2017]

Q3. The chemical name of the nitrification inhibitor 'N-Serve' used to improve nitrogen use efficiency is:

- A. Dicyandiamide (DCD)
- B. Thiourea
- C. 2-chloro-6-(trichloromethyl) pyridine
- D. Neem oil cake extract
- **Correct Answer: C**
- **Explanation:** N-Serve is the commercial trade name of Nitrapyrin, which is chemically 2-chloro-6-(trichloromethyl) pyridine. It suppresses Nitrosomonas bacteria to prevent the rapid oxidation of ammonium to nitrite. [KPSC AAO 2019]

Q4. Which crop is utilized as an indicator plant for detecting Boron deficiency in agricultural soils?

- A. Maize
- B. Sunflower
- C. Potato
- D. Mustard

- **Correct Answer: B**

- **Explanation:** Sunflower (*Helianthus annuus*) is highly sensitive to Boron deficiency. Symptoms like chaffy heads, poor seed set, and hollow stems appear rapidly, making it a reliable indicator plant. [KPSC AO 2014]

Q5. What is the equivalent acidity of Ammonium chloride fertilizer?

- A. 80
- B. 93
- C. 110
- D. 128

- **Correct Answer: D**

- **Explanation:** Ammonium chloride (26% N) has an equivalent acidity of 128, which is the highest among common nitrogenous fertilizers. It requires 128 kg of calcium carbonate to neutralize the acidity produced by 100 kg of this fertilizer. [KPSC AAO 2015]

Q6. The pre-emergence herbicide Pendimethalin belongs to which chemical group?

- A. Sulfonylureas
- B. Triazines
- C. Dinitroanilines
- D. Phenoxy acetic acids

- **Correct Answer: C**

- **Explanation:** Pendimethalin belongs to the dinitroaniline class of herbicides. It works by inhibiting cell division (mitosis) in growing root and shoot tips of weeds before they emerge. [KPSC AO 2017]

Q7. Which of the following is a stomatal-closing type of antitranspirant?

- A. Kaolin
- B. Phenylmercuric acetate (PMA)
- C. Mobileaf
- D. Silicone

- **Correct Answer: B**

- **Explanation:** Phenylmercuric acetate (PMA) at low concentrations and Absciscic acid (ABA) act as stomatal-closing antitranspirants. They induce stomatal closure to reduce water loss while allowing limited carbon assimilation. [KPSC AAO 2017]

Q8. The permanent wilting point (PWP) of soil corresponds to a soil moisture tension of:

- A. -0.33 bar
- B. -1.0 bar
- C. -15 bars
- D. -31 bars

- **Correct Answer: C**

- **Explanation:** At the permanent wilting point, the soil moisture tension is approximately -15 bars (or -1.5 MPa). Plants cannot exert enough suction to extract water from soil pores at this tension. [KPSC AO 2014]

Q9. Under dryland conditions of Karnataka, what is the recommended intercropping system and row ratio for Rabi Jowar + Chickpea?

- A. 1:1

- B. 2:1
- C. 3:1
- D. 4:2
- **Correct Answer: B**
- **Explanation:** In the Northern Dry Zone of Karnataka, Rabi Jowar + Chickpea intercropping in a 2:1 row ratio is highly recommended for Vertisols to optimize yield stability and soil resource use. [KPSC AO 2019]

Q10. The enzyme nitrogenase, which catalyzes biological nitrogen fixation in legume root nodules, requires which two metallic cofactors?

- A. Iron and Zinc
- B. Iron and Molybdenum
- C. Copper and Manganese
- D. Molybdenum and Boron
- **Correct Answer: B**
- **Explanation:** The nitrogenase enzyme is a complex metalloprotein consisting of two components: the Fe-protein (dinitrogenase reductase) and the MoFe-protein (dinitrogenase), requiring both Iron (Fe) and Molybdenum (Mo) cofactors. [KPSC AO 2019]

Q11. The plant water use efficiency (WUE) follows which of the order under dry environments?

- A. C3 > C4 > CAM
- B. C4 > C3 > CAM
- C. CAM > C4 > C3
- D. CAM > C3 > C4
- **Correct Answer: C**
- **Explanation:** CAM plants (like pineapple, agave) have the highest WUE (100-150 g water/g dry matter) because they open stomata at night. C4 plants (like maize, sorghum) are intermediate, and C3 plants (like rice, wheat) have the lowest WUE. [KPSC AO 2017]

Q12. Which of the following is a selective, systemic, post-emergence herbicide used to control broadleaf weeds?

- A. Paraquat
- B. Glyphosate
- C. 2,4-Dichlorophenoxyacetic acid
- D. Alachlor
- **Correct Answer: C**
- **Explanation:** 2,4-D is a selective, systemic herbicide that targets broadleaf (dicot) weeds. It mimics auxins, causing uncontrolled growth (epinasty) and eventual death in dicots while sparing monocot cereals. [KPSC AO 2014]

Q13. The critical growth stage for water in Groundnut is:

- A. Seedling stage
- B. Pegging to pod formation stage
- C. Maturity stage
- D. Stooling stage
- **Correct Answer: B**

- **Explanation:** For groundnut, the pegging and pod development stage (occurring 40 to 70 days after sowing) is the most critical for soil moisture. Moisture stress at this stage leads to poor peg penetration and unfilled pods ('pop pods'). [KPSC AO 2017]

Q14. The standard meteorological week (SMW) corresponding to the onset of the South-West Monsoon in Karnataka (around June 1st week) is usually:

- A. Week 18
- B. Week 22
- C. Week 26
- D. Week 30
- **Correct Answer: B**
- **Explanation:** Week 22 in the Standard Meteorological Calendar corresponds to May 28 to June 3, which is the historical onset period of the South-West Monsoon over coastal and southern Karnataka. [KPSC AAO 2015]

Q15. What is the biuret content limit in urea fertilizer meant for foliar spray applications?

- A. 0.5%
- B. 1.0%
- C. 1.5%
- D. 2.0%
- **Correct Answer: B**
- **Explanation:** While the standard biuret limit for soil application of urea is 1.5%, for foliar spray applications, the biuret content should not exceed 1.0% to avoid phytotoxic burning of leaves. [KPSC AAO 2015]

Q16. The biological control of carrot grass (*Parthenium hysterophorus*) is successfully achieved using which agent?

- A. *Neochetina eichhorniae*
- B. *Zygogramma bicolorata*
- C. *Dactylopius opuntiae*
- D. *Cyrtobagous salviniae*
- **Correct Answer: B**
- **Explanation:** *Zygogramma bicolorata*, commonly known as the Mexican beetle, is a leaf-feeding chrysomelid beetle that defoliates *Parthenium* plants, suppressing its growth and seed production. [KPSC AO 2014]

Q17. Which of the following cropping systems is characterized by planting a second crop before the harvest of the standing first crop?

- A. Sequence cropping
- B. Relay cropping
- C. Intercropping
- D. Mixed cropping
- **Correct Answer: B**
- **Explanation:** Relay cropping involves sowing the second crop when the first crop has reached its reproductive stage but before harvest. An example is sowing chickpea/lathyrus in standing rice fields (paira/utera cropping). [KPSC AAO 2019]

Q18. The crop coefficient (Kc) of a crop is mathematically defined as the ratio of:

- A. Reference ET to Crop ET
- B. Crop ET to Reference ET
- C. Transpiration to Evaporation
- D. Yield to Consumptive Water Use
- **Correct Answer: B**
- **Explanation:** The crop coefficient (K_c) is the ratio of crop potential evapotranspiration (ET_c) to the reference crop evapotranspiration (ET_o). It varies with the crop type, growth stage, and climate. [KPSC AO 2017]

Q19. In the System of Rice Intensification (SRI), the age of seedlings used for transplanting is:

- A. 10 to 12 days
- B. 21 to 25 days
- C. 30 to 35 days
- D. 40 to 45 days
- **Correct Answer: A**
- **Explanation:** SRI emphasizes transplanting very young, single seedlings (with seed sac attached) aged between 10 and 12 days (2-leaf stage) to preserve their high tillering and root growth potential. [KPSC AO 2019]

Q20. What is the nitrogen content (%) in Ammonium Sulfate fertilizer?

- A. 18%
- B. 20.6%
- C. 25%
- D. 46%
- **Correct Answer: B**
- **Explanation:** Ammonium sulfate contains 20.6% nitrogen and 24% sulfur. It is an excellent source of both nitrogen and sulfur, especially suitable for sulfur-loving crops like oilseeds and tea. [KPSC AAO 2017]

Q21. The weed Echinochloa crus-galli is a major morphological mimic in the fields of:

- A. Wheat
- B. Rice
- C. Sorghum
- D. Sugarcane
- **Correct Answer: B**
- **Explanation:** Echinochloa crus-galli (barnyard grass) is a primary mimic weed in rice fields. In its vegetative stage, it looks identical to rice seedlings, making manual weeding difficult. [KPSC AO 2019]

Q22. Which clay mineral is responsible for the high cracking and swelling properties of Vertisols in Northern Karnataka?

- A. Kaolinite
- B. Illite
- C. Montmorillonite
- D. Gibbsite
- **Correct Answer: C**

- **Explanation:** Montmorillonite (a 2:1 expanding silicate clay mineral of the smectite group) is dominant in Vertisols. Its structure allows water molecules to enter between layers, causing swelling and cracking upon drying. [KPSC AAO 2019]

Q23. The primary objective of 'nipping' in chickpea (Bengal gram) is to:

- A. Prevent vegetative growth to conserve water
- B. Promote lateral branching and increase pod bearing sites
- C. Induce early flowering
- D. Facilitate mechanical harvesting

• **Correct Answer: B**

- **Explanation:** Nipping involves plucking the apical buds of chickpea plants when they are 30-40 days old. This removes apical dominance, encouraging lateral branching and more flowers and pods. [KPSC AAO 2017]

Q24. The pre-plant incorporation (PPI) herbicide commonly used to control weeds in pulses and oilseeds is:

- A. Pendimethalin
- B. Fluchloralin
- C. Atrazine
- D. Glyphosate

• **Correct Answer: B**

- **Explanation:** Fluchloralin (trade name Basalin) is applied as a pre-plant incorporation herbicide. It is incorporated into the top soil layers before sowing to prevent volatilization loss from solar exposure. [KPSC AO 2017]

Q25. Which type of water is held between field capacity (-0.33 bar) and hygroscopic coefficient (-31 bar) in soil?

- A. Gravitational water
- B. Capillary water
- C. Hygroscopic water
- D. Runoff water

• **Correct Answer: B**

- **Explanation:** Capillary water is held between -0.33 bar (field capacity) and -31 bar. The portion of capillary water held between -0.33 bar and -15 bar (wilting point) is available to plants. [KPSC AAO 2017]

Q26. The moisture content at field capacity is measured at a soil suction pressure of:

- A. -0.1 bar
- B. -0.33 bar
- C. -1.0 bar
- D. -15 bars

• **Correct Answer: B**

- **Explanation:** Field capacity refers to the amount of water remaining in the soil after excess gravitational water has drained away, representing a soil suction of -0.33 bar (or -0.1 bar for sandy soils). [KPSC AO 2014]

Q27. Which of the following is a C4 plant?

- A. Rice
- B. Wheat

- C. Sugarcane
- D. Potato

• **Correct Answer: C**

- **Explanation:** Sugarcane is a C₄ plant. It uses the Hatch-Slack pathway for carbon dioxide fixation, which provides high efficiency under high light and temperature conditions. Rice, wheat, and potato are C₃ plants. [KPSC AAO 2019]

Q28. The critical limit of organic carbon (%) for Indian soils to categorize them as 'low soil organic carbon status' is:

- A. < 0.2%
- B. < 0.5%
- C. < 0.75%
- D. < 1.0%

• **Correct Answer: B**

- **Explanation:** Soils containing less than 0.5% organic carbon are categorized as low in organic carbon status, while those with 0.5% to 0.75% are medium, and >0.75% are high. [KPSC AAO 2015]

Q29. Which of the following nitrogenous fertilizers is a neutral fertilizer and is commercially known as 'Kisan Khad'?

- A. Urea
- B. Ammonium sulfate
- C. Calcium ammonium nitrate (CAN)
- D. Sodium nitrate

• **Correct Answer: C**

- **Explanation:** Calcium ammonium nitrate (CAN) contains 25% Nitrogen. Half of the nitrogen is in ammonium form and half in nitrate form. It is neutral because the calcium carbonate component neutralizes its acidifying effect. [KPSC AO 2017]

Q30. Which weed is a root total parasite that commonly attacks solanaceous crops like tobacco and tomato?

- A. Cuscuta
- B. Orobancha
- C. Striga
- D. Loranthus

• **Correct Answer: B**

- **Explanation:** Orobancha (broomrape) is a chlorophyll-free, root total parasite. It attaches to the roots of host plants, particularly solanaceous (tobacco, tomato, brinjal) and brassica crops, depleting nutrients. [KPSC AO 2017]

Q31. The crop variety 'GPU-28' is a blast-resistant variety of which crop?

- A. Rice
- B. Finger Millet (Ragi)
- C. Sorghum
- D. Pearl Millet

• **Correct Answer: B**

- **Explanation:** GPU-28 is a popular, blast-resistant, high-yielding variety of finger millet (Ragi) developed by UAS Bangalore, widely cultivated in Southern Karnataka. [KPSC AO 2014]

Q32. The process of nitrogen transformation from inorganic ammonium to nitrate forms by soil microbes is known as:

- A. Ammonification
- B. Nitrification
- C. Denitrification
- D. Nitrogen fixation
- **Correct Answer: B**
- **Explanation:** Nitrification is a two-step aerobic biological process: first, ammonium is converted to nitrite by Nitrosomonas, and second, nitrite is converted to nitrate by Nitrobacter. [KPSC AAO 2019]

Q33. Which of the following instruments is used to measure solar radiation intensity?

- A. Pyranometer
- B. Anemometer
- C. Psychrometer
- D. Barometer
- **Correct Answer: A**
- **Explanation:** A pyranometer is an instrument designed to measure solar irradiance on a planar surface, capturing global solar radiation (both direct and diffuse). [KPSC AO 2019]

Q34. The type of drought that occurs due to the prolonged absence of rainfall leading to depletion of ground water and drying of surface reservoirs is:

- A. Meteorological drought
- B. Agricultural drought
- C. Hydrological drought
- D. Socio-economic drought
- **Correct Answer: C**
- **Explanation:** Hydrological drought is associated with the effects of precipitation periods on water supply, marked by depletion of surface water, streamflow, and groundwater reserves. [KPSC AO 2017]

Q35. For rainfed sowing of Ragi, what is the recommended seed rate when using a seed drill?

- A. 5 kg/ha
- B. 10 kg/ha
- C. 15 kg/ha
- D. 20 kg/ha
- **Correct Answer: B**
- **Explanation:** Under rainfed conditions, drilling or broadcasting ragi requires a seed rate of 10 kg/ha to ensure optimal plant stand, whereas transplanting requires only 5 kg/ha. [KPSC AAO 2017]

Q36. Which of the following micro-irrigation systems is most suitable for close-spaced field crops like sugarcane and vegetables?

- A. Drip irrigation
- B. Sprinkler irrigation
- C. Border strip irrigation

- D. Surge irrigation
- **Correct Answer: A**

• **Explanation:** Drip/trickle irrigation is highly efficient (90-95% efficiency) and applies water directly to the base of the crop root zones, making it ideal for high-value rows and closely spaced horticultural crops. [KPSC AAO 2019]

Q37. The principal weed of wheat fields, *Phalaris minor*, developed resistance against which herbicide in India?

- A. 2,4-D
- B. Atrazine
- C. Isoproturon
- D. Sulfosulfuron
- **Correct Answer: C**

• **Explanation:** *Phalaris minor* developed widespread resistance to Isoproturon in the rice-wheat cropping systems of Indo-Gangetic plains, first documented by Dr. R.K. Malik in 1993. [KPSC AO 2019]

Q38. The fertilizer Single Super Phosphate (SSP) contains what percentage of Phosphorus (P_2O_5)?

- A. 16%
- B. 32%
- C. 46%
- D. 60%
- **Correct Answer: A**

• **Explanation:** Single Super Phosphate (SSP) contains 16% water-soluble P_2O_5 , 12% sulfur, and 21% calcium, acting as an important phosphorus and sulfur source for crops. [KPSC AAO 2017]

Q39. The weed *Parthenium hysterophorus* is native to which region?

- A. East Africa
- B. Tropical America / Mexico
- C. South-East Asia
- D. Mediterranean region
- **Correct Answer: B**

• **Explanation:** *Parthenium hysterophorus* (carrot grass) is an invasive weed native to Mexico and tropical America. It entered India as a contaminant in wheat imported under the PL-480 scheme in the 1950s. [KPSC AAO 2015]

Q40. Which of the following is a non-selective, systemic, post-emergence herbicide?

- A. Paraquat
- B. Atrazine
- C. Glyphosate
- D. Fluchloralin
- **Correct Answer: C**

• **Explanation:** Glyphosate is a non-selective, systemic, post-emergence herbicide. It is translocated throughout the plant to kill roots and shoots. Paraquat is non-selective but is a contact herbicide. [KPSC AAO 2015]

Q41. The relative humidity of the air is measured using which instrument?

- A. Pyranometer
- B. Psychrometer
- C. Tensiometer
- D. Lysimeter
- **Correct Answer: B**
- **Explanation:** A psychrometer (a type of hygrometer with dry and wet bulb thermometers) is used to measure the relative humidity of the atmosphere. [KPSC AAO 2017]

Q42. What is the primary product of denitrification under highly anaerobic waterlogged soil conditions?

- A. Nitric oxide (NO)
- B. Nitrogen gas (N₂)
- C. Nitrous oxide (N₂O)
- D. Ammonia gas (NH₃)
- **Correct Answer: B**
- **Explanation:** Under strict anaerobic conditions, denitrifying bacteria reduce nitrate completely to nitrogen gas (N₂), which is lost to the atmosphere. Under partially anaerobic conditions, nitrous oxide (N₂O) may escape. [KPSC AO 2017]

Q43. The crop variety 'DCH-32' is a famous interspecific cotton hybrid developed in Karnataka. What are its parents?

- A. G. hirsutum x G. barbadense
- B. G. arboreum x G. herbaceum
- C. G. hirsutum x G. herbaceum
- D. G. barbadense x G. arboreum
- **Correct Answer: A**
- **Explanation:** DCH-32 is a popular interspecific hybrid between Gossypium hirsutum (DS-28) and Gossypium barbadense (SB-289-E) developed by UAS Dharwad. [KPSC AAO 2017]

Q44. The soil moisture tension at field capacity in sandy soils is around:

- A. -0.1 bar
- B. -0.33 bar
- C. -1.0 bar
- D. -15 bars
- **Correct Answer: A**
- **Explanation:** For sandy/coarse-textured soils, field capacity is reached at a lower soil moisture tension of -0.1 bar, whereas for medium to fine-textured clay soils it is -0.33 bar. [KPSC AO 2019]

Q45. Which of the following organic acids is primarily secreted by Phosphate Solubilizing Bacteria (PSB) to solubilize soil phosphorus?

- A. Acetic acid
- B. Gluconic acid
- C. Sulfuric acid
- D. Hydrochloric acid

- **Correct Answer: B**

- **Explanation:** PSB (like *Pseudomonas* and *Bacillus* species) secrete organic acids, primarily gluconic acid and 2-ketogluconic acid, which chelate cations bound to phosphate, releasing soluble orthophosphate. [KPSC AAO 2017]

Q46. The crop water requirement of Sugarcane under Karnataka conditions is approximately:

- A. 500 - 800 mm
- B. 900 - 1200 mm
- C. 1500 - 2500 mm
- D. 3000 - 4000 mm

- **Correct Answer: C**

- **Explanation:** Sugarcane is a long-duration crop (12-18 months) with a high total water requirement of 1500 to 2500 mm, needing regular irrigation intervals of 10-15 days. [KPSC AO 2019]

Q47. The nitrification inhibitor 'AM' refers to:

- A. 2-amino-4-chloro-6-methyl pyrimidine
- B. Allyl thiourea
- C. Ammonium thiosulfate
- D. Azadirachtin compound

- **Correct Answer: A**

- **Explanation:** 'AM' is a synthetic nitrification inhibitor chemically known as 2-amino-4-chloro-6-methyl pyrimidine, used to suppress the activity of *Nitrosomonas*. [KPSC AAO 2019]

Q48. Which type of soil water is held between field capacity (-0.33 bar) and permanent wilting point (-15 bars)?

- A. Superfluous water
- B. Available water
- C. Unavailable water
- D. Hygroscopic water

- **Correct Answer: B**

- **Explanation:** Available water is the portion of soil water that can be readily absorbed by plant roots, held between field capacity (-0.33 bar) and the permanent wilting point (-15 bars). [KPSC AAO 2017]

Q49. The principal agronomic practice of 'nipping' in Chickpea should be ideally performed at:

- A. 15-20 days after sowing
- B. 30-40 days after sowing
- C. 60-70 days after sowing
- D. Just before harvesting

- **Correct Answer: B**

- **Explanation:** Nipping is done 30-40 days after sowing when the plants are about 15-20 cm high. Apical bud removal forces lateral growth, leading to more flowers and higher yield. [KPSC AAO 2017]

Q50. The thermal conductivity of soils is affected by moisture. Which statement is correct?

- A. Thermal conductivity decreases with soil moisture

- B. Thermal conductivity increases with soil moisture
- C. Thermal conductivity remains unaffected by soil moisture
- D. Dry soils have higher thermal conductivity than wet soils
- **Correct Answer: B**
- **Explanation:** Water has a thermal conductivity about 25 times greater than air. Therefore, replacing air in soil pores with water increases the thermal conductivity of the soil. [KPSC AO 2019]

Q51. The indicator plant used for detecting Boron deficiency in soil is:

- A. Tomato
- B. Sunflower
- C. Potato
- D. Maize
- **Correct Answer: B**
- **Explanation:** Sunflower acts as an indicator plant for boron because it is highly sensitive to boron deficiency. [KPSC AAO 2015]

Q52. Which of the following is the predominant soil order found in Karnataka?

- A. Inceptisols
- B. Alfisols
- C. Vertisols
- D. Entisols
- **Correct Answer: B**
- **Explanation:** Alfisols (Red soils) are the most predominant soil order in Karnataka, covering more than 50% of the state's area, especially in the southern districts. [KPSC AO 2019]

Q53. The plant pigment Phytochrome is involved in which physiological process?

- A. Transpiration
- B. Photoperiodism and Germination
- C. Respiration
- D. Stomatal movement
- **Correct Answer: B**
- **Explanation:** Phytochrome is a photoreceptor pigment that controls photoperiodic responses (flowering), seed germination, and seedling development. [KPSC AO 2017]

Q54. The average annual rainfall of Karnataka is approximately:

- A. 850 mm
- B. 1150 mm
- C. 1450 mm
- D. 1850 mm
- **Correct Answer: B**
- **Explanation:** Karnataka receives an average annual rainfall of about 1150 mm, with wide variations across the coastal, malnad, and dry zones. [KPSC AAO 2017]

Q55. The nitrogen use efficiency of urea can be improved by coating it with:

- A. Gypsum
- B. Neem Oil Cake
- C. Rock phosphate
- D. Clay mineral
- **Correct Answer: B**
- **Explanation:** Neem oil cake contains epinimbin and nimbin which act as natural nitrification inhibitors, slowing down nitrogen release and improving efficiency. [KPSC AO 2017]

Q56. What is the critical growth stage for irrigation in finger millet (Ragi)?

- A. Seedling stage
- B. Flowering and grain-filling stage
- C. Tillering stage
- D. Physiological maturity
- **Correct Answer: B**
- **Explanation:** Flowering and grain-filling are the most critical stages for water in Ragi. Moisture stress at this stage leads to poor grain set and low yields. [KPSC AO 2017]

Q57. Which of the following is a total stem parasite?

- A. Loranthus
- B. Cuscuta
- C. Orobanche
- D. Striga
- **Correct Answer: B**
- **Explanation:** Cuscuta (dodder) is a total stem parasite that lacks chlorophyll and leaves, wrapping around the host stems and drawing nutrients through haustoria. [KPSC AAO 2019]

Q58. The biological control of water hyacinth is achieved using which insect?

- A. Neochetina eichhorniae
- B. Zygogramma bicolorata
- C. Dactylopius opuntiae
- D. Cyrtobagous salviniae
- **Correct Answer: A**
- **Explanation:** Neochetina eichhorniae is a weevil that feeds on water hyacinth, acting as an effective biological control agent. [KPSC AAO 2015]

Q59. The crop water requirement is lowest in which of the following crops?

- A. Rice
- B. Finger Millet (Ragi)
- C. Sugarcane
- D. Wheat

- **Correct Answer: B**

- **Explanation:** Ragi is a highly drought-tolerant crop with a low water requirement (350-400 mm) compared to rice, wheat, and sugarcane. [KPSC AO 2014]

Q60. The weed Striga is a parasite on which crop?

- A. Lucerne
- B. Sorghum
- C. Tobacco
- D. Mustard

- **Correct Answer: B**

- **Explanation:** Striga (witchweed) is a root semi-parasite that infects cereal crops like sorghum, maize, sugarcane, and pearl millet. [KPSC AO 2019]

Q61. The relative humidity of the air is measured using a:

- A. Pyranometer
- B. Psychrometer
- C. Barometer
- D. Anemometer

- **Correct Answer: B**

- **Explanation:** A psychrometer or hygrometer is used to measure atmospheric humidity. [KPSC AAO 2017]

Q62. Which of the following acts as a reflectant-type antitranspirant?

- A. PMA
- B. Kaolin
- C. Mobileaf
- D. Cycocel

- **Correct Answer: B**

- **Explanation:** Kaolin forms a white reflective film on leaves that reduces the absorption of solar radiation, lowering transpiration. [KPSC AO 2017]

Q63. The vegetative barrier commonly used for soil and water conservation in drylands is:

- A. Vetiver Grass
- B. Napier Grass
- C. Guinea Grass
- D. Cynodon dactylon

- **Correct Answer: A**

- **Explanation:** Vetiver grass (*Vetiveria zizanioides*) is widely used as a vegetative barrier due to its deep root system and ability to slow runoff water. [KPSC AO 2019]

Q64. The nitrogen fixing biofertilizer Azospirillum is recommended for:

- A. Leguminous crops

- B. Non-leguminous crops like Ragi and Sorghum
- C. Horticultural crops only
- D. Oilseeds only

• **Correct Answer: B**

- **Explanation:** Azospirillum is an associative symbiotic nitrogen fixer recommended for cereals and non-legume crops. [KPSC AAO 2017]

Q65. The first chemical weed control in Phalaris minor in India was challenged by resistance to:

- A. 2,4-D
- B. Isoproturon
- C. Atrazine
- D. Butachlor

• **Correct Answer: B**

- **Explanation:** Phalaris minor developed resistance to Isoproturon in wheat fields in northern India, first reported by Dr. R.K. Malik. [KPSC AO 2019]

Q66. What is the oil content of groundnut seeds?

- A. 20-25%
- B. 45-50%
- C. 60-65%
- D. 15-18%

• **Correct Answer: B**

- **Explanation:** Groundnut seeds contain approximately 45-50% oil and 25-28% protein. [KPSC AO 2017]

Q67. The nitrification inhibitor 'N-Serve' chemically refers to:

- A. Dicyandiamide
- B. 2-chloro-6-(trichloromethyl) pyridine
- C. Thiourea
- D. Neem cake

• **Correct Answer: B**

- **Explanation:** N-Serve contains active Nitrapyrin (2-chloro-6-(trichloromethyl) pyridine) as a nitrification inhibitor. [KPSC AAO 2019]

Q68. A dryland area is defined when the annual rainfall is:

- A. < 750 mm
- B. 750 - 1150 mm
- C. > 1150 mm
- D. < 500 mm

• **Correct Answer: A**

- **Explanation:** Dryland farming areas receive an annual rainfall of less than 750 mm and experience frequent crop failures. [KPSC AO 2017]

Q69. The fertilizer that has the highest equivalent acidity is:

- A. Urea
- B. Ammonium chloride
- C. Ammonium sulfate
- D. Anhydrous ammonia

• **Correct Answer: B**

• **Explanation:** Ammonium chloride has the highest equivalent acidity of 128 among the common nitrogenous fertilizers. [KPSC AAO 2015]

Q70. The plant growth retardant Cycocel (CCC) acts as which type of antitranspirant?

- A. Film-forming
- B. Growth retardant / physiological
- C. Stomatal closing
- D. Reflectant

• **Correct Answer: B**

• **Explanation:** Cycocel restricts vegetative growth and increases root depth, acting as a growth retardant to improve water use. [KPSC AAO 2015]

Q71. Which class of seed is the progeny of nucleus seed and is characterized by a golden yellow tag?

- A. Breeder Seed
- B. Foundation Seed
- C. Registered Seed
- D. Certified Seed

• **Correct Answer: A**

• **Explanation:** Breeder seed is the direct progeny of nucleus seed. It is produced by the breeder and has 100% genetic purity. It is labeled with a golden yellow certificate tag. [KPSC AAO 2015]

Q72. What is the color of the certification tag issued for Foundation Seed?

- A. Golden Yellow
- B. Azure Blue
- C. White
- D. Opal Green

• **Correct Answer: C**

• **Explanation:** Foundation seed is the progeny of breeder seed and is labeled with a white-colored tag. It has a minimum genetic purity of 99.5%. [KPSC AO 2017]

Q73. The Azure Blue tag is associated with which class of seed?

- A. Breeder Seed
- B. Foundation Seed
- C. Certified Seed
- D. Registered Seed

- **Correct Answer: C**

- **Explanation:** Certified seed is the progeny of foundation seed and is distributed to farmers. It is identified by an azure blue certificate tag. [KPSC AO 2019]

Q74. The Tetrazolium (TZ) test determines seed viability by staining living tissues with which color?

- A. Dark Blue
- B. Brilliant Green
- C. Red / Pink
- D. Yellow

- **Correct Answer: C**

- **Explanation:** In the TZ test, active dehydrogenase enzymes in living seed embryos reduce 2,3,5-triphenyl tetrazolium chloride to triphenyl formazan, which is a stable, non-diffusible red compound. [KPSC AO 2014]

Q75. What is the isolation distance (meters) required for producing Foundation Seed of hybrid maize?

- A. 100 m
- B. 200 m
- C. 400 m
- D. 600 m

- **Correct Answer: C**

- **Explanation:** To prevent cross-pollination from other maize fields, the minimum isolation distance required for foundation seed production of hybrid maize is 400 meters. [KPSC AAO 2015]

Q76. Which class of seed is not commercially produced in India?

- A. Breeder Seed
- B. Foundation Seed
- C. Registered Seed
- D. Certified Seed

- **Correct Answer: C**

- **Explanation:** Registered seed is the progeny of foundation seed. Although recognized under the seed classes, it is rarely produced in India, where foundation seed is directly multiplied to produce certified seed. [KPSC AAO 2017]

Q77. The Seed Act of India was enacted in which year?

- A. 1963
- B. 1966
- C. 1968
- D. 1974

- **Correct Answer: B**

- **Explanation:** The Seed Act was passed by the Indian Parliament in 1966 to regulate the quality of seeds sold in the market. It came into force on October 2, 1969. [KPSC AO 2019]

Q78. The Seed Rules were framed under the Seed Act in the year:

- A. 1966

- B. 1968
- C. 1970
- D. 1972
- **Correct Answer: B**
- **Explanation:** The Seed Rules, 1968 were notified to outline the concrete procedures for seed certification, labeling, inspection, and the role of seed analysts. [KPSC AO 2017]

Q79. Which of the following is a recalcitrant seed?

- A. Wheat
- B. Rice
- C. Mango
- D. Mustard
- **Correct Answer: C**
- **Explanation:** Mango seeds are recalcitrant. They cannot survive drying below critical moisture content (usually 15-20%) or exposure to freezing temperatures, and must be sown soon after extraction. [KPSC AAO 2017]

Q80. The biochemical reaction in the Tetrazolium test involves which group of enzymes?

- A. Hydrolases
- B. Dehydrogenases
- C. Amylases
- D. Proteases
- **Correct Answer: B**
- **Explanation:** Dehydrogenase enzymes (active in respiring tissues of the seed embryo) transfer hydrogen ions to the colorless tetrazolium salt solution, reducing it to red triphenyl formazan. [KPSC AAO 2017]

Q81. The International Seed Testing Association (ISTA) has its headquarters in which country?

- A. Italy
- B. Switzerland
- C. Netherlands
- D. USA
- **Correct Answer: B**
- **Explanation:** ISTA was founded in 1924 during the International Seed Testing Congress in Cambridge, UK. Its current secretariat headquarters is located in Wallisellen, Switzerland. [KPSC AAO 2017]

Q82. Which seed processing machine separates seeds of the same size based on differences in their density/specific weight?

- A. Air-Screen Cleaner
- B. Specific Gravity Separator
- C. Indented Cylinder Separator
- D. Spiral Separator
- **Correct Answer: B**
- **Explanation:** A Specific Gravity Separator uses a combination of air blast and mechanical oscillation to stratify seeds, separating

heavy, viable seeds from light, damaged ones of identical size. [KPSC AO 2019]

Q83. The minimum genetic purity required for certified seed of hybrid cotton is:

- A. 90%
- B. 95%
- C. 99%
- D. 100%
- **Correct Answer: A**
- **Explanation:** The Indian Minimum Seed Certification Standards (IMSCS) prescribe a minimum genetic purity of 90% for certified seed of cotton hybrids (and 95% for foundation seeds). [KPSC AO 2017]

Q84. The germination type where the cotyledons are raised above the ground due to hypocotyl elongation is called:

- A. Epigeal
- B. Hypogeal
- C. Vivipary
- D. Apomixis
- **Correct Answer: A**
- **Explanation:** In epigeal germination (e.g., French bean, castor, mustard), the hypocotyl elongates rapidly and arches, pulling the cotyledons out of the soil. [KPSC AO 2019]

Q85. The method of exposing seeds to cold, moist conditions to overcome physiological embryo dormancy is:

- A. Scarification
- B. Stratification
- C. Priming
- D. Vernalization
- **Correct Answer: B**
- **Explanation:** Stratification is the placement of seeds in a moist medium (like sand or peat) at low temperatures (2-10°C) for a specific period to trigger biochemical changes that break dormancy. [KPSC AAO 2017]

Q86. What is the minimum germination percentage required for certified seed of Finger Millet (Ragi)?

- A. 65%
- B. 70%
- C. 75%
- D. 80%
- **Correct Answer: C**
- **Explanation:** According to the Indian Minimum Seed Certification Standards, certified seed of finger millet (Ragi) must have a minimum germination capacity of 75%. [KPSC AO 2017]

Q87. The 'Grow Out Test' (GOT) is primarily conducted to determine which seed quality parameter?

- A. Physical purity
- B. Genetic purity
- C. Seed viability

- D. Seed moisture content

- **Correct Answer: B**

- **Explanation:** The Grow Out Test (GOT) is the standard field evaluation method used to determine the genetic purity of a seed lot by raising plants and verifying morphological characteristics. [KPSC AO 2017]

Q88. Which of the following chemicals is commonly used to break seed dormancy by acting as a substrate treatment at 0.2% concentration?

- A. Gibberellic Acid
- B. Potassium nitrate (KNO₃)
- C. Thiourea
- D. Sodium chloride

- **Correct Answer: B**

- **Explanation:** ISTA recommends using 0.2% Potassium Nitrate (KNO₃) as a moistening agent for the germination substrate to break physiological dormancy in many grass and crop seeds. [KPSC AO 2017]

Q89. The National Seeds Corporation (NSC) was established in which year?

- A. 1956
- B. 1963
- C. 1966
- D. 1969

- **Correct Answer: B**

- **Explanation:** The National Seeds Corporation (NSC) was established in March 1963 under the Companies Act, 1956, to implement seed multiplication programs and set up quality control labs. [KPSC AO 2014]

Q90. The Karnataka State Seed Certification Agency (KSSCA) was established in the year:

- A. 1974
- B. 1978
- C. 1982
- D. 1986

- **Correct Answer: A**

- **Explanation:** KSSCA was registered as an autonomous body under the Karnataka Societies Registration Act in 1974 to carry out seed certification work as per the Seed Act, 1966. [KPSC AAO 2015]

Q91. The seed vigour test that measures the leakage of electrolytes from seed tissues is:

- A. Paper piercing test
- B. Tetrazolium vigor test
- C. Electrical Conductivity (EC) test
- D. Brick gravel test

- **Correct Answer: C**

- **Explanation:** The EC test measures the electrical conductivity of seed steep water. Low-vigour seeds have degraded cell membranes and leak more electrolytes, resulting in high EC readings. [KPSC AAO 2017]

Q92. For vacuum storage of agricultural seeds, the moisture content of the seeds should be reduced below:

- A. 8%
- B. 12%
- C. 14%
- D. 16%
- **Correct Answer: A**
- **Explanation:** For moisture-vapor proof packaging or vacuum storage, seeds must be dried below 8.0% moisture (compared to 10-12% for normal bags) to prevent internal respiration and mold growth. [KPSC AO 2019]

Q93. The separation of crop seeds based on their length differences is best done using a:

- A. Specific Gravity Separator
- B. Disc Separator
- C. Air-Screen Cleaner
- D. Spiral Separator
- **Correct Answer: B**
- **Explanation:** A Disc Separator or an Indented Cylinder Separator separates seeds based on length. Pockets or indents lift short seeds while allowing long seeds to pass through. [KPSC AAO 2019]

Q94. The formula for calculating the Real Value of Seed (RVS) is:

- A. $(\text{Purity \%} \times \text{Germination \%}) / 100$
- B. $(\text{Purity \%} + \text{Germination \%}) / 2$
- C. $(\text{Purity \%} / \text{Germination \%}) \times 100$
- D. $(\text{Purity \%} \times \text{Moisture \%}) / \text{Germination \%}$
- **Correct Answer: A**
- **Explanation:** Real Value of Seed (RVS) or Utility Value represents the actual quality of a seed lot. It is calculated by multiplying physical purity percentage by germination percentage and dividing by 100. [KPSC AO 2017]

Q95. Rogueing is an essential operation in seed production plots. It refers to:

- A. Harvesting the crop at physiological maturity
- B. Removing off-type plants and objectionable weeds
- C. Treating seeds with fungicides before sowing
- D. Grading seeds using an air-screen cleaner
- **Correct Answer: B**
- **Explanation:** Rogueing is the manual removal of off-types (plants deviating from the varietal characteristics), diseased plants, and objectionable weeds from the seed crop. [KPSC AO 2014]

Q96. Which of the following is the commercial method of vegetative propagation in Mango?

- A. Air layering
- B. Epicotyl (Stone) grafting
- C. Stooling
- D. Hardwood cuttings

- **Correct Answer: B**

- **Explanation:** Epicotyl grafting (stone grafting) and veneer grafting are the primary commercial vegetative propagation methods in mango. Epicotyl grafting uses very young seedlings (8-15 days old) as rootstocks. [KPSC AO 2014]

Q97. Which rootstock is commercially used in Sapota propagation in India?

- A. Rangpur Lime
- B. Dogridge
- C. Rayan / Khirni (Manilkara hexandra)
- D. Trifoliate Orange

- **Correct Answer: C**

- **Explanation:** Rayan (also known as Khirni) is the standard rootstock used for approach grafting or softwood grafting of Sapota. It provides a strong, drought-tolerant root system. [KPSC AAO 2015]

Q98. The grape rootstock 'Dogridge' is widely used in Northern Karnataka to impart resistance against:

- A. Powdery mildew
- B. Salinity and Drought
- C. Anthracnose
- D. Cold injury

- **Correct Answer: B**

- **Explanation:** Dogridge (*Vitis champinii*) is a vigorous rootstock used in grape cultivation. It has deep root systems that tolerate high salinity and drought conditions common in Northern dry zones of Karnataka. [KPSC AAO 2019]

Q99. The physiological disorder 'Black Tip' of Mango is primarily caused by which nutritional deficiency combined with brick kiln gases?

- A. Calcium
- B. Zinc
- C. Boron
- D. Iron

- **Correct Answer: C**

- **Explanation:** Black tip is caused by Boron deficiency and toxicity from sulfur dioxide, carbon monoxide, and ethylene gases emitted by nearby brick kilns. Borax sprays are used as a control measure. [KPSC AAO 2017]

Q100. The chemical Paclobutrazol (Cultar) is applied to the soil of mango trees to:

- A. Control mango hoppers
- B. Induce regular and early flowering
- C. Accelerate vegetative growth
- D. Control anthracnose disease

- **Correct Answer: B**

- **Explanation:** Paclobutrazol is a gibberellin biosynthesis inhibitor. Applying it to the soil around the collar zone of mango trees 2 to 3 months before expected flowering restricts vegetative flushes and promotes floral bud differentiation. [KPSC AO 2017]

Q101. The commercial vegetative propagation method 'Stooling' is widely used in:

- A. Mango
- B. Guava
- C. Sapota
- D. Citrus
- **Correct Answer: B**
- **Explanation:** Stooling (mound layering) is the commercial method for guava propagation. Mother plants are cut back near ground level, and soil is mounded over the emerging shoots to induce rooting. [KPSC AO 2017]

Q102. The training system of grapes where vines are spread over an overhead horizontal trellis, highly popular in Karnataka, is:

- A. Bower / Pergola system
- B. Kniffin system
- C. Telephone system
- D. Head system
- **Correct Answer: A**
- **Explanation:** The Bower (Pergola) system is the most popular system for commercial grape varieties (like Thompson Seedless, Anab-e-Shahi) in Southern India due to high yields and canopy exposure. [KPSC AAO 2019]

Q103. The physiological disorder 'Blossom End Rot' in Tomato is caused by the deficiency of:

- A. Boron
- B. Calcium
- C. Nitrogen
- D. Potassium
- **Correct Answer: B**
- **Explanation:** Blossom End Rot is a physiological disorder of tomato fruits caused by calcium deficiency in the distal end of the fruit, often triggered by irregular irrigation. [KPSC AO 2017]

Q104. The whipping symptom (Whiptail) of Cauliflower is caused by the deficiency of Molybdenum under which soil condition?

- A. Alkaline soils
- B. Acid soils
- C. Saline soils
- D. Waterlogged soils
- **Correct Answer: B**
- **Explanation:** Molybdenum becomes highly unavailable to plants in acid soils ($\text{pH} < 5.5$). Its deficiency leads to the whiptail disorder in cauliflower, where the leaf blades fail to develop. [KPSC AAO 2015]

Q105. Which pigment is responsible for the red color of tomato fruits, and at what temperature is its synthesis inhibited?

- A. Carotene, $>25^{\circ}\text{C}$
- B. Lycopene, $>30^{\circ}\text{C}$
- C. Anthocyanin, $>35^{\circ}\text{C}$

- D. Xanthophyll, >20°C

- **Correct Answer: B**

- **Explanation:** Lycopene is the carotenoid pigment responsible for the red color in tomatoes. Its synthesis is optimal between 21°C and 24°C and is completely inhibited at temperatures above 30°C. [KPSC AAO 2019]

Q106. The bitterness in cucurbitaceous vegetables is due to the presence of:

- A. Solanine
- B. Cucurbitacin
- C. Sinigrin
- D. Capsaicin

- **Correct Answer: B**

- **Explanation:** Bitterness in cucurbits is caused by cucurbitacins, a class of tetracyclic triterpenoid compounds. They are synthesized under stress conditions. [KPSC AAO 2017]

Q107. The organic chemical compound responsible for the characteristic pungency of onion bulbs is:

- A. Allyl propyl disulfide
- B. Diallyl disulfide
- C. Allyl isothiocyanate
- D. Capsaicin

- **Correct Answer: A**

- **Explanation:** Allyl propyl disulfide is the volatile sulfur compound responsible for the pungency and aroma of onions. [KPSC AO 2019]

Q108. The plant growth regulator widely used to promote rooting in hardwood stem cuttings of grapes and pomegranate is:

- A. Gibberellic Acid (GA3)
- B. Indole-3-Butyric Acid (IBA)
- C. Kinetin
- D. Absciscic Acid (ABA)

- **Correct Answer: B**

- **Explanation:** IBA is a synthetic auxin used to induce rooting in stem cuttings. Cuttings are dipped in IBA solution (1000-3000 ppm) before planting. [KPSC AAO 2017]

Q109. Which chemical is used to break the physiological bud dormancy in potato tubers?

- A. Thiourea
- B. Maleic Hydrazide (MH)
- C. Calcium Carbonate
- D. Potassium Permanganate

- **Correct Answer: A**

- **Explanation:** Thiourea (1% solution) or Gibberellic acid (GA3) are used to break the dormancy of potato seed tubers. [KPSC AO 2019]

Q110. The commercial method of propagation in Rose is:

- A. Air layering
- B. Shield/T-Budding
- C. Tongue grafting
- D. Inarching

• **Correct Answer: B**

- **Explanation:** T-budding (also called shield budding) is the commercial method of propagating roses. *Rosa multiflora* or *Rosa indica* are commonly used as rootstocks. [KPSC AAO 2017]

Q111. The process of removing side buds to promote a single, large terminal flower in commercial Chrysanthemum is known as:

- A. Pinching
- B. Disbudding
- C. Desuckering
- D. Training

• **Correct Answer: B**

- **Explanation:** Disbudding is the removal of side/lateral buds to divert nutrients to the central terminal bud, producing a larger bloom. [KPSC AAO 2019]

Q112. The maturity index of commercial banana is determined by observing:

- A. The color of the skin
- B. The disappearance of angularity (roundness of fingers)
- C. The size of the bunch
- D. The moisture content of the pulp

• **Correct Answer: B**

- **Explanation:** As banana fingers mature, the sharp angles on the fruit disappear and the fruit becomes rounded. [KPSC AAO 2015]

Q113. Which gaseous chemical compound is used as a post-harvest treatment to inhibit ethylene action and extend the shelf life of fruits?

- A. Ethylene gas
- B. 1-Methylcyclopropene (1-MCP)
- C. Calcium carbide
- D. Sulfur dioxide

• **Correct Answer: B**

- **Explanation:** 1-MCP is a synthetic gaseous compound that binds to ethylene receptors in plant tissues, blocking ethylene action and delaying ripening. [KPSC AO 2017]

Q114. Modified Atmosphere Packaging (MAP) preserves fresh horticultural produce by:

- A. Decreasing CO₂ and increasing O₂ levels
- B. Increasing CO₂ and decreasing O₂ levels
- C. Maintaining ambient atmospheric gases
- D. Removing all air to create a vacuum

- **Correct Answer: B**

- **Explanation:** MAP involves modifying the gas composition inside the package by reducing oxygen (O₂) and increasing carbon dioxide (CO₂) levels, which slows respiration. [KPSC AAO 2017]

Q115. The yellow color of onion flesh is due to the pigment:

- A. Anthocyanin
- B. Quercetin
- C. Lycopene
- D. Carotenoid

- **Correct Answer: B**

- **Explanation:** Quercetin (a flavonoid pigment) is responsible for the yellow skin color of yellow onions. [KPSC AO 2017]

Q116. Which crop is known as the 'King of Spices'?

- A. Cardamom
- B. Black Pepper
- C. Nutmeg
- D. Clove

- **Correct Answer: B**

- **Explanation:** Black Pepper (*Piper nigrum*) is called the 'King of Spices' due to its dominant role in international trade. [KPSC AAO 2017]

Q117. The chemical used in flower vase solutions to extend cut flower longevity by inhibiting microbial blockages in stems is:

- A. Silver Thiosulfate (STS)
- B. Gibberellic Acid
- C. Maleic Hydrazide
- D. Ethylene

- **Correct Answer: A**

- **Explanation:** Silver Thiosulfate (STS) acts as an anti-ethylene agent and floral preservative. It prevents bacterial growth in the vase water. [KPSC AO 2019]

Q118. What is the recommended nursery seed rate (g/ha) for raising hybrid tomato seedlings?

- A. 100-150 g/ha
- B. 400-500 g/ha
- C. 800-1000 g/ha
- D. 1500-2000 g/ha

- **Correct Answer: A**

- **Explanation:** For hybrid tomatoes, the seed rate is 100-150 g/ha due to high germination percentages. [KPSC AO 2017]

Q119. Tapioca (Cassava) tubers contain a toxic glucoside called:

- A. Solanine
- B. Linamarin

- C. Sinigrin
- D. Cucurbitacin
- **Correct Answer: B**
- **Explanation:** Tapioca roots contain the cyanogenic glucoside Linamarin, which releases toxic hydrocyanic acid (HCN) upon tissue damage. [KPSC AAO 2015]

Q120. The spice vanilla is obtained from which botanical part of the *Vanilla planifolia* plant?

- A. Dried flower buds
- B. Cured pods (fruits)
- C. Rhizomes
- D. Bark
- **Correct Answer: B**
- **Explanation:** Vanilla is extracted from the fully grown, cured pods (often called beans or fruits) of the orchid *Vanilla planifolia*. [KPSC AAO 2015]

Q121. Which of the following clay minerals exhibits the highest Cation Exchange Capacity (CEC)?

- A. Kaolinite
- B. Illite
- C. Montmorillonite
- D. Vermiculite
- **Correct Answer: D**
- **Explanation:** Vermiculite has a very high Cation Exchange Capacity (CEC) ranging from 100 to 150 cmol(p+)/kg, which is higher than Montmorillonite (80-100 cmol(p+)/kg), Illite (15-40 cmol(p+)/kg), and Kaolinite (3-15 cmol(p+)/kg) due to extensive isomorphous substitution in its tetrahedral sheet (KPSC AO 2017).

Q122. The settling velocity of a soil particle in a suspension during mechanical analysis is calculated using Stokes' Law.

Which parameter is settling velocity directly proportional to?

- A. Radius of the particle
- B. Square of the radius of the particle
- C. Viscosity of the liquid
- D. Density of the liquid
- **Correct Answer: B**
- **Explanation:** According to Stokes' Law, the terminal settling velocity (v) of a spherical particle is given by $v = \frac{(2/9) * g * r^2 * (dp - dl)}{n}$, showing that velocity is directly proportional to the square of the radius (r^2) of the particle (KPSC AO 2017).

Q123. What is the primary mineral source of Boron in soils?

- A. Apatite
- B. Tourmaline
- C. Orthoclase
- D. Olivine
- **Correct Answer: B**

- **Explanation:** Tourmaline is a complex borosilicate mineral and is the main primary mineral source of the micronutrient Boron in soil profiles (KPSC AAO 2017).

Q124. Under the ISSS soil classification system, what is the size range of silt particles?

- A. 0.2 to 0.02 mm
- B. 0.02 to 0.002 mm
- C. Less than 0.002 mm
- D. 2.0 to 0.2 mm
- **Correct Answer: B**

- **Explanation:** The International Soil Science Society (ISSS) classifies silt as particles having a diameter between 0.02 and 0.002 mm. Clay is < 0.002 mm, fine sand is 0.2-0.02 mm, and coarse sand is 2.0-0.2 mm (KPSC AO 2014).

Q125. A soil sample has a bulk density of 1.325 g/cm³ and a particle density of 2.65 g/cm³. What is the porosity percentage of this soil?

- A. 30%
- B. 40%
- C. 50%
- D. 60%
- **Correct Answer: C**

- **Explanation:** Porosity (%) = $[1 - (BD / PD)] * 100$. Substituting the values: $[1 - (1.325 / 2.65)] * 100 = [1 - 0.5] * 100 = 50\%$ (KPSC AAO 2017).

Q126. Which type of soil structure is characterized by horizontal dimensions being much larger than vertical dimensions and is commonly found in compacted horizons?

- A. Granular
- B. Columnar
- C. Prismatic
- D. Platy
- **Correct Answer: D**

- **Explanation:** Platy soil structure consists of thin, flat, horizontal plates. It limits downward water movement and root growth due to its horizontal orientation and compaction (KPSC AO 2017).

Q127. In the Munsell color notation '7.5YR 5/4', what does the number '5' represent?

- A. Hue
- B. Value
- C. Chroma
- D. Soil texture
- **Correct Answer: B**

- **Explanation:** The Munsell system specifies color in terms of three variables: Hue (dominant spectral color, e.g., 7.5YR), Value (lightness/darkness, from 0 to 10, e.g., 5), and Chroma (purity or strength of color, e.g., 4) (KPSC AO 2017).

Q128. Which soil moisture constant represents water held at a suction pressure of -15 bars (pF 4.2)?

- A. Field capacity

- B. Permanent wilting point
- C. Hygroscopic coefficient
- D. Maximum water holding capacity

• **Correct Answer: B**

- **Explanation:** Permanent wilting point (PWP) is the soil moisture content at which plants cannot exert enough suction to extract water, occurring at -15 bars (-1.5 MPa) of soil water potential (KPSC AAO 2015).

Q129. Which instrument is most suitable for measuring soil moisture tension in the field within the range of 0 to 0.85 bars?

- A. Tensiometer
- B. Gypsum block
- C. Neutron probe
- D. Pressure membrane apparatus

• **Correct Answer: A**

- **Explanation:** Tensiometers measure the physical tension of soil water directly. They are highly accurate in wet ranges but fail above 0.85 bars due to air entering the porous cup (vacuum break) (KPSC AO 2017).

Q130. What does active acidity in soil refer to?

- A. H^+ and Al^{3+} ions adsorbed on soil colloids
- B. H^+ and Al^{3+} ions present in the soil solution
- C. Non-exchangeable acidity
- D. Acidity due to organic matter decay

• **Correct Answer: B**

- **Explanation:** Active acidity is due to hydrogen (H^+) and aluminum (Al^{3+}) ions in the soil solution. It is measured when determining soil pH. Adsorbed ions represent exchangeable or reserve acidity (KPSC AAO 2017).

Q131. Which equation is used to calculate the Sodium Adsorption Ratio (SAR) of irrigation water or soil extract?

- A. $SAR = Na^+ / \sqrt{Ca^{2+} + Mg^{2+}}$
- B. $SAR = Na^+ / \sqrt{((Ca^{2+} + Mg^{2+})/2)}$
- C. $SAR = (Ca^{2+} + Mg^{2+}) / \sqrt{Na^+}$
- D. $SAR = Na^+ / (Ca^{2+} + Mg^{2+})$

• **Correct Answer: B**

- **Explanation:** The Sodium Adsorption Ratio is calculated as $SAR = [Na^+] / \sqrt{([Ca^{2+}] + [Mg^{2+}])/2}$, where all ion concentrations are expressed in milliequivalents per liter (meq/L) (KPSC AAO 2015).

Q132. Reclamation of sodic soils is chemically achieved by applying which of the following amendments?

- A. Lime
- B. Calcium carbonate
- C. Gypsum
- D. Sodium chloride

• **Correct Answer: C**

- **Explanation:** Gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) is the most common amendment for sodic soils. Calcium replaces exchangeable sodium on the clay complex, forming soluble sodium sulphate, which is then leached out (KPSC AO 2017).

Q133. Which method is standard for estimating the gypsum requirement of sodic soils?

- A. Walkley and Black method
- B. Schoonover's method
- C. Bray's method
- D. Olsen's method

• **Correct Answer: B**

- **Explanation:** Schoonover's method determines the gypsum requirement by shaking soil with saturated calcium chloride solution and determining the loss of calcium from the solution by EDTA titration (KPSC AO 2017).

Q134. What are the chemical characteristics of a typical saline soil?

- A. $\text{EC} > 4 \text{ dS/m}$, $\text{ESP} < 15$, $\text{SAR} < 13$
- B. $\text{EC} < 4 \text{ dS/m}$, $\text{ESP} > 15$, $\text{SAR} > 13$
- C. $\text{EC} > 4 \text{ dS/m}$, $\text{ESP} > 15$, $\text{SAR} > 13$
- D. $\text{EC} < 4 \text{ dS/m}$, $\text{ESP} < 15$, $\text{SAR} < 13$

• **Correct Answer: A**

- **Explanation:** Saline soils have an electrical conductivity (EC) $> 4 \text{ dS/m}$, exchangeable sodium percentage (ESP) < 15 , and sodium adsorption ratio (SAR) < 13 . The pH is usually less than 8.5 (KPSC AAO 2017).

Q135. The Walkley and Black method is widely used to estimate soil organic carbon. What is the primary oxidizing agent used in this method?

- A. Potassium permanganate
- B. Hydrogen peroxide
- C. Potassium dichromate
- D. Sodium hydroxide

• **Correct Answer: C**

- **Explanation:** The Walkley-Black wet oxidation method uses 1N Potassium Dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$) in the presence of concentrated Sulphuric Acid (H_2SO_4) to oxidize organic carbon (KPSC AO 2017).

Q136. What is the standard correction factor used in the Walkley and Black method because only about 77% of organic carbon is oxidized?

- A. 1.724
- B. 1.3
- C. 0.58
- D. 2.0

• **Correct Answer: B**

- **Explanation:** Since the recovery of organic carbon by the Walkley-Black method is about 77%, a correction factor of 1.3 ($100/77$) is applied. Note that 1.724 is the 'Bemmelen factor' used to convert organic carbon to total organic matter (assuming OM contains 58% carbon) (KPSC AAO 2017).

Q137. Which extractant is used in Olsen's method to estimate available phosphorus in neutral to alkaline soils?

- A. 0.03 M NH_4F + 0.025 M HCl
- B. 0.5 M NaHCO_3 at pH 8.5
- C. 1 N NH_4OAc at pH 7.0
- D. 0.1 N HCl
- **Correct Answer: B**
- **Explanation:** Olsen's reagent is 0.5 M Sodium Bicarbonate (NaHCO_3) adjusted to pH 8.5. It is designed to extract phosphorus from neutral and alkaline soils by precipitating calcium as calcium carbonate, thereby releasing phosphate (KPSC AO 2017).

Q138. Which method is most appropriate for determining available nitrogen in soil?

- A. Alkaline permanganate method
- B. Olsen's method
- C. Bray and Kurtz method
- D. Ammonium acetate extraction
- **Correct Answer: A**
- **Explanation:** The alkaline permanganate method (Subbiah and Asija, 1956) is used to estimate available nitrogen in soil by distilling the soil with alkaline potassium permanganate (KMnO_4) to release ammonia (KPSC AAO 2015).

Q139. Which of the following is the standard extractant for available potassium in soils?

- A. 0.5 M NaHCO_3
- B. Neutral normal ammonium acetate (1N NH_4OAc)
- C. 0.1 N HCl
- D. Bray's reagent
- **Correct Answer: B**
- **Explanation:** Neutral normal ammonium acetate (1N NH_4OAc , pH 7.0) is the standard extractant for exchangeable and water-soluble potassium (available K) in soils, measured using a flame photometer (KPSC AO 2017).

Q140. Which nutrient deficiency causes the 'white bud' symptom in maize crops?

- A. Boron
- B. Iron
- C. Zinc
- D. Copper
- **Correct Answer: C**
- **Explanation:** Zinc deficiency in maize causes the diagnostic 'white bud' symptom, characterized by chlorosis or whitening of the growing tips and young leaves (KPSC AAO 2017).

Q141. The whip tail disease in cauliflower is an indicator of which nutrient deficiency?

- A. Boron
- B. Manganese
- C. Molybdenum
- D. Iron
- **Correct Answer: C**

- **Explanation:** Whip tail in cauliflower is caused by Molybdenum deficiency, which leads to restricted leaf blade development, leaving only the midrib resembling a whip (KPSC AO 2017).

Q142. Which nutrient is absorbed by plants as an uncharged boric acid (H_3BO_3) molecule?

- A. Boron
- B. Silicon
- C. Chlorine
- D. Molybdenum

• **Correct Answer: A**

- **Explanation:** Boron is unique because it is absorbed primarily as the uncharged molecular species, boric acid (H_3BO_3), at normal soil pH values rather than as an ion (KPSC AAO 2017).

Q143. Which fertilizer has the highest equivalent acidity among common nitrogenous fertilizers?

- A. Urea
- B. Ammonium sulphate
- C. Ammonium chloride
- D. Anhydrous ammonia

• **Correct Answer: C**

- **Explanation:** Ammonium chloride has an equivalent acidity of 128, which is higher than Ammonium sulphate (110) and Urea (80). It means 128 kg of $CaCO_3$ is needed to neutralize the acidity produced by 100 kg of ammonium chloride (KPSC AO 2017).

Q144. As per the Fertilizer Control Order (FCO), the biuret content in urea should not exceed what percentage for soil application?

- A. 0.5%
- B. 1.5%
- C. 2.5%
- D. 3.0%

• **Correct Answer: B**

- **Explanation:** According to the Indian Fertilizer Control Order (FCO), the maximum limit of biuret in urea is 1.5% by weight for soil application and 0.5% for foliar spray to prevent phytotoxicity (KPSC AO 2017).

Q145. Which of the following compounds is a widely used chemical nitrification inhibitor?

- A. NBPT
- B. Dicyandiamide (DCD)
- C. PPD
- D. Neem cake

• **Correct Answer: B**

- **Explanation:** Dicyandiamide (DCD) and Nitrapyrin (N-Serve) are chemical nitrification inhibitors that delay the conversion of ammonium to nitrite by depressing *Nitrosomonas* activity. NBPT is a urease inhibitor (KPSC AO 2017).

Q146. What type of clay mineral is Kaolinite?

- A. 1:1 non-expanding silicate clay

- B. 2:1 expanding silicate clay
- C. 2:1 non-expanding silicate clay
- D. 2:2 or 2:1:1 silicate clay

• **Correct Answer: A**

- **Explanation:** Kaolinite is a 1:1 type clay mineral consisting of one tetrahedral silica sheet bonded to one octahedral alumina sheet. It has no expanding properties and low CEC due to hydrogen bonding between layers (KPSC AO 2014).

Q147. Which soil order is characterized by high clay content, dominant shrink-swell behavior, and deep cracks during the dry season?

- A. Alfisols
- B. Vertisols
- C. Inceptisols
- D. Aridisols

• **Correct Answer: B**

- **Explanation:** Vertisols are heavy clay soils (containing >30% clay) with high content of expanding 2:1 clays (montmorillonite). They crack deeply when dry and swell when wet, causing self-ploughing (KPSC AO 2017).

Q148. What is the dominant soil order in the southern dry zone of Karnataka?

- A. Vertisols
- B. Alfisols
- C. Histosols
- D. Spodosols

• **Correct Answer: B**

- **Explanation:** Alfisols (Red soils) are the dominant soils in southern Karnataka, including the southern dry zone, characterized by a clay accumulation horizon (argillic horizon) and moderate base saturation (KPSC AAO 2017).

Q149. The organic matter fraction that is soluble in both acid and alkali solutions is:

- A. Humic acid
- B. Fulvic acid
- C. Humin
- D. Humus

• **Correct Answer: B**

- **Explanation:** Fulvic acid is the fraction of humus that remains in solution after acidification of the alkaline extract; it is soluble in both acid and alkali (KPSC AAO 2017).

Q150. Which diagnostic subsurface horizon is characterized by the illuvial accumulation of silicate clay?

- A. Cambic
- B. Argillic
- C. Spodic
- D. Oxic

• **Correct Answer: B**

- **Explanation:** An argillic horizon is a diagnostic subsurface B horizon that has a significantly higher silicate clay content than the overlying A or E horizon, formed by illuviation (clay translocation) (KPSC AO 2014).

Q151. The C:N ratio of humified organic matter in most cultivated agricultural soils is approximately:

- A. 10:1
- B. 30:1
- C. 100:1
- D. 5:1
- **Correct Answer: A**
- **Explanation:** In typical arable soils, the carbon-to-nitrogen (C:N) ratio of the stable organic matter (humus) converges to a narrow range of 10:1 to 12:1 (KPSC AAO 2015).

Q152. How does the cation exchange capacity (CEC) of a soil with pH-dependent charge change as the pH increases?

- A. Decreases
- B. Increases
- C. Remains unchanged
- D. First increases, then decreases
- **Correct Answer: B**
- **Explanation:** As soil pH increases, H⁺ ions dissociate from the edges of clay minerals and functional groups of organic matter (like carboxyl and phenolic groups), creating more negative charges and thus increasing the CEC (KPSC AAO 2017).

Q153. What is the critical limit of DTPA-extractable Zinc (Zn) in soils below which response to zinc fertilization is expected?

- A. 0.6 mg/kg
- B. 1.2 mg/kg
- C. 2.0 mg/kg
- D. 4.5 mg/kg
- **Correct Answer: A**
- **Explanation:** The widely accepted critical limit of DTPA-extractable zinc in soils is 0.6 ppm (or 0.6 mg/kg of soil). Below this, zinc deficiency is highly probable (KPSC AO 2017).

Q154. Which reagent is used in the turbidimetric estimation of available sulphur in soils?

- A. Ammonium acetate
- B. Sodium bicarbonate
- C. Barium chloride
- D. Silver nitrate
- **Correct Answer: C**
- **Explanation:** Available sulphur (sulphate-S) is extracted from soils (often using calcium chloride or monocalcium phosphate) and precipitated as barium sulphate (BaSO₄) using Barium Chloride (BaCl₂) to measure turbidity spectrophotometrically (KPSC AAO 2017).

Q155. What is the primary source of potassium in soils?

- A. Pyrite
- B. Micas and Feldspars

- C. Calcite
- D. Apatite
- **Correct Answer: B**
- **Explanation:** Primary potassium-bearing minerals include orthoclase feldspar, muscovite mica, and biotite mica. Their weathering releases potassium into the exchangeable and soil solution phases (KPSC AAO 2015).

Q156. What is the specific heat of a typical dry mineral soil compared to water?

- A. 0.2 cal/g
- B. 0.5 cal/g
- C. 1.0 cal/g
- D. 2.0 cal/g
- **Correct Answer: A**
- **Explanation:** The specific heat of dry mineral soil is approximately 0.2 calories per gram per °C, which is only one-fifth of the specific heat of water (1.0 cal/g/°C). Wet soils require much more energy to heat up (KPSC AAO 2015).

Q157. Which soil order represents organic soils containing more than 20% organic matter by weight?

- A. Spodosols
- B. Histosols
- C. Oxisols
- D. Andisols
- **Correct Answer: B**
- **Explanation:** Histosols are organic soils that form in waterlogged or cold conditions where organic matter decomposition is very slow. They must contain at least 20-30% organic matter by weight depending on clay content (KPSC AO 2017).

Q158. Which soil-forming process involves the accumulation of calcium carbonate (CaCO₃) in the B or C horizon of a soil profile?

- A. Laterization
- B. Calcification
- C. Podzocalization
- D. Gleization
- **Correct Answer: B**
- **Explanation:** Calcification is the deposition and accumulation of calcium carbonate in soil horizons. It is common in arid, semi-arid, and sub-humid regions where evapotranspiration exceeds precipitation (KPSC AO 2017).

Q159. Which soil physical property is defined as the relative ease with which water, air, and roots can penetrate the soil?

- A. Soil consistency
- B. Soil permeability
- C. Soil texture
- D. Soil density
- **Correct Answer: B**
- **Explanation:** Permeability is the characteristic of soil that governs the rate at which water and air move through it. It is strongly

influenced by soil structure and texture (KPSC AAO 2017).

Q160. What is the diagnostic surface horizon that is dark-colored, has high base saturation (>50%), and is rich in organic matter?

- A. Ochric epipedon
- B. Mollic epipedon
- C. Umbric epipedon
- D. Histic epipedon

• **Correct Answer: B**

• **Explanation:** A mollic epipedon is a thick, dark, organic-rich surface horizon with a base saturation of 50% or more. Umbric is similar but has less than 50% base saturation (KPSC AAO 2017).

Q161. The SMP (Shoemaker-McLean-Pratt) buffer method is used to determine which of the following requirements in soil testing?

- A. Gypsum requirement of sodic soils
- B. Lime requirement of acid soils
- C. Nitrogen requirement of rice
- D. Potash requirement of sugarcane

• **Correct Answer: B**

• **Explanation:** The SMP buffer method is a standard laboratory procedure used to determine the lime requirement of acid soils by measuring the decrease in pH of a buffer solution when mixed with soil (KPSC AO 2017).

Q162. Which of the following elements is mobile in plants but immobile in soils, leading to specific fertilizer placement requirements?

- A. Nitrogen
- B. Phosphorus
- C. Potassium
- D. Boron

• **Correct Answer: B**

• **Explanation:** Phosphorus is highly immobile in soil due to chemical fixation (as Fe/Al phosphates in acid soils and Ca phosphates in alkaline soils) but is mobile within the plant, moving from older to younger tissues (KPSC AAO 2017).

Q163. What is the mineral name for the primary source of Phosphorus in soils?

- A. Pyrite
- B. Hematite
- C. Apatite
- D. Gypsum

• **Correct Answer: C**

• **Explanation:** Apatite $[\text{Ca}_5(\text{PO}_4)_3(\text{F}, \text{Cl}, \text{OH})]$ is the primary crystalline phosphate mineral found in rocks, from which most soil phosphorus is derived by weathering (KPSC AAO 2015).

Q164. Which type of soil acidity is measured directly when we perform a standard soil pH measurement in a 1:2.5 soil-water suspension?

- A. Active acidity
- B. Exchangeable acidity
- C. Reserve acidity
- D. Residual acidity
- **Correct Answer: A**
- **Explanation:** Measuring soil pH using a glass electrode in a soil-water suspension measures active acidity, which is the concentration of free H^+ ions in the soil solution phase (KPSC AO 2017).

Q165. Which clay mineral is a 2:1:1 (or 2:2) non-expanding silicate clay mineral with an interlayer brucite sheet?

- A. Illite
- B. Vermiculite
- C. Chlorite
- D. Kaolinite
- **Correct Answer: C**
- **Explanation:** Chlorite consists of a 2:1 silicate layer alternating with a brucite-like $[Mg(OH)_2]$ sheet in the interlayer space, giving it a 2:1:1 (or 2:2) structure (KPSC AAO 2017).

Q166. The moisture tension at the hygroscopic coefficient of a soil is approximately:

- A. -0.33 bar
- B. -15 bars
- C. -31 bars
- D. -10000 bars
- **Correct Answer: C**
- **Explanation:** Hygroscopic water is the water held tightly as a thin film on the surface of soil particles at a tension of -31 bars (-3.1 MPa) up to -10,000 bars (KPSC AO 2017).

Q167. Under waterlogged conditions, which form of iron becomes dominant, imparting a bluish-grey color to the soil profile?

- A. Ferric iron (Fe^{3+})
- B. Ferrous iron (Fe^{2+})
- C. Iron oxide (Fe_3O_4)
- D. Pyrite (FeS_2)
- **Correct Answer: B**
- **Explanation:** In anaerobic waterlogged soils, microorganisms use ferric iron (Fe^{3+}) as an electron acceptor, reducing it to soluble ferrous iron (Fe^{2+}). This process is known as gleying or gleization (KPSC AAO 2017).

Q168. Which nitrogenous fertilizer is physiologically acidic and leaves a high acidic residue in the soil?

- A. Sodium nitrate
- B. Calcium nitrate
- C. Ammonium sulphate
- D. Potassium nitrate

- **Correct Answer: C**

- **Explanation:** Ammonium sulphate is a physiologically acidic fertilizer because plants absorb ammonium (NH_4^+) ions more rapidly than sulphate (SO_4^{2-}) ions, and the nitrification of ammonium releases H^+ ions into the soil (KPSC AO 2017).

Q169. Which micronutrient is highly deficient in calcareous soils because it precipitates at high pH (> 7.5)?

- A. Molybdenum
- B. Iron
- C. Chlorine
- D. Boron

- **Correct Answer: B**

- **Explanation:** Iron solubility decreases dramatically as pH increases. In calcareous soils (rich in CaCO_3 , pH 7.5-8.5), iron is precipitated as insoluble oxides/hydroxides, causing lime-induced chlorosis (KPSC AO 2014).

Q170. Which soil order represents weakly developed soils that show early stages of profile development and have a cambic subsurface horizon?

- A. Entisols
- B. Inceptisols
- C. Alfisols
- D. Mollisols

- **Correct Answer: B**

- **Explanation:** Inceptisols are soils with minimal profile development, showing some alteration of parent material (like structural development or color changes) but lacking clay accumulation. They typically possess a cambic horizon (KPSC AO 2017).

Q171. Who is considered the Father of Soil Microbiology due to his discovery of chemoautotrophy and nitrification?

- A. Louis Pasteur
- B. Robert Koch
- C. Sergei Winogradsky
- D. Martinus Beijerinck

- **Correct Answer: C**

- **Explanation:** Sergei Winogradsky discovered that some soil bacteria obtain energy by oxidizing inorganic compounds (chemoautotrophy) and identified the bacterial steps of nitrification, founding soil microbiology (KPSC AAO 2017).

Q172. What unique compound is responsible for the heat resistance of bacterial endospores?

- A. Peptidoglycan
- B. Dipicolinic acid
- C. Lipopolysaccharide
- D. Teichoic acid

- **Correct Answer: B**

- **Explanation:** Calcium dipicolinate (dipicolinic acid complexed with calcium) is found in the core of bacterial endospores. It dehydrates the spore and stabilizes DNA, conferring high heat and chemical resistance (KPSC AAO 2015).

Q173. Which component is present in the cell walls of Gram-negative bacteria but absent in Gram-positive bacteria?

- A. Peptidoglycan
- B. Teichoic acid
- C. Lipopolysaccharide (LPS) outer membrane
- D. Cytoplasmic membrane

• **Correct Answer: C**

- **Explanation:** Gram-negative bacteria have a thin peptidoglycan layer surrounded by an outer membrane composed of lipopolysaccharides (LPS) and lipoproteins. Gram-positive cell walls have thick peptidoglycan and teichoic acids (KPSC AO 2017).

Q174. Which protein contains iron and acts as an oxygen carrier/scavenger to protect the nitrogenase enzyme from oxygen toxicity in root nodules?

- A. Ferredoxin
- B. Leghemoglobin
- C. Cytochrome c
- D. Nitrogenase reductase

• **Correct Answer: B**

- **Explanation:** Leghemoglobin is a pink, oxygen-binding hemoprotein synthesized symbiotically in legume nodules. It regulates oxygen concentration, supplying enough for bacteroid respiration while keeping free oxygen low to protect nitrogenase (KPSC AO 2017).

Q175. Which Rhizobium species is specifically matched as the symbiont for Alfalfa (Medicago sativa)?

- A. Rhizobium leguminosarum
- B. Rhizobium meliloti
- C. Rhizobium trifolii
- D. Rhizobium phaseoli

• **Correct Answer: B**

- **Explanation:** Rhizobium meliloti is the specific symbiotic partner for Medicago (Alfalfa), Melilotus (sweet clover), and Trigonella (fenugreek) (KPSC AO 2017).

Q176. Which of the following is a free-living, aerobic, heterotrophic nitrogen-fixing bacterium widely used as a biofertilizer for non-leguminous crops?

- A. Rhizobium leguminosarum
- B. Azotobacter chroococcum
- C. Azospirillum brasilense
- D. Beijerinckia indica

• **Correct Answer: B**

- **Explanation:** Azotobacter chroococcum is a free-living, aerobic, heterotrophic diazotroph. It fixes nitrogen in the soil and produces growth-promoting substances, making it a popular biofertilizer for cereals and vegetables (KPSC AAO 2017).

Q177. The symbiotic association between the water fern Azolla and an nitrogen-fixing cyanobacterium is utilized in wetland rice fields. What is the partner cyanobacterium?

- A. Nostoc muscorum
- B. Anabaena azollae

- C. Aulosira fertilissima
- D. Oscillatoria limosa
- **Correct Answer: B**

• **Explanation:** Anabaena azollae is a nitrogen-fixing blue-green alga (cyanobacterium) that lives symbiotically within the leaf cavities of the floating water fern Azolla pinnata, providing nitrogen in flooded paddy fields (KPSC AO 2017).

Q178. Which non-leguminous plant forms a nitrogen-fixing symbiosis with the actinomycete genus Frankia?

- A. Soybean
- B. Casuarina
- C. Sesbania
- D. Crotalaria
- **Correct Answer: B**

• **Explanation:** Frankia is a filamentous actinomycete that forms root nodules on actinorhizal plants such as Casuarina equisetifolia and Alnus species, fixing atmospheric nitrogen (KPSC AAO 2017).

Q179. What is the primary mechanism by which Phosphate Solubilizing Bacteria (PSB) solubilize insoluble mineral phosphates in soil?

- A. Production of nitrogenase enzyme
- B. Secretion of organic acids like gluconic acid
- C. Secretion of alkaline phosphatase
- D. Absorbing phosphorus and releasing it upon death
- **Correct Answer: B**

• **Explanation:** PSB (like Bacillus megaterium) secrete organic acids (e.g., gluconic, citric, lactic acids) which lower soil pH and chelate divalent cations (like Ca^{2+}), releasing orthophosphate ions into the soil solution (KPSC AO 2017).

Q180. Arbuscular Mycorrhizal (AM) fungi belong to which taxonomic phylum?

- A. Ascomycota
- B. Basidiomycota
- C. Glomeromycota
- D. Zygomycota
- **Correct Answer: C**

• **Explanation:** All arbuscular mycorrhizal fungi, which form obligate symbiotic associations with the roots of about 80% of terrestrial plants, are classified under the phylum Glomeromycota (KPSC AO 2014).

Q181. What glycoprotein, produced by AM fungi, plays a major role in binding soil particles together to stabilize microaggregates?

- A. Glomalin
- B. Chitin
- C. Callose
- D. Suberin
- **Correct Answer: A**

- **Explanation:** Glomalin is a hydrophobic glycoprotein produced abundantly by the hyphae and spores of AM fungi. It is highly resistant to degradation and acts as a biological cement to stabilize soil structure (KPSC AAO 2017).

Q182. Which organism is responsible for the oxidation of nitrite (NO₂-) to nitrate (NO₃-) in the second step of nitrification?

- A. Nitrosomonas
- B. Nitrobacter
- C. Pseudomonas
- D. Azotobacter
- **Correct Answer: B**

- **Explanation:** Nitrification is a two-step process: Nitrosomonas oxidizes ammonium (NH₄⁺) to nitrite (NO₂-), and Nitrobacter oxidizes nitrite (NO₂-) to nitrate (NO₃-). Both are chemoautotrophic obligate aerobes (KPSC AAO 2017).

Q183. Under which condition does biological denitrification occur rapidly in agricultural soils?

- A. High oxygen, dry soil
- B. Anaerobic, waterlogged soil
- C. Highly acidic, dry soil
- D. Low organic matter soil
- **Correct Answer: B**

- **Explanation:** Denitrification is an anaerobic respiratory process where facultative anaerobic bacteria (e.g., Pseudomonas, Paracoccus) use nitrate (NO₃-) as a terminal electron acceptor in the absence of oxygen, reducing it to N₂O and N₂ gases (KPSC AO 2017).

Q184. What is the C:N ratio threshold above which net immobilization of nitrogen occurs in soil when crop residues are added?

- A. < 20:1
- B. > 30:1
- C. 10:1
- D. 15:1
- **Correct Answer: B**

- **Explanation:** Residues with C:N > 30:1 do not contain enough nitrogen to satisfy the requirements of decomposing microbes. The microbes absorb inorganic nitrogen from the soil solution, causing net N immobilization. Below 20:1, net mineralization occurs (KPSC AO 2017).

Q185. Which group of microorganisms is responsible for the degradation of lignin, the most recalcitrant organic component of plant residues?

- A. Spore-forming bacteria
- B. White-rot Basidiomycete fungi
- C. Green algae
- D. Methanogenic archaeobacteria
- **Correct Answer: B**

- **Explanation:** Lignin degradation is slow and aerobic. White-rot fungi (Basidiomycetes) are the most efficient lignin decomposers, using extracellular enzymes like lignin peroxidase, manganese peroxidase, and laccase (KPSC AAO 2017).

Q186. During the composting process, which microbial group dominates during the high-temperature 'thermophilic' phase (50-60°C)?

- A. Mesophilic fungi
- B. Yeasts
- C. Thermophilic actinomycetes and Bacillus species
- D. Pseudomonas species
- **Correct Answer: C**
- **Explanation:** As composting temperature rises above 45°C, mesophiles die or go dormant. Thermophilic bacteria (such as Bacillus subtilis) and thermophilic actinomycetes (like Thermoactinomyces) take over, breaking down complex proteins, cellulose, and hemicellulose (KPSC AO 2017).

Q187. Lactic acid bacteria preserve green forage as silage by anaerobically fermenting sugars to lower the pH to which target range?

- A. 5.5 to 6.0
- B. 3.8 to 4.2
- C. 2.0 to 2.5
- D. 7.0 to 7.5
- **Correct Answer: B**
- **Explanation:** Silage preservation depends on rapid acidification under anaerobic conditions. Lactic acid bacteria ferment water-soluble carbohydrates to lactic acid, dropping the pH to 3.8-4.2, which inhibits spoilage microbes (KPSC AAO 2017).

Q188. Which of the following is the most widely used carrier material for commercial biofertilizers in India?

- A. Vermiculite
- B. Lignite
- C. Charcoal
- D. Talc
- **Correct Answer: B**
- **Explanation:** Powdered lignite is the most common carrier for bacterial biofertilizers (Rhizobium, Azotobacter, PSB) in India due to its high water-holding capacity, organic matter content, and availability (KPSC AO 2017).

Q189. As per BIS standards, what is the minimum viable count of Rhizobium cells per gram of carrier-based biofertilizer at the time of manufacture?

- A. 10^5 cells/g
- B. 10^6 cells/g
- C. 10^8 cells/g
- D. 10^{10} cells/g
- **Correct Answer: C**
- **Explanation:** According to Bureau of Indian Standards (BIS) and FCO specifications, carrier-based biofertilizers must contain a minimum viable count of 10^8 cells per gram of carrier at the time of manufacture, and 10^7 cells/g at the time of expiry (KPSC AAO 2017).

Q190. What is the function of siderophores produced by biocontrol agents like Pseudomonas fluorescens?

- A. Fixing nitrogen in the rhizosphere
- B. Solubilizing rock phosphate

- C. Chelating ferric iron to suppress pathogens
- D. Synthesizing plant growth hormones

• **Correct Answer: C**

- **Explanation:** Siderophores are low-molecular-weight iron-chelating ligands. Biocontrol bacteria release them to bind ferric iron (Fe^{3+}), making it unavailable to soil-borne fungal pathogens (like *Pythium*), thereby suppressing disease (KPSC AO 2017).

Q191. Which gene cluster encodes the nitrogenase enzyme and other proteins required for biological nitrogen fixation in diazotrophs?

- A. nod genes
- B. nif genes
- C. fix genes
- D. vir genes

• **Correct Answer: B**

- **Explanation:** The nif (nitrogen fixation) gene cluster contains structural genes (nifH, nifD, nifK) encoding nitrogenase and regulatory/processing genes involved in cofactor synthesis and electron transport (KPSC AO 2017).

Q192. Fungal cell walls differ from bacterial cell walls because they contain which polymer?

- A. Peptidoglycan
- B. Chitin
- C. Cellulose
- D. Teichoic acid

• **Correct Answer: B**

- **Explanation:** Fungal cell walls are composed of chitin (a polymer of N-acetylglucosamine) and glucans. Bacterial cell walls contain peptidoglycan (KPSC AAO 2017).

Q193. The enzyme nitrogenase complex, which reduces N_2 to NH_3 , is composed of which two metalloproteins?

- A. Iron protein (Fe-protein) and Molybdenum-Iron protein (MoFe-protein)
- B. Copper protein and Zinc protein
- C. Cytochrome and Plastocyanin
- D. Ferritin and Leghemoglobin

• **Correct Answer: A**

- **Explanation:** Nitrogenase consists of Component II (dinitrogenase reductase, an Fe-protein that provides electrons) and Component I (dinitrogenase, a MoFe-protein that binds and reduces nitrogen gas) (KPSC AO 2017).

Q194. Which free-living nitrogen-fixing bacterium is acidophilic and dominant in acidic tropical soils?

- A. *Azotobacter chroococcum*
- B. *Beijerinckia indica*
- C. *Azospirillum lipoferum*
- D. *Derris gummosa*

• **Correct Answer: B**

- **Explanation:** *Beijerinckia indica* is characteristically acid-tolerant (acidophilic) and occurs primarily in tropical soils with low pH,

whereas Azotobacter is highly sensitive to acid and prefers neutral/alkaline pH (KPSC AAO 2017).

Q195. Which type of mycorrhizae forms a mantle (sheath) of hyphae around the root surface and a intercellular network called the Hartig net within the root cortex?

- A. Endomycorrhizae
- B. Ectomycorrhizae
- C. Ericoid mycorrhizae
- D. Orchid mycorrhizae
- **Correct Answer: B**
- **Explanation:** Ectomycorrhizae do not penetrate the host root cells. Instead, they form an external fungal mantle around the root tip and a network of hyphae (Hartig net) between the cortical cells. They are common in woody forest trees (Pinaceae, Fagaceae) (KPSC AAO 2017).

Q196. If a map is drawn to a scale where 1 cm represents 20 meters on the ground, what is its Representative Fraction (RF)?

- A. 1:20
- B. 1:200
- C. 1:2000
- D. 1:20000
- **Correct Answer: C**
- **Explanation:** Representative Fraction is the ratio of map distance to ground distance in the same units. $1 \text{ cm} / 20 \text{ m} = 1 \text{ cm} / (20 * 100) \text{ cm} = 1:2000$ (KPSC AO 2017).

Q197. What is the standard length of a Gunter's chain used in surveying?

- A. 100 feet
- B. 66 feet
- C. 33 feet
- D. 50 feet
- **Correct Answer: B**
- **Explanation:** Gunter's chain is 66 feet long and contains 100 links (each link is 0.66 feet or 7.92 inches). It is widely used because 10 square chains equal 1 acre (KPSC AO 2014).

Q198. In leveling operations, what is the first reading taken on a benchmark of known elevation after setting up the instrument?

- A. Fore Sight (FS)
- B. Intermediate Sight (IS)
- C. Back Sight (BS)
- D. Reduced Level (RL)
- **Correct Answer: C**
- **Explanation:** A Back Sight (BS) is the first staff reading taken on a point of known elevation (such as a Benchmark or Change Point) to determine the Height of Instrument (HI = Elevation + BS) (KPSC AO 2017).

Q199. Given: Benchmark elevation = 100.00 m, Back Sight = 1.50 m, and Fore Sight on a point A = 2.50 m. What is the Reduced Level (RL) of point A?

- A. 99.00 m
- B. 101.00 m
- C. 104.00 m
- D. 97.50 m

• **Correct Answer: A**

- **Explanation:** Height of Instrument (HI) = BM Elevation + BS = 100.00 + 1.50 = 101.50 m. Reduced Level of point A = HI - FS = 101.50 - 2.50 = 99.00 m (KPSC AO 2017).

Q200. In the Universal Soil Loss Equation (USLE), $A = R * K * LS * C * P$, what does the factor 'K' represent?

- A. Rainfall erosivity factor
- B. Soil erodibility factor
- C. Cover management factor
- D. Slope length factor

• **Correct Answer: B**

- **Explanation:** In USLE, R represents rainfall runoff erosivity, K represents soil erodibility (inherent susceptibility of soil to erosion), LS represents slope length and steepness, C represents cover management, and P represents support practices (KPSC AAO 2015).

Q201. Which stage of wind erosion is responsible for the movement of 50% to 75% of soil particles, specifically those with diameters between 0.1 and 0.5 mm?

- A. Suspension
- B. Saltation
- C. Surface creep
- D. Detachment

• **Correct Answer: B**

- **Explanation:** Saltation is the primary mode of wind erosion where medium-sized soil particles (0.1 to 0.5 mm) bounce along the surface. Suspension moves fine particles (<0.1 mm) into the air, and surface creep rolls larger particles (>0.5 mm) along the ground (KPSC AO 2017).

Q202. For which conditions is contour bunding recommended as a soil and water conservation measure?

- A. Rainfall > 800 mm, land slope > 10%
- B. Rainfall < 600 mm, land slope up to 6%
- C. Rainfall 600-800 mm, land slope 6-10%
- D. Undulating sandy areas with high rainfall

• **Correct Answer: B**

- **Explanation:** Contour bunding is practiced in low rainfall areas (<600 mm) on slopes up to 6%. The bunds store runoff water behind them, promoting infiltration. In higher rainfall areas, graded bunds are used (KPSC AAO 2017).

Q203. The Broad Bed Furrow (BBF) system developed by ICRISAT for deep black soils consists of a bed and a furrow of what standard total width?

- A. 100 cm bed and 50 cm furrow (total 150 cm)
- B. 75 cm bed and 25 cm furrow (total 100 cm)
- C. 120 cm bed and 30 cm furrow (total 150 cm)

- D. 150 cm bed and 50 cm furrow (total 200 cm)

• **Correct Answer: A**

- **Explanation:** The BBF system consists of a 100 cm wide semi-permanent bed and a 50 cm wide furrow (total spacing 150 cm). It is highly effective for drainage and moisture conservation in Vertisols (KPSC AAO 2017).

Q204. A four-stroke internal combustion engine completes one power stroke in how many revolutions of the crankshaft?

- A. One
- B. Two
- C. Four
- D. Half

• **Correct Answer: B**

- **Explanation:** In a four-stroke engine, a complete cycle consists of four piston strokes (suction, compression, power, exhaust) corresponding to two full revolutions (720 degrees) of the crankshaft (KPSC AO 2017).

Q205. What is the typical compression ratio range of a agricultural diesel engine?

- A. 6:1 to 10:1
- B. 14:1 to 20:1
- C. 22:1 to 30:1
- D. 4:1 to 6:1

• **Correct Answer: B**

- **Explanation:** Diesel engines are Compression Ignition (CI) engines and require high compression ratios (typically 14:1 to 20:1) to heat the air sufficiently to ignite the injected fuel. Petrol engines (SI) range from 6:1 to 10:1 (KPSC AO 2017).

Q206. The power developed inside the engine cylinder of an internal combustion engine is called:

- A. Brake Horse Power (BHP)
- B. Indicated Horse Power (IHP)
- C. Frictional Horse Power (FHP)
- D. Drawbar Horse Power (DHP)

• **Correct Answer: B**

- **Explanation:** Indicated Horse Power (IHP) is the theoretical power generated inside the engine cylinder by combustion. Brake Horse Power (BHP) is the actual power available at the crankshaft ($BHP = IHP - \text{Frictional HP}$) (KPSC AAO 2017).

Q207. Which part of the mouldboard plough is responsible for cutting the furrow slice horizontally from the ground?

- A. Share
- B. Mouldboard
- C. Landside
- D. Frog

• **Correct Answer: A**

- **Explanation:** The share is the sharp, cutting edge of the plough bottom that makes the horizontal cut of the furrow slice. The mouldboard lifts and turns it, and the landside counteracts lateral pressure (KPSC AAO 2015).

Q208. In a disc plough, what is the angle between the plane of the cutting edge of the disc and the direction of travel called?

- A. Tilt angle
- B. Disc angle
- C. Rake angle
- D. Clearance angle
- **Correct Answer: B**
- **Explanation:** The disc angle is the angle between the plane of the cutting edge of the disc and the line of travel, typically 42° to 45°. The tilt angle is the inclination from the vertical, typically 15° to 25° (KPSC AAO 2017).

Q209. Which type of seed metering mechanism is most commonly found in standard Indian seed-cum-fertilizer drills?

- A. Inclined plate
- B. Fluted feed roller
- C. Cup feed mechanism
- D. Brush feed mechanism
- **Correct Answer: B**
- **Explanation:** The fluted feed roller is the most widely used seed metering mechanism. It consists of a fluted cylinder that rotates to deliver seeds. The seed rate is adjusted by shifting the fluted roller axially (KPSC AO 2017).

Q210. During calibration of a 9-row seed drill, the row-to-row spacing is 20 cm. What is the effective working width of the seed drill?

- A. 1.80 meters
- B. 1.60 meters
- C. 2.00 meters
- D. 1.50 meters
- **Correct Answer: A**
- **Explanation:** Working width (W) = Number of furrows (N) * Row spacing (S). $W = 9 * 20 \text{ cm} = 180 \text{ cm} = 1.80 \text{ meters}$ (KPSC AAO 2015).

Q211. Which type of sprayer is hand-operated, carried on the operator's back, and requires continuous pumping to maintain spray pressure?

- A. Compression sprayer
- B. Knapsack sprayer
- C. Power sprayer
- D. Ultra-Low Volume (ULV) sprayer
- **Correct Answer: B**
- **Explanation:** The knapsack sprayer has a hydraulic pump operated by a lever, requiring continuous pumping during spraying. A compression sprayer is pumped beforehand to build air pressure inside the tank (KPSC AO 2017).

Q212. Which irrigation method has the highest water application efficiency (90% to 95%) by delivering water directly to the plant root zone?

- A. Sprinkler irrigation
- B. Drip irrigation

- C. Border strip irrigation
- D. Check basin irrigation

• **Correct Answer: B**

- **Explanation:** Drip (or trickle) irrigation delivers water slowly and directly to the plant root zone through emitters, minimizing evaporation and percolation losses, achieving 90-95% efficiency (KPSC AAO 2017).

Q213. What is a trapezoidal weir with side slopes of 1 horizontal to 4 vertical used for water measurement in open channels called?

- A. Rectangular weir
- B. Cippoletti weir
- C. V-notch weir
- D. Parshall flume

• **Correct Answer: B**

- **Explanation:** The Cippoletti weir is a trapezoidal weir where the sides have a slope of 1:4 (horizontal:vertical). This specific slope compensates for end contractions of the water flow (KPSC AO 2017).

Q214. Which water measuring device is highly preferred for channels carrying sediment-laden water because it resists clogging and operates under low head loss?

- A. V-notch weir
- B. Rectangular weir
- C. Parshall flume
- D. Orifice plate

• **Correct Answer: C**

- **Explanation:** A Parshall flume is a self-cleaning venturi-type flume with a contracting throat, depressing floor, and diverging exit. It is ideal for sediment-laden agricultural runoff and irrigation channels (KPSC AAO 2017).

Q215. For the safe storage of cereal grains like paddy and wheat, what is the maximum recommended moisture content?

- A. 8% to 10%
- B. 12% to 14%
- C. 16% to 18%
- D. 5% to 6%

• **Correct Answer: B**

- **Explanation:** Cereal grains can be safely stored at 12% to 14% moisture content. For oilseeds, which contain higher lipids, the moisture content must be lower (8% to 10%) to prevent rancidity and microbial growth (KPSC AAO 2017).

Q216. Deep bin drying differs from shallow bin drying because in deep bins, the depth of the grain mass is:

- A. Less than its diameter
- B. Greater than its diameter
- C. Equal to its diameter
- D. Restricted to less than 1 meter

• **Correct Answer: B**

- **Explanation:** Deep bin drying applies to bins where the grain depth is greater than the bin's diameter (often >2.5 m). Airflow resistance is high, requiring high static pressure fans. In shallow bins, depth is less than diameter (KPSC AO 2017).

Q217. What is the primary objective of parboiling paddy before milling?

- A. To decrease the milling yield
- B. To reduce head rice recovery
- C. To gelatinize starch, reducing grain breakage during milling
- D. To bleach the grain and make it white
- **Correct Answer: C**

- **Explanation:** Parboiling (soaking, steaming, and drying) gelatinizes the starch granules inside the rice kernel, filling cracks and hardening the grain. This significantly reduces breakage (improving head rice recovery) and retains B-vitamins (KPSC AAO 2017).

Q218. Which type of sub-surface drainage system layout consists of lateral drains entering the main drain from both sides at right angles?

- A. Herringbone system
- B. Gridiron system
- C. Random system
- D. Double main system
- **Correct Answer: B**
- **Explanation:** In a gridiron system, parallel laterals enter the main drain from one side or both sides at right angles, suitable for uniform flat fields. Herringbone has laterals entering the main at acute angles from both sides (KPSC AAO 2017).

Q219. What is the primary source of farm power in Indian agriculture that has shown a declining trend over the past few decades?

- A. Mechanical power
- B. Electrical power
- C. Draught animal power
- D. Human power
- **Correct Answer: C**
- **Explanation:** Draught animal power (bullocks, buffalos) was once the mainstay of Indian farming, but has declined significantly due to rapid mechanization (tractors, power tillers, diesel engines) (KPSC AO 2017).

Q220. The speed ratio of the camshaft to the crankshaft in a four-stroke internal combustion engine is:

- A. 1:1
- B. 1:2
- C. 2:1
- D. 1:4
- **Correct Answer: B**
- **Explanation:** In a four-stroke engine, each valve opens once every two revolutions of the crankshaft. Therefore, the camshaft (which operates the valves) must rotate at half the speed of the crankshaft, a ratio of 1:2 (KPSC AAO 2017).

Q221. Which of the following histones is NOT a component of the nucleosome core particle but functions as a linker histone to stabilize chromatin structure?

- A. Histone H2A
- B. Histone H2B
- C. Histone H3
- D. Histone H1

• **Correct Answer: D**

- **Explanation:** The nucleosome core particle consists of an octamer of core histones (two each of H2A, H2B, H3, and H4) and 146 bp of DNA wrapped around it. Histone H1 acts as a linker histone that binds to the linker DNA and the entry/exit site on the core particle, stabilizing the chromatin fiber. (KPSC AO 2017)

Q222. Meiotic crossing over, which leads to genetic recombination, occurs during which specific stage of Prophase I?

- A. Leptotene
- B. Zygotene
- C. Pachytene
- D. Diplotene

• **Correct Answer: C**

- **Explanation:** Crossing over occurs in the pachytene stage of Prophase I in meiosis. During this stage, homologous chromosomes are fully synapsed, and non-sister chromatids of homologous chromosomes exchange genetic segments. (KPSC AO 2014)

Q223. In complementary gene interaction, two non-allelic genes are required to produce the wild-type phenotype. What is the characteristic F2 phenotypic ratio?

- A. 9:7
- B. 15:1
- C. 9:3:4
- D. 12:3:1

• **Correct Answer: A**

- **Explanation:** Complementary gene action yields an F2 phenotypic ratio of 9:7. It occurs when a dominant allele from both loci (e.g., A_B_) is required to express the wild-type phenotype, while any other combination results in the mutant/alternative phenotype. (KPSC AO 2014)

Q224. Which F2 phenotypic ratio is observed in duplicate dominant epistasis, where either of the dominant genes alone or together produces the same phenotype?

- A. 9:6:1
- B. 15:1
- C. 13:3
- D. 12:3:1

• **Correct Answer: B**

- **Explanation:** Duplicate dominant epistasis produces a 15:1 F2 ratio. A single dominant allele of either gene (A_ or B_ or A_B_) is sufficient to produce the dominant phenotype. Only the double homozygous recessive genotype (aabb) produces the recessive phenotype. (KPSC AO 2017)

Q225. In recessive epistasis (supplementary gene action), the homozygous recessive genotype at one locus masks the expression of alleles at another locus. What is the F2 ratio?

- A. 9:7

- B. 12:3:1
- C. 9:3:4
- D. 9:6:1
- **Correct Answer: C**
- **Explanation:** Recessive epistasis results in a 9:3:4 F₂ ratio. This occurs when the recessive homozygote at one locus (e.g., aa) masks the phenotypic expression of the alleles at a second locus (B or b), as observed in onion bulb color. (KPSC AAO 2015)

Q226. What F₂ phenotypic ratio is produced by dominant epistasis, where a dominant allele at one locus masks the expression of both dominant and recessive alleles at a second locus?

- A. 12:3:1
- B. 13:3
- C. 9:7
- D. 9:6:1
- **Correct Answer: A**
- **Explanation:** Dominant epistasis yields a 12:3:1 F₂ ratio. Here, a dominant allele at the first locus (A₋) masks the expression of alleles at the second locus (B or b). The alleles at the second locus are expressed only when the first locus is homozygous recessive (aa). (KPSC AO 2014)

Q227. A cross between plants segregating for polymeric gene interaction (duplicate genes with cumulative effect) will yield which F₂ phenotypic ratio?

- A. 9:7
- B. 9:6:1
- C. 15:1
- D. 13:3
- **Correct Answer: B**
- **Explanation:** Polymeric gene interaction displays a 9:6:1 ratio in the F₂ generation. If both genes are dominant (A₋B₋), they produce one phenotype; if either gene is dominant alone (A₋bb or aaB₋), they produce a second intermediate phenotype; and if both are recessive (aabb), they produce a third phenotype. (KPSC AAO 2017)

Q228. In inhibitory gene action, a dominant allele of one gene prevents the phenotypic expression of a dominant allele at another locus. What is the F₂ phenotypic ratio?

- A. 13:3
- B. 12:3:1
- C. 9:7
- D. 15:1
- **Correct Answer: A**
- **Explanation:** Inhibitory gene action yields a 13:3 ratio in F₂. This happens when a dominant inhibitor allele (I₋) masks the expression of a dominant trait-producing allele (A₋). Thus, only aaB₋ (or similar) will express the alternative phenotype. (KPSC AO 2017)

Q229. Wilhelm Johannsen formulated the 'Pure-line Theory' in 1903. Which self-pollinated crop did he use for his studies?

- A. *Pisum sativum*
- B. *Phaseolus vulgaris*

- C. *Hordeum vulgare*
- D. *Triticum aestivum*
- **Correct Answer: B**

• **Explanation:** Wilhelm Johannsen conducted his experiments on the 'Princess' variety of *Phaseolus vulgaris* (common bean), demonstrating that selection within a pure line is ineffective because the variation is environmental and not heritable. (KPSC AO 2014)

Q230. Who first proposed the single-cross hybrid method in maize in 1909 using inbred lines?

- A. D.F. Jones
- B. G.H. Shull
- C. W. Johannsen
- D. H. Nilsson-Ehle
- **Correct Answer: B**

• **Explanation:** G.H. Shull proposed the single-cross hybrid method in maize in 1909. It involves crossing two distinct inbred lines (A x B) to produce an F1 hybrid that exhibits maximum heterosis. (KPSC AAO 2015)

Q231. The double-cross hybrid method in maize was proposed by D.F. Jones in 1918 to overcome which major limitation of single crosses?

- A. Inbreeding depression in F1
- B. Low seed yields and poor quality of seeds produced on weak inbred female parents
- C. Lack of genetic uniformity in the commercial crop
- D. Poor adaptability of single-cross hybrids
- **Correct Answer: B**

• **Explanation:** D.F. Jones proposed the double-cross hybrid system (A x B) x (C x D) in 1918 because early maize inbreds were highly weak and yielded very low amounts of seed. Crossing two single-cross hybrids (which are vigorous F1s) as parents allowed high commercial seed yields. (KPSC AO 2017)

Q232. The term 'Heterosis' was coined by which scientist in 1914 to replace the phrase 'heterozygosis'?

- A. G.H. Shull
- B. E.M. East
- C. W. Bateson
- D. T.H. Morgan
- **Correct Answer: A**

• **Explanation:** The term 'Heterosis' was coined by G.H. Shull in 1914 to describe the increased vigor, size, speed of development, and fertility of F1 hybrids over their parents. (KPSC AO 2017)

Q233. According to the dominance hypothesis of heterosis (Davenport, 1908), hybrid vigor is due to which genetic mechanism?

- A. Absolute superiority of the heterozygous genotype over both homozygotes
- B. Masking of deleterious recessive alleles by dominant alleles accumulated from both parents
- C. Cumulative effect of multiple genes governing a quantitative trait
- D. Non-allelic gene interactions like epistasis
- **Correct Answer: B**

- **Explanation:** The dominance hypothesis of heterosis suggests that heterosis results from the accumulation of dominant favorable alleles from both parents, which mask the expression of harmful, deleterious recessive alleles. (KPSC AO 2014)

Q234. The overdominance hypothesis of heterosis (Shull and East) states that:

- A. Favorable dominant alleles mask unfavorable recessives
- B. The heterozygous genotype (Aa) has a higher physiological value than either homozygous genotype (AA or aa)
- C. Polygenes act in a purely additive manner
- D. Mitochondria from the female parent interact with chloroplasts from the male parent
- **Correct Answer: B**
- **Explanation:** The overdominance hypothesis suggests that the heterozygote (Aa) is superior to both homozygotes (AA and aa) at a given locus. This is also called single-locus heterosis or cumulative action of divergent alleles. (KPSC AAO 2015)

Q235. Triticale is an allopolyploid (specifically an amphidiploid) developed by crossing which two crop species?

- A. Triticum durum x Triticum aestivum
- B. Triticum aestivum x Secale cereale
- C. Oryza sativa x Secale cereale
- D. Hordeum vulgare x Secale cereale
- **Correct Answer: B**
- **Explanation:** Triticale is the first man-made cereal crop, synthesized by crossing wheat (Triticum aestivum or Triticum turgidum) with rye (Secale cereale), followed by chromosome doubling using colchicine. (KPSC AO 2014)

Q236. In U's Triangle (Woo Jang-choon, 1935), Brassica napus (n=19) is shown to be an allopolyploid derived from which parents?

- A. Brassica nigra (n=8) and Brassica rapa (n=10)
- B. Brassica nigra (n=8) and Brassica oleracea (n=9)
- C. Brassica oleracea (n=9) and Brassica rapa (n=10)
- D. Brassica juncea (n=18) and Brassica nigra (n=8)
- **Correct Answer: C**
- **Explanation:** According to U's Triangle, Brassica napus (rape, $2n=38$) is an allotetraploid derived from hybridization between Brassica oleracea (cabbage, $n=9$) and Brassica rapa (turnip rape, $n=10$) followed by chromosome doubling. (KPSC AAO 2017)

Q237. Which species combination and chromosome doubling leads to the production of the amphidiploid Brassica juncea (Indian mustard, n=18)?

- A. Brassica nigra (n=8) x Brassica rapa (n=10)
- B. Brassica nigra (n=8) x Brassica oleracea (n=9)
- C. Brassica carinata (n=17) x Brassica rapa (n=10)
- D. Brassica oleracea (n=9) x Brassica rapa (n=10)
- **Correct Answer: A**
- **Explanation:** Brassica juncea ($2n=36$) is an amphidiploid arising from the hybridization of Brassica nigra (black mustard, $n=8$) and Brassica rapa ($n=10$), followed by doubling of chromosomes. (KPSC AO 2017)

Q238. Brassica carinata (Abyssinian mustard, n=17) is an allopolyploid species derived from which two Brassica diploids?

- A. B. nigra ($n=8$) and B. rapa ($n=10$)
- B. B. nigra ($n=8$) and B. oleracea ($n=9$)
- C. B. oleracea ($n=9$) and B. rapa ($n=10$)
- D. B. juncea ($n=18$) and B. napus ($n=19$)
- **Correct Answer: B**
- **Explanation:** Brassica carinata ($2n=34$) is an allopolyploid species developed from natural crossing between Brassica nigra ($n=8$) and Brassica oleracea ($n=9$). (KPSC AAO 2015)

Q239. A chromosome mutation resulting in the loss of a single chromosome from a diploid set ($2n-1$) is classified as:

- A. Nullisomy
- B. Monosomy
- C. Trisomy
- D. Haploidy
- **Correct Answer: B**
- **Explanation:** Monosomy is a form of aneuploidy where a diploid cell lacks one chromosome ($2n-1$). Nullisomy is $2n-2$, trisomy is $2n+1$, and tetrasomy is $2n+2$. (KPSC AO 2017)

Q240. Which chromosome aberration is particularly useful for allocating newly discovered genes to specific chromosomes in hexaploid bread wheat (*Triticum aestivum*)?

- A. Inversions
- B. Monosomics
- C. Duplications
- D. Translocations
- **Correct Answer: B**
- **Explanation:** Monosomic series ($2n-1$) in hexaploid bread wheat (developed by Ernest Sears) are highly stable and are used as a powerful cytogenetic tool to locate genes to specific chromosomes. (KPSC AAO 2017)

Q241. What is the biochemical mode of action of colchicine, which is used to induce polyploidy in plant breeding?

- A. Inhibits DNA polymerase activity during DNA replication
- B. Prevents spindle fiber assembly by binding to tubulin heterodimers
- C. Excises nitrogenous bases from DNA strands
- D. Blocks transcription of mitotic cyclin genes
- **Correct Answer: B**
- **Explanation:** Colchicine is an alkaloid that binds to tubulin dimers, preventing their polymerization into microtubules. This inhibits the formation of spindle fibers during mitosis, halting cell division at metaphase and resulting in a doubled chromosome number in the cell. (KPSC AAO 2015)

Q242. Seedless triploid watermelon ($3x$) is produced by crossing a tetraploid ($4x$) line and a diploid ($2x$) line. Which parent must be the tetraploid?

- A. The pollen parent (male)
- B. The seed parent (female)
- C. Either male or female parent can be tetraploid

- D. A cytoplasmic male-sterile line

- **Correct Answer: B**

- **Explanation:** In triploid watermelon seed production, the tetraploid (4x) must be the female (seed) parent, and the diploid (2x) must be the male (pollen) parent. A reciprocal cross (diploid female x tetraploid male) results in poor seed development and low viability. (KPSC AO 2017)

Q243. A point mutation in DNA involving the substitution of adenine (purine) for guanine (purine) is classified as a:

- A. Transition mutation
- B. Transversion mutation
- C. Frameshift mutation
- D. Deletion mutation

- **Correct Answer: A**

- **Explanation:** Transition mutations are point mutations that interchange a purine for another purine (A $\leftrightarrow$ G) or a pyrimidine for another pyrimidine (C $\leftrightarrow$ T). Transversions exchange a purine for a pyrimidine, or vice versa. (KPSC AAO 2017)

Q244. A point mutation where a cytosine (pyrimidine) is replaced by an adenine (purine) is a:

- A. Transition
- B. Transversion
- C. Frameshift
- D. Insertion

- **Correct Answer: B**

- **Explanation:** This is a transversion mutation because a pyrimidine (cytosine) is replaced by a purine (adenine). Transversions represent substitutions across chemical groups. (KPSC AO 2017)

Q245. The chemical mutagen Ethyl Methane Sulfonate (EMS) acts primarily by which mechanism to induce mutations in crop plants?

- A. Deaminating cytosine to uracil
- B. Intercalating between DNA bases, causing frameshifts
- C. Alkylating guanine, leading to G:C to A:T transition mutations
- D. Forming thymine dimers

- **Correct Answer: C**

- **Explanation:** Ethyl Methane Sulfonate (EMS) is an alkylating agent that adds an ethyl group to guanine, forming O6-ethylguanine. During DNA replication, this modified guanine pairs with thymine instead of cytosine, causing a G:C to A:T transition mutation. (KPSC AAO 2017)

Q246. How do acridine dyes like proflavin and acridine orange induce mutations in DNA?

- A. By causing alkylation of adenine
- B. By intercalating between adjacent base pairs, causing insertions or deletions during replication
- C. By producing free radicals that break phosphodiester bonds
- D. By acting as base analogs

- **Correct Answer: B**

- **Explanation:** Acridine dyes are planar molecules that insert (intercalate) between adjacent base pairs of the DNA double helix. This

distorts the helix, causing DNA polymerase to slip and insert or delete single bases, leading to frameshift mutations. (KPSC AO 2014)

Q247. Who was the pioneering scientist who first demonstrated in 1927 that X-rays could induce artificial mutations in *Drosophila melanogaster*?

- A. L.J. Stadler
- B. H.J. Muller
- C. T.H. Morgan
- D. Hugo de Vries
- **Correct Answer: B**
- **Explanation:** H.J. Muller discovered the mutagenic effect of X-rays in *Drosophila* in 1927, which earned him the Nobel Prize. L.J. Stadler demonstrated the mutagenic effects of X-rays in plants (barley and maize) shortly after. (KPSC AO 2014)

Q248. Which scientist demonstrated the mutagenic effects of X-rays in crop plants (specifically barley and maize) in 1928?

- A. H.J. Muller
- B. L.J. Stadler
- C. G.H. Shull
- D. Barbara McClintock
- **Correct Answer: B**
- **Explanation:** L.J. Stadler independently demonstrated that X-rays and ultraviolet radiation were highly mutagenic in plants using maize and barley in 1928. (KPSC AAO 2015)

Q249. A mutation that changes a codon specifying the amino acid glutamine (CAA) into a termination codon (UAA) is known as a:

- A. Missense mutation
- B. Silent mutation
- C. Nonsense mutation
- D. Frameshift mutation
- **Correct Answer: C**
- **Explanation:** A nonsense mutation is a point mutation that converts a codon that specifies an amino acid into a premature stop codon (UAG, UAA, or UGA), leading to the premature termination of translation and a truncated protein. (KPSC AO 2017)

Q250. A mutation that changes a codon from GAG (coding for glutamic acid) to GTG (coding for valine) is classified as a:

- A. Missense mutation
- B. Silent mutation
- C. Nonsense mutation
- D. Synonymous mutation
- **Correct Answer: A**
- **Explanation:** A missense mutation is a single nucleotide substitution that changes the codon to specify a different amino acid (glutamic acid to valine), which can significantly alter the function of the translated protein. (KPSC AAO 2015)

Q251. Due to the degeneracy of the genetic code, some base substitutions do not change the amino acid sequence of the protein. These are known as:

- A. Missense mutations
- B. Silent mutations
- C. Nonsense mutations
- D. Frameshift mutations
- **Correct Answer: B**
- **Explanation:** Silent (or synonymous) mutations are point mutations that alter a codon but do not change the amino acid incorporated into the polypeptide chain because multiple codons specify the same amino acid. (KPSC AO 2014)

Q252. Cytoplasmic male sterility (CMS), which is widely used in hybrid seed production, is governed by genes residing in which cellular organelle?

- A. Plastids
- B. Mitochondria
- C. Nucleolus
- D. Ribosomes
- **Correct Answer: B**
- **Explanation:** CMS is governed by mitochondrial genes (mitochondrial DNA). Since cytoplasm is inherited maternally through the female parent's egg cell, CMS exhibits maternal inheritance. (KPSC AO 2017)

Q253. In Cytoplasmic-Genetic Male Sterility (CGMS), how is pollen fertility restored in the F1 hybrid crop sold to farmers?

- A. By treating the F1 seeds with gibberellic acid
- B. By nuclear restorer (Rf) genes provided by the male parent (R-line)
- C. By cytoplasmic factors present in the maintainer parent
- D. By spontaneous reversion during meiosis
- **Correct Answer: B**
- **Explanation:** In CGMS, fertility is restored in the commercial F1 hybrid by nuclear restorer (Rf) genes. The female line (A-line) has sterile cytoplasm and recessive non-restoring nuclear alleles. The restorer line (R-line) provides dominant nuclear Rf alleles that suppress the mitochondrial sterility, making the F1 hybrid fully fertile. (KPSC AAO 2017)

Q254. In Gametophytic Self-Incompatibility (GSI), the incompatibility reaction is determined by:

- A. The diploid genotype of the maternal tissue of the pistil
- B. The haploid S-allele genotype of the pollen grain itself
- C. The diploid genotype of the parent plant that produced the pollen
- D. The environmental temperature during pollen tube growth
- **Correct Answer: B**
- **Explanation:** In gametophytic self-incompatibility, the phenotype of the pollen is determined by its own haploid S-genotype. If the pollen grain carries an S-allele (e.g., S1) that matches either S-allele in the diploid pistil (e.g., S1S2), the pollen tube growth will be arrested in the style. (KPSC AAO 2017)

Q255. In Sporophytic Self-Incompatibility (SSI), pollen tube germination or penetration is blocked at the stigma surface.

This reaction is determined by:

- A. The haploid S-allele of the pollen grain
- B. The diploid S-allele genotype of the parent plant that produced the pollen

- C. The genotype of the egg cell
- D. The maternal chloroplast DNA of the stigma
- **Correct Answer: B**
- **Explanation:** In SSI, the pollen phenotype is determined by the diploid S-genotype of the parent plant that produced the pollen. If the pollen parent is S1S2, all pollen grains behave as S1S2, regardless of their individual S-allele, due to the deposition of S-gene products from the tapetum onto the pollen coat. (KPSC AO 2014)

Q256. In commercial hybrid seed production using the CMS system, what is the role of the B-line (maintainer line)?

- A. It is used as the male parent to produce fertile hybrid seeds for the market
- B. It is isogenic to the A-line but is male-fertile, used to multiply seeds of the male-sterile A-line
- C. It is used to restore fertility in the commercial hybrid crop
- D. It is a source of new mutations for disease resistance
- **Correct Answer: B**
- **Explanation:** The B-line is the maintainer line. It has normal cytoplasm (fertile) but has the same nuclear genotype as the A-line (male-sterile). When A-line (female) is pollinated by B-line (male), the progeny is 100% male-sterile (A-line), which maintains and multiplies the A-line seeds. (KPSC AAO 2017)

Q257. A synthetic variety is produced by crossing which parents in all possible combinations?

- A. A group of open-pollinated varieties of unknown combining ability
- B. A set of inbred lines or clones that have been tested and shown to have high general combining ability (GCA)
- C. Two inbred lines exhibiting maximum specific combining ability (SCA)
- D. Multi-lines carrying vertical resistance genes
- **Correct Answer: B**
- **Explanation:** Synthetic varieties are created by intercrossing a set of inbred lines, clones, or open-pollinated lines that have been pre-tested for high general combining ability (GCA). The variety is then maintained by open pollination. (KPSC AO 2014)

Q258. How does a composite variety differ primarily from a synthetic variety in plant breeding?

- A. A composite variety is genetically homogeneous, whereas a synthetic variety is heterogeneous
- B. A composite variety is developed by mixing seeds of phenotypically superior lines and letting them open pollinate, without testing for combining ability
- C. A composite variety has much lower genetic flexibility than a synthetic variety
- D. Composite varieties require annual seed replacement by the farmer
- **Correct Answer: B**
- **Explanation:** A composite variety is developed by mixing seeds of several outstanding lines/varieties and letting them open pollinate without formal testing for combining ability. Synthetic varieties require testing of parents for GCA prior to synthesis. (KPSC AAO 2015)

Q259. Broad-sense heritability (H^2) is calculated as the ratio of which two variance components?

- A. Additive genetic variance to phenotypic variance
- B. Total genetic variance to phenotypic variance
- C. Dominance variance to additive variance
- D. Environmental variance to genotypic variance
- **Correct Answer: B**

- **Explanation:** Broad-sense heritability (H^2) is the proportion of total phenotypic variance (V_p) that is due to genetic differences (V_g) among individuals: $H^2 = V_g / V_p$. (KPSC AAO 2015)

Q260. Narrow-sense heritability (h^2) is mathematically defined as:

- A. V_g / V_p
- B. V_a / V_p
- C. V_d / V_p
- D. V_e / V_p

• **Correct Answer: B**

- **Explanation:** Narrow-sense heritability (h^2) is the ratio of additive genetic variance (V_a) to phenotypic variance (V_p): $h^2 = V_a / V_p$. It is highly valued in breeding because additive variance is the component that directly responds to selection. (KPSC AO 2017)

Q261. According to N.I. Vavilov, what is the primary center of origin of bread wheat (*Triticum aestivum*)?

- A. Hindustan Center
- B. South-West Asian Center
- C. Mediterranean Center
- D. Chinese Center

• **Correct Answer: B**

- **Explanation:** Vavilov designated the South-West Asian primary center (comprising Asia Minor, Iran, Afghanistan) as the primary center of origin for bread wheat. (KPSC AO 2014)

Q262. The primary center of origin for cultivated rice (*Oryza sativa*) is identified as the:

- A. Chinese Center
- B. Hindustan Center
- C. South American Center
- D. Abyssinian Center

• **Correct Answer: B**

- **Explanation:** The Hindustan primary center of origin (which includes India and Burma) is recognized as the center of origin for cultivated rice. (KPSC AO 2014)

Q263. Acclimatization, which occurs during plant introduction, is defined as:

- A. Genetic engineering of the plant to tolerate heat stress
- B. The adaptation of an introduced plant variety to its new environment over several generations
- C. The application of growth regulators to induce cold tolerance
- D. Testing the introduced plant for seed-borne pathogens

• **Correct Answer: B**

- **Explanation:** Acclimatization is the process of natural selection operating on an introduced plant population, adapting it to its new climatic conditions over several generations of cultivation. (KPSC AAO 2015)

Q264. In the pedigree breeding method of self-pollinated crops, in which generation does individual plant selection and record-keeping begin?

- A. F1 generation
- B. F2 generation

- C. F5 generation
- D. F8 generation
- **Correct Answer: B**
- **Explanation:** In the pedigree method, individual plant selection begins in the F2 generation because it represents the peak of segregation and genetic variability. The selected plants are harvested individually, and their progenies are grown separately in F3. (KPSC AAO 2015)

Q265. What is the primary advantage of the bulk breeding method over the pedigree method in self-pollinated crops?

- A. Allows evaluation of individual plants in F2
- B. It is simple, cheap, and permits natural selection to act on the population under field stress
- C. Leads to complete homozygosity in the F2 generation
- D. Requires extensive record-keeping of parental lineages
- **Correct Answer: B**
- **Explanation:** The bulk method is highly simple and inexpensive compared to the pedigree method because segregating generations are grown in bulk without individual plant selection or pedigree records until F5 or F6, allowing natural selection to filter out undesirable genotypes. (KPSC AO 2017)

Q266. The Single Seed Descent (SSD) method is characterized by which procedure to advance segregating generations?

- A. Bulk harvesting of all seeds from the best F2 plants
- B. Advancing the population by taking a single seed from every plant in the population and bulking them to grow the next generation
- C. Selecting only homozygous recessive plants in each generation
- D. Backcrossing the selected F2 plants to the recurrent parent
- **Correct Answer: B**
- **Explanation:** The SSD method involves harvesting exactly one seed from each plant in the F2 and subsequent generations, mixing these seeds, and growing them to advance generations. It is used to rapidly reach homozygosity under glasshouse conditions without applying selection pressure. (KPSC AAO 2017)

Q267. In the backcross breeding method, the commercial variety that receives the gene of interest is called the:

- A. Non-recurrent parent
- B. Recurrent parent
- C. Donor parent
- D. Tester parent
- **Correct Answer: B**
- **Explanation:** The commercial variety being improved is the recurrent parent (or recipient parent) because it is repeatedly crossed back with the hybrid progeny to recover its elite genetic background. The parent donating the gene is the non-recurrent or donor parent. (KPSC AO 2017)

Q268. Recurrent selection is a breeding procedure designed primarily to:

- A. Develop complete homozygosity in a single generation
- B. Systematically increase the frequency of desirable alleles in a population while maintaining genetic variability
- C. Introduce cytoplasmic male sterility into cross-pollinated crops
- D. Produce F1 hybrid seeds directly
- **Correct Answer: B**

- **Explanation:** Recurrent selection aims to increase the frequency of favorable alleles in a population through repeated cycles of selection, evaluation, and intermating of selected individuals, which avoids the rapid depletion of genetic variability. (KPSC AAO 2017)

Q269. Polygenic inheritance (multiple factor hypothesis) was first demonstrated by H. Nilsson-Ehle in 1909 using which character?

- A. Plant height in pea
- B. Kernel color in wheat
- C. Oil content in corn
- D. Sugar content in sugar beet

• **Correct Answer: B**

- **Explanation:** H. Nilsson-Ehle demonstrated polygenic inheritance in 1909 using the kernel color of wheat, which showed a continuous variation from red to white. He explained this using a model of multiple independent loci having an additive effect on the phenotype. (KPSC AO 2014)

Q270. General combining ability (GCA) is a measure of the average performance of a line in hybrid combinations. It is primarily associated with which type of genetic variance?

- A. Dominance variance
- B. Additive genetic variance
- C. Non-allelic epistatic variance
- D. Environmental variance

• **Correct Answer: B**

- **Explanation:** GCA is primarily due to additive genetic variance and additive x additive gene action, which are fixable components of genetic variance. Specific combining ability (SCA) is due to non-additive gene action (dominance and epistasis). (KPSC AAO 2015)

Q271. What cytological abnormalities are typically formed during meiotic anaphase I in a heterozygote for a paracentric inversion?

- A. A dicentric bridge and an acentric fragment
- B. A ring chromosome and a linear fragment
- C. Univalents and trivalents
- D. Two centromeres on a single chromosome and no centromere on the other

• **Correct Answer: A**

- **Explanation:** A paracentric inversion does not involve the centromere. Crossing over within the inversion loop in a heterozygote leads to the formation of a dicentric chromosome (which forms a dicentric bridge as it is pulled to opposite poles during anaphase I) and an acentric fragment (which lacks a centromere and is lost). (KPSC AAO 2017)

Q272. A pericentric inversion differs from a paracentric inversion in that it:

- A. Excludes the centromere from the inverted segment
- B. Includes the centromere in the inverted segment, which can alter the chromosome arm ratio
- C. Always results in a dicentric bridge
- D. Occurs only in B-chromosomes

• **Correct Answer: B**

- **Explanation:** A pericentric inversion includes the centromere within the inverted region. Consequently, if the break points are

asymmetric relative to the centromere, the arm ratio changes (e.g., metacentric to submetacentric), which can be identified cytologically under the microscope. (KPSC AO 2017)

Q273. If two genes are linked in the coupling phase (cis), a testcross of the dihybrid (AaBb x aabb) will yield a progeny distribution that:

- A. Conforms to a 1:1:1:1 ratio
- B. Shows a significantly higher frequency of parental combinations (AB and ab) compared to recombinant combinations (Ab and aB)
- C. Shows a higher frequency of recombinant combinations (Ab and aB)
- D. Yields only dominant phenotypes
- **Correct Answer: B**
- **Explanation:** In the coupling phase, dominant alleles (A and B) are on one chromosome and recessive alleles (a and b) are on the homologous chromosome. Linkage limits recombination, so the parental gametes (AB and ab) are produced at a much higher frequency than recombinant gametes (Ab and aB). (KPSC AO 2014)

Q274. Pleiotropy is defined as the genetic phenomenon where:

- A. A single gene influences multiple, seemingly unrelated phenotypic traits
- B. Multiple genes contribute to the expression of a single trait
- C. A dominant allele at one locus masks another locus
- D. Cytoplasmic genes modify nuclear gene expression
- **Correct Answer: A**
- **Explanation:** Pleiotropy occurs when a single gene mutation results in multiple distinct phenotypic effects. A classic example is the gene controlling flower color in peas, which also affects seed coat color and axillary pigmentation. (KPSC AAO 2015)

Q275. An environmentally induced phenotype that mimics a genotype-controlled phenotype, but is not genetically transmissible, is called a:

- A. Genocopy
- B. Phenocopy
- C. Pleiotropic mimic
- D. Mutant phenocopy
- **Correct Answer: B**
- **Explanation:** A phenocopy is an individual whose phenotype is modified by environmental factors during development to mimic a phenotype that is otherwise determined by a specific genotype (e.g., heat-shock causing a phenotype that resembles a genetic mutant, but the offspring are normal). (KPSC AO 2017)

Q276. Which term describes the percentage of individuals with a given genotype who actually exhibit the expected phenotype?

- A. Expressivity
- B. Penetrance
- C. Pleiotropy
- D. Epistasis
- **Correct Answer: B**
- **Explanation:** Penetrance refers to the proportion of individuals carrying a particular variant of a gene (genotype) that also express the associated trait (phenotype). Complete penetrance means 100% of carriers express the trait. (KPSC AAO 2017)

Q277. During meiotic Prophase I, the synaptonemal complex begins to assemble, aligning homologous chromosomes. In which stage does this synapsis start?

- A. Leptotene
- B. Zygotene
- C. Pachytene
- D. Diplotene
- **Correct Answer: B**
- **Explanation:** Synapsis, or the highly specific pairing of homologous chromosomes, begins during the zygotene stage of Prophase I. The synaptonemal complex is fully formed by the pachytene stage. (KPSC AO 2017)

Q278. Polytene chromosomes, which are giant chromosomes formed by repeated rounds of replication without cell division, are classically found in which tissues?

- A. Amphibian oocytes
- B. Salivary glands of Dipteran insects
- C. Root tips of onion plants
- D. Pollen mother cells of maize
- **Correct Answer: B**
- **Explanation:** Polytene chromosomes are giant chromosomes first discovered by Balbiani in 1881. They are found in the salivary glands of Dipteran larvae (like *Drosophila*). They are formed by endoreduplication (repeated DNA replication without chromatid separation or cell division). (KPSC AO 2014)

Q279. Inbreeding depression is characterized by a loss of vigor and fertility due to selfing. Its severity is highest in which of the following categories of crops?

- A. Autogamous crops (e.g., rice, wheat)
- B. Allogamous crops (e.g., maize, alfalfa)
- C. Asexually propagated crops
- D. Polyploid autogamous crops
- **Correct Answer: B**
- **Explanation:** Inbreeding depression is highly severe in cross-pollinated (allogamous) crops because they naturally carry a high frequency of deleterious recessive alleles in a heterozygous state. Selfing forces these recessives into homozygosity, causing a severe drop in vigor. (KPSC AO 2014)

Q280. Which of the following is a true characteristic of B-chromosomes (supernumerary chromosomes)?

- A. They are essential for the survival and reproduction of the plant
- B. They pair regularly with A-chromosomes during meiotic metaphase I
- C. They are extra, non-vital chromosomes that do not show homology with normal A-chromosomes
- D. They are found only in animal cells
- **Correct Answer: C**
- **Explanation:** B-chromosomes are supernumerary chromosomes that are extra, dispensable (non-vital) chromosomes present in some individuals of a species (e.g., rye, maize). They do not pair with the normal chromosomes (A-chromosomes) during meiosis. (KPSC AO 2017)

Q281. Who first conceptualized and proposed the theory of cellular totipotency in plants in 1902?

- A. Gottlieb Haberlandt
- B. Toshio Murashige
- C. Folke Skoog
- D. Georges Morel

• **Correct Answer: A**

- **Explanation:** Gottlieb Haberlandt in 1902 proposed that every individual plant cell possesses all the genetic information necessary to divide, differentiate, and regenerate into a complete plant. For this, he is recognized as the father of plant tissue culture. (KPSC AO 2017)

Q282. Which two nitrogen sources are present in high concentrations in Murashige and Skoog (MS) tissue culture medium?

- A. Ammonium nitrate and Potassium nitrate
- B. Urea and Sodium nitrate
- C. Ammonium sulfate and Calcium nitrate
- D. Ammonium chloride and Amino acids

• **Correct Answer: A**

- **Explanation:** MS medium is characterized by a very high level of nitrogen, provided as both ammonium ions (NH_4^+ from ammonium nitrate) and nitrate ions (NO_3^- from potassium nitrate and ammonium nitrate). This specific balance is critical for organogenesis. (KPSC AAO 2017)

Q283. In plant tissue culture, the regeneration of shoots vs. roots is controlled by the ratio of auxins to cytokinins. What hormonal balance induces shoot differentiation?

- A. High auxin to low cytokinin ratio
- B. Low auxin to high cytokinin ratio
- C. Equal concentrations of auxin and gibberellin
- D. High abscisic acid and low auxin

• **Correct Answer: B**

- **Explanation:** Skoog and Miller (1957) demonstrated that organogenesis is controlled by the auxin-to-cytokinin ratio. A low auxin to high cytokinin ratio induces shoot development, while a high auxin to low cytokinin ratio induces root development. (KPSC AO 2014)

Q284. Which plant growth regulator is most commonly used in tissue culture to induce callus formation from explants?

- A. Indole-3-acetic acid (IAA)
- B. 2,4-Dichlorophenoxyacetic acid (2,4-D)
- C. Benzylaminopurine (BAP)
- D. Gibberellic acid (GA3)

• **Correct Answer: B**

- **Explanation:** 2,4-D is a strong synthetic auxin widely used to induce callus formation. It promotes cell division and dedifferentiation of explant tissues into a mass of unorganized parenchymatous cells (callus). (KPSC AAO 2015)

Q285. What is the primary anatomical difference between somatic embryogenesis and organogenesis during plant regeneration?

- A. Somatic embryos are unipolar, whereas organogenesis yields bipolar structures
- B. Somatic embryos are bipolar structures (possessing both root and shoot meristems), while organogenesis yields unipolar shoots or

roots

- C. Somatic embryos do not contain vascular connections
- D. Organogenesis only occurs from haploid cells

• **Correct Answer: B**

- **Explanation:** Somatic embryogenesis is the development of bipolar structures containing both shoot and root apices. Organogenesis produces unipolar structures (either shoots or roots) that must be placed on secondary media to induce the missing pole. (KPSC AO 2017)

Q286. Why is the shoot apical meristem culture widely used as the primary technique to produce virus-free plantlets in vegetatively propagated crops?

- A. The meristem has a high rate of mutation that kills viruses
- B. The meristem lacks vascular connections, and rapid cell division outpaces viral replication and transport
- C. Cytokinins present in the meristem act as antiviral agents
- D. Meristem tissue is treated with high concentrations of antibiotics

• **Correct Answer: B**

- **Explanation:** The shoot apical meristem is generally free of viruses because it lacks vascular bundles (phloem/xylem), which are the pathways for viral transport. Furthermore, cells in the meristem divide extremely rapidly, outpacing viral replication. (KPSC AO 2014)

Q287. Haploid plants were first obtained in vitro through anther culture of *Datura innoxia* in 1964 by which pair of scientists?

- A. Murashige and Skoog
- B. Guha and Maheshwari
- C. Cocking and Takebe
- D. Morel and Martin

• **Correct Answer: B**

- **Explanation:** Sipra Guha and Satish C. Maheshwari discovered in 1964 that haploid plants could be successfully regenerated from microspores of *Datura innoxia* through anther culture. (KPSC AAO 2015)

Q288. Doubled haploids (dihaploids) are produced by treating haploids with colchicine. What is their main value in plant breeding?

- A. They are highly resistant to viral diseases
- B. They achieve complete homozygosity at all loci in a single generation, saving years of selfing
- C. They exhibit high levels of heterosis
- D. They exhibit male sterility, preventing selfing

• **Correct Answer: B**

- **Explanation:** Haploid plants (n) have only one set of chromosomes. Treating them with colchicine doubles their chromosome set (2n), producing doubled haploids that are completely homozygous at all loci. This bypasses the 6-8 generations of selfing normally required to produce inbred lines. (KPSC AO 2017)

Q289. Which enzyme combination is standard for isolating viable protoplasts by digesting the cell walls of plant tissues?

- A. Amylase and Protease
- B. Cellulase and Pectinase
- C. Lipase and Hemicellulase

- D. Restriction endonuclease and Ligase

- **Correct Answer: B**

- **Explanation:** The plant cell wall consists primarily of cellulose fibers embedded in a pectin matrix. Isolating protoplasts (cells without cell walls) requires enzymatic digestion using cellulase (to break down cellulose) and pectinase (to break down the middle lamella pectin). (KPSC AO 2017)

Q290. Which chemical agent is most commonly used as a fusogen to induce protoplast fusion in somatic hybridization experiments?

- A. Polyethylene glycol (PEG)
- B. Colchicine
- C. Acetosyringone
- D. Calcium chloride only

- **Correct Answer: A**

- **Explanation:** Polyethylene glycol (PEG) is a chemical fusogen. It dehydrates the protoplast membrane, causing them to aggregate and fuse. When combined with high pH and high calcium ions, it yields high rates of somatic hybridization. (KPSC AAO 2017)

Q291. In somatic hybridization, cybrids (cytoplasmic hybrids) are defined as cells that contain:

- A. Nuclear genomes from both parents, but organelles from only one parent
- B. The nuclear genome of only one parent, and cytoplasmic organelles (mitochondria/chloroplasts) from both or the other parent
- C. Nuclear genomes from neither parent, but organelles from both
- D. Dual nuclei and no cytoplasm

- **Correct Answer: B**

- **Explanation:** Cybrids are cytoplasmic hybrids. They carry the nucleus of only one parent, but their cytoplasm contains organelles (mitochondria and/or chloroplasts) from both parents or from the non-nuclear donor parent. This is highly useful for transferring CMS traits. (KPSC AO 2014)

Q292. The term 'Somaclonal Variation', describing genetic and epigenetic variations arising during in vitro plant regeneration, was coined in 1981 by:

- A. Larkin and Scowcroft
- B. Murashige and Skoog
- C. Watson and Crick
- D. Southern and Alwine

- **Correct Answer: A**

- **Explanation:** Larkin and Scowcroft coined the term 'Somaclonal Variation' in 1981 to describe the genetic and epigenetic variations observed in plants regenerated from any form of cell/tissue culture. (KPSC AAO 2015)

Q293. Type II restriction endonucleases are preferred in recombinant DNA technology because they:

- A. Cleave DNA at a non-specific distance from their recognition site and require ATP
- B. Recognize specific palindromic sequences and cleave the DNA at precise sites within or close to these sequences, requiring only Mg^{2+}
- C. Cut only single-stranded RNA
- D. Join sticky ends of DNA fragments

- **Correct Answer: B**

- **Explanation:** Type II restriction enzymes recognize a specific palindromic sequence (usually 4-8 bp) and cut the DNA at a fixed, precise position within or immediately adjacent to that sequence. They do not require ATP (only Mg^{2+} as cofactor), making them ideal for DNA cloning. (KPSC AAO 2017)

Q294. T4 DNA ligase, used to join DNA fragments, catalyzes the formation of which bond?

- A. Hydrogen bond between complementary base pairs
- B. Phosphodiester bond between adjacent 3'-OH and 5'-phosphate ends of DNA
- C. Glycosidic bond between sugar and base
- D. Peptide bond between amino acids

- **Correct Answer: B**

- **Explanation:** T4 DNA ligase joins DNA fragments by catalyzing the synthesis of a covalent phosphodiester bond between the 3'-hydroxyl group of one nucleotide and the 5'-phosphate group of the adjacent nucleotide. (KPSC AO 2014)

Q295. What is the function of the enzyme alkaline phosphatase in plasmid cloning?

- A. Degrades foreign DNA in the host cell
- B. Removes the 5'-phosphate group from linearized plasmid vectors to prevent their self-ligation
- C. Adds methyl groups to bacterial DNA to protect it from restriction enzymes
- D. Synthesizes primers for DNA replication

- **Correct Answer: B**

- **Explanation:** Alkaline phosphatase catalyzes the removal of 5'-phosphate groups from DNA. Treating a linearized vector with alkaline phosphatase prevents it from recircularizing (self-ligating) in the presence of ligase because ligase requires a 5'-phosphate to join DNA ends. (KPSC AAO 2015)

Q296. Which enzyme is used in genetic engineering to synthesize a complementary DNA (cDNA) strand using a messenger RNA (mRNA) template?

- A. DNA polymerase I
- B. Reverse transcriptase
- C. RNA polymerase II
- D. Terminal transferase

- **Correct Answer: B**

- **Explanation:** Reverse transcriptase is an RNA-dependent DNA polymerase that uses an RNA template (such as mRNA) to synthesize a complementary DNA (cDNA) strand, which is essential for cloning eukaryotic genes without introns. (KPSC AO 2017)

Q297. The plasmid pBR322 is a cloning vector that carries selectable marker genes for resistance to which two antibiotics?

- A. Kanamycin and Neomycin
- B. Ampicillin and Tetracycline
- C. Chloramphenicol and Streptomycin
- D. Erythromycin and Penicillin

- **Correct Answer: B**

- **Explanation:** pBR322 contains two antibiotic resistance genes: ampR (conferring resistance to ampicillin) and tetR (conferring resistance to tetracycline). Cloning a gene into one of these restriction sites results in insertional inactivation, allowing selection of recombinants. (KPSC AAO 2017)

Q298. In Agrobacterium-mediated gene transfer, which region of the Ti plasmid is actually excised and integrated into the plant nuclear genome?

- A. The virulence (vir) region
- B. The T-DNA region
- C. The origin of replication (ori)
- D. The opine catabolism genes

• **Correct Answer: B**

• **Explanation:** The T-DNA (transfer DNA) is the only segment of the Ti plasmid that is transferred from Agrobacterium into the host plant cell nucleus, where it integrates into the host genome. (KPSC AO 2014)

Q299. What is the function of the phenolic compound acetosyringone, which is released by wounded plant cells during Agrobacterium infection?

- A. It acts as an antibiotic that kills non-transformed cells
- B. It acts as a chemoattractant and induces the transcription of virulence (vir) genes on the Ti plasmid
- C. It forms the cell wall of regenerated shoots
- D. It inhibits the transcription of T-DNA genes

• **Correct Answer: B**

• **Explanation:** Acetosyringone is a phenolic compound released by wounded plant cells. It binds to the VirA sensory protein on the Agrobacterium membrane, activating the VirA-VirG two-component system, which induces the expression of other vir genes needed for T-DNA excision and transfer. (KPSC AO 2017)

Q300. A binary vector system in Agrobacterium-mediated transformation consists of:

- A. A single plasmid containing both T-DNA and vir genes
- B. Two plasmids: a disarmed helper plasmid containing vir genes, and a separate cloning plasmid containing the T-DNA with the gene of interest
- C. Two different strains of Agrobacterium infecting the same cell
- D. A plasmid and a bacteriophage working together

• **Correct Answer: B**

• **Explanation:** The binary vector system uses two plasmids. The helper plasmid contains the vir genes but lacks T-DNA (disarmed). The cloning plasmid is smaller, replicates in both E. coli and Agrobacterium, and contains the T-DNA borders enclosing the gene of interest and selectable markers. (KPSC AAO 2017)

Q301. Who developed the biolistics (particle bombardment or gene gun) method of gene transfer in 1987, using tungsten or gold microparticles?

- A. John Sanford
- B. Marc van Montagu
- C. Mary-Dell Chilton
- D. Kary Mullis

• **Correct Answer: A**

• **Explanation:** John Sanford and colleagues at Cornell University developed the biolistic gene transfer method in 1987. It uses high-pressure helium gas to shoot gold or tungsten microparticles coated with DNA directly into plant tissues. (KPSC AAO 2015)

Q302. Which method of gene transfer utilizes short, high-voltage electrical pulses to introduce foreign DNA into plant

protoplasts?

- A. Biolistics
- B. Electroporation
- C. Microinjection
- D. Liposome fusion
- **Correct Answer: B**
- **Explanation:** Electroporation uses high-voltage electrical pulses to create transient, sub-microscopic pores in the plasma membrane of protoplasts, allowing external DNA molecules to diffuse into the cell. (KPSC AO 2017)

Q303. Which of the following molecular markers is hybridization-based and exhibits co-dominance?

- A. Random Amplified Polymorphic DNA (RAPD)
- B. Restriction Fragment Length Polymorphism (RFLP)
- C. Amplified Fragment Length Polymorphism (AFLP)
- D. Inter Simple Sequence Repeats (ISSR)
- **Correct Answer: B**
- **Explanation:** RFLP is a classic hybridization-based marker that uses restriction enzymes and southern blotting with labeled probes. It is co-dominant (can distinguish homozygotes from heterozygotes) and highly reproducible. (KPSC AAO 2017)

Q304. RAPD is a PCR-based marker. What is a key feature of the primers and PCR conditions used in RAPD?

- A. Uses two long (25 bp) sequence-specific primers under high annealing temperatures
- B. Uses a single, short (usually 10 bp) arbitrary primer under low annealing temperatures (~35°C)
- C. Requires radioactive probes for detection
- D. Amplifies only mitochondrial DNA
- **Correct Answer: B**
- **Explanation:** RAPD uses a single, short (10 bp) arbitrary primer. Because the primer is short and has no sequence-specific target, a low annealing temperature (35-37°C) is used to allow non-specific binding, which amplifies multiple genomic loci (dominant marker). (KPSC AO 2014)

Q305. Why are Simple Sequence Repeats (SSRs or microsatellites) highly favored in plant genetic mapping and marker-assisted selection?

- A. They are dominant, non-PCR-based markers
- B. They are highly polymorphic, PCR-based, co-dominant markers that are locus-specific and reproducible
- C. They do not require gel electrophoresis for analysis
- D. They are only present in chloroplast genomes
- **Correct Answer: B**
- **Explanation:** SSRs consist of tandemly repeated short DNA motifs (e.g., AT, GATA). Due to replication slippage, the number of repeats varies, making them highly polymorphic. They are amplified by PCR using locus-specific flanking primers, yielding reproducible, co-dominant results. (KPSC AAO 2015)

Q306. The Amplified Fragment Length Polymorphism (AFLP) marker system works by:

- A. Hybridizing genomic DNA directly to a chip containing synthetic oligonucleotides
- B. Digesting genomic DNA with restriction enzymes, ligating double-stranded adapters, and amplifying the fragments using selective

primers

- C. Sequencing the entire genome using next-generation sequencing
- D. Amplifying random segments using a single primer

• **Correct Answer: B**

- **Explanation:** AFLP combines restriction digestion and PCR. Genomic DNA is digested with two enzymes (usually EcoRI and MseI), adapters are ligated to the ends, and fragments are amplified via PCR using primers containing selective nucleotide bases (yielding high multiplex ratios). (KPSC AO 2017)

Q307. Which type of molecular marker represents a single base-pair change in a DNA sequence and is the most abundant type of sequence marker in genomes?

- A. SSR
- B. SNP
- C. RAPD
- D. SCAR

• **Correct Answer: B**

- **Explanation:** Single Nucleotide Polymorphisms (SNPs) represent changes in a single nucleotide base pair (A, T, C, or G) at a specific locus. They are highly abundant throughout the genome and are analyzed using high-throughput chip assays. (KPSC AAO 2017)

Q308. In Marker-Assisted Backcrossing (MABC), what does background selection refer to?

- A. Selecting for the donor parent's target gene
- B. Using molecular markers to select for the maximum recovery of the recurrent parent's genome
- C. Selecting against seedling diseases in the nursery
- D. Selecting for cytoplasmic male sterility

• **Correct Answer: B**

- **Explanation:** Background selection in MABC involves using molecular markers across the genome to select offspring that have the highest percentage of the recurrent parent's genome. This accelerates the recovery of the elite parental background from 6-8 generations to 3-4 generations. (KPSC AAO 2015)

Q309. Bt cotton contains crystalline endotoxin (Cry) genes from *Bacillus thuringiensis*. What is the mode of action of these Cry proteins in susceptible insect larvae?

- A. They inhibit neurotransmitter release at synapses
- B. They bind to specific midgut receptors, causing pore formation, cell lysis, and starvation
- C. They degrade the insect's cuticle, leading to dehydration
- D. They block chitin synthesis during molting

• **Correct Answer: B**

- **Explanation:** Cry proteins are ingested by the insect larva and solubilized in the alkaline environment of its midgut. The active toxin binds to specific cadherin-like receptors on the midgut epithelial cell membrane, creating pores that cause osmotic lysis of the cells, gut paralysis, and death. (KPSC AO 2017)

Q310. Golden Rice was engineered to synthesize beta-carotene (provitamin A) in the endosperm. Which two genes were introduced to complete the pathway?

- A. Phytoene synthase (psy) from daffodil and Phytoene desaturase (crtI) from *Erwinia uredovora*
- B. EPSP synthase (epsps) and glyphosate oxidoreductase

- C. Polygalacturonase (pg) and ACC synthase
- D. Cry1Ac and Cry2Ab from *Bacillus thuringiensis*

• **Correct Answer: A**

- **Explanation:** Golden Rice was developed by Ingo Potrykus and Peter Beyer. Eukaryotic endosperm has geranylgeranyl diphosphate (GGPP) but lacks enzymes to make beta-carotene. They introduced the phytoene synthase (psy) gene from daffodil (*Narcissus pseudonarcissus*) and the phytoene desaturase (crtI) gene from the bacterium *Erwinia uredovora* to complete the pathway. (KPSC AAO 2017)

Q311. Roundup Ready crops are resistant to glyphosate because they express which transgene?

- A. The bar gene from *Streptomyces*
- B. The cp4 epsps gene from *Agrobacterium* sp. strain CP4, which is insensitive to glyphosate
- C. The pat gene for phosphinothricin acetyltransferase
- D. The PG gene in antisense orientation

• **Correct Answer: B**

- **Explanation:** Glyphosate inhibits the plant enzyme 5-enolpyruvylshikimate-3-phosphate synthase (EPSPS), blocking aromatic amino acid synthesis. Roundup Ready crops express the cp4 epsps gene from *Agrobacterium* sp. CP4, which encodes an EPSPS enzyme that is highly active even in the presence of glyphosate. (KPSC AO 2014)

Q312. The Flavr Savr tomato was the first commercially grown GM food crop. Ripening delay was achieved by targeting the polygalacturonase (PG) gene using which technology?

- A. CRISPR-Cas9 knockout
- B. Antisense RNA technology
- C. Overexpression of ethylene receptors
- D. Translocation of chromosomal segments

• **Correct Answer: B**

- **Explanation:** Flavr Savr tomato (Calgene, 1994) used antisense RNA technology. An antisense copy of the polygalacturonase (PG) gene was introduced, which bound to the endogenous PG mRNA, blocking its translation. This reduced PG enzyme levels, which breaks down pectin in the cell wall, delaying softening. (KPSC AAO 2015)

Q313. In CRISPR-Cas9 genome editing, the Cas9 endonuclease is guided to a specific genomic target by a customized:

- A. Zinc finger protein
- B. Single guide RNA (sgRNA)
- C. Transcription activator-like effector
- D. DNA probe

• **Correct Answer: B**

- **Explanation:** The CRISPR-Cas9 system uses a single guide RNA (sgRNA) consisting of a targeting sequence (complementary to the target DNA) and a scaffold sequence (which binds to Cas9). This directs Cas9 to introduce a double-strand break at the target genomic site. (KPSC AO 2017)

Q314. What is the critical target sequence required for Cas9 endonuclease to recognize and cleave target DNA in plants?

- A. A poly-A tail
- B. Protospacer Adjacent Motif (PAM) sequence, typically 5'-NGG-3'
- C. A TATA box promoter
- D. A 5'-cap structure

- **Correct Answer: B**

- **Explanation:** For Cas9 to cleave DNA, the target sequence must be immediately adjacent to a Protospacer Adjacent Motif (PAM). For the standard SpCas9, the PAM is 5'-NGG-3'. If the PAM is absent, Cas9 will not bind or cleave, preventing off-target activity. (KPSC AAO 2017)

Q315. Polymerase Chain Reaction (PCR) consists of cyclic temperature changes. What is the typical temperature sequence for Denaturation, Annealing, and Extension?

- A. 94°C -> 55°C -> 72°C
- B. 72°C -> 55°C -> 94°C
- C. 55°C -> 94°C -> 72°C
- D. 94°C -> 72°C -> 55°C

- **Correct Answer: A**

- **Explanation:** The three steps in a PCR cycle are: 1) Denaturation at ~94°C (separating double-stranded DNA), 2) Annealing at ~50-60°C (primers binding to templates), and 3) Extension at ~72°C (Taq polymerase synthesizing new strands). (KPSC AAO 2017)

Q316. Taq DNA polymerase is widely used in PCR. What is its main limitation in molecular biology applications?

- A. It is highly unstable at 94°C
- B. It lacks 3'-to-5' proofreading exonuclease activity, resulting in a higher error rate compared to Pfu polymerase
- C. It cannot synthesize DNA from a thymine template
- D. It works only on RNA templates

- **Correct Answer: B**

- **Explanation:** Taq DNA polymerase (isolated from *Thermus aquaticus*) is highly thermostable but lacks 3'-to-5' proofreading exonuclease activity. This leads to a higher rate of misincorporated nucleotides (~1 in 9000 bases) compared to proofreading polymerases like Pfu. (KPSC AO 2014)

Q317. In Quantitative PCR (qPCR), how is the accumulation of DNA monitored in real time during the reaction?

- A. By running a gel after each PCR cycle
- B. By measuring the increase in fluorescence from dsDNA-binding dyes (like SYBR Green) or fluorophore-labeled probes
- C. By measuring the absorption of UV light by single-stranded primers
- D. By radioactive labeling of dNTPs

- **Correct Answer: B**

- **Explanation:** qPCR uses fluorescent dyes (like SYBR Green, which fluoresces strongly when bound to double-stranded DNA) or fluorescently labeled sequence-specific probes (like TaqMan). The instrument measures the fluorescent signal during the extension phase of each cycle, reflecting the amount of DNA amplified. (KPSC AAO 2015)

Q318. Sanger DNA sequencing (dideoxy method) relies on the incorporation of chain-terminating dideoxynucleoside triphosphates (ddNTPs). Why do ddNTPs terminate chain elongation?

- A. They lack the 2'-H atom
- B. They lack the 3'-OH group necessary for phosphodiester bond formation
- C. They contain a methyl group at the 5'-position
- D. They bind irreversibly to DNA polymerase

- **Correct Answer: B**

- **Explanation:** ddNTPs lack a hydroxyl group (-OH) at the 3'-carbon of the ribose sugar. When a ddNTP is incorporated into the

growing DNA strand, DNA polymerase cannot form a phosphodiester bond with the next incoming dNTP, terminating chain synthesis. (KPSC AO 2017)

Q319. Synthetic seeds (artificial seeds) are produced by encapsulating somatic embryos or shoot buds in which protective gelling matrix?

- A. Agar-agar
- B. Sodium alginate, which is complexed with calcium chloride
- C. Gelrite
- D. Polyacrylamide gel
- **Correct Answer: B**
- **Explanation:** Synthetic seeds are produced by mixing somatic embryos, shoot tips, or nodal segments with sodium alginate solution (usually 2-3%), then dropping them into a calcium chloride solution (100 mM). This forms round, semi-solid calcium alginate beads that act as artificial seed coats. (KPSC AO 2014)

Q320. Which regulatory body in India is the apex authority that evaluates biosafety data and approves the commercial cultivation of Genetically Modified (GM) crops?

- A. National Biodiversity Authority (NBA)
- B. Genetic Engineering Appraisal Committee (GEAC) under the MoEFCC
- C. Indian Council of Agricultural Research (ICAR)
- D. Food Safety and Standards Authority of India (FSSAI)
- **Correct Answer: B**
- **Explanation:** The Genetic Engineering Appraisal Committee (GEAC), operating under the Ministry of Environment, Forest and Climate Change (MoEFCC), is the statutory apex body responsible for approvals of proposals relating to the release of genetically engineered organisms and products into the environment, including commercial cultivation of GM crops. (KPSC AO 2014)

Q321. Which plant pathogen was responsible for the historic Irish Famine in 1845?

- A. Phytophthora infestans
- B. Helminthosporium oryzae
- C. Plasmopara viticola
- D. Puccinia graminis
- **Correct Answer: A**
- **Explanation:** The Irish Famine (1845) was caused by Phytophthora infestans, an oomycete causing Late Blight of Potato, leading to complete crop devastation and massive human migration (KPSC AAO 2017).

Q322. Bordeaux mixture was discovered by Millardet in 1885 to control which disease?

- A. Downy mildew of grapes
- B. Powdery mildew of mango
- C. Late blight of potato
- D. Citrus canker
- **Correct Answer: A**
- **Explanation:** Bordeaux mixture (Copper sulphate + Lime + Water) was discovered by PMA Millardet in 1885 to control Downy Mildew of Grapevine (Plasmopara viticola) in France (KPSC AO 2017).

Q323. The Bengal Famine of 1943 was caused by which pathogen?

- A. *Bipolaris oryzae*
- B. *Pyricularia oryzae*
- C. *Xanthomonas oryzae*
- D. *Tilletia indica*

• **Correct Answer: A**

- **Explanation:** The Bengal Famine of 1943 was caused by *Helminthosporium oryzae* (*Bipolaris oryzae*), which causes Brown Spot of Rice, leading to widespread food shortages (KPSC AO 2014).

Q324. Which pathogen caused the devastation of Arabica coffee plantations in Sri Lanka in 1869?

- A. *Hemileia vastatrix*
- B. Coffee ringspot virus
- C. *Colletotrichum coffeanum*
- D. *Cercospora coffeicola*

• **Correct Answer: A**

- **Explanation:** Coffee rust, caused by *Hemileia vastatrix*, destroyed the coffee industry in Ceylon (Sri Lanka) starting in 1869, causing the country to switch from coffee to tea cultivation (KPSC AAO 2015).

Q325. Which physiological phenomena are associated with the club root of crucifers caused by *Plasmodiophora brassicae*?

- A. Hypertrophy and Hyperplasia
- B. Necrosis and Chlorosis
- C. Abscission and Epinasty
- D. Plasmolysis and Wilting

• **Correct Answer: A**

- **Explanation:** Club root of crucifers (*Plasmodiophora brassicae*) causes abnormal cell division (hyperplasia) and cell enlargement (hypertrophy) in roots, forming club-like galls (KPSC AO 2017).

Q326. Which of the following is an internally seed-borne fungal disease?

- A. Loose smut of wheat
- B. Grain smut of sorghum
- C. Covered smut of barley
- D. Whip smut of sugarcane

• **Correct Answer: A**

- **Explanation:** Loose smut of wheat (*Ustilago nuda tritici*) is internally seed-borne, where mycelium remains dormant in the embryo. It is controlled by hot water or solar heat treatment (KPSC AAO 2017).

Q327. Grain smut of sorghum (*Sporisorium sorghi*) is externally seed-borne and can be managed by treating seeds with:

- A. Sulfur dust at 4 g/kg
- B. Streptomycin at 100 ppm
- C. Hot water at 65°C
- D. Bordeaux paste

• **Correct Answer: A**

- **Explanation:** Grain smut of sorghum is externally seed-borne and is managed effectively by seed dressing with elemental sulfur dust at 3-4 g/kg of seed (KPSC AO 2014).

Q328. Citrus canker is caused by which Gram-negative bacterium?

- A. *Xanthomonas axonopodis* pv. *citri*
- B. *Pseudomonas syringae*
- C. *Ralstonia solanacearum*
- D. *Erwinia amylovora*

• **Correct Answer: A**

- **Explanation:** Citrus canker is caused by *Xanthomonas axonopodis* pv. *citri*, which causes raised, corky, cancrroid lesions on leaves, twigs, and fruits of acid lime (KPSC AO 2017).

Q329. Rice tungro virus (RTV) is biologically transmitted in nature by which insect vector?

- A. *Nephotettix virescens*
- B. *Nilaparvata lugens*
- C. *Sogatella furcifera*
- D. *Leptocorisa acuta*

• **Correct Answer: A**

- **Explanation:** Rice tungro virus is transmitted by the Green Leafhopper (*Nephotettix virescens*) in a non-persistent, semi-persistent manner (KPSC AO 2017).

Q330. Which vector transmits the Tomato Spotted Wilt Virus (TSWV) in a circulative propagative manner?

- A. *Frankliniella occidentalis*
- B. *Bemisia tabaci*
- C. *Aphis gossypii*
- D. *Hishimonus phycitis*

• **Correct Answer: A**

- **Explanation:** Tomato spotted wilt virus is transmitted by Thrips, specifically the Western flower thrips (*Frankliniella occidentalis*) (KPSC AAO 2017).

Q331. Bunchy top of banana is caused by a Babuvirus and is transmitted by which insect vector?

- A. *Pentalonia nigronervosa*
- B. *Odoiporus longicollis*
- C. *Cosmopolites sordidus*
- D. *Bemisia tabaci*

• **Correct Answer: A**

- **Explanation:** Bunchy top of banana (BBTV) is transmitted by the banana aphid, *Pentalonia nigronervosa* (KPSC AO 2017).

Q332. Sugarcane grassy shoot disease (GSD) is caused by which type of pathogen?

- A. Phytoplasma
- B. Viroid

- C. Fastidious bacterium
- D. Virus
- **Correct Answer: A**

• **Explanation:** Grassy shoot disease of sugarcane is caused by a phytoplasma (pleiomorphic prokaryote lacking a cell wall) and is transmitted by *Pyrilla perpusilla* and aphids (KPSC AAO 2015).

Q333. What is the insect vector for little leaf of brinjal?

- A. *Hishimonus phycitis*
- B. *Amrasca biguttula*
- C. *Bemisia tabaci*
- D. *Myzus persicae*
- **Correct Answer: A**

• **Explanation:** Little leaf of brinjal (caused by phytoplasma) is transmitted by the leafhopper *Hishimonus phycitis* (KPSC AO 2014).

Q334. Sandal spike disease is caused by phytoplasma and transmitted by which vector in India?

- A. *Coelidia indica*
- B. *Stephanitis typica*
- C. *Hishimonus phycitis*
- D. *Proutista moesta*
- **Correct Answer: A**

• **Explanation:** Sandal spike disease is transmitted by the leafhopper vector *Coelidia indica* (KPSC AO 2017).

Q335. The sclerotia of *Claviceps fusiformis* causing Ergot of Pearl Millet contain toxic alkaloids that cause:

- A. Ergotism
- B. Lathyrism
- C. Aflatoxicosis
- D. Favism
- **Correct Answer: A**

• **Explanation:** Ergot of pearl millet (*Claviceps fusiformis*) sclerotia contain alkaloids (ergometrine, ergotamine) that cause ergotism in animals and humans upon consumption (KPSC AAO 2015).

Q336. Whip smut of sugarcane is caused by which fungus?

- A. *Sporisorium scitamineum*
- B. *Colletotrichum falcatum*
- C. *Fusarium sacchari*
- D. *Ustilago maydis*
- **Correct Answer: A**

• **Explanation:** Whip smut of sugarcane is caused by the fungus *Sporisorium scitamineum* (formerly *Ustilago scitaminea*), producing a black whip-like structure (KPSC AO 2017).

Q337. Which vascular wilt of banana is caused by soil-borne *Fusarium oxysporum* f. sp. *cubense*?

- A. Panama wilt
- B. Moko disease
- C. Sigatoka leaf spot
- D. Bunchy top
- **Correct Answer: A**
- **Explanation:** Panama wilt of banana is caused by *Fusarium oxysporum* f. sp. *cubense*, which invades roots and blocks xylem vessels (KPSC AAO 2017).

Q338. Which test is used for rapid field diagnosis of Solanaceous bacterial wilt caused by *Ralstonia solanacearum*?

- A. Ooze test
- B. Gram staining
- C. Starch iodine test
- D. Tetrazolium test
- **Correct Answer: A**
- **Explanation:** The ooze test is used for *Ralstonia solanacearum*. Suspending a cut stem in water reveals streaming of milky bacterial ooze from vascular bundles (KPSC AO 2017).

Q339. Black chaff of wheat is caused by which bacterial pathogen?

- A. *Xanthomonas translucens* pv. *undulosa*
- B. *Pseudomonas syringae*
- C. *Clavibacter tritici*
- D. *Erwinia amylovora*
- **Correct Answer: A**
- **Explanation:** Black chaff of wheat is caused by *Xanthomonas translucens* pv. *undulosa*, producing dark brown/black stripes on glumes (KPSC AAO 2017).

Q340. What is the key chemical difference between Bordeaux mixture and Burgundy mixture?

- A. Burgundy mixture uses sodium carbonate instead of lime
- B. Bordeaux mixture uses sodium carbonate instead of lime
- C. Burgundy mixture contains ammonium carbonate
- D. Bordeaux mixture does not contain copper sulphate
- **Correct Answer: A**
- **Explanation:** Burgundy mixture uses Sodium Carbonate (washing soda) instead of Calcium Hydroxide (lime) along with Copper Sulphate (KPSC AAO 2015).

Q341. Chestnut compound is prepared by mixing copper sulphate and ammonium carbonate in what ratio?

- A. 2:11
- B. 5:5
- C. 1:1.25
- D. 1:10
- **Correct Answer: A**

- **Explanation:** Chestnut compound is prepared by mixing 2 parts of copper sulphate and 11 parts of ammonium carbonate by weight, used to prevent damping-off (KPSC AO 2017).

Q342. Who introduced the concept of vertical and horizontal resistance and apparent infection rate (r) in plant pathology?

- A. J.E. Vanderplank
- B. Anton de Bary
- C. E.C. Stakman
- D. K.C. Mehta
- **Correct Answer: A**

- **Explanation:** J.E. Vanderplank is the Father of Plant Disease Epidemiology, who defined vertical/horizontal resistance and infection rate parameters (KPSC AO 2017).

Q343. Carboxin (Vitavax) is a systemic fungicide belonging to the oxathiin group which specifically inhibits:

- A. Succinate dehydrogenase in Basidiomycetes
- B. RNA polymerase I in Oomycetes
- C. Ergosterol biosynthesis in Ascomycetes
- D. Chitin synthase in Zygomycetes
- **Correct Answer: A**

- **Explanation:** Carboxin specifically targets Basidiomycetes (like loose smut) by inhibiting succinate dehydrogenase in the mitochondrial respiratory chain (KPSC AAO 2017).

Q344. Metalaxyl (Ridomil) is a systemic fungicide highly effective against Downy mildews because it inhibits:

- A. RNA Polymerase I in Oomycetes
- B. Fungal cell wall synthesis
- C. Tubulin polymerization
- D. Mitochondrial complex III
- **Correct Answer: A**

- **Explanation:** Metalaxyl specifically controls Oomycetes (like Pythium, Phytophthora, downy mildews) by inhibiting RNA Polymerase I, stopping rRNA synthesis (KPSC AO 2017).

Q345. Which biocontrol fungus acts by producing chitinases and glucanases to degrade cell walls of soil-borne pathogens?

- A. Trichoderma harzianum
- B. Pseudomonas fluorescens
- C. Bacillus subtilis
- D. Beauveria bassiana
- **Correct Answer: A**

- **Explanation:** Trichoderma harzianum is a mycoparasite that secretes cell-wall-degrading enzymes (chitinases, beta-1,3-glucanases) to control pathogens (KPSC AO 2017).

Q346. Pseudomonas fluorescens suppresses plant pathogens primarily by producing iron-chelating compounds called:

- A. Siderophores
- B. Phytoalexins

- C. Pyrethrins
- D. Mycotoxins
- **Correct Answer: A**
- **Explanation:** *Pseudomonas fluorescens* produces siderophores (pyoverdine) which bind ferric iron (Fe^{3+}), making it unavailable to plant pathogens (KPSC AO 2017).

Q347. What is the characteristic sexual fruiting body of powdery mildews (Erysiphales)?

- A. Cleistothecium
- B. Perithecium
- C. Apothecium
- D. Pseudothecium
- **Correct Answer: A**
- **Explanation:** Powdery mildews produce closed sexual fruiting bodies called cleistothecia (or chasmothecia) with diagnostic appendages (KPSC AAO 2017).

Q348. *Puccinia graminis tritici* is a heteroecious rust. What is its alternate host?

- A. *Berberis vulgaris*
- B. *Solanum tuberosum*
- C. *Hordeum vulgare*
- D. *Triticum aestivum*
- **Correct Answer: A**
- **Explanation:** *Puccinia graminis tritici* requires *Berberis vulgaris* (Barberry) to complete its sexual cycle, making it heteroecious (KPSC AO 2017).

Q349. Which spore stage of *Puccinia graminis tritici* serves as the repeating asexual stage on wheat?

- A. Urediniospore
- B. Teliospore
- C. Aeciospore
- D. Basidiospore
- **Correct Answer: A**
- **Explanation:** Urediniospores (Stage II) are dikaryotic, repeating spores that infect wheat stems throughout the season (KPSC AO 2017).

Q350. White rust of crucifers is caused by *Albugo candida*, which belongs to which class of organisms?

- A. Oomycetes
- B. Basidiomycetes
- C. Ascomycetes
- D. Zygomycetes
- **Correct Answer: A**
- **Explanation:** *Albugo candida* is an Oomycete (Kingdom Chromista/Stramenopila) and is not a true fungus (KPSC AAO 2015).

Q351. Which chemical treatment is recommended for soil drenching in nursery beds to control damping-off?

- A. Copper Oxychloride 0.3%
- B. Streptomycin 10%
- C. Sulfur dust
- D. Methyl Demeton
- **Correct Answer: A**
- **Explanation:** Damping-off (*Pythium* spp.) is managed by soil drenching with contact fungicides like Copper Oxychloride at 0.3% (KPSC AO 2014).

Q352. Boat-shaped (spindle-shaped) lesions on rice leaves are diagnostic of which disease?

- A. Rice Blast
- B. Bacterial Leaf Blight
- C. Sheath Blight
- D. Brown Spot
- **Correct Answer: A**
- **Explanation:** Rice blast (*Pyricularia oryzae*) produces spindle/boat-shaped necrotic lesions with grey centers and brown borders (KPSC AO 2017).

Q353. Sheath blight of rice, caused by *Rhizoctonia solani*, spreads in the field primarily through:

- A. Sclerotia
- B. Conidia
- C. Basidiospores
- D. Zoospores
- **Correct Answer: A**
- **Explanation:** *Rhizoctonia solani* is a sclerotial fungus that survives and spreads via hard survival structures called sclerotia in soil/water (KPSC AAO 2017).

Q354. Which seedling wilt stage is observed in Rice due to systemic bacterial leaf blight?

- A. Kresek
- B. Hopper burn
- C. Silver shoot
- D. Rosette
- **Correct Answer: A**
- **Explanation:** Kresek is the systemic seedling wilt stage of rice leaf blight (*Xanthomonas oryzae* pv. *oryzae*) resulting from early infection (KPSC AO 2017).

Q355. The causal agent of false smut of rice, which forms greenish velvety spore balls, is:

- A. *Ustilaginoidea virens*
- B. *Tilletia barclayana*
- C. *Sporisorium sorghi*
- D. *Pyricularia oryzae*
- **Correct Answer: A**

- **Explanation:** False smut of rice is caused by *Ustilaginoidea virens* (Ascomycota), turning individual grains into velvety balls of chlamydospores (KPSC AAO 2017).

Q356. Yellow vein mosaic of Okra is transmitted by the insect vector:

- A. *Bemisia tabaci*
- B. *Amrasca biguttula*
- C. *Aphis gossypii*
- D. *Thrips tabaci*

• **Correct Answer: A**

- **Explanation:** Yellow vein mosaic of okra (Begomovirus) is transmitted by the whitefly *Bemisia tabaci* (KPSC AO 2017).

Q357. Internal red discoloration of sugarcane stalk with white transverse bands is a symptom of:

- A. Red rot
- B. Whip smut
- C. Grassy shoot
- D. Ratoon stunting

• **Correct Answer: A**

- **Explanation:** Red rot of sugarcane (*Colletotrichum falcatum*) split stalks show red vascular tissues interrupted by white transverse patches (KPSC AO 2017).

Q358. Tikka disease of groundnut consists of early and late leaf spots. Late leaf spot is caused by:

- A. *Phaeoisariopsis personata*
- B. *Cercospora arachidicola*
- C. *Puccinia arachidis*
- D. *Aspergillus niger*

• **Correct Answer: A**

- **Explanation:** Late leaf spot (Tikka) is caused by *Phaeoisariopsis personata* (*Cercosporidium personatum*), producing darker, blackish spots (KPSC AO 2014).

Q359. Which nematode genus is responsible for root knot and giant cell formation in Solanaceous crops?

- A. *Meloidogyne*
- B. *Pratylenchus*
- C. *Globodera*
- D. *Tylenchulus*

• **Correct Answer: A**

- **Explanation:** Root-knot nematode is *Meloidogyne incognita*, which induces hypertrophy and hyperplasia forming giant cells for feeding (KPSC AO 2017).

Q360. Which potato pest/pathogen is a major quarantine concern restricted to the Nilgiris in India?

- A. *Globodera rostochiensis*
- B. *Phytophthora infestans*

- C. *Ralstonia solanacearum*
- D. *Phthorimaea operculella*
- **Correct Answer: A**
- **Explanation:** Golden nematode of potato (*Globodera rostochiensis*) is a cyst nematode under strict domestic quarantine in Nilgiri hills (KPSC AO 2017).

Q361. Which order does the yellow stem borer of rice belong to?

- A. Lepidoptera
- B. Coleoptera
- C. Hemiptera
- D. Diptera
- **Correct Answer: A**
- **Explanation:** Yellow stem borer (*Scirpophaga incertulas*) is a lepidopteran moth whose caterpillars bore into rice stems (KPSC AO 2017).

Q362. Complete metamorphosis (holometabolism) is characteristic of which insect order?

- A. Coleoptera
- B. Hemiptera
- C. Orthoptera
- D. Thysanoptera
- **Correct Answer: A**
- **Explanation:** Coleoptera, Lepidoptera, Diptera, and Hymenoptera undergo complete metamorphosis (Egg-Larva-Pupa-Adult) (KPSC AO 2017).

Q363. Which sclerite forms the forehead region of the insect head?

- A. Frons
- B. Vertex
- C. Clypeus
- D. Gena
- **Correct Answer: A**
- **Explanation:** The frons is the large sclerite on the front of the head (forehead) between the eyes (KPSC AAO 2017).

Q364. Aristate antennae are characteristic of which insect?

- A. Housefly
- B. Honeybee
- C. Cockroach
- D. Termite
- **Correct Answer: A**
- **Explanation:** Aristate antennae are found in the housefly (*Musca domestica*), where the terminal antennal segment bears a bristle called arista (KPSC AO 2017).

Q365. Plumose antennae, which are highly feathery, are found in:

- A. Male mosquito
- B. Female mosquito
- C. Butterfly
- D. Housefly
- **Correct Answer: A**
- **Explanation:** Plumose (feathery) antennae are found in male mosquitoes, acting as auditory receptors to locate female wing vibrations (KPSC AO 2017).

Q366. Moniliform antennae, resembling a string of beads, are characteristic of:

- A. Termites
- B. Cockroaches
- C. Grasshoppers
- D. Ants
- **Correct Answer: A**
- **Explanation:** Moniliform antennae, where segments are globose and resemble beads, are found in termites (Isoptera) (KPSC AO 2017).

Q367. Chewing and lapping mouthparts are characteristic of which insect?

- A. Honeybee
- B. Butterfly
- C. Mosquito
- D. Housefly
- **Correct Answer: A**
- **Explanation:** Honeybees (*Apis* spp.) possess chewing and lapping mouthparts, where mandibles chew/mold wax and maxillae/labium form a lapping tongue (KPSC AO 2017).

Q368. Siphoning mouthparts are found in which group of insects?

- A. Butterflies and Moths
- B. Bees and Wasps
- C. Mosquitoes and Bugs
- D. Thrips and Aphids
- **Correct Answer: A**
- **Explanation:** Butterflies and moths (Lepidoptera) have siphoning mouthparts where the galeae of maxillae form a long coiled tube (proboscis) (KPSC AAO 2017).

Q369. Asymmetrical mouthparts (lacking the right mandible) adapted for rasping and sucking are found in:

- A. Thrips
- B. Aphids
- C. Whitefly
- D. Jassids
- **Correct Answer: A**

- **Explanation:** Thrips (Thysanoptera) have asymmetrical rasping-sucking mouthparts, where the right mandible is vestigial or absent (KPSC AAO 2017).

Q370. Halteres, the modified hind wings acting as balancing organs, are found in which order?

- A. Diptera
- B. Coleoptera
- C. Lepidoptera
- D. Hymenoptera
- **Correct Answer: A**

- **Explanation:** Diptera (true flies) have only two wings; the hind wings are reduced to small club-shaped balancing organs called halteres (KPSC AO 2017).

Q371. Raptorial legs, modified for seizing and holding prey, are found in:

- A. Praying mantis
- B. Grasshopper
- C. Mole cricket
- D. Honeybee
- **Correct Answer: A**

- **Explanation:** Praying mantids have forelegs modified into raptorial claws with spiny inner edges to grasp and secure prey (KPSC AAO 2017).

Q372. On which leg segment is the pollen basket (corbicula) of worker honeybees located?

- A. Tibia of hind leg
- B. Femur of middle leg
- C. Tarsus of hind leg
- D. Coxa of foreleg
- **Correct Answer: A**

- **Explanation:** The corbicula is a smooth, hair-fringed concave area on the outer surface of the tibia of the worker honeybee's hind leg (KPSC AO 2017).

Q373. Which part of the insect alimentary canal is NOT lined with cuticular intima?

- A. Mesenteron (Midgut)
- B. Stomodeum (Foregut)
- C. Proctodeum (Hindgut)
- D. Pharynx
- **Correct Answer: A**

- **Explanation:** The midgut (mesenteron) is of endodermal origin and lacks a cuticular lining. Instead, it is protected by a peritrophic membrane (KPSC AAO 2017).

Q374. Which group of insects is unique in completely lacking Malpighian tubules?

- A. Aphids
- B. Termites

- C. Locusts
- D. Beetles
- **Correct Answer: A**
- **Explanation:** Aphids (Aphididae) lack Malpighian tubules, relying on rapid sap transport and specialized gut structures for waste management (KPSC AO 2017).

Q375. How many pairs of spiracles are typically present in a standard insect respiratory system?

- A. 10 pairs
- B. 8 pairs
- C. 12 pairs
- D. 6 pairs
- **Correct Answer: A**
- **Explanation:** A typical insect has 10 pairs of spiracles: 2 thoracic and 8 abdominal pairs (KPSC AAO 2017).

Q376. Which endocrine gland secretes Juvenile Hormone (JH) to maintain larval characteristics?

- A. Corpora allata
- B. Prothoracic gland
- C. Corpora cardiaca
- D. Brain neurosecretory cells
- **Correct Answer: A**
- **Explanation:** Juvenile Hormone is secreted by the corpora allata. It keeps the insect in a larval/nymphal stage during molting (KPSC AO 2017).

Q377. Prothoracic glands secrete which hormone responsible for initiating molting?

- A. Ecdysone
- B. Juvenile Hormone
- C. Bursicon
- D. Prothoracicotrophic hormone
- **Correct Answer: A**
- **Explanation:** Prothoracic glands secrete ecdysone (molting hormone) which triggers apolysis and cuticle synthesis (KPSC AAO 2017).

Q378. The European honeybee (*Apis mellifera*) was successfully introduced to India in 1962 by:

- A. A.S. Atwal
- B. M.S. Swaminathan
- C. H.S. Pruthi
- D. Ramachandra Rao
- **Correct Answer: A**
- **Explanation:** *Apis mellifera* was successfully introduced to India (Punjab) in 1962 by Dr. A.S. Atwal, which transformed commercial beekeeping (KPSC AAO 2017).

Q379. American foulbrood disease of honeybee larvae is caused by which pathogen?

- A. Paenibacillus larvae
- B. Melissococcus plutonius
- C. Ascosphaera apis
- D. Nosema apis
- **Correct Answer: A**
- **Explanation:** American foulbrood is a highly destructive bacterial disease caused by the spore-forming bacterium Paenibacillus larvae (KPSC AO 2017).

Q380. Which silkworm species is endemic to Assam and produces golden yellow silk?

- A. Antheraea assamensis
- B. Bombyx mori
- C. Antheraea mylitta
- D. Samia cynthia ricini
- **Correct Answer: A**
- **Explanation:** Muga silkworm (Antheraea assamensis) is endemic to Assam and produces highly valued golden-yellow silk (KPSC AAO 2017).

Q381. Pebrine disease of silkworm is caused by which microsporidian pathogen?

- A. Nosema bombycis
- B. Paenibacillus larvae
- C. Beauveria bassiana
- D. Saccharomyces cerevisiae
- **Correct Answer: A**
- **Explanation:** Pebrine is caused by the microsporidian Nosema bombycis, which is transmitted transovarially (KPSC AAO 2017).

Q382. Which tree species is considered the premium host for cultivating kusmi lac?

- A. Schleicheria oleosa
- B. Butea monosperma
- C. Ziziphus mauritiana
- D. Acacia nilotica
- **Correct Answer: A**
- **Explanation:** Kusum (Schleicheria oleosa) is the premium host tree for kusmi lac cultivation, yielding high-quality light-colored lac (KPSC AO 2017).

Q383. Dead hearts and white ears in rice are the classic injury symptoms of which pest?

- A. Scirpophaga incertulas
- B. Nilaparvata lugens
- C. Orseolia oryzae
- D. Leptocorisa acuta
- **Correct Answer: A**

- **Explanation:** Yellow stem borer (*Scirpophaga incertulas*) larvae bore into the stem, cutting off vascular bundles, causing dead hearts at vegetative and white ears at reproductive stages (KPSC AO 2017).

Q384. Hopper burn in rice fields is caused by the feeding damage of which pest?

- A. *Nilaparvata lugens*
- B. *Nephotettix virescens*
- C. *Orseolia oryzae*
- D. *Cnaphalocrocis medinalis*
- **Correct Answer: A**

- **Explanation:** Brown planthopper (*Nilaparvata lugens*) feeds on phloem sap at the base of rice plants, causing circular patches of dry, burnt crops called hopper burn (KPSC AAO 2017).

Q385. Silver shoot or onion leaf gall in rice is caused by the larvae of:

- A. *Orseolia oryzae*
- B. *Scirpophaga incertulas*
- C. *Cnaphalocrocis medinalis*
- D. *Leptocorisa acuta*
- **Correct Answer: A**

- **Explanation:** Rice gall midge (*Orseolia oryzae*) larvae feed on the apical meristem, stimulating the leaf sheath to grow into a silver/dirty-white tubular gall (KPSC AO 2014).

Q386. Rosette flowers in cotton are caused by the infestation of which pest?

- A. *Pectinophora gossypiella*
- B. *Helicoverpa armigera*
- C. *Earias vitella*
- D. *Bemisia tabaci*
- **Correct Answer: A**

- **Explanation:** Pink bollworm (*Pectinophora gossypiella*) caterpillars feed on buds/flowers, webbing petals together into a rosette shape (KPSC AO 2017).

Q387. Which cotton bollworm feeds by keeping its body outside the boll and inserting only its head?

- A. *Helicoverpa armigera*
- B. *Pectinophora gossypiella*
- C. *Earias vitella*
- D. *Earias insulana*
- **Correct Answer: A**

- **Explanation:** American bollworm (*Helicoverpa armigera*) is a polyphagous pest that characteristically feeds with its body outside the cotton boll, inserting only its head (KPSC AO 2017).

Q388. Early shoot borer of sugarcane causes dead hearts that can be easily pulled out, caused by:

- A. *Chilo infuscatellus*
- B. *Scirpophaga excerptalis*

- C. *Pyrilla perpusilla*
- D. *Chilo sacchariphagus indicus*
- **Correct Answer: A**
- **Explanation:** Early shoot borer (*Chilo infuscatellus*) bores into young sugarcane shoots (up to 3 months), producing foul-smelling, easily extractable dead hearts (KPSC AO 2014).

Q389. Which borer causes the 'bunchy top' symptom in sugarcane due to damage to the apical meristem?

- A. *Scirpophaga excerptalis*
- B. *Chilo infuscatellus*
- C. *Chilo sacchariphagus indicus*
- D. *Sesamia inferens*
- **Correct Answer: A**
- **Explanation:** Top shoot borer (*Scirpophaga excerptalis*) bores down the midrib into the growing point, causing parallel shot holes and terminal branching (bunchy top) (KPSC AAO 2017).

Q390. *Epiricania melanoleuca* is a highly effective lepidopteran ectoparasitoid of which pest?

- A. *Pyrilla perpusilla*
- B. *Nilaparvata lugens*
- C. *Chilo infuscatellus*
- D. *Helicoverpa armigera*
- **Correct Answer: A**
- **Explanation:** *Epiricania melanoleuca* is a moth whose larvae ectoparasitize sugarcane planthoppers (*Pyrilla perpusilla*) nymphs and adults, sucking their body fluids (KPSC AO 2017).

Q391. What is the lowest pest population density that will cause economic damage, requiring control measures?

- A. Economic Injury Level (EIL)
- B. Economic Threshold Level (ETL)
- C. General Equilibrium Position (GEP)
- D. Carrying Capacity
- **Correct Answer: A**
- **Explanation:** EIL is the lowest pest density that will cause economic damage (where cost of control equals value of yield saved) (KPSC AO 2017).

Q392. Organophosphate insecticides act as neurotoxins by inhibiting which enzyme?

- A. Acetylcholinesterase
- B. Succinate dehydrogenase
- C. Chitin synthase
- D. Cytochrome oxidase
- **Correct Answer: A**
- **Explanation:** Organophosphates inhibit acetylcholinesterase (AChE) at nerve synapses, preventing acetylcholine breakdown and causing muscle spasms and death (KPSC AO 2017).

Q393. To which chemical class does the systemic insecticide Carbofuran belong?

- A. Carbamates
- B. Organophosphates
- C. Organochlorines
- D. Synthetic Pyrethroids
- **Correct Answer: A**
- **Explanation:** Carbofuran (Furadan) is a systemic carbamate insecticide that reversibly inhibits acetylcholinesterase (KPSC AAO 2017).

Q394. Which class of insecticides acts as agonists on postsynaptic Nicotinic Acetylcholine Receptors?

- A. Neonicotinoids
- B. Organophosphates
- C. Carbamates
- D. Synthetic Pyrethroids
- **Correct Answer: A**
- **Explanation:** Neonicotinoids (e.g., Imidacloprid, Thiamethoxam) mimic acetylcholine, binding to postsynaptic nicotinic receptors, causing constant excitation (KPSC AAO 2017).

Q395. What is the target site of diamide insecticides like Chlorantraniliprole?

- A. Ryanodine receptors
- B. Sodium channels
- C. Acetylcholinesterase
- D. GABA receptors
- **Correct Answer: A**
- **Explanation:** Diamides target insect ryanodine receptors, causing unregulated calcium release from muscle stores, causing rapid contraction and paralysis (KPSC AO 2017).

Q396. Trichogramma chilonis is a widely utilized biological control agent which acts as a/an:

- A. Egg parasitoid
- B. Larval parasitoid
- C. Pupal parasitoid
- D. Adult predator
- **Correct Answer: A**
- **Explanation:** Trichogramma chilonis is an extremely small hymenopteran wasp that parasitizes and destroys eggs of Lepidoptera (KPSC AO 2017).

Q397. Green lacewing (Chrysoperla carnea) larvae are generalist predators feeding on:

- A. Sucking pests
- B. Stem borers
- C. Root grubs
- D. Soil nematodes

- **Correct Answer: A**

- **Explanation:** Chrysoperla carnea (Green lacewing) larvae are voracious predators of soft-bodied sucking insects like aphids, jassids, and thrips (KPSC AO 2017).

Q398. Where does the Bt delta-endotoxin bind to exert its insecticidal activity?

- A. Midgut epithelium cells
- B. Nervous system synapses
- C. Salivary glands
- D. Malpighian tubules

- **Correct Answer: A**

- **Explanation:** Bt delta-endotoxins dissolve in the alkaline midgut, are cleaved by proteases, and bind to cadherin receptors on midgut cells, creating lethal pores (KPSC AO 2017).

Q399. White muscardine disease of insects is caused by which entomopathogenic fungus?

- A. Beauveria bassiana
- B. Metarhizium anisopliae
- C. Nomuraea rileyi
- D. Verticillium lecanii

- **Correct Answer: A**

- **Explanation:** Beauveria bassiana causes white muscardine disease, characterized by white, powdery mycelial growth covering the dead insect body (KPSC AO 2017).

Q400. Which chemical is the primary component of pheromone traps used for Helicoverpa armigera?

- A. Z-11-hexadecenal
- B. Z-9-tetradecenal
- C. Grandlure
- D. Methyl eugenol

- **Correct Answer: A**

- **Explanation:** Pheromone traps for Helicoverpa armigera use synthetic female sex pheromones, with Z-11-hexadecenal as the major component (KPSC AO 2017).

Q401. Which draft cattle breed of Karnataka was bred by Tippu Sultan for military transport?

- A. Amritmahal
- B. Hallikar
- C. Khillar
- D. Deoni

- **Correct Answer: A**

- **Explanation:** Amritmahal was bred and maintained by the Mysore Kings, including Tippu Sultan, as quick-moving draft animals for transporting military supplies (KPSC AAO 2017).

Q402. Deoni breed of cattle originated from which district of Karnataka?

- A. Bidar

- B. Vijayapura
- C. Belagavi
- D. Mandya
- **Correct Answer: A**
- **Explanation:** Deoni is a dual-purpose cattle breed with white coats and black/red spots that originated in the Bidar district of Karnataka (KPSC AO 2017).

Q403. Which dwarf, disease-resistant cattle breed is native to the Western Ghats of Karnataka?

- A. Malnad Gidda
- B. Krishna Valley
- C. Hallikar
- D. Ongole
- **Correct Answer: A**
- **Explanation:** Malnad Gidda is a dwarf cattle breed of the Western Ghats of Karnataka, highly adapted to survive on poor forest grazing and humid climates (KPSC AO 2017).

Q404. Which indigenous dairy cattle breed is known as 'Lola' due to its characteristic loose skin?

- A. Sahiwal
- B. Gir
- C. Red Sindhi
- D. Tharparkar
- **Correct Answer: A**
- **Explanation:** Sahiwal, the highest milk-yielding indigenous cow, has very loose skin, leading to the local name 'Lola' (KPSC AAO 2017).

Q405. Which exotic dairy cattle breed produces milk with the highest fat percentage (~5% to 5.5%)?

- A. Jersey
- B. Holstein Friesian
- C. Brown Swiss
- D. Guernsey
- **Correct Answer: A**
- **Explanation:** Jersey cattle produce milk with the highest fat content (around 5% to 5.5%) among common exotic dairy breeds (KPSC AAO 2017).

Q406. Jet-black skin and tightly curled horns are characteristic of which buffalo breed?

- A. Murrah
- B. Surti
- C. Jaffarabadi
- D. Bhadawari
- **Correct Answer: A**
- **Explanation:** Murrah buffaloes from Haryana/Punjab are characterized by a deep jet-black body and tightly curled horns, yielding the highest milk (KPSC AO 2017).

Q407. Which buffalo breed is famous for its copper-colored hair coat and high milk fat content?

- A. Bhadawari
- B. Murrah
- C. Surti
- D. Nili-Ravi
- **Correct Answer: A**
- **Explanation:** Bhadawari buffaloes of Uttar Pradesh are known for their copper-colored coat and high milk fat content ranging from 8.5% to 13% (KPSC AO 2017).

Q408. Sickle-shaped, flat horns and two white collars on the brisket are characteristic of:

- A. Surti
- B. Murrah
- C. Jaffarabadi
- D. Bhadawari
- **Correct Answer: A**
- **Explanation:** Surti buffaloes of Gujarat have flat, sickle-shaped horns and two distinct white collars (stripes) on the brisket area (KPSC AAO 2017).

Q409. Which sheep breed of Karnataka is globally famous for its meaty hindquarters and superior mutton?

- A. Bannur / Mandya
- B. Deccani
- C. Hassan
- D. Bellary
- **Correct Answer: A**
- **Explanation:** Mandya (or Bannur) sheep is a highly prized mutton breed in Karnataka, known for its compact, wedge-shaped body and high dressing percentage (KPSC AO 2017).

Q410. Which goat breed is commonly known as the 'city goat' because it is highly suited for stall-feeding?

- A. Barbari
- B. Jamunapari
- C. Beetle
- D. Osmanabadi
- **Correct Answer: A**
- **Explanation:** Barbari is a small, prolific goat breed well-suited for urban environments and intensive stall-feeding (KPSC AAO 2017).

Q411. Which goat breed is famous for high prolificacy (multiple births) and excellent skin quality?

- A. Black Bengal
- B. Jamunapari
- C. Barbari
- D. Beetle

- **Correct Answer: A**

- **Explanation:** Black Bengal goats are highly prolific (frequent twins/triplets), producing high-quality meat (chevon) and prized soft leather (KPSC AO 2017).

Q412. Which chicken breed is famous for its black plumage, black bones, and black meat?

- A. Kadaknath
- B. Aseel
- C. Giriraja
- D. White Leghorn

- **Correct Answer: A**

- **Explanation:** Kadaknath (Kalamasi) is a breed from Madhya Pradesh with high melanin deposition, resulting in black feathers, meat, and bones (KPSC AO 2017).

Q413. Which backyard poultry breeds were developed by KVAFSU, Bangalore for rural rearing?

- A. Giriraja and Swarnadhara
- B. Kadaknath and Aseel
- C. White Leghorn and RIR
- D. Swarnadhara and Kadaknath

- **Correct Answer: A**

- **Explanation:** Giriraja and Swarnadhara are high-yield, disease-resistant backyard poultry breeds developed by KVAFSU, Bangalore (KPSC AO 2017).

Q414. Colostrum is rich in which immunoglobulin, providing passive immunity to the newborn calf?

- A. IgG
- B. IgM
- C. IgA
- D. IgE

- **Correct Answer: A**

- **Explanation:** Colostrum contains high amounts of immunoglobulins, particularly IgG, which are absorbed intact by the calf's intestine to provide passive immunity (KPSC AO 2017).

Q415. What is the normal specific gravity range of pure cow milk at 15.5°C?

- A. 1.028 to 1.030
- B. 1.030 to 1.032
- C. 1.018 to 1.022
- D. 1.035 to 1.040

- **Correct Answer: A**

- **Explanation:** Cow milk specific gravity is 1.028-1.030. Buffalo milk is slightly heavier, ranging from 1.030-1.032 (KPSC AAO 2017).

Q416. Which method is the standard volumetric test used to estimate fat percentage in milk?

- A. Gerber method
- B. Lactometer method

- C. Kjeldahl method
- D. Babcock method
- **Correct Answer: A**
- **Explanation:** The Gerber method uses sulfuric acid to digest milk protein and amyl alcohol to separate the fat layer, read directly on a butyrometer (KPSC AO 2014).

Q417. High Temperature Short Time (HTST) pasteurization of milk requires heating to:

- A. 72°C for 15 seconds
- B. 63°C for 30 minutes
- C. 100°C for 10 minutes
- D. 85°C for 5 seconds
- **Correct Answer: A**
- **Explanation:** HTST pasteurization involves heating milk to 72°C for 15 seconds followed by rapid cooling to below 4°C (KPSC AO 2017).

Q418. What is the ideal pH range of high-quality, well-preserved silage?

- A. 3.5 to 4.2
- B. 5.0 to 5.8
- C. 6.2 to 6.8
- D. 2.0 to 2.8
- **Correct Answer: A**
- **Explanation:** Silage preservation is successful when lactic acid fermentation lowers the pH to 3.5-4.2, inhibiting spoilage microbes (KPSC AO 2017).

Q419. Which disease is a zoonotic bacterial infection causing sudden death and tarry blood discharge?

- A. Anthrax
- B. Black Quarter
- C. Mastitis
- D. Foot and Mouth Disease
- **Correct Answer: A**
- **Explanation:** Anthrax (*Bacillus anthracis*) causes rapid septicemia and death, characterized by non-clotting, dark tarry blood oozing from natural orifices (KPSC AO 2017).

Q420. Mastitis in dairy cattle is diagnosed in fields using which test?

- A. California Mastitis Test (CMT)
- B. Gerber test
- C. CMT and Gerber
- D. Methylene Blue Reduction Test
- **Correct Answer: A**
- **Explanation:** CMT is a rapid field test for subclinical mastitis, where a reagent reacts with DNA in somatic cells to form a gel (KPSC AO 2017).

Q421. Which Agro-Climatic Zone of Karnataka is the largest in area and is characterized by the lowest average annual rainfall?

- A. Northern Dry Zone (Zone 3)
- B. Southern Dry Zone (Zone 6)
- C. Central Dry Zone (Zone 4)
- D. Northeast Dry Zone (Zone 2)
- **Correct Answer: A**
- **Explanation:** Zone 3 (Northern Dry Zone), headquartered in Vijayapura, is the largest agro-climatic zone in Karnataka, receiving the lowest average annual rainfall of 460-585 mm. (KPSC AO 2017)

Q422. The flagship water harvesting scheme 'Krishi Bhagya' was launched by the Government of Karnataka in which year?

- A. 2012
- B. 2014
- C. 2016
- D. 2018
- **Correct Answer: B**
- **Explanation:** Krishi Bhagya was launched in the year 2014-15 to improve rainwater harvesting and conservation in dryland regions of Karnataka. (KPSC AO 2017)

Q423. Which organization developed and maintains the 'FRUITS' portal, a unified farmer registration system in Karnataka?

- A. National Informatics Centre (NIC) and DPAR e-Governance
- B. Karnataka State Agricultural Marketing Board
- C. UAS Bangalore
- D. NABARD Karnataka Regional Office
- **Correct Answer: A**
- **Explanation:** The FRUITS (Farmer Registration and Unified Beneficiary Information System) portal was developed by NIC in coordination with the Department of Personnel and Administrative Reforms (DPAR) e-Governance, Government of Karnataka. (Latest 2020)

Q424. What is the primary cash incentive provided per hectare under the 'Raitha Siri' scheme to encourage the cultivation of minor millets in Karnataka?

- A. Rs. 5,000
- B. Rs. 10,000
- C. Rs. 15,000
- D. Rs. 20,000
- **Correct Answer: B**
- **Explanation:** Raitha Siri provides an incentive of Rs. 10,000 per hectare (up to a limit of 2 hectares) directly to the bank accounts of farmers cultivating minor millets. (Latest 2019)

Q425. In which year was the University of Agricultural Sciences, Dharwad established?

- A. 1963
- B. 1986
- C. 2008
- D. 2012
- **Correct Answer: B**
- **Explanation:** UAS Dharwad was established in the year 1986 by splitting from UAS Bangalore to cater to the northern districts of Karnataka. (KPSC AAO 2017)

Q426. Which of the following is the first specialized Horticultural University in Karnataka and where is it headquartered?

- A. UAS Raichur
- B. UHS Bagalkot
- C. KSNUAHS Shivamogga
- D. UHS Kolar
- **Correct Answer: B**
- **Explanation:** The University of Horticultural Sciences (UHS), Bagalkot was established in 2008 and is the first exclusive horticultural university in the state. (KPSC AO 2017)

Q427. The 'Samrakshane' portal in Karnataka is specifically designed to facilitate and monitor which agricultural program?

- A. Krishi Bhagya
- B. Pradhan Mantri Fasal Bima Yojana (PMFBY)
- C. Soil Health Card Scheme
- D. Custom Hiring Centers
- **Correct Answer: B**
- **Explanation:** Samrakshane is the online portal of Karnataka used for the administration, enrollment, and claim settlement of crop insurance under PMFBY. (KPSC AO 2017)

Q428. Under the 'Ksheera Dhare' scheme, what is the incentive amount paid by the Karnataka government per liter of milk supplied by dairy farmers?

- A. Rs. 2
- B. Rs. 3
- C. Rs. 5
- D. Rs. 7
- **Correct Answer: C**
- **Explanation:** The Ksheera Dhare scheme provides a financial incentive of Rs. 5 per liter of milk supplied to milk cooperative societies. (KPSC AO 2017)

Q429. What is the thickness of the polythene lining sheet recommended for farm ponds under the 'Krishi Bhagya' scheme to prevent seepage loss?

- A. 100-200 microns
- B. 300-500 microns
- C. 600-800 microns

- D. 1000 microns
- **Correct Answer: B**
- **Explanation:** To check seepage in farm ponds (Krishi Honda), the polythene lining sheets of 300 to 500 microns thickness are recommended. (KPSC AAO 2017)

Q430. The GI-tagged agricultural product 'Sirsi Supari' belongs to which category of plantation crops?

- A. Arecanut
- B. Coconut
- C. Cashew
- D. Black Pepper
- **Correct Answer: A**
- **Explanation:** Sirsi Supari is a GI-tagged arecanut variety grown in Sirsi, Siddapur, and Yellapur taluks of Uttara Kannada district. (Latest 2019)

Q431. Which Agro-Climatic Zone of Karnataka is the wettest, receiving an annual rainfall ranging between 3000 mm and 4700 mm?

- A. Hill Zone (Zone 9)
- B. Coastal Zone (Zone 10)
- C. Northeast Dry Zone (Zone 2)
- D. Southern Transition Zone (Zone 7)
- **Correct Answer: B**
- **Explanation:** Zone 10 (Coastal Zone), headquartered at Brahmavar, is the wettest zone in Karnataka with annual rainfall of 3000 to 4700 mm. (KPSC AO 2017)

Q432. The 'Bhoo Siri' scheme launched in Karnataka aims to provide financial assistance for which of the following agricultural activities?

- A. Purchasing farm machinery
- B. Purchasing organic inputs and bio-fertilizers
- C. Boring new borewells
- D. Buying high-yielding seed varieties
- **Correct Answer: B**
- **Explanation:** Bhoo Siri provides a financial assistance/subsidy of Rs. 2,500 per acre to farmers to encourage organic and natural farming inputs. (Latest 2020)

Q433. What is the name of the salt-tolerant traditional paddy variety grown in the estuarine 'Gajni' lands of Uttara Kannada district?

- A. Intan
- B. Kagga
- C. Jaya
- D. Tellahamsa
- **Correct Answer: B**

- **Explanation:** Kagga rice is a traditional, salt-tolerant variety grown in the saline, waterlogged coastal tracts (Gajni lands) of Kumta, Uttara Kannada. (KPSC AAO 2017)

Q434. In which year was the first state organic farming policy implemented in Karnataka, making it the first state in India to do so?

- A. 2004
- B. 2008
- C. 2012
- D. 2017
- **Correct Answer: A**

- **Explanation:** Karnataka formulated and implemented the organic farming policy in 2004, being the pioneer state in India. It was later revised in 2017. (KPSC AO 2017)

Q435. The 'Raita Vidya Nidhi' scholarship scheme was initially launched for children of which category, and later expanded to others?

- A. Government employees
- B. Farmers
- C. Agricultural laborers
- D. Weavers
- **Correct Answer: B**

- **Explanation:** Raita Vidya Nidhi was launched in 2021 for the children of farmers and was later extended to children of weavers, agricultural laborers, taxi/auto drivers, etc. (Latest 2021)

Q436. The 'Krishi Yantra Dhare' scheme is executed through which model of service delivery?

- A. 100% government ownership
- B. Public-Private Partnership (PPP) through Custom Hiring Centers (CHCs)
- C. Cooperative marketing societies
- D. Self-Help Groups (SHGs)
- **Correct Answer: B**

- **Explanation:** Krishi Yantra Dhare operates on a PPP model, establishing Custom Hiring Centers (CHCs) at the hobli level to rent out farm machinery. (KPSC AO 2017)

Q437. Which district in Karnataka is the leading producer of turmeric in the southern dry zone, utilizing deep red soils?

- A. Mandya
- B. Chamarajanagar
- C. Mysuru
- D. Hassan
- **Correct Answer: B**

- **Explanation:** Chamarajanagar is the leading producer of turmeric in southern Karnataka, benefiting from suitable soil and climate. (KPSC AAO 2015)

Q438. The 'REWARD' project implemented by the Watershed Development Department of Karnataka is funded/assisted by which international agency?

- A. Asian Development Bank
- B. World Bank
- C. International Monetary Fund
- D. Japan International Cooperation Agency
- **Correct Answer: B**
- **Explanation:** The REWARD (Rejuvenating Watersheds for Agricultural Resilience through Innovative Development) project is assisted by the World Bank. (KPSC AO 2017)

Q439. The headquarters of the Hill Zone (Zone 9) of Karnataka is located at:

- A. Brahmavar
- B. Mudigere
- C. Shivamogga
- D. Madikeri
- **Correct Answer: B**
- **Explanation:** Zone 9 (Hill Zone) has its research headquarters at Mudigere in Chikkamagaluru district. (KPSC AAO 2015)

Q440. Under the 'Krishi Bhagya' scheme, what is the storage capacity of a standard 10m x 10m x 3m farm pond (Krishi Honda)?

- A. 150 cubic meters
- B. 200 cubic meters
- C. 300 cubic meters
- D. 450 cubic meters
- **Correct Answer: C**
- **Explanation:** A 10m x 10m x 3m farm pond has a volume of $10 * 10 * 3 = 300$ cubic meters (or 3,00,000 liters) capacity. (KPSC AAO 2015)

Q441. The unique GI-tagged 'Kamalapur Red Banana' is primarily grown in which district of Karnataka?

- A. Kalaburagi
- B. Bidar
- C. Mysuru
- D. Ramanagara
- **Correct Answer: A**
- **Explanation:** Kamalapur Red Banana is grown in Kamalapur, Kalaburagi district, known for its high mineral content and red peel. (Latest 2008)

Q442. Which local cattle breed is known as the dwarf draft breed of Western Ghats in Karnataka, characterized by high disease resistance?

- A. Hallikar
- B. Amritmahal
- C. Malnad Gidda

- D. Deoni
- **Correct Answer: C**
- **Explanation:** Malnad Gidda is a native dwarf cattle breed of the Western Ghats (Malnad region) of Karnataka, known for its adaptability and resilience. (KPSC AO 2017)

Q443. The 'Krishi Sanjeevini' program in Karnataka uses mobile laboratories for:

- A. Artificial insemination
- B. On-farm testing of soil, water, and diagnosis of pests/diseases
- C. Selling seeds
- D. Direct procurement of crops
- **Correct Answer: B**
- **Explanation:** Krishi Sanjeevini vans are mobile soil and plant clinic units deployed to test soil and water quality, and diagnose pests/diseases on-site. (Latest 2021)

Q444. In which year was the Keladi Shivappa Nayaka University of Agricultural and Horticultural Sciences (KSNUAHS), Shivamogga established?

- A. 2008
- B. 2010
- C. 2012
- D. 2014
- **Correct Answer: C**
- **Explanation:** KSNUAHS Shivamogga was established in 2012 by separating the Shimoga and Chikmagalur areas from UAS Bangalore and UHS Bagalkot. (KPSC AAO 2017)

Q445. The 'Dharwad Pedha' which received a GI tag is a popular dairy product. Which dairy cattle or buffalo breed is historically associated with the Dharwad region's milk?

- A. Murrah
- B. Dharwari / Pandharpuri buffalo
- C. Surti
- D. Jaffrabadi
- **Correct Answer: B**
- **Explanation:** The local buffaloes (often referred to as Dharwari or Pandharpuri types) provide the high-fat milk traditional to the Dharwad Pedha industry. (KPSC AO 2017)

Q446. The headquarters of the Northeast Transition Zone (Zone 1) of Karnataka is located at:

- A. Gulbarga
- B. Raichur
- C. Bidar
- D. Yadgir
- **Correct Answer: C**
- **Explanation:** Zone 1 (Northeast Transition Zone) has its headquarters at Bidar. (KPSC AAO 2017)

Q447. What is the main objective of the 'Raita Sanjeevini' scheme implemented by the Karnataka Department of Agriculture?

- A. Free distribution of organic seeds
- B. Financial assistance/ex-gratia for accidental death or injury during farm operations
- C. Solar pump subsidy
- D. Drip irrigation subsidy
- **Correct Answer: B**
- **Explanation:** Raita Sanjeevini provides ex-gratia financial help to farmers who suffer accidental death or permanent disablement while performing farm work. (KPSC AO 2017)

Q448. The 'Bhoomi' portal, which digitizes land records in Karnataka, is linked with agricultural credit institutions to enable:

- A. Instant loan disbursement without land security
- B. Online creation of charge/mortgage on land for crop loans
- C. Direct seed subsidy
- D. Online purchase of farm tractors
- **Correct Answer: B**
- **Explanation:** The Bhoomi database is integrated with banks so that banks can create an online charge on a farmer's land register for securing loans. (KPSC AO 2014)

Q449. The GI-tagged flower 'Hadagali Mallige' is grown in which of the following districts of Karnataka?

- A. Mysuru
- B. Vijayanagara (formerly Ballari)
- C. Udupi
- D. Kolar
- **Correct Answer: B**
- **Explanation:** Hadagali Mallige is grown in Huvina Hadagali taluk of Vijayanagara district (previously Ballari). (KPSC AO 2017)

Q450. Under the revised Karnataka Organic Farming Policy 2017, the state aims to convert what percentage of cultivable land in each district to organic farming?

- A. 5%
- B. 10%
- C. 15%
- D. 20%
- **Correct Answer: B**
- **Explanation:** The 2017 policy set a target of converting 10% of cultivable land in each district into organic farming over the next few years. (KPSC AO 2017)

Q451. The 'Upper Krishna Project' (UKP) utilizes water from the Krishna River. Which dam is the main reservoir under UKP Phase I and II?

- A. Almatti Dam (Lal Bahadur Shastri Dam)

- B. Narayanpur Dam (Basava Sagar Dam)
- C. Tungabhadra Dam
- D. Bhadra Dam

• **Correct Answer: A**

- **Explanation:** The Almatti Dam in Bagalkot is the primary reservoir of the Upper Krishna Project for Phase I and II. (KPSC AO 2014)

Q452. The 'Byadagi Chilli' (GI tagged) is world-famous because of its:

- A. Extremely high pungency
- B. High color value (oleoresin) and low pungency
- C. Short shelf life
- D. Large seed size

• **Correct Answer: B**

- **Explanation:** Byadagi Chilli is known for its high deep-red color value (due to capsanthin) and very low pungency (capsaicin content), making it perfect for food industry extraction. (KPSC AAO 2017)

Q453. The 'Coorg Green Cardamom' GI tag is associated with which geographical region in Karnataka?

- A. Coastal plains
- B. Western Ghats (Malnad)
- C. Northern dry plains
- D. Eastern dry plains

• **Correct Answer: B**

- **Explanation:** Coorg Green Cardamom is grown in the high-rainfall hilly tracts of Western Ghats in Kodagu and adjoining districts. (KPSC AO 2017)

Q454. In Karnataka's Digital Crop Survey system, who is primarily responsible for recording and uploading the crop details in the mobile app?

- A. Village Accountants
- B. Farmers or trained Private Crop Surveyors
- C. Agriculture Officers
- D. Gram Panchayat Presidents

• **Correct Answer: B**

- **Explanation:** The system allows farmers themselves or designated private crop surveyors (young rural youth) to walk the fields, geotag the crop, and upload it. (Latest 2020)

Q455. The headquarters of the Eastern Dry Zone (Zone 5) of Karnataka is located at:

- A. Hebbal (Bangalore)
- B. Chintamani
- C. Kolar
- D. Ramanagara

• **Correct Answer: B**

- **Explanation:** Zone 5 (Eastern Dry Zone) has its zonal agricultural research station at Chintamani in Chikkaballapura district. (KPSC AO 2017)

Q456. The 'Kari Ishad' mango variety which recently received a GI tag is native to which taluk of Uttara Kannada?

- A. Kumta
- B. Sirsi
- C. Ankola
- D. Karwar
- **Correct Answer: C**

- **Explanation:** The Kari Ishad mango, celebrated for its superb taste and pulp yield, is native to Ankola taluk of Uttara Kannada. (Latest 2023)

Q457. What is the primary source of irrigation water for the 'Malaprabha' command area development?

- A. Almatti Dam
- B. Renuka Sagar Dam (Saundatti)
- C. Lakkavalli Dam
- D. Hosapete Dam
- **Correct Answer: B**

- **Explanation:** The Renuka Sagar Dam across the Malaprabha River at Saundatti is the source of irrigation for this command area. (KPSC AO 2014)

Q458. The 'Nanjangud Banana' (Rasabale) has a unique taste and texture. Which soil type is crucial for expression of these quality traits in Nanjangud?

- A. Laterite soil
- B. High organic clayey black soil
- C. Sandy desert soil
- D. Coastal saline soil
- **Correct Answer: B**

- **Explanation:** High organic matter black clay soils of Nanjangud, combined with local climatic factors, give the fruit its unique aroma and taste. (KPSC AAO 2015)

Q459. The 'Siri Dhanya' scheme in Karnataka refers to the promotion of:

- A. High-yielding rice varieties
- B. Organic pulses
- C. Nutri-cereals (Millets)
- D. Export-quality spices
- **Correct Answer: C**

- **Explanation:** Siri Dhanya is the collective term used by the Government of Karnataka to promote and brand millets (ragi, jowar, minor millets). (Latest 2018)

Q460. The headquarters of the Southern Dry Zone (Zone 6) of Karnataka is located at Mandya (VC Farm). Which soil order dominates this zone?

- A. Vertisols
- B. Alfisols
- C. Aridisols
- D. Histosols

• **Correct Answer: B**

• **Explanation:** Zone 6 (Southern Dry Zone) is dominated by Alfisols (Red sandy and red loam soils). (KPSC AAO 2017)

Q461. Which cost concept represents the actual out-of-pocket expenses incurred by an owner-operator plus the rental value of owned land, rent paid for leased-in land, and interest on owned capital?

- A. Cost A2
- B. Cost B2
- C. Cost C1
- D. Cost C2

• **Correct Answer: B**

• **Explanation:** Cost B2 is calculated by adding the rental value of owned land and rent paid for leased-in land to Cost B1 (Cost A1 + interest on value of owned capital). (KPSC AO 2017)

Q462. In production economics, the 'rational stage' of production is:

- A. Stage I
- B. Stage II
- C. Stage III
- D. Both Stage I and II

• **Correct Answer: B**

• **Explanation:** Stage II is the rational stage because here the Marginal Physical Product (MPP) is positive but declining, and the Average Physical Product (APP) is also declining. It ends where $MPP = 0$. (KPSC AO 2017)

Q463. The 'least-cost combination' of inputs in factor-factor relationship is achieved when:

- A. $MRTS_{xy} = P_x / P_y$
- B. $MRTS_{xy} = P_y / P_x$
- C. $MRPT_{xy} = P_x / P_y$
- D. $MVP = MIC$

• **Correct Answer: A**

• **Explanation:** The least-cost combination is achieved where the Marginal Rate of Technical Substitution of X for Y ($MRTS_{xy}$) equals the ratio of input prices (P_x/P_y). (KPSC AAO 2015)

Q464. Which agency recommends the Minimum Support Prices (MSP) for mandated agricultural crops in India?

- A. NABARD
- B. Commission for Agricultural Costs and Prices (CACP)
- C. Food Corporation of India (FCI)
- D. NITI Aayog

• **Correct Answer: B**

- **Explanation:** CACP is the agency that recommends MSPs to the Cabinet Committee on Economic Affairs (CCEA), which makes the final decision. (KPSC AO 2017)

Q465. The 'Kisan Credit Card' (KCC) scheme was introduced in 1998 based on the recommendations of which committee?

- A. M. Narasimham Committee
- B. R.V. Gupta Committee
- C. B. Sivaraman Committee
- D. Nariman Committee

• **Correct Answer: B**

- **Explanation:** The KCC scheme was formulated based on the recommendations of the R.V. Gupta Committee to provide timely and flexible short-term credit. (KPSC AO 2017)

Q466. When the market structure consists of a single buyer and many sellers, it is known as:

- A. Monopoly
- B. Monopsony
- C. Oligopoly
- D. Duopoly

• **Correct Answer: B**

- **Explanation:** Monopsony refers to a market situation where there is only one buyer for a product or service. (KPSC AAO 2017)

Q467. Marketed surplus is defined as:

- A. Total production minus retention for family consumption, seed, feed, and wages
- B. Actual quantity sold in the market
- C. Total production minus marketable surplus
- D. Expected production minus retention

• **Correct Answer: B**

- **Explanation:** Marketed surplus is the actual quantity of agricultural produce sold by the farmer in the market, regardless of whether requirements are met. (KPSC AO 2017)

Q468. Under WTO's Agreement on Agriculture (AoA), direct income support payments linked to production-limiting programs are categorized under:

- A. Green Box
- B. Blue Box
- C. Amber Box
- D. Red Box

• **Correct Answer: B**

- **Explanation:** The Blue Box contains Amber Box support measures that require farmers to limit production. Currently, there are no limits on Blue Box spending. (KPSC AAO 2015)

Q469. The apex bank for agricultural and rural development in India, NABARD, was established on which date?

- A. 2nd October 1975
- B. 12th July 1982

- C. 1st January 1985
- D. 14th April 2016
- **Correct Answer: B**
- **Explanation:** NABARD was established on July 12, 1982, under Act 61 of 1981, following the recommendations of the B. Sivaraman Committee. (KPSC AO 2014)

Q470. What is the formula for calculating the 'Producer's Share in Consumer Rupee' (Ps)?

- A. $Ps = (\text{Consumer Price} / \text{Producer Price}) * 100$
- B. $Ps = (\text{Producer Price} / \text{Consumer Price}) * 100$
- C. $Ps = (\text{Price Spread} / \text{Consumer Price}) * 100$
- D. $Ps = (\text{Producer Price} / \text{Price Spread}) * 100$
- **Correct Answer: B**
- **Explanation:** $Ps = (Pp / Pc) * 100$, where Pp is the price received by the producer and Pc is the price paid by the consumer. (KPSC AAO 2017)

Q471. The Lead Bank Scheme was launched in 1969 following the recommendations of:

- A. Gadgil Study Group and Nariman Committee
- B. Narasimham Committee
- C. Sivaraman Committee
- D. R.V. Gupta Committee
- **Correct Answer: A**
- **Explanation:** The Nariman Committee recommended the Lead Bank Scheme in 1969 to assign districts to commercial banks to act as leaders in credit extension. (KPSC AO 2014)

Q472. The slope of an Isoquant curve represents the:

- A. Price ratio of inputs
- B. Marginal Rate of Technical Substitution (MRTS)
- C. Marginal Rate of Product Transformation (MRPT)
- D. Elasticity of substitution
- **Correct Answer: B**
- **Explanation:** The slope of an isoquant is the Marginal Rate of Technical Substitution ($MRTS = dY/dX$), which indicates how much of input Y must be replaced by X. (KPSC AAO 2015)

Q473. According to Engel's Law, as a family's income increases:

- A. The absolute expenditure on food decreases
- B. The percentage of income spent on food decreases
- C. The percentage of income spent on food increases
- D. The expenditure on luxury items decreases
- **Correct Answer: B**
- **Explanation:** Engel's Law states that the proportion (percentage) of family income spent on food decreases as family income rises, even though absolute food expenditure may rise. (KPSC AO 2014)

Q474. The first Regional Rural Bank (RRB) in India was established on 2nd October 1975 under the recommendation of:

- A. Nariman Committee
- B. M. Narasimham Committee
- C. Sivaraman Committee
- D. R.V. Gupta Committee
- **Correct Answer: B**
- **Explanation:** RRBs were established on October 2, 1975, on the recommendations of the Narasimham Committee on Rural Credit. (KPSC AAO 2017)

Q475. If the cross-elasticity of demand between two agricultural inputs (e.g., tractor and diesel) is negative, the two inputs are:

- A. Substitutes
- B. Complements
- C. Independent
- D. Giffen goods
- **Correct Answer: B**
- **Explanation:** A negative cross-elasticity means that as the price of tractors rises, the demand for diesel falls, indicating they are complementary inputs. (KPSC AAO 2015)

Q476. In production economics, the factor-product relationship deals with:

- A. Optimization of input levels (How much to produce)
- B. Least-cost combination of inputs (How to produce)
- C. Enterprise selection (What to produce)
- D. Pricing of inputs
- **Correct Answer: A**
- **Explanation:** The factor-product relationship deals with the allocation of a single input to produce a single product, determining the optimum level of input (How much to produce). (KPSC AO 2017)

Q477. The CACP recommends MSP for how many mandated agricultural crops?

- A. 20 crops
- B. 22 crops
- C. 25 crops
- D. 28 crops
- **Correct Answer: B**
- **Explanation:** CACP recommends MSP for 22 mandated crops (plus Fair and Remunerative Price, FRP, for sugarcane, making 23 crops in total). (KPSC AO 2017)

Q478. The 'Opportunity Cost' concept is also known as:

- A. Prime cost
- B. Alternative cost
- C. Out-of-pocket cost

- D. Sunk cost
- **Correct Answer: B**
- **Explanation:** Opportunity cost is the value of the next best alternative forgone, representing the alternative cost of using a resource. (KPSC AO 2014)

Q479. When the elasticity of demand is zero ($E_d = 0$), the demand curve is:

- A. Horizontal
- B. Vertical
- C. Downward sloping
- D. Upward sloping
- **Correct Answer: B**
- **Explanation:** An elasticity of zero means quantity demanded does not change with price, resulting in a vertical demand curve (perfectly inelastic). (KPSC AO 2014)

Q480. In a Cobb-Douglas production function, if the sum of input elasticities ($\alpha + \beta$) is greater than 1, the production process exhibits:

- A. Constant returns to scale
- B. Increasing returns to scale
- C. Decreasing returns to scale
- D. Negative returns to scale
- **Correct Answer: B**
- **Explanation:** If $\alpha + \beta > 1$, doubling the inputs results in more than double the output, representing increasing returns to scale. (KPSC AAO 2017)

Q481. The 'AGMARK' quality certification mark was introduced under the Agricultural Produce (Grading and Marking) Act of:

- A. 1937
- B. 1955
- C. 1965
- D. 1986
- **Correct Answer: A**
- **Explanation:** AGMARK was introduced under the Agricultural Produce (Grading and Marking) Act, 1937. (KPSC AO 2017)

Q482. The value of marginal product (VMP) is calculated by multiplying:

- A. Total product by product price
- B. Marginal physical product by product price
- C. Average product by product price
- D. Marginal physical product by input price
- **Correct Answer: B**
- **Explanation:** $VMP = MPP \times P_y$, where MPP is the marginal physical product and P_y is the price of the output. Under perfect competition, $VMP = MVP$. (KPSC AO 2017)

Q483. The market where goods are transacted on the spot and delivered immediately is called:

- A. Forward market
- B. Spot or Cash market
- C. Future market
- D. Terminal market
- **Correct Answer: B**
- **Explanation:** Spot or cash markets are those where goods are sold and bought on the spot and delivered immediately. (KPSC AAO 2017)

Q484. The difference between Cost C2 and Cost A1 is:

- A. Rental value of owned land + interest on owned fixed capital + imputed value of family labor
- B. Rent paid for leased land + cost of seeds
- C. Only the imputed value of family labor
- D. Managerial cost of the farmer
- **Correct Answer: A**
- **Explanation:** Cost C2 = Cost B2 + family labor. Cost A1 is the out-of-pocket expenses. The difference includes rent/rental value of owned land, interest on owned fixed capital, and family labor value. (KPSC AO 2017)

Q485. In a partial budget, which of the following is recorded?

- A. Total costs and total returns of the entire farm
- B. Only the added costs, reduced returns, reduced costs, and added returns of the proposed change
- C. Only depreciation and interest
- D. Cash transactions only
- **Correct Answer: B**
- **Explanation:** Partial budgeting is a tool to assess the profitability of a minor change by considering only the four elements: added costs, reduced returns, reduced costs, and added returns. (KPSC AO 2014)

Q486. What is the discount rate at which the Net Present Value (NPV) of a project is exactly zero?

- A. Benefit-Cost Ratio
- B. Internal Rate of Return (IRR)
- C. Payback Period
- D. Prime Interest Rate
- **Correct Answer: B**
- **Explanation:** The IRR is defined as the discount rate that equates the present value of cash inflows with the present value of cash outflows (NPV = 0). (KPSC AO 2014)

Q487. The slope of the production frontier in product-product relationship is known as:

- A. MRTS
- B. MRPT (Marginal Rate of Product Transformation)
- C. Elasticity of substitution
- D. Output elasticity

- **Correct Answer: B**

- **Explanation:** The slope of the Product Transformation Curve (Production Possibility Curve) is the Marginal Rate of Product Transformation (MRPT), showing how much of one product must be given up to produce more of another. (KPSC AAO 2015)

Q488. Under the law of variable proportions, the point where Marginal Physical Product (MPP) reaches its maximum is called:

- A. Point of Inflection
- B. Optimum point
- C. Boundary point
- D. Ridge point

- **Correct Answer: A**

- **Explanation:** The Point of Inflection on the TPP curve corresponds to the point where MPP is at its maximum. (KPSC AO 2014)

Q489. Which type of market integration involves a merger of two firms that are at completely different stages of production but in the same industry chain, moving towards the raw material source?

- A. Forward Vertical Integration
- B. Backward Vertical Integration
- C. Horizontal Integration
- D. Conglomerate Integration

- **Correct Answer: B**

- **Explanation:** Backward vertical integration occurs when a firm merges with a supplier at an earlier stage in the supply chain (closer to raw materials). (KPSC AO 2017)

Q490. The ratio of Total Assets to Total Liabilities is a measure of:

- A. Liquidity
- B. Solvency
- C. Profitability
- D. Efficiency

- **Correct Answer: B**

- **Explanation:** Solvency ratios, like the asset-liability ratio, measure the firm's ability to cover all liabilities in the long run. (KPSC AAO 2017)

Q491. Which cost concept is used by the CACP as the basic benchmark for recommending MSPs, aiming to cover C2 cost plus 50%?

- A. Cost A2
- B. Cost A2 + FL (imputed value of family labor)
- C. Cost B2
- D. Cost C3

- **Correct Answer: B**

- **Explanation:** CACP recommends MSP based on a minimum of 50% return over the Cost A2 + FL (FL = Family Labor) concept. (KPSC AO 2017)

Q492. In a market, the 'Price Spread' increases when:

- A. Marketing efficiency increases
- B. Middlemen margins increase
- C. Producer's share increases
- D. Marketing channels are shortened

• **Correct Answer: B**

• **Explanation:** Price spread is $P_c - P_p$. It increases when the costs and margins of intermediaries (middlemen) increase. (KPSC AO 2014)

Q493. The 'Simplex Method' of solving linear programming problems was developed by:

- A. Wassily Leontief
- B. George B. Dantzig
- C. J.R. Hicks
- D. Alfred Marshall

• **Correct Answer: B**

• **Explanation:** George Dantzig developed the Simplex Method in 1947 to solve linear programming models. (KPSC AAO 2015)

Q494. The 'Law of Equi-Marginal Returns' is also known as the:

- A. Law of diminishing returns
- B. Law of opportunity cost
- C. Law of substitution
- D. Law of returns to scale

• **Correct Answer: C**

• **Explanation:** The Law of Equi-Marginal Returns is called the Law of Substitution because it involves substituting a less productive activity with a more productive one until marginal returns are equalized. (KPSC AO 2017)

Q495. If the price of a good increases and the total revenue also increases, the price elasticity of demand for that good is:

- A. Inelastic ($E_p < 1$)
- B. Elastic ($E_p > 1$)
- C. Unitary elastic ($E_p = 1$)
- D. Perfectly elastic ($E_p = \text{infinity}$)

• **Correct Answer: A**

• **Explanation:** When demand is inelastic, a price increase leads to a less than proportionate decrease in quantity demanded, so total revenue increases. (KPSC AAO 2017)

Q496. A market where agricultural produce is brought for first-time sale by farmers to wholesalers is termed as:

- A. Primary market
- B. Secondary market
- C. Retail market
- D. Terminal market

- **Correct Answer: A**

• **Explanation:** Primary agricultural markets are those situated in agricultural areas where farmers sell their produce for the first time. (KPSC AO 2014)

Q497. Which of the following is a non-recurring or fixed cost on a farm?

- A. Wages of temporary labor
- B. Cost of diesel fuel
- C. Annual land tax
- D. Cost of nitrogenous fertilizers

- **Correct Answer: C**

• **Explanation:** Annual land tax is a fixed cost that must be paid regardless of the level of output or farm operations. (KPSC AAO 2015)

Q498. The straight-line method of calculating depreciation uses which formula?

- A. $(\text{Original Cost} - \text{Salvage Value}) / \text{Useful Life}$
- B. $(\text{Original Cost} + \text{Salvage Value}) / \text{Useful Life}$
- C. $(\text{Original Cost} * \text{Useful Life}) - \text{Salvage Value}$
- D. $\text{Salvage Value} / \text{Useful Life}$

- **Correct Answer: A**

• **Explanation:** Straight Line Depreciation = $(C - S) / L$, where C is cost, S is salvage value, and L is the useful life of the asset. (KPSC AAO 2017)

Q499. The locus of points of tangency between isoquants and isocost lines represents the:

- A. Ridge line
- B. Isocline
- C. Expansion path
- D. Production frontier

- **Correct Answer: C**

• **Explanation:** The expansion path is the locus of points where the marginal rate of technical substitution (slope of isoquant) equals the input price ratio (slope of isocost). (KPSC AO 2014)

Q500. Under the WTO's Agreement on Agriculture, what box includes subsidies that are deemed to be trade-distorting but are exempted due to development programs for low-income farmers?

- A. Green Box
- B. S&D (Special and Differential Treatment) Box
- C. Blue Box
- D. Amber Box

- **Correct Answer: B**

• **Explanation:** Special and Differential (S&D) treatment box allows developing countries to provide agricultural support for low-income or resource-poor farmers, exempted from reductions. (KPSC AO 2017)

[HIGH PROBABILITY] Top 200 High Probability Questions

Review these high-probability Q&As based on the last 10 years KPSC patterns (KPSC AO 2014, KPSC AO 2017):

Q1. What is the physiological basis of drought tolerance in CAM plants compared to C3 and C4 plants?

- **Correct Answer: CAM (Crassulacean Acid Metabolism) plants tolerate drought by opening their stomata at night (scotoactive opening) to fix CO₂ into malic acid, which (stored in vacuoles; During the day, stomata remain closed to prevent transpiration, and malic acid is decarboxylated to release CO₂ internally for photosynthesis). [KPSC AO 2017]**

Q2. Explain the difference between meteorological drought and agricultural drought.

- **Correct Answer: Meteorological drought** (defined purely by rainfall deficiency, occurring when the seasonal rainfall over an area is less than 75% of its long-term climatological normal; Agricultural drought occurs when soil moisture and rainfall are inadequate during the crop growing season to support healthy crop growth to maturity, causing extreme soil moisture stress and crop wilting). [KPSC AAO 2015]

Q3. Why is the use of chlorine-containing fertilizers like Muriate of Potash (KCl) restricted in tobacco and potato cultivation?

- **Correct Answer: Chlorine ions (Cl⁻) accumulate in the leaves of tobacco and tubers of potato when KCl (used; In tobacco, chlorine reduces the burning quality and fire-holding capacity of the leaf, causing it to burn unevenly). [KPSC AO 2014]**

Q4. Describe the critical growth stage for irrigation in wheat and the consequences of moisture stress at this stage.

- **Correct Answer: Crown Root Initiation (CRI) stage, occurring 21 days after sowing (DAS), (the most critical growth stage for irrigation in wheat; The transition from seminal roots to crown roots occurs during this period). [KPSC AO 2014]**

Q5. How does the nitrification inhibitor 'N-Serve' function in agricultural soils?

- **Correct Answer: N-Serve (2-chloro-6-(trichloromethyl) pyridine) (inhibiting the activity of Nitrosomonas bacteria in the soil; These bacteria are responsible for the first step of nitrification (converting ammonium ions to nitrite)). [KPSC AAO 2019]**

Q6. Describe the diagnostic deficiency symptoms of Zinc in rice (Khaira disease) and its management.

- **Correct Answer: Khaira disease appears in young rice seedlings as chlorosis and bronze/dusty-brown spots on the older leaves; Growth is stunted, and root development is poor. [KPSC AAO 2015]**

Q7. What is the System of Rice Intensification (SRI) and what are its core principles?

- **Correct Answer: SRI (a rice cultivation methodology designed to increase grain yields while using less water and seed; Its core principles are: 1) Transplanting young seedlings (10-12 days old at the 2-leaf stage); 2) Planting a single seedling per hill with wide spacing (25 x 25 cm); 3) Keeping soil moist but not continuously flooded (alternate wetting and drying); and 4) Using a rotary hoe/cono-weeder to control weeds and aerate the soil). [KPSC AO 2019]**

Q8. Explain the role of Potassium in plant stress tolerance and stomatal regulation.

- **Correct Answer: Potassium (K⁺) acts as an osmoticum; It regulates the opening and closing of stomata by actively pumping K⁺ ions into guard cells, which lowers water potential, causing water to enter, guard cells to swell, and the stomata to open. [KPSC AAO 2017]**

Q9. What is the significance of the biuret content in urea, and what are its regulatory limits?

- **Correct Answer: Biuret (NH₂-CO-NH-CO-NH₂) (a toxic byproduct formed during the high-temperature manufacturing of urea; When present in high concentrations, it interferes with protein synthesis and causes phytotoxicity in plants). [KPSC AAO 2015]**

Q10. Differentiate between the mechanism of action of Glyphosate and Paraquat.

- **Correct Answer: Glyphosate (a non-selective, systemic, post-emergence herbicide that inhibits the enzyme EPSP synthase in the shikimic acid pathway, blocking the synthesis of aromatic amino acids (phenylalanine, tyrosine, tryptophan); Paraquat is a non-selective, contact,**

post-emergence herbicide that accepts electrons from Photosystem I to form superoxide radicals, which destroy cell membranes and cause rapid desiccation of foliage within hours). [KPSC AO 2017]

Q11. What is the critical weed-free period for Finger Millet (Ragi) in Karnataka and why is it important?

- **Correct Answer: Critical weed-free period for finger millet** (25 to 35 days after transplanting (DAT) or 30 to 45 days after sowing (DAS) in direct-seeded crops; Finger millet grows slowly in its early stages, making it highly vulnerable to weed competition). [KPSC AAO 2017]

Q12. State the parents of the interspecific cotton hybrid 'DCH-32' and its significance in Karnataka.

- **Correct Answer: DCH-32** (an interspecific hybrid developed by crossing *Gossypium hirsutum* (DS-28) as the female parent and *Gossypium barbadense* (SB-289-E) as the male parent at UAS Dharwad; It is widely cultivated in Karnataka for its extra-long staple length, high spinning potential, and suitability for transitional zones under irrigation). [KPSC AAO 2017]

Q13. Describe the physiological disorder 'poppod' in groundnut, its cause, and control measures.

- **Correct Answer: Poppod** (a physiological disorder in groundnut characterized by empty or unfilled shells (lack of kernels inside the pods); It is caused by calcium deficiency in the pod zone during the pegging and pod development stages). [KPSC AAO 2017]

Q14. How does the reflective type of antitranspirant Kaolin reduce transpiration?

- **Correct Answer: Kaolin** (a clay mineral that, when sprayed as a 2% to 6% aqueous suspension, forms a thin, white, reflective film on the leaf surface; This film increases the leaf's albedo (reflectance of solar radiation), lowering leaf temperature and reducing the vapor pressure gradient between the leaf interior and the atmosphere, thereby decreasing transpiration). [KPSC AO 2017]

Q15. Why is the first week of June historically considered critical for KPSC agricultural weather planning in Karnataka?

- **Correct Answer: First week of June corresponds to Standard Meteorological Week (SMW) 22**, which (the historical period for the onset of the South-West Monsoon over coastal and southern Karnataka; Agricultural planning, including land preparation, nursery sowing for paddy, and direct sowing of kharif crops like ragi and red gram, is timed around this week to utilize monsoon rains). [KPSC AAO 2015]

Q16. What is the equivalent acidity of nitrogenous fertilizers, and list the values for Ammonium Sulfate and Urea.

- **Correct Answer: Equivalent acidity** (the amount of calcium carbonate (CaCO_3) required to neutralize the acidity produced in the soil by applying 100 kg of a particular fertilizer; For Ammonium Sulfate, the equivalent acidity is 110 (requires 110 kg CaCO_3), and for Urea, it is 80 (requires 80 kg CaCO_3)). [KPSC AO 2019]

Q17. Describe the biological control of Water Hyacinth (*Eichhornia crassipes*).

- **Correct Answer: Water Hyacinth** (biologically controlled using two species of host-specific chevron-patterned weevils: *Neochetina eichhorniae* and *Neochetina bruchi*; The adults feed on the leaves, creating scars, while the larvae tunnel through the petioles and crown, causing the plant to rot, lose buoyancy, and sink). [KPSC AAO 2015]

Q18. Explain the concept of 'Surge Irrigation' and its benefits.

- **Correct Answer: Surge irrigation** (the intermittent application of water to furrows or borders in a series of on-off cycles (surges) of predetermined time intervals (e.g., 20 minutes on, 20 minutes off); During the off period, the soil surface seals slightly, reducing infiltration rates). [KPSC AO 2017]

Q19. What are the characteristics of the Northern Dry Zone (Zone 3) of Karnataka?

- **Correct Answer: Northern Dry Zone** (the largest agro-climatic zone in Karnataka; It has a dry semi-arid climate with an annual rainfall of 464 to 785 mm, received primarily during the late monsoon (September-October)). [KPSC AO 2014]

Q20. What is the chemical composition and agricultural utility of Single Super Phosphate (SSP)?

- **Correct Answer: SSP contains 16% phosphorus** (as monocalcium phosphate, P_2O_5 equivalent), 12%

sulfur (as calcium sulfate), and 21% calcium; It is highly valued in oilseed (groundnut, sunflower) and pulse production because it supplies phosphorus for root development, calcium for pod filling, and sulfur for oil and protein synthesis. [KPSC AAO 2017]

Q21. Explain the seed certification classes in India and their identifying tag colors.

- **Correct Answer:** In India, seed certification involves four main classes: 1) Breeder Seed: Produced by the originating breeder, 100% pure, identified by a Golden Yellow tag; 2) Foundation Seed: Progeny of breeder seed, 99.5% pure, identified by a White tag; 3) Registered Seed: Progeny of foundation seed (rare in India), identified by a Purple or Opal Green tag; and 4) Certified Seed: Progeny of foundation seed, sold to farmers, identified by an Azure Blue tag; Nucleus seed is the starting material, has 100% purity, but has no tag. [KPSC AAO 2017]

Q22. Describe the biochemical principle and steps of the Tetrazolium (TZ) test.

- **Correct Answer:** The TZ test determines seed viability; Seeds are pre-conditioned by soaking in water, bisected to expose the embryo, and immersed in a 0.1% to 1.0% aqueous solution of 2,3,5-triphenyl tetrazolium chloride (pH 6.5-7.5) at 30-35°C in the dark. [KPSC AAO 2014]

Q23. Differentiate between orthodox and recalcitrant seeds, providing examples.

- **Correct Answer:** Orthodox seeds can tolerate desiccation down to low moisture contents (5% to 7%) and can be stored at sub-zero temperatures without losing viability (e.g., rice, wheat, ragi, pulses); Recalcitrant seeds lose viability if dried below a high critical moisture threshold (12% to 30%) or if stored at low temperatures, as their embryos are sensitive to desiccation and chilling (e.g., mango, jackfruit, rubber, coconut). [KPSC AAO 2017]

Q24. Define 'isolation distance' and explain its significance in seed production.

- **Correct Answer:** Isolation distance (the minimum physical distance required between a seed production crop field and any neighboring field of the same crop variety (or related varieties/weeds) to prevent genetic contamination through cross-pollination; It also prevents mechanical mixtures during harvest). [KPSC AAO 2015]

Q25. What is seed priming and what are its physiological benefits?

- **Correct Answer:** Seed priming (a pre-sowing hydration treatment where seeds are soaked in water or osmotic solutions (like PEG, KNO₃) to initiate early physiological and metabolic processes (imbibition, enzyme activation) without allowing radicle protrusion; The seeds are then dried back to their original moisture content before sowing). [KPSC AAO 2017]

Q26. State the key provisions of the Seed Act, 1966.

- **Correct Answer:** The Seed Act of 1966 regulates the quality of seeds sold in India; Key provisions include: 1) Establishment of the Central Seed Committee and Central Seed Certification Board; 2) Notification of crop varieties for which seed sales will be regulated; 3) Prescription of minimum limits for germination and physical purity; 4) Compulsory labeling of notified seed lots; 5) Appointment of Seed Inspectors with power to inspect, sample, and seize non-compliant seeds. [KPSC AAO 2019]

Q27. What is the purpose of the Grow Out Test (GOT) and how is it conducted?

- **Correct Answer:** Grow Out Test (GOT) (conducted to verify the genetic purity of a seed lot; A representative sample of the seed lot is sown alongside authentic control seeds in a field). [KPSC AAO 2017]

Q28. How does a Specific Gravity Separator function in seed processing?

- **Correct Answer:** A specific gravity separator separates seeds of the same size based on differences in their density or specific weight; It consists of an oscillating deck through which air is blown from below. [KPSC AAO 2019]

Q29. Explain the electrical conductivity (EC) test for seed vigour.

- **Correct Answer:** EC test (a rapid physiological vigour test; A weighed sample of seeds is soaked in deionized water for 24 hours). [KPSC AAO 2017]

Q30. What are the rules for safe seed storage under Harrington's guidelines?

- **Correct Answer:** Harrington's rules of thumb for orthodox seed storage state: 1) For every 1%

decrease in seed moisture content, the storage life of the seed (doubled (applicable between 5% and 14% moisture); 2) For every 10°F (approx; 5.6°C) decrease in storage temperature, the storage life of the seed is doubled; 3) The sum of the relative humidity (%) of the storage air and the storage temperature in °F should not exceed 100 for safe storage). [KPSC AO 2017]

Q31. Describe the physiological disorder 'Spongy Tissue' in Alphonso mango and its mitigation.

- **Correct Answer: Spongy Tissue** (a physiological disorder affecting the mesocarp (pulp) of Alphonso mangoes, characterized by a sour, yellow, sponge-like patch that develops near the stone; It is caused by convective heat radiating from the soil surface after harvest or during fruit maturity, which inactivates amylase enzymes and causes starch breakdown anomalies). [KPSC AO 2014]

Q32. Explain the role of Paclobutrazol in mango canopy and flower management.

- **Correct Answer: Paclobutrazol (Cultar)** (a synthetic plant growth retardant that acts by inhibiting gibberellin biosynthesis; It is applied to the soil around the collar zone of mango trees 2 to 3 months before expected flowering). [KPSC AO 2017]

Q33. Describe the causes and control of Blossom End Rot (BER) in tomato.

- **Correct Answer: Blossom End Rot (BER)** (a physiological disorder of tomato characterized by water-soaked, sunken, black necrotic lesions at the distal (blossom) end of the fruit; It is caused by a localized calcium deficiency in the developing fruit). [KPSC AO 2017]

Q34. Explain the pruning cycle of commercial grape cultivation in Karnataka.

- **Correct Answer: Karnataka, grapevines** (pruned twice a year: 1) Foundation/Back Pruning (April): Vines are pruned to 1-2 buds to encourage vegetative growth, canopy development, and nutrient accumulation during the monsoon; 2) Fruit/Forward Pruning (October): Shoots are pruned to 5-10 buds (depending on the variety) to induce flowering and fruit set; This timing ensures fruit development and harvesting occur during the dry, warm winter months). [KPSC AO 2017]

Q35. What is the physiological basis of using 1-Methylcyclopropene (1-MCP) in post-harvest technology?

- **Correct Answer: 1-MCP** (a gaseous synthetic compound that has a high affinity for ethylene receptors in plant cells; By binding to these receptors, 1-MCP blocks the action of both endogenous and exogenous ethylene). [KPSC AO 2017]

Q36. Detail the differences between 'Pinching' and 'Disbudding' in commercial flower cultivation.

- **Correct Answer: Pinching** (the removal of the growing apical shoot tip (soft tip) of a plant to break apical dominance, which stimulates the growth of lateral branches and results in a bushier plant with more flowers (e.g., marigold, carnation); Disbudding is the selective removal of lateral flower buds to divert nutrients to a single terminal flower bud, producing a larger bloom (e.g., standard chrysanthemum, carnation)). [KPSC AA0 2019]

Q37. Explain the maturity indices for commercial harvesting of banana.

- **Correct Answer: Maturity indices for banana** include: 1) Disappearance of angularity (fingers become rounded in cross-section); 2) Light green color of the skin (turning from dark green); 3) Drying of the floral ends (style and stigma); 4) Pulp-to-peel ratio (increases from 1.2 to 1.6 as the pulp accumulates starch); and 5) Computation of days from shooting (approx; 90-120 days depending on the variety and season). [KPSC AA0 2015]

Q38. What is the botanical nature of the edible parts of cauliflower and broccoli?

- **Correct Answer: In cauliflower (*Brassica oleracea* var; *botrytis*)**, the edible part is the 'curd', which is a compact, terminal shoot system consisting of thousands of hypertrophied, pre-floral, immature meristems. [KPSC AO 2014 / KPSC AA0 2015]

Q39. Describe the physiological disorder 'Whiptail' in cauliflower.

- **Correct Answer: Whiptail** (a physiological disorder of cauliflower caused by molybdenum deficiency; Molybdenum is a cofactor for nitrate reductase; its absence prevents normal nitrogen assimilation). [KPSC AA0 2015]

Q40. How does Modified Atmosphere Packaging (MAP) extend the shelf life of fresh fruits?

- **Correct Answer: MAP** (enclosing fresh produce in polymeric films with selective gas permeability; The natural respiration of the fruits consumes oxygen (O₂) and produces carbon dioxide (CO₂) inside the package). [KPSC AAO 2017]

Q41. Explain the difference between Soil Texture and Soil Structure. Which one can be modified by tillage?

- **Correct Answer: Soil texture** (the relative proportions of sand, silt, and clay particles in a soil mass; It is an inherent physical property that cannot be changed by normal agronomic practices or tillage).

Q42. What is isomorphous substitution? State its significance in clay mineral chemistry.

- **Correct Answer: Isomorphous substitution** (the replacement of one atom by another of similar size in a crystal lattice without changing the basic crystal structure; For example, Al³⁺ replacing Si⁴⁺ in tetrahedral sheets, or Mg²⁺/Fe²⁺ replacing Al³⁺ in octahedral sheets).

Q43. Define Cation Exchange Capacity (CEC) and write down the descending order of CEC for major clay minerals and organic colloids.

- **Correct Answer: Cation Exchange Capacity (CEC)** (the total quantity of exchangeable cations that a soil colloid can adsorb per unit dry mass, expressed in cmol(p+)/kg; The descending order of CEC is: Organic matter/Humus (150-300) > Vermiculite (100-150) > Montmorillonite (80-100) > Illite (15-40) > Kaolinite (3-15 cmol(p+)/kg) (KPSC AO 2017)).

Q44. How does lime reclaim acidic soils? Write the chemical reaction involved.

- **Correct Answer: Lime (CaCO₃) neutralizes** acidity by reacting with soil water to produce bicarbonate and calcium ions; Calcium ions (Ca²⁺) displace exchangeable H⁺ and Al³⁺ from the soil colloid surface.

Q45. Distinguish between Saline and Sodic soils based on EC, pH, ESP, and SAR.

- **Correct Answer:** `**1. Saline Soils: Electrical Conductivity (EC) > 4 dS/m; pH < 8.5; Exchangeable Sodium Percentage (ESP) < 15%; Sodium Adsorption Ratio (SAR) < 13.

2. Sodic (Alkali) Soils: EC < 4 dS/m; pH > 8.5; ESP > 15%; SAR > 13 (KPSC AO 2017).**.`

Q46. What is the principle of the Walkley and Black method for soil organic carbon estimation? Why is a correction factor used?

- **Correct Answer: Principle** (wet oxidation of organic carbon by potassium dichromate (K₂Cr₂O₇) in the presence of concentrated H₂SO₄; The heat of dilution of H₂SO₄ is used for the reaction).

Q47. How is available phosphorus estimated in acidic soils versus alkaline soils? Name the reagents.

- **Correct Answer:** `**1. Acidic Soils: Estimated using Bray's No. 1 method. Reagent: 0.03 M NH₄F + 0.025 M HCl. Fluoride ions complex Fe/Al, releasing phosphate.

2. Alkaline Soils: Estimated using Olsen's method. Reagent: 0.5 M NaHCO₃ (pH 8.5). Bicarbonate precipitates Ca as CaCO₃, releasing phosphate (KPSC AO 2017).**.`

Q48. What are the soil moisture constants at Field Capacity, Permanent Wilting Point, and Hygroscopic Coefficient? Provide their suction pressures.

- **Correct Answer:** `**1. Field Capacity: -0.33 bar (-33 kPa or pF 2.5).

2. Permanent Wilting Point: -15 bars (-1.5 MPa or pF 4.2).

3. Hygroscopic Coefficient: -31 bars (-3.1 MPa or pF 4.5) (KPSC AAO 2015).**.`

Q49. Explain the role of Leghemoglobin in root nodules of leguminous plants.

- **Correct Answer: Leghemoglobin** (a pink iron-containing pigment found in the cytoplasm of infected nodule cells; Its function is to regulate oxygen supply).

Q50. Define bulk density and particle density. Why is bulk density generally lower in clayey soils than sandy soils?

- **Correct Answer: Bulk density (BD)** (the mass of dry soil divided by its total volume (solids + pores); Particle density (PD) is the mass of dry soil divided by the volume of solid particles only).

Q51. List the diagnostic criteria for identifying a Vertisol in Soil Taxonomy.

- **Correct Answer: Soil** (classified as a Vertisol if it has: 1; A layer 25 cm or more thick with slickensides or wedge-shaped aggregates).

Q52. What is the equivalent acidity of a fertilizer? Which nitrogenous fertilizer has the highest equivalent acidity?

- **Correct Answer: Equivalent acidity** (the amount of calcium carbonate (CaCO₃) required to neutralize the acidic residue left in the soil by the application of 100 kg of a fertilizer; Among common solid nitrogenous fertilizers, Ammonium chloride has the highest equivalent acidity of 128, followed by Ammonium sulphate at 110 (KPSC AO 2017)).

Q53. What are the common visual symptoms of Nitrogen and Potassium deficiency in plants?

- **Correct Answer:** 1. Nitrogen deficiency: Uniform yellowing (chlorosis) of older leaves first, progressing upwards. V-shaped chlorosis starting from leaf tip along the midrib in cereals.

2. Potassium deficiency: Marginal chlorosis and scorching/burning of tips and margins of older leaves, making them appear ragged (KPSC AAO 2017).

Q54. Explain the difference between Active, Exchangeable, and Reserve acidity in soils.

- **Correct Answer:** 1. Active Acidity: Due to H⁺ and Al³⁺ ions in the soil solution. It is small in magnitude but determines soil pH.

2. Exchangeable Acidity: Due to exchangeable H⁺ and Al³⁺ ions on soil colloids, displaceable by neutral salts.

3. Reserve (Residual) Acidity: Bound H⁺ and Al³⁺ ions on clay and organic matter that cannot be displaced easily. It is the largest fraction (KPSC AO 2017).

Q55. What is the critical limit of DTPA-extractable Zinc and Iron in agricultural soils?

- **Correct Answer: Critical limit for DTPA-extractable Zinc (Zn)** (0.6 mg/kg (or ppm) of soil; The critical limit for DTPA-extractable Iron (Fe) is 4.5 mg/kg (or ppm) of soil).

Q56. Explain the term 'N-fixation' and name two free-living, two symbiotic, and two associative nitrogen-fixing microorganisms.

- **Correct Answer: N-fixation** (the enzymatic conversion of atmospheric gaseous nitrogen (N₂) into ammonia (NH₃) by diazotrophic microorganisms; 1).

Q57. What is the role of mycorrhizae in plant nutrition? How do ectomycorrhizae and endomycorrhizae differ structurally?

- **Correct Answer: Mycorrhizae form a** symbiotic association with plant roots, expanding the effective surface area for nutrient and water absorption, particularly for immobile elements like Phosphorus and Zinc; Structurally, Ectomycorrhizae form an external mantle (sheath) around root tips and an intercellular Hartig net without penetrating cortical cell walls.

Q58. State Stokes' Law equation and explain its parameters. What are its major assumptions in soil mechanical analysis?

- **Correct Answer: Equation:** $v = \frac{2 \cdot g \cdot r^2 \cdot (dp - dl)}{9 \cdot n}$, where v = settling velocity, g = acceleration due to gravity, r = particle radius, dp = particle density, dl = liquid density, n = viscosity of liquid; Key assumptions: 1.

Q59. What is the Bemmelen factor? Explain its use.

- **Correct Answer: Bemmelen factor** (1.724; It is used to convert estimated soil organic carbon (OC) content to total soil organic matter (SOM) based on the assumption that soil organic matter contains approximately 58% carbon by weight (SOM = OC * 100/58 = OC * 1.724) (KPSC AAO 2017)).

Q60. How does the Schoonover method calculate the Gypsum Requirement (GR) of a sodic soil?

- **Correct Answer: Schoonover's method involves** shaking a soil sample with saturated gypsum (calcium sulphate) solution; The calcium ions displace exchangeable sodium from the soil colloid.

Q61. Describe the steps and organisms involved in the biological Nitrification process.

- **Correct Answer:** **Nitrification** (a two-step aerobic process carried out by chemoautotrophic bacteria:

1; Ammonium oxidation: Ammonium (NH₄⁺) is oxidized to Nitrite (NO₂⁻) by Nitrosomonas species (2NH₄⁺ + 3O₂ → 2NO₂⁻ + 4H⁺ + 2H₂O)).

Q62. What is biological Denitrification? Under what conditions does it occur, and what are the gaseous end products?

- **Correct Answer: Denitrification** (the microbial reduction of nitrate (NO₃⁻) to gaseous nitrogen forms under anaerobic or waterlogged soil conditions; Facultative anaerobic bacteria (e.g., *Pseudomonas denitrificans*, *Paracoccus*) use nitrate as an electron acceptor).

Q63. Explain the difference between nitrogen Mineralization and Immobilization. How does the C:N ratio of crop residues affect these processes?

- **Correct Answer: Mineralization** (the conversion of organic nitrogen compounds (proteins, amino acids) into inorganic nitrogen (NH₄⁺, NO₃⁻) by microbes; Immobilization is the reverse process, where microbes uptake inorganic nitrogen to build their cell proteins).

Q64. What are Phosphate Solubilizing Microorganisms (PSMs)? How do they solubilize inorganic phosphates in soil?

- **Correct Answer: PSMs** (soil bacteria (e.g., *Bacillus megaterium*, *Pseudomonas putida*) and fungi (e.g., *Aspergillus niger*) that convert insoluble mineral phosphates (like tricalcium phosphate) into plant-available orthophosphate forms; They do this primarily by producing organic acids (gluconic, citric, oxalic acids) that lower soil pH and chelate calcium, iron, or aluminum, releasing phosphate (KPSC AO 2017)).

Q65. Explain the significance of Glomalin in agricultural soils.

- **Correct Answer: Glomalin** (a highly stable, hydrophobic glycoprotein secreted by the hyphae and spores of Arbuscular Mycorrhizal (AM) fungi; It acts as an organic 'glue' that binds soil mineral particles and organic matter together to form stable water-resistant macroaggregates).

Q66. What is the biological role of the nitrogenase enzyme complex? Why is it sensitive to oxygen?

- **Correct Answer: The nitrogenase complex** catalyzes the reduction of atmospheric nitrogen gas to ammonia (N₂ + 8H⁺ + 8e⁻ + 16ATP → 2NH₃ + H₂ + 16ADP + 16Pi); It consists of two proteins: Fe-protein and MoFe-protein.

Q67. List the quality control standards specified by BIS/FCO for commercial carrier-based Rhizobium biofertilizers.

- **Correct Answer: Standards include:** 1. No contamination at 10⁻⁵ dilution. 2. Minimum viable count of 10⁸ cells/g of carrier on dry basis at manufacture, and 10⁷ cells/g at expiry. 3. Carrier pH must be between 6.0 and 7.5. 4. Minimum efficiency showing nodulation on host plant. 5. Carrier particle size must pass through a 75-100 mesh sieve (KPSC AAO 2017)..

Q68. What is the silage-making process? Explain the role of lactic acid bacteria in silage preservation.

- **Correct Answer: Silage making involves** preserving high-moisture green forage under anaerobic conditions; Lactic acid bacteria (e.g., *Lactobacillus plantarum*, *Enterococcus faecalis*) multiply anaerobically and ferment water-soluble carbohydrates (sugars) in the forage into lactic acid.

Q69. How do Siderophores act as biocontrol agents against soil-borne plant pathogens?

- **Correct Answer: Siderophores** (low-molecular-weight, high-affinity iron-chelating ligands produced by rhizosphere bacteria like *Pseudomonas fluorescens*; Under iron-limiting conditions, these bacteria secrete siderophores to bind ferric iron (Fe³⁺) and uptake it).

Q70. What are Nod factors and flavonoids? How do they coordinate the Rhizobium-legume symbiosis?

- **Correct Answer: Flavonoids** (chemical signaling compounds secreted by the roots of host legumes under nitrogen-limiting conditions; Rhizobium bacteria in the rhizosphere detect these flavonoids, which trigger the transcription of bacterial nodulation (nod) genes).

Q71. Differentiate between Back Sight (BS), Fore Sight (FS), and Intermediate Sight (IS) in surveying and leveling.

- **Correct Answer:** 1. Back Sight (BS): The first staff reading taken after setting up the level on a point of known elevation (like a Benchmark). Used to establish Height of Instrument (HI).

2. Fore Sight (FS): The last staff reading taken on a point before shifting the instrument or ending operations. Used to find the elevation of a new point or change point.

3. Intermediate Sight (IS): Any staff reading taken between the BS and FS from a single instrument setup (KPSC AO 2017).

Q72. Write down the Universal Soil Loss Equation (USLE) and define each factor.

- **Correct Answer:** **USLE** ($A = R * K * LS * C * P$; - A: Computed average annual soil loss per unit area).

Q73. Explain the three modes of wind erosion: Saltation, Suspension, and Surface Creep.

- **Correct Answer:** 1. Saltation (50-75% of erosion): Bouncing of medium-sized soil particles (0.1 to 0.5 mm) along the ground, caused by wind pressure.

2. Suspension (3-15% of erosion): Fine particles (<0.1 mm) are lifted high into the air and carried long distances.

3. Surface Creep (5-25% of erosion): Rolling or sliding of larger particles (0.5 to 2.0 mm) along the soil surface by wind and impact of saltating particles (KPSC AO 2017).

Q74. Compare the performance of a two-stroke engine and a four-stroke engine based on power strokes and thermal efficiency.

- **Correct Answer:** A two-stroke engine completes one power stroke in every one revolution (360°) of the crankshaft; A four-stroke engine completes one power stroke in every two revolutions (720°) of the crankshaft.

Q75. What is engine calibration? Define Indicated Horse Power (IHP), Brake Horse Power (BHP), and Mechanical Efficiency.

- **Correct Answer:** 1. IHP: Theoretical power developed inside the engine cylinder by gas expansion.

2. BHP: The actual useful power available at the engine crankshaft, measured using a dynamometer or brake test.

3. Mechanical Efficiency: The ratio of BHP to IHP, expressed as a percentage: Mech Eff (%) = $(BHP / IHP) * 100$. (Typically 80-90%) (KPSC AO 2017).

Q76. List the four main components of a mouldboard plough bottom and describe their functions.

- **Correct Answer:** 1. Share: The sharp edge that makes the horizontal cut of the furrow slice.

2. Mouldboard: The curved plate that lifts, turns, pulverizes, and inverts the cut furrow slice.

3. Landside: The flat plate that presses against the furrow wall, counteracting lateral thrust and stabilizing the plough.

4. Frog: The central metal casting to which the share, mouldboard, and landside are bolted (KPSC AAO 2015).

Q77. Explain the terms 'Disc Angle' and 'Tilt Angle' in a disc plough. What are their typical values?

- **Correct Answer:** 1. Disc Angle: The angle between the plane of the cutting edge of the disc and the direction of travel. It affects the width of cut and penetration. Typical value: 42° to 45°.

2. Tilt Angle: The angle of inclination of the disc plane from a vertical line. It affects the depth of cut and soil turning. Typical value: 15° to 25° (KPSC AAO 2017).

Q78. Explain how a seed drill is calibrated. Why is calibration necessary?

- **Correct Answer:** **Calibration** (the process of adjusting the seed metering mechanism to deliver the desired seed rate (kg/ha) before planting; Step 1: Measure width (W) and wheel diameter (D)).

Q79. What is the difference between a knapsack sprayer and a compression sprayer?

- **Correct Answer:** **Knapsack sprayer** (worn on the back and has a built-in hydraulic pump operated continuously by a hand lever during spraying, maintaining constant pressure (3-5 kg/cm²); A compression (pneumatic) sprayer requires pumping air into the sealed tank before spraying to build pressure).

Q80. Describe the main components and water efficiency of Drip Irrigation.

- **Correct Answer:** **Components include:** Control head (pump, valves, fertilizer injector/venturi), filters (sand/media, screen, disc), mains, sub-mains, laterals, and emitters/drippers; Drip irrigation has a high water application efficiency of 90-95% because water is applied slowly (1-8 liters/hr) and directly to the root zone, minimizing evaporation, wind drift, and deep percolation losses (KPSC AO 2017).

Q81. What is the genetic basis of complementary gene interaction (9:7)?

- **Correct Answer: Complementary gene interaction occurs when two non-allelic genes** (both required to produce a single wild-type phenotype; In the F₂ generation, this results in a phenotypic ratio of 9:7, where individuals with at least one dominant allele at each locus (A_B_) express the trait, while individuals that are homozygous recessive at either locus (A_bb, aaB_, or aabb) show the alternative/mutant phenotype).

Q82. What is the difference between gametophytic and sporophytic self-incompatibility?

- **Correct Answer: Gametophytic Self-Incompatibility (GSI), the incompatibility reaction** (determined by the haploid S-allele genotype of the pollen grain itself, and pollen tube growth is inhibited in the style; In Sporophytic Self-Incompatibility (SSI), the reaction is determined by the diploid S-allele genotype of the parent plant that produced the pollen, and pollen germination or stigma penetration is inhibited at the stigma surface).

Q83. Explain the role of the 'A', 'B', and 'R' lines in CGMS hybrid seed production.

- **Correct Answer: 'A' line** (the male-sterile line (used as the female parent for hybrid seed production); the 'B' line is the maintainer line (isogenic to the A-line but with fertile cytoplasm, used to maintain and multiply A-line seed by crossing A x B); and the 'R' line is the restorer line (possesses dominant nuclear restorer genes, used as the male parent to cross with the A-line to produce fertile F₁ hybrid seed for farmers); (KPSC AA0 2017)).

Q84. What is the cytological difference between paracentric and pericentric inversions?

- **Correct Answer: A paracentric inversion** does not include the centromere within the inverted chromosomal segment, and heterozygotes form a dicentric bridge and an acentric fragment during meiotic anaphase I; A pericentric inversion includes the centromere, which alters the relative arm lengths (arm ratio) of the chromosome, and crossing over does not result in a bridge or fragment, but leads to duplicate/deficient chromatids.

Q85. Explain the dominance vs. overdominance hypotheses of heterosis.

- **Correct Answer: The dominance hypothesis** (Davenport, Bruce) attributes heterosis to the masking of harmful recessive alleles by favorable dominant alleles accumulated from both parents in the F₁ hybrid; The overdominance hypothesis (Shull, East) suggests that the heterozygote (Aa) is superior to either homozygote (AA or aa) due to a greater physiological buffering capacity or complementary functions of divergent alleles.

Q86. What is the Brassica U's Triangle and its chromosomal implications?

- **Correct Answer: U's Triangle (Woo Jang-choon, 1935)** shows the relationship between three basic diploid Brassica species: B; nigra (n=8, B genome), B.

Q87. Why is narrow-sense heritability more useful to plant breeders than broad-sense heritability?

- **Correct Answer: Narrow-sense heritability ($h^2 = V_a / V_p$)** represents the proportion of phenotypic variance due only to additive genetic variance (V_a); Unlike dominance or epistatic variance, additive genetic variance is directly transmissible and fixable in self-pollinated crops, making narrow-sense heritability the key predictor of response to selection (genetic advance).

Q88. What is the difference between vertical and horizontal disease resistance in crops?

- **Correct Answer: Vertical resistance** (pathogen race-specific, controlled by one or few oligogenes (major genes), and typically operates on a gene-for-gene basis, providing complete but unstable resistance; Horizontal resistance is race-non-specific, controlled by polygenes (minor genes), and is durable, providing stable partial protection in the field by slowing down epidemic development).

Q89. Explain the gene-for-gene hypothesis proposed by H.H. Flor.

- **Correct Answer: Proposed by H.H. Flor in 1942 based on flax rust (Melampsora lini), the hypothesis states that for each resistance gene (R) in the host plant, there is a complementary avirulence gene (Avr) in the pathogen. Disease resistance occurs only when the host carries the dominant R gene and the pathogen carries the corresponding dominant Avr gene; all other combinations result in disease. (KPSC AO 2017).**

Q90. What are the main steps in the pedigree method of plant breeding?

- **Correct Answer: The pedigree method** starts by crossing parents to produce F₁ seeds; F₁ plants are grown to produce F₂.

Q91. Explain the single seed descent (SSD) method and its advantage.

- **Correct Answer: The SSD method** involves advancing segregating generations (F₂ to F₅/F₆) by harvesting exactly one seed from each plant in the population and mixing them to grow the next generation; It maintains the pedigree population with minimal selection pressure, requiring very little space, and allows rapid advancement of generations (typically 2-3 generations per year in greenhouses) to achieve homozygosity before selection.

Q92. Why is the backcross method highly effective for transferring oligogenic traits?

- **Correct Answer: Backcross method** (highly effective because it repeatedly crosses the hybrid progeny with the commercial variety (recurrent parent), recovering ~99% of its genome after 6-8 backcrosses; It allows the transfer of a single oligogenic trait (such as disease resistance) from a donor parent while restoring the yield and quality characteristics of the elite recurrent parent).

Q93. What is the role of colchicine in polyploidy breeding and how does it act?

- **Correct Answer: Colchicine** (an alkaloid isolated from *Colchicum autumnale* used to induce polyploidy; It binds to tubulin heterodimers, preventing their assembly into microtubules, which completely blocks spindle fiber formation during mitosis).

Q94. How does triploid seedless watermelon occur, and why is it sterile?

- **Correct Answer: Triploid watermelon (3x=33)** (produced by crossing a tetraploid (4x) female parent with a diploid (2x) male parent; It is seedless and sterile because the three sets of homologous chromosomes cannot pair properly during meiotic prophase I, leading to unequal chromosome distribution (aneuploidy) in the gametes).

Q95. Define inbreeding depression and its intensity in self- vs. cross-pollinated crops.

- **Correct Answer: Inbreeding depression** (the reduction in fitness, vigor, and yield resulting from inbreeding (selfing); It is highly severe in cross-pollinated crops (e.g., maize, carrot, alfalfa) because they carry many deleterious recessive alleles in a heterozygous state).

Q96. What is the difference between a synthetic variety and a composite variety?

- **Correct Answer: Synthetic variety** (developed by intercrossing a set of inbred lines or clones that have been tested and shown to possess high general combining ability (GCA); A composite variety is developed by mixing seeds of several phenotypically outstanding open-pollinated lines and allowing them to intercross, without testing them for combining ability).

Q97. Explain the genetic inheritance of cytoplasmic male sterility (CMS).

- **Correct Answer: CMS** (governed by mitochondrial genes located in the cytoplasm; Because cytoplasm is contributed almost exclusively by the female parent via the egg cell, CMS is inherited maternally).

Q98. What is the genetic basis of polygenic inheritance, and what classic example is used?

- **Correct Answer: Polygenic inheritance (quantitative inheritance)** (governed by several genes (polygenes) that act in an additive or non-additive manner, where each gene has a small, cumulative effect on the trait; The phenotypic expression shows a continuous range of variation and is highly influenced by the environment).

Q99. Explain transgressive segregation and its utility in breeding.

- **Correct Answer: Transgressive segregation** (the appearance of individuals in the F₂ or subsequent generations that exhibit a phenotype that falls outside the range of both parents; It is caused by the recombination of complementary alleles from both parents, allowing breeders to select transgressive segregants that are superior in traits like height, yield, or stress tolerance to either parent).

Q100. What are the cytological characteristics of B-chromosomes?

- **Correct Answer: B-chromosomes (supernumerary or extra chromosomes)** (non-essential, dispensable chromosomes found in some plant populations (e.g., maize, rye); Cytologically, they are smaller than normal (A) chromosomes, highly heterochromatic, replicate late in the S-phase, and do not synapse or pair with A-chromosomes during meiotic division).

Q101. Explain cellular totipotency and who first conceptualized it.

- **Correct Answer: Cellular totipotency** (the inherent capacity of a single plant cell to divide, differentiate, and regenerate into a complete, fertile plant when provided with appropriate nutrient media and environmental conditions; This concept was first proposed by Gottlieb Haberlandt in 1902).

Q102. What is the physiological role of the auxin-to-cytokinin ratio in tissue culture?

- **Correct Answer: The auxin-to-cytokinin ratio** regulates morphogenesis in tissue culture; A high auxin to low cytokinin ratio promotes root differentiation (rhizogenesis).

Q103. Why is meristem culture used to produce virus-free plants?

- **Correct Answer: Meristem culture (using the apical dome)** (used to eliminate viruses because the meristem is devoid of vascular tissues (xylem and phloem) through which viruses migrate; Additionally, active division of cells in the meristem dome outpaces viral replication, and high endogenous auxin concentrations in the shoot tip inhibit viral multiplication).

Q104. What are doubled haploids, and what is their advantage in plant breeding?

- **Correct Answer: Doubled haploids (dihaploids)** (diploid plants produced by doubling the chromosomes of haploid plants (obtained from anther/pollen culture) using colchicine; Their primary advantage is that they achieve complete, 100% homozygosity at all loci in a single generation, saving 6-8 years of self-pollination).

Q105. Explain somatic hybridization and the role of chemical fusogens like PEG.

- **Correct Answer: Somatic hybridization** (the fusion of isolated protoplasts from two distinct plant species to produce hybrid cells, bypassing sexual barriers; Chemical fusogens like Polyethylene Glycol (PEG) alter the surface charge of protoplasts and dehydrate the plasma membrane, bringing them into close contact and inducing membrane fusion).

Q106. What are cybrids and how are they useful in crop improvement?

- **Correct Answer: Cybrids (cytoplasmic hybrids)** (cells containing the nuclear genome of only one parent and the cytoplasmic organelles (mitochondria or chloroplasts) from both or just the other parent; They are useful for transferring organelle-encoded traits, like cytoplasmic male sterility (CMS) or herbicide resistance, from a donor parent without altering the elite nuclear background).

Q107. Define somaclonal variation and its primary causes.

- **Correct Answer: Somaclonal variation** (the genetic, chromosomal, or epigenetic variation observed in plants regenerated from in vitro tissue culture; Its primary causes include chromosomal rearrangements (translocations, deletions), changes in ploidy level, point mutations, gene amplification, transposon activation, and alterations in DNA methylation patterns induced by tissue culture stress and high concentrations of growth regulators).

Q108. How do Type II restriction endonucleases differ from Type I and Type III?

- **Correct Answer: Type II restriction** endonucleases recognize specific palindromic sequences and cleave double-stranded DNA at precise sites within or immediately adjacent to their recognition sequence; Unlike Type I and Type III enzymes, they do not require ATP, only magnesium ions (Mg^{2+}), and do not cut DNA at random, non-specific locations, making them essential for recombinant DNA cloning.

Q109. What is the function of alkaline phosphatase in recombinant DNA cloning?

- **Correct Answer: Alkaline phosphatase catalyzes** the removal of the 5'-phosphate group from the ends of linearized double-stranded DNA vectors; This prevents the vector from self-ligating (recircularizing) in the presence of DNA ligase, forcing the insertion of the target foreign DNA fragment, which retains its 5'-phosphate groups.

Q110. Explain how *Agrobacterium tumefaciens* acts as a natural genetic engineer.

- **Correct Answer: *Agrobacterium tumefaciens*** (a soil bacterium that naturally transfers a specific segment of its Ti (tumor-inducing) plasmid, called the T-DNA, into the plant cell nucleus; The T-DNA integrates into the plant chromosome and expresses genes that direct the plant to synthesize opines (nutrients for the bacteria) and hormones (causing crown gall

tumors)).

Q111. What is the role of acetosyringone in Agrobacterium-mediated gene transfer?

- **Correct Answer: Acetosyringone** (a phenolic compound released by wounded plant cells; It acts as a chemical signal that binds to the VirA protein on the Agrobacterium membrane, activating a signal transduction cascade that induces the transcription of virulence (vir) genes on the Ti plasmid).

Q112. Describe the binary vector system used in plant transformation.

- **Correct Answer: The binary vector** system uses two plasmids in a single Agrobacterium cell: 1) a helper plasmid that contains the virulence (vir) genes but lacks the T-DNA region, and 2) a cloning vector (disarmed Ti plasmid) that contains the T-DNA border sequences enclosing the gene of interest and selectable markers; The vir proteins synthesized from the helper plasmid act in trans to transfer the T-DNA from the cloning vector into the host genome.

Q113. How does the gene gun (particle bombardment) method transfer genes into plants?

- **Correct Answer: The gene gun** (biolistics) method uses high-pressure helium gas to accelerate heavy microparticles (gold or tungsten, 1-2 micrometers in diameter) coated with DNA; These microparticles penetrate the plant cell wall and membrane, delivering the DNA directly into the cytoplasm or nucleus, where it can integrate into the host genome, bypassing biological host range limitations.

Q114. Contrast co-dominant markers with dominant molecular markers.

- **Correct Answer: Co-dominant markers** (e.g., RFLP, SSR, SNP) can distinguish between homozygous dominant, homozygous recessive, and heterozygous individuals because they detect different alleles as distinct bands or signals; Dominant markers (e.g., RAPD, AFLP) cannot distinguish heterozygotes from homozygous dominant individuals, showing only the presence or absence of a single band.

Q115. Explain the role of foreground and background selection in marker-assisted backcrossing (MABC).

- **Correct Answer: In MABC, foreground** selection uses molecular markers tightly linked to the target gene (e.g., disease resistance) to confirm its presence in backcross generations; Background selection uses genome-wide markers to select individuals that show the highest recovery of the recurrent parent's genome, reducing the number of backcrosses required to restore the elite genetic background from 6-8 to 3-4 generations.

Q116. What is gene pyramiding, and how is it used in breeding for disease resistance?

- **Correct Answer: Gene pyramiding** (the assembly of multiple genes governing a single trait (e.g., different resistance genes against rice blast) into a single elite variety using marker-assisted selection; This prevents pathogens from easily overcoming resistance, providing broad-spectrum and highly durable protection in the field).

Q117. What is the mode of action of Bt endotoxins in transgenic crops?

- **Correct Answer: Transgenic Bt crops** express crystalline delta-endotoxins (Cry proteins); When ingested by susceptible insect larvae, the alkaline environment of the midgut solubilizes the toxin.

Q118. How was Golden Rice engineered to produce beta-carotene in the endosperm?

- **Correct Answer: Golden Rice** (engineered by introducing three genes into the endosperm pathway to convert geranylgeranyl diphosphate (GGPP) to beta-carotene: the phytoene synthase (psy) gene from daffodil, and the phytoene desaturase/carotene desaturase (crtI) gene from the soil bacterium *Erwinia uredovora*; A third gene, lycopene beta-cyclase (lcy), was also evaluated but found to be already present in endosperm tissues).

Q119. Explain the mechanism of glyphosate resistance in Roundup Ready crops.

- **Correct Answer: Glyphosate** (a herbicide that binds and inhibits the plant enzyme 5-enolpyruvylshikimate-3-phosphate synthase (EPSPS), which blocks the synthesis of aromatic amino acids (phenylalanine, tyrosine, tryptophan); Roundup Ready crops are engineered with the cp4 epsps gene from *Agrobacterium* sp).

Q120. What is the mechanism of CRISPR-Cas9 genome editing?

- **Correct Answer: CRISPR-Cas9 genome editing** uses a single guide RNA (sgRNA) to guide the Cas9 endonuclease to a specific genomic target sequence located adjacent to a Protospacer Adjacent Motif (PAM); Cas9 introduces a double-strand break (DSB) at the target site.

Q121. Explain the physiological basis of Bordeaux mixture's fungicidal action.

- **Correct Answer: Bordeaux mixture** (a combination of copper sulphate (CuSO₄) and calcium hydroxide (lime); When dissolved in water, it forms a gelatinous precipitate of basic copper sulphates).

Q122. Differentiate between systemic acquired resistance (SAR) and induced systemic resistance (ISR).

- **Correct Answer: SAR** (triggered by necrotizing pathogens or chemical elicitors (like salicylic acid) and is mediated by the salicylic acid (SA) signaling pathway, leading to the accumulation of Pathogenesis-Related (PR) proteins (e.g., chitinases, beta-1,3-glucanases); ISR is initiated by colonization of roots by beneficial plant growth-promoting rhizobacteria (PGPR) (e.g., *Pseudomonas fluorescens*) and is mediated by jasmonic acid (JA) and ethylene (ET) signaling pathways, without the accumulation of PR proteins (KPSC AAO 2017)).

Q123. What are phytoalexins and how do they function in plant defense?

- **Correct Answer: Phytoalexins** (low-molecular-weight, antimicrobial secondary metabolites synthesized de novo by plants in response to physical injury, chemical injury, or pathogen infection; Examples include pisatin in peas, phaseollin in French beans, and rishitin in potatoes).

Q124. What is the 'Kresek' stage of bacterial leaf blight (BLB) of rice and how does it occur?

- **Correct Answer: 'Kresek'** (the acute, systemic wilt phase of BLB caused by *Xanthomonas oryzae* pv; *oryzae*).

Q125. Describe the life cycle and host alteration of *Puccinia graminis tritici*.

- **Correct Answer: *Puccinia graminis tritici*** (a heteroecious rust requiring two hosts to complete its life cycle: wheat (primary host) and barberry (*Berberis vulgaris*, alternate host); It is macrocyclic, producing five distinct spore stages: Stage 0 (Pycniospores/Spermatia) and Stage I (Aeciospores) on barberry leaves; Stage II (Urediniospores - repeating dikaryotic stage) and Stage III (Teliospores - overwintering diploid stage) on wheat stems; and Stage IV (Basidiospores - haploid stage) produced by germination of teliospores in soil, which then infect barberry (KPSC AO 2017)).

Q126. What is the difference between hyperplasia and hypertrophy? Cite a plant disease example.

- **Correct Answer: Hyperplasia** (the abnormal increase in the number of cells due to accelerated cell division, whereas hypertrophy is the abnormal increase in the size of cells; Both phenomena are stimulated by pathogen-induced hormonal imbalances (specifically auxins and cytokinins)).

Q127. What is the causal agent and vector of little leaf of brinjal, and how is it managed?

- **Correct Answer: Little leaf of brinjal** (a wall-less, phloem-restricted prokaryote called phytoplasma; It is transmitted in nature by the leafhopper vector *Hishimonus phycitis*).

Q128. Explain the principle of hot water treatment and solar heat treatment of wheat seeds for controlling loose smut.

- **Correct Answer: Loose smut of wheat (*Ustilago nuda tritici*)** (an internally seed-borne pathogen dormant inside the embryo; 1) Hot water treatment (Jensen): Seeds are soaked in cold water (20-30°C) for 4-5 hours to activate the dormant mycelium, then immersed in hot water at 54°C for exactly 10 minutes to kill the fungus without damaging seed viability).

Q129. What are viroids and how do they differ from viruses?

- **Correct Answer: Viroids** (the smallest known infectious pathogens, consisting of a short, circular, single-stranded RNA molecule without a protein coat (capsid) or envelope; In contrast, viruses consist of a nucleic acid core (DNA or RNA) enclosed in a protein capsid).

Q130. Describe the symptoms of Red rot of sugarcane and name its causal organism.

- **Correct Answer: Red rot of sugarcane** (the fungus *Colletotrichum falcatum*; Key symptoms

include: 1) Yellowing and shriveling of third and fourth leaves, 2) Red discoloration of the internal pith of the stalk when split longitudinally, with diagnostic white transverse bands, and 3) An acidic, alcoholic, sour odor emitted from the split stems due to fermentation (KPSC AO 2017)).

Q131. How do Benzimidazole fungicides like Carbendazim control fungal pathogens?

- **Correct Answer: Benzimidazole fungicides (e.g., Carbendazim, Benomyl)** (systemic fungicides that act as mitotic inhibitors; They bind specifically to the monomeric beta-tubulin protein of the fungus, preventing its polymerization into microtubules).

Q132. Explain the significance of the 'ooze test' in plant disease diagnosis.

- **Correct Answer: Ooze test** (a rapid diagnostic tool used to differentiate bacterial wilt (caused by *Ralstonia solanacearum*) from fungal wilts (caused by *Fusarium* or *Verticillium*) in Solanaceous crops; A small section of the infected stem is cut and suspended in a glass of clean water).

Q133. What is Sandal Spike disease? State its causal agent, vector, and diagnostic symptoms.

- **Correct Answer: Sandal Spike disease** (a destructive disease of Sandalwood (*Santalum album*) caused by phytoplasma and transmitted by the leafhopper vector *Coelidia indica*; Diagnostic symptoms include extreme reduction in leaf size, shortening of internodes, leaves becoming stiff, erect, and pointed like spikes, and complete suppression of reproductive organs, eventually leading to the death of the tree in 2-3 years (KPSC AO 2017)).

Q134. Detail the morphological differences between early and late leaf spots of groundnut (Tikka disease).

- **Correct Answer: 1) Early Leaf Spot (*Cercospora arachidicola*): Spots** (circular, light brown to dark brown, and are surrounded by a prominent, bright yellow chlorotic halo; Conidia are formed mostly on the upper leaf surface).

Q135. What are fastidious vascular bacteria (FVBs) and how do they differ from normal plant-pathogenic bacteria?

- **Correct Answer: FVBs (formerly rickettsia-like organisms)** (specialized Gram-negative bacteria that reside exclusively in the xylem or phloem of plants and are transmitted by sap-feeding insect vectors; Unlike normal plant-pathogenic bacteria (like *Xanthomonas* or *Pseudomonas*), FVBs are difficult or impossible to culture on standard synthetic agar media because they have complex nutritional requirements (hence 'fastidious')).

Q136. Define EIL and ETL, and explain their relationship in IPM decision making.

- **Correct Answer: Economic Injury Level (EIL)** (the lowest pest population density that will cause economic damage (where cost of control equals the value of yield saved); Formula: $EIL = C / (V * I * D)$).

Q137. Explain the mechanism of action of Organophosphate and Carbamate insecticides.

- **Correct Answer: Both Organophosphates (OPs, e.g., Chlorpyrifos) and Carbamates (e.g., Carbofuran)** target the nervous system by inhibiting the enzyme acetylcholinesterase (AChE) at synaptic junctions; AChE normally breaks down the neurotransmitter acetylcholine (ACh) to terminate the nerve impulse.

Q138. What is the biological control of Sugarcane Pyrilla (*Pyrilla perpusilla*) using *Epiricania melanoleuca*?

- **Correct Answer: Sugarcane Pyrilla** (a major sucking pest controlled biologically by the lepidopteran ectoparasitoid *Epiricania melanoleuca*; The female parasitoid lays eggs on sugarcane leaves).

Q139. Describe the different types of insect wing modifications with examples.

- **Correct Answer: 1) Elytra: Hardened**, heavily sclerotized forewings of Coleoptera (beetles) that protect the delicate hindwings and do not assist in flight; 2) Tegmina: Leathery, parchment-like forewings of Orthoptera (grasshoppers, cockroaches) that protect the hindwings.

Q140. What are the endocrine glands in insects that regulate growth and metamorphosis, and what do they secrete?

- **Correct Answer: 1) Neurosecretory cells (NSC)** in the brain: Secrete Prothoracicotropic Hormone (PTTH); 2) Corpora Allata: Endocrine glands located behind the brain that secrete Juvenile Hormone (JH).

Q141. Explain the mode of action of *Bacillus thuringiensis* (Bt) delta-endotoxin in insects.

- **Correct Answer: Bt produces crystalline (Cry) proteins** during sporulation; When an insect ingests these crystals, they dissolve in the highly alkaline environment (pH > 9) of the insect midgut.

Q142. Differentiate between holometabolous and hemimetabolous metamorphosis in insects, with examples.

- **Correct Answer: Holometabolous (Complete Metamorphosis):** Life cycle includes four distinct stages: Egg, Larva, Pupa, and Adult; Larvae look completely different from adults and feed on different food sources.

Q143. How does the cotton whitefly (*Bemisia tabaci*) cause damage and act as a vector?

- **Correct Answer: Bemisia tabaci damages** plants in two ways: 1) Direct feeding: Nymphs and adults suck sap from the phloem of leaves, reducing plant vigor and causing chlorosis; They excrete sticky honeydew, which leads to the growth of sooty mould, blocking photosynthesis.

Q144. Detail the symptoms of infestation by the Rice Gall Midge (*Orseolia oryzae*).

- **Correct Answer: The larvae of** the Rice Gall Midge migrate to the growing point of the rice shoot and feed on the apical meristem; This feeding stimulates the leaf sheath to grow into a hollow, tubular, dirty-white or silver-colored gall known as a 'silver shoot' or 'onion leaf' (onion shoot).

Q145. What are the main differences between the four honeybee species found in India?

- **Correct Answer: 1) Apis dorsata (Rock bee):** Largest, open-nesting (single large comb on cliffs/trees), highly aggressive, migratory, cannot be domesticated; 2) *Apis florea* (Little bee): Smallest, single comb in bushes, low honey yield, gentle.

Q146. What is the cause, transmission, and management of Pebrine disease in Silkworms?

- **Correct Answer: Pebrine** (a fatal disease of *Bombyx mori* caused by the microsporidian parasite *Nosema bombycis*; It is transmitted both horizontally (ingestion of contaminated feed/feces) and vertically (transovarian transmission from mother moth to eggs)).

Q147. Define the structural and functional modifications of the hind leg of a worker honeybee.

- **Correct Answer: Hind leg of a worker honeybee** (highly modified for pollen collection and transport: 1) Corbicula (Pollen Basket): A smooth, concave outer surface of the hind tibia, lined with stiff curving hairs, used to hold packed pollen; 2) Pollen Comb: Rows of stiff bristles on the inner surface of the basitarsus used to scrape pollen from the body).

Q148. Explain the nature of damage caused by Yellow Stem Borer (*Scirpophaga incertulas*) in Rice.

- **Correct Answer: Scirpophaga incertulas** (a monophagous pest of rice; The larva bores into the stem, feeding on the internal tissues and vascular bundles).

Q149. What are synthetic pyrethroids? Mention their advantages and disadvantages in agriculture.

- **Correct Answer: Synthetic pyrethroids (e.g., Cypermethrin, Deltamethrin)** (synthetic analogs of natural pyrethrins; Advantages: Extremely high insecticidal activity at very low doses, quick knockdown effect, and relatively low mammalian toxicity compared to OPs).

Q150. Describe the lac insect's life cycle and how lac is secreted.

- **Correct Answer: Kerria lacca nymphs** (crawlers) settle on young succulent twigs of host trees (Kusum, Palas, Ber) and insert their mouthparts to suck sap; They start secreting a resinous substance from cutaneous glands located all over their body.

Q151. What is the origin, characteristics, and historical significance of the Amritmahal cattle breed?

- **Correct Answer: Amritmahal** (an elite draught cattle breed of Karnataka, originating from the dry districts of Chitradurga, Tumakuru, and Chikkamagaluru; It was historically bred and patronized by the Kings of Mysore, particularly Tippu Sultan, who created a dedicated department (Amrit Mahal Department) to breed these cattle for rapid military transport of artillery).

Q152. Describe the Malnad Gidda breed of cattle. Why is it gaining importance?

- **Correct Answer: Malnad Gidda** (a dwarf cattle breed native to the Western Ghats (hilly and coastal areas) of Karnataka; They are very small, weighing around 80-120 kg).

Q153. Differentiate between Gerber method and Lactometer method in milk analysis.

- **Correct Answer: Gerber Method:** A chemical, volumetric method used to determine the fat percentage in milk; It uses sulfuric acid to digest milk proteins (casein) and release fat, and amyl alcohol to separate the fat column which is read on a butyrometer scale.

Q154. Explain the process of silage making and the chemical changes during preservation.

- **Correct Answer: Silage** (green fodder preserved under anaerobic conditions; Green fodder (e.g., maize, sorghum) is harvested at the milk/dough stage (30-35% dry matter), chopped, and tightly packed into a silo to exclude air).

Q155. What is milk fever in dairy cows? Mention its etiology, symptoms, and treatment.

- **Correct Answer: Milk fever (parturient paresis)** (a metabolic disease of high-producing dairy cows occurring shortly after calving; Etiology: Caused by a sudden drop in blood calcium levels (hypocalcemia) due to the rapid drain of calcium into colostrum).

Q156. Why is post-mortem examination strictly prohibited in animals suspected of dying from Anthrax?

- **Correct Answer: Anthrax** (the spore-forming, Gram-positive bacterium *Bacillus anthracis*; Inside the host body, the bacteria exist in a vegetative form).

Q157. Describe the characteristics of the Mandya (Bannur) sheep breed of Karnataka.

- **Correct Answer: Mandya sheep (also known as Bannur)** (a premier mutton breed originating from the Mandya district of Karnataka; Characteristics include a compact, wedge-shaped body, short and drooping ears, and a white coat with occasional light brown patches).

Q158. What are the differences between cow milk and colostrum?

- **Correct Answer: Colostrum** (the first milk secreted after calving, whereas normal milk is secreted later; Colostrum is thick, yellow, and contains up to 4-5 times more protein (mainly immunoglobulins like IgG for passive immunity), more minerals (especially calcium and phosphorus), more Vitamin A, and lower fat and lactose than normal milk).

Q159. Explain the pasteurization of milk and compare LTLT and HTST methods.

- **Correct Answer: Pasteurization** (the process of heating milk to a specific temperature for a set time to destroy all pathogenic microorganisms (specifically *Coxiella burnetii*) without significantly altering its nutritive value or taste; Low Temperature Long Time (LTLT): Milk is heated to 63°C (145°F) and held for 30 minutes).

Q160. Discuss Brucellosis in cattle: causal agent, transmission, symptoms, and public health importance.

- **Correct Answer: Brucellosis** (*Brucella abortus*; Transmission: Ingestion of contaminated feed, water, uterine discharges, or through breeding).

Q161. What is the significance of the FRUITS database in Karnataka's agricultural administration?

- **Correct Answer: FRUITS (Farmer Registration and Unified Beneficiary Information System)** assigns a unique Farmer ID (FID) to every registered farmer by linking their profile with Bhoomi land records, bank details, and crop surveys, eliminating physical paperwork for subsidies; (Latest 2020).

Q162. Name the ten Agro-Climatic Zones of Karnataka and their respective headquarters.

- **Correct Answer: Zone 1: Northeast Transition** (Bidar), **Zone 2: Northeast Dry** (Raichur), **Zone 3: Northern Dry** (Vijayapura), **Zone 4: Central Dry** (Babbur, Chitradurga), **Zone 5: Eastern Dry** (Chintamani), **Zone 6: Southern Dry** (Mandya), **Zone 7: Southern Transition** (Shivamogga), **Zone 8: Northern Transition** (Dharwad), **Zone 9: Hill Zone** (Mudigere), **Zone 10: Coastal Zone** (Brahmavar); (KPSC AO 2017).

Q163. How does the 'Krishi Bhagya' scheme support dryland agriculture in Karnataka?

- **Correct Answer: It funds the construction of farm ponds (Krishi Honda) to collect runoff water, provides polythene sheets (300-500 microns) for lining, diesel/solar pumpsets for**

lifting water, and micro-irrigation systems (drip/sprinkler) for protective watering during dry spells; (KPSC AO 2017).

Q164. What are the components and financial provisions of the 'Raitha Siri' scheme?

- **Correct Answer: Raitha Siri promotes** minor millets by providing Rs; 10,000 per hectare (capped at 2 hectares) directly to farmers growing Foxtail, Little, Kodo, Barnyard, Proso, or Browntop millets.

Q165. When were the five agricultural/horticultural universities of Karnataka established?

- **Correct Answer: UAS Bangalore: 1963;** UAS Dharwad: 1986; UAS Raichur: 2008; UHS Bagalkot: 2008; and KSNUAHS Shivamogga: 2012; (KPSC AAO 2017).

Q166. What is the 'Ksheera Dhare' scheme and its incentive structure?

- **Correct Answer: Ksheera Dhare** (an incentive program for Karnataka's dairy farmers, providing Rs; 5 per liter of milk supplied to registered cooperative societies, paid directly to their bank accounts).

Q167. Explain the role of the 'Samrakshane' portal in Karnataka's implementation of PMFBY.

- **Correct Answer: Samrakshane** (a centralized web application for enrolling farmers, collecting premium shares, tracking crop cutting experiments, and disbursing insurance claims directly to Aadhaar-linked accounts; (KPSC AO 2017)).

Q168. What is the GI-tagged 'Sirsi Supari' and what makes it unique?

- **Correct Answer: Sirsi Supari** (a unique arecanut from Uttara Kannada, characterized by a rounded structure, heavy weight, low tannin content, and high chewing quality, which received a GI tag in 2019; (Latest 2019)).

Q169. Describe the 'Krishi Yantra Dhare' scheme.

- **Correct Answer: It** (a Public-Private Partnership (PPP) initiative that establishes agricultural machinery custom hiring centers at the hobli level, helping small farmers rent expensive machinery at subsidized rates; (KPSC AO 2017)).

Q170. What is the 'REWARD' project and which agency assists it in Karnataka?

- **Correct Answer: REWARD (Rejuvenating Watersheds for Agricultural Resilience through Innovative Development) project** (a soil and water conservation program assisted by the World Bank; (KPSC AO 2017)).

Q171. Explain the significance of 'Kagga Rice' and its production ecology.

- **Correct Answer: Kagga** (a traditional, salt-tolerant red rice variety grown in the saline, tide-prone 'Gajni' estuarine lands of Kumta, Uttara Kannada, without synthetic fertilizers; (KPSC AAO 2017)).

Q172. What was the focus of the Karnataka State Organic Farming Policy 2017?

- **Correct Answer: It revised the** 2004 policy, aiming to bring 10% of cultivable area in each district under organic farming, establishing organic clusters, and facilitating certification; (KPSC AO 2017).

Q173. What is the 'Raita Vidya Nidhi' scheme?

- **Correct Answer: It** (a scholarship scheme launched in 2021 by the Karnataka government to provide Rs; 2,000 to Rs).

Q174. Explain the role of 'Krishi Sanjeevini' vans in Karnataka.

- **Correct Answer: These** (mobile agricultural testing laboratories that travel to farmers' fields to analyze soil and water samples, and diagnose crop pests, diseases, and nutrient deficiencies; (Latest 2021)).

Q175. What are the key details of the GI-tagged 'Byadagi Chilli'?

- **Correct Answer: Byadagi Chilli** (grown in Haveri and Dharwad districts, valued globally for its

bright red color (high oleoresin content) and mild heat (low capsaicin); (KPSC AAO 2017)).

Q176. What is the 'Raita Sanjeevini' scheme in Karnataka?

- **Correct Answer: It** (a state insurance scheme providing ex-gratia compensation of up to Rs; 2 lakh to farmers or their family members in case of death or permanent disability during agricultural work).

Q177. Differentiate between Cost A1 and Cost A2 in agricultural cost estimation.

- **Correct Answer: Cost A1 includes** all paid-out cash and kind expenses (labor, seed, fertilizer, irrigation, depreciation, interest); Cost A2 is Cost A1 plus rent paid for leased-in land.

Q178. Define Cost B2 and Cost C2.

- **Correct Answer: Cost B2** (Cost B1 (Cost A1 + interest on owned fixed capital) plus rental value of owned land and rent for leased land; Cost C2 is Cost B2 plus the imputed value of family labor).

Q179. Explain the economic significance of Stage II of the Law of Variable Proportions.

- **Correct Answer: Stage II** (the rational stage of production where the Average Product (AP) is declining, and the Marginal Product (MP) is positive but declining; Profit is maximized in this stage).

Q180. State the optimization rule for the Factor-Product relationship.

- **Correct Answer: Profit-maximizing level of a single variable input** (reached when its Marginal Value Product (MVP) equals its Marginal Input Cost (MIC), i.e., $MVP = MIC$; (KPSC AO 2017)).

Q181. How is the least-cost combination of inputs determined in the Factor-Factor relationship?

- **Correct Answer: It** (achieved at the point of tangency between the isoquant and the isocost line, where the Marginal Rate of Technical Substitution (MRTS) equals the input price ratio (P_x/P_y); (KPSC AAO 2015)).

Q182. Explain the difference between Marketed Surplus and Marketable Surplus.

- **Correct Answer: Marketable surplus** (the theoretical surplus left after satisfying family requirements; Marketed surplus is the actual quantity sold in the market).

Q183. What is 'distress sale' and which category of farmers does it affect?

- **Correct Answer: Distress sale occurs** when small and marginal farmers sell their crops immediately after harvest at low prices due to urgent cash needs, making marketed surplus exceed marketable surplus; (KPSC AO 2014).

Q184. Explain the three boxes of domestic support under the WTO Agreement on Agriculture (AoA).

- **Correct Answer: Green Box: Non-distorting** subsidies (research, extension, disaster relief); Blue Box: Production-limiting subsidies.

Q185. What is the de minimis limit under the WTO Amber Box for developing and developed countries?

- **Correct Answer: De minimis allows** trade-distorting support up to 5% of the total value of agricultural production for developed countries, and up to 10% for developing countries; (KPSC AO 2017).

Q186. When was NABARD established and on whose recommendations?

- **Correct Answer: NABARD** (established on July 12, 1982, under the National Bank for Agriculture and Rural Development Act, 1981, recommended by the B; Sivaraman Committee).

Q187. What is the Kisan Credit Card (KCC) scheme and which committee recommended it?

- **Correct Answer: KCC** (launched in 1998 on the R.V; Gupta Committee's recommendation to provide short-term crop loans up to Rs).

Q188. What is the 'Lead Bank Scheme' and when was it introduced?

- **Correct Answer: Introduced in 1969** on the Nariman Committee's recommendation, it assigns districts to commercial banks to lead credit planning and coordination for rural development; (KPSC AO 2014).

Q189. Explain horizontal and vertical market integration with examples.

- **Correct Answer: Horizontal integration** (the merger of firms at the same stage of the market (e.g., two grain wholesalers); Vertical integration is the merger of firms at different stages (e.g., a sugar mill buying sugarcane farms)).

Q190. Define 'Price Spread' and the 'Producer's Share in Consumer Rupee'.

- **Correct Answer: Price spread** (the difference between the consumer's price (P_c) and the producer's price (P_p); Producer's share is $(P_p / P_c) \times 100$, measuring marketing efficiency).

Q191. What is 'Opportunity Cost' and how is it used in farm decision-making?

- **Correct Answer: Opportunity cost** (the value of the next best alternative foregone; It helps farmers choose the most profitable enterprise when resources are limited).

Q192. Define 'Engel's Law' and its implication for agricultural demand.

- **Correct Answer: Engel's Law states that as income rises, the proportion of income spent on food declines, meaning the demand for food** (income-inelastic as a nation grows wealthier; (KPSC AO 2014)).

Q193. What are 'Ridge Lines' on an isoquant map?

- **Correct Answer: Ridge lines represent the boundaries of the economic region of production, where the marginal products of the inputs** (zero; Outside these lines, inputs are redundant).

Q194. What is the CACP and what is its role in MSP formulation?

- **Correct Answer: Commission for Agricultural Costs and Prices (CACP)** (a statutory body under the Ministry of Agriculture that recommends MSPs for 22 mandated crops; (KPSC AO 2017)).

Q195. What is the 'AgriSURE' fund and who launched it?

- **Correct Answer: AgriSURE** (a Rs. 750 crore venture capital fund launched by NABARD in 2024 to support agritech startups and rural enterprises).

Q196. What is the 'PM-PRANAM' scheme and what are its key features?

- **Correct Answer: PM-PRANAM incentivizes states** to reduce chemical fertilizer use and promote organic nutrients; The state receives 50% of the fertilizer subsidy savings as a grant.

Q197. What are the key properties and benefits of Nano Urea and Nano DAP developed by IFFCO?

- **Correct Answer: They have a particle size of 20-50 nm and a nutrient use efficiency of over 80%; One 500 ml bottle replaces one 45 kg bag of conventional urea/DAP.**

Q198. Detail the financial support and payment frequency under the 'PM-KISAN' scheme.

- **Correct Answer: It provides Rs. 6,000 per year in three equal installments of Rs. 2,000 every four months to all landholding farmer families. (Latest 2019).**

Q199. What is the 'Namo Drone Didi' scheme?

- **Correct Answer: Launched in late 2023,** it provides agricultural drones and pilot training to 15,000 women Self-Help Groups (SHGs) to spray liquid nano-fertilizers and pesticides; (Latest 2023).

Q200. Explain the 'One Nation One Fertilizer' scheme.

- **Correct Answer: The scheme mandates** all fertilizer companies to market their subsidized fertilizers under the single brand name BHARAT to eliminate brand confusion; (Latest 2024).

[NOTE] Top 100 Must-Revise Questions

These 100 questions cover critical Karnataka policies, agro-climatic zones, and statistics:

Q1. What is the soil moisture tension at the permanent wilting point and field capacity?

- **Correct Answer: Permanent wilting point** occurs at -15 bars (1.5 MPa) of soil moisture tension; Field capacity occurs at -0.33 bar (33 kPa) in medium-to-fine textured soils, and -0.1 bar (10 kPa) in sandy soils. [KPSC AO 2014]

Q2. List the nitrogen content and equivalent acidity of Ammonium Chloride.

- **Correct Answer: Ammonium chloride contains** 26% Nitrogen; It has an equivalent acidity of 128, meaning it requires 128 kg of calcium carbonate (CaCO_3) to neutralize the acidity produced in the soil by 100 kg of the fertilizer. [KPSC AAO 2015]

Q3. Define the critical weed-free period for finger millet (ragi) under transplanted conditions.

- **Correct Answer: Critical weed-free period for transplanted ragi** (25 to 35 days after transplanting (DAT)); Effective weed control during this window is essential to prevent significant yield losses). [KPSC AAO 2017]

Q4. What is the chemical name and target of the herbicide Glyphosate?

- **Correct Answer: Glyphosate** (N-(phosphonomethyl) glycine; It is a non-selective, systemic, post-emergence herbicide that targets the EPSP synthase enzyme in the shikimic acid pathway, blocking aromatic amino acid synthesis). [KPSC AAO 2015]

Q5. Name the parents of the cotton hybrid variety 'DCH-32'.

- **Correct Answer: 'DCH-32'** (an interspecific cotton hybrid developed by crossing *Gossypium hirsutum* (DS-28) as the female parent and *Gossypium barbadense* (SB-289-E) as the male parent). [KPSC AAO 2017]

Q6. State the biuret limit in urea for soil and foliar applications.

- **Correct Answer: According to the Fertilizer Control Order (FCO)**, the biuret content in urea must not exceed 1.5% by weight for soil application and should be restricted to less than 1.0% for foliar spray applications. [KPSC AAO 2015]

Q7. What is the largest agro-climatic zone in Karnataka?

- **Correct Answer: Northern Dry Zone (Zone 3)** (the largest agro-climatic zone in Karnataka, covering districts like Bagalkot, Vijayapura, Gadag, Dharwad, Koppal, and parts of Belagavi). [KPSC AO 2014]

Q8. Which nutrient deficiency causes 'poppod' in groundnut, and how is it managed?

- **Correct Answer: Poppod (empty shells)** (calcium deficiency in the pod zone; It is managed by applying Gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) at 500 kg/ha at the pegging stage (40-45 DAS)). [KPSC AAO 2017]

Q9. What are the core seedling age and spacing guidelines in the SRI system of rice cultivation?

- **Correct Answer: In the System of Rice Intensification (SRI)**, seedlings should be transplanted at 10-12 days of age (2-leaf stage) at a spacing of 25 cm x 25 cm with a single seedling per hill. [KPSC AO 2019]

Q10. Explain the difference in potometer and lysimeter applications.

- **Correct Answer: Potometer** (used to measure transpiration from a plant shoot, whereas a lysimeter is used to measure total evapotranspiration (evaporation + transpiration) from a cropped field). [KPSC AAO 2017]

Q11. What are the tag colors for Breeder, Foundation, and Certified seeds?

- **Correct Answer: Breeder Seed** has a Golden Yellow tag; Foundation Seed has a White tag; and Certified Seed has an Azure Blue tag. [KPSC AO 2017]

Q12. Explain the biochemical basis of the Tetrazolium (TZ) test staining.

- **Correct Answer: Dehydrogenase enzymes** in living embryonic tissues reduce the colorless 2,3,5-triphenyl tetrazolium chloride solution to red triphenyl formazan; Dead tissues lack active enzymes and remain unstained. [KPSC AO 2014]

Q13. What are the isolation distance requirements for foundation seed of self-pollinated rice and hybrid maize?

- **Correct Answer: Foundation seed production** of self-pollinated rice requires a minimum isolation distance of 3 meters, whereas hybrid maize requires 400 meters. [KPSC AAO 2015 / 2017]

Q14. State the formula for the Real Value of Seed (RVS).

- **Correct Answer: Real Value of Seed (RVS)** (calculated as: (Purity % x Germination %) / 100; It determines the utility value of a seed lot). [KPSC AO 2017]

Q15. Differentiate between epigeal and hypogeal seed germination.

- **Correct Answer: In epigeal germination**, the hypocotyl elongates and pushes the cotyledons above the soil surface (e.g., beans, castor); In hypogeal germination, the epicotyl elongates while the cotyledons remain below the soil surface (e.g., chickpea, wheat). [KPSC AO 2019]

Q16. Name the physiological disorders in mango caused by Boron deficiency and convective heat.

- **Correct Answer: Boron deficiency** (combined with toxic gases from brick kilns) causes 'Black Tip' of mango; Convective soil heat during fruit development causes 'Spongy Tissue' in Alphonso mangoes. [KPSC AO 2014 / KPSC AAO 2017]

Q17. What are the commercial propagation methods for Mango and Sapota?

- **Correct Answer: Mango** (commercially propagated by Epicotyl (stone) grafting or veneer grafting; Sapota is propagated by Approach grafting or softwood grafting on Rayan / Khirni (Manilkara hexandra) rootstock). [KPSC AO 2014 / KPSC AAO 2015]

Q18. Which growth regulator is used to overcome alternate bearing in mango, and what is its mode of action?

- **Correct Answer: Paclobutrazol (Cultar)** (applied to the soil; It inhibits gibberellin biosynthesis, restricting vegetative growth and promoting early, regular flower bud differentiation). [KPSC AO 2017]

Q19. What causes Blossom End Rot in tomato and Whiptail in cauliflower?

- **Correct Answer: Blossom End Rot in tomato** (calcium deficiency; Whiptail in cauliflower is caused by molybdenum deficiency, which is common in acidic soils (pH < 5.5)). [KPSC AAO 2015 / KPSC AO 2017]

Q20. How does 1-Methylcyclopropene (1-MCP) extend the post-harvest shelf life of fruits?

- **Correct Answer: 1-MCP** (a gaseous compound that binds to ethylene receptors in plant cells, blocking ethylene action; This delays ripening, reduces respiration, and slows down softening and senescence). [KPSC AO 2017]

Q21. What are the pH ranges for Saline, Sodic, and Acid soils?

- **Correct Answer: Acid soils: pH < 6.5; Saline soils: pH > 8.5** (usually neutral to slightly alkaline).

Q22. Write down the electrical conductivity (EC) threshold for saline soils in dS/m at 25°C.

- **Correct Answer: Critical EC threshold for saline soils** (> 4.0 dS/m (KPSC AAO 2015)).

Q23. What is the C:N ratio of normal arable agricultural soils?

- **Correct Answer: The C:N ratio** of normal arable agricultural soils ranges from 10:1 to 12:1 (KPSC AAO 2015).

Q24. Name the primary clay mineral in red soils and state its structure type.

- **Correct Answer: Kaolinite** (the primary clay mineral in red soils, and it is a 1:1 type non-expanding silicate clay (KPSC AO 2014)).

Q25. What are the dominant soil orders of Karnataka's Red and Black soils respectively?

- **Correct Answer: Red soils belong** to Alfisols (and Inceptisols); Black soils belong to Vertisols (KPSC AAO 2017).

Q26. What is the standard particle density of mineral soils used for porosity calculations?

- **Correct Answer: Standard average particle density of mineral soils** (2.65 g/cm³ (KPSC AAO 2017)).

Q27. Which nutrient deficiency causes 'whip tail' in cauliflower?

- **Correct Answer: Molybdenum (Mo) deficiency** causes whip tail in cauliflower (KPSC AO 2017).

Q28. What is the biuret limit in urea for foliar application according to the FCO?

- **Correct Answer: Biuret limit in urea for foliar spray** (0.5% (soil application limit is 1.5%) (KPSC AO 2017)).

Q29. What is the equivalent basicity of calcium cyanamide?

- **Correct Answer: Equivalent basicity of calcium cyanamide** (63 (KPSC AAO 2015)).

Q30. Which indicator plant is used to detect Boron deficiency in fields?

- **Correct Answer: Sunflower** (the classic indicator plant used to detect Boron deficiency (KPSC AO 2017)).

Q31. What is the minimum viable cell count for Rhizobium in carrier-based biofertilizers at the time of expiry?

- **Correct Answer: Minimum viable count** (10⁷ cells per gram of carrier (10⁸ cells/g at manufacture) (KPSC AAO 2017)).

Q32. Which bacteria are responsible for the oxidation steps of Nitrification?

- **Correct Answer: Nitrosomonas oxidizes ammonium** (NH₄⁺) to nitrite (NO₂⁻), and Nitrobacter oxidizes nitrite (NO₂⁻) to nitrate (NO₃⁻) (KPSC AAO 2017).

Q33. What fern-algae symbiotic combination is used as a biofertilizer in low-land rice?

- **Correct Answer: The symbiotic association** between the water fern Azolla and the cyanobacterium Anabaena azollae (KPSC AO 2017).

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- **Correct Answer: Gluconic acid (and 2-ketogluconic acid)** (the major organic acid produced by PSB (KPSC AO 2017)).

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- **Correct Answer: AM fungi belong** to the phylum Glomeromycota (KPSC AO 2014).

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- **Correct Answer: Complementary epistasis exhibits** a 9:7 ratio; Duplicate dominant epistasis exhibits a 15:1 ratio.

Q42. What cytological structures represent the physical crossover points during meiosis?

- **Correct Answer: Chiasmata (singular: chiasma)** (the X-shaped cytological configurations representing the physical sites of crossing over between non-sister chromatids of homologous chromosomes during meiotic prophase I; (KPSC AAO 2017)).

Q43. What is the first man-made cereal crop, and what is its parentage?

- **Correct Answer: Triticale** (the first man-made cereal crop, developed by crossing wheat (*Triticum durum* or *T. aestivum*) and rye (*Secale cereale*) followed by chromosome doubling with colchicine).

Q44. What is the chromosome composition of the nucleosome core particle?

- **Correct Answer: The nucleosome core** particle consists of 146 base pairs of DNA wrapped 1.65 times around an octamer of core histones, which contains two molecules each of H₂A, H₂B, H₃, and H₄; (KPSC AO 2017).

Q45. How do transition and transversion mutations differ at the nucleotide level?

- **Correct Answer: A transition mutation** substitutes a purine for another purine (A ↔ G) or a pyrimidine for a pyrimidine (C ↔ T); A transversion mutation substitutes a purine for a pyrimidine, or vice versa.

Q46. What chemical mutagen is most commonly used to induce transition mutations in crops?

- **Correct Answer: Ethyl Methane Sulfonate (EMS)** (the most widely used chemical mutagen in plant breeding, acting as an alkylating agent that induces G:C to A:T transition mutations; (KPSC AAO 2017)).

Q47. What are the three lines involved in the CGMS system of hybrid seed production?

- **Correct Answer: A-line** (the male-sterile line; the B-line is the maintainer line (used to maintain the A-line); and the R-line is the restorer line (used to restore fertility in the F₁ hybrid); (KPSC AAO 2017)).

Q48. How many backcrosses are typically required to recover over 99% of the recurrent parent's genome?

- **Correct Answer: Typically, 6 to 8 backcrosses (BC₆ to BC₈)** (required to recover approximately 99.2% to 99.6% of the recurrent parent's genetic background; (KPSC AAO 2017)).

Q49. What is the genetic control of maternal inheritance?

- **Correct Answer: Maternal inheritance (cytoplasmic inheritance)** (controlled by genes located in the organelle genomes, namely mitochondrial DNA (mtDNA) and chloroplast DNA (cpDNA), and is inherited through the female gamete; (KPSC AAO 2015)).

Q50. What is the difference between General Combining Ability (GCA) and Specific Combining Ability (SCA) in terms of gene action?

- **Correct Answer: GCA** (governed by additive genetic variance and additive × additive gene action, which is fixable; SCA is governed by non-additive gene action, including dominance and epistatic variances, which is non-fixable but exploitable in F₁ hybrids).

Q51. What is the primary difference between somatic embryogenesis and organogenesis?

- **Correct Answer: Somatic embryogenesis regenerates** plants via bipolar structures containing both root and shoot apices, which develop directly or indirectly; Organogenesis produces unipolar shoots or roots that require separate media to develop the missing root or shoot pole.

Q52. What enzymes are used for plant protoplast isolation?

- **Correct Answer: Mixture of cellulase (to digest cellulose) and pectinase (to digest pectin)** (used to enzymatically remove the plant cell wall and isolate viable protoplasts; (KPSC AO 2017)).

Q53. What is the function of T4 DNA ligase in genetic engineering?

- **Correct Answer: T4 DNA ligase** joins double-stranded DNA fragments by catalyzing the formation of a phosphodiester bond between the 3'-OH and 5'-phosphate ends of the DNA strands; (KPSC AO 2014).

Q54. What plasmid is the classic cloning vector containing ampicillin and tetracycline resistance genes?

- **Correct Answer: pBR322** (the classic plasmid cloning vector containing two selectable marker genes: bla (ampicillin resistance) and tetR (tetracycline resistance); (KPSC AA0 2017)).

Q55. What is the role of the PAM sequence in CRISPR-Cas9 gene editing?

- **Correct Answer: Protospacer Adjacent Motif (PAM, typically 5'-NGG-3' for Cas9)** (a short DNA sequence next to the target site; It is required for Cas9 binding and cleavage; without it, Cas9 cannot cut the target DNA).

Q56. What is the thermostable DNA polymerase used in PCR, and what is its limitation?

- **Correct Answer: Taq DNA polymerase, isolated from Thermus aquaticus**, (the primary enzyme used in PCR; Its main limitation is that it lacks 3'-to-5' proofreading exonuclease activity, leading to a higher mutation error rate during amplification).

Q57. How does Sanger DNA sequencing terminate chain elongation?

- **Correct Answer: Sanger sequencing uses** dideoxynucleotide triphosphates (ddNTPs) as chain terminators because they lack the 3'-OH group needed for DNA polymerase to form the next phosphodiester bond; (KPSC AO 2017).

Q58. What is the encapsulating agent used to produce synthetic seeds?

- **Correct Answer: Sodium alginate (which polymerizes into calcium alginate beads upon exposure to calcium chloride)** (the standard matrix used to encapsulate somatic embryos or shoot buds; (KPSC AO 2014)).

Q59. Which apex regulatory body in India approves the commercial release of GM crops?

- **Correct Answer: Genetic Engineering Appraisal Committee (GEAC), which functions under the Ministry of Environment, Forest and Climate Change (MoEFCC)**, (the statutory body in India; (KPSC AO 2014)).

Q60. What technology was used to develop the first commercial GM food crop, the Flavr Savr tomato?

- **Correct Answer: Antisense RNA technology** (used to downregulate the polygalacturonase (PG) enzyme, which degrades pectin in cell walls, thereby delaying fruit softening during ripening; (KPSC AA0 2015)).

Q61. What is the composition and ratio of Bordeaux mixture?

- **Correct Answer: Bordeaux mixture** (composed of Copper Sulphate, Quick Lime (Calcium Oxide), and Water; The standard concentration is 1% (ratio of 1 kg Copper Sulphate : 1 kg Quick Lime : 100 liters of Water) or the historic 5:5:50 formula (5 lbs Copper Sulphate : 5 lbs Lime : 50 gallons Water) (KPSC AO 2017)).

Q62. Which fungus causes the late blight of potato, responsible for the Irish Famine?

- **Correct Answer: Late blight of potato** (the oomycete pathogen *Phytophthora infestans* (KPSC AA0 2017)).

Q63. Which spore stage of *Puccinia graminis tritici* is the dikaryotic, repeating asexual stage on wheat?

- **Correct Answer: Urediniospore (Stage II)** (the repeating asexual dikaryotic spore stage on wheat stems (KPSC AO 2017)).

Q64. What is the causal agent and vector of little leaf of brinjal?

- **Correct Answer: Causal agent** (phytoplasma (a wall-less prokaryote) and the vector is the leafhopper *Hishimonus phycitis* (KPSC AO 2014)).

Q65. Explain the Luthra and Sattar solar heat treatment for loose smut of wheat.

- **Correct Answer: Seeds** (soaked in water for 4 hours in a summer morning to activate dormant internally seed-borne mycelium (*Ustilago nuda tritici*), then spread on concrete in the hot sun for 4 hours to kill the activated mycelium without affecting seed germination (KPSC AO 2017)).

Q66. Which systemic fungicide specifically controls smuts (Basidiomycetes)?

- **Correct Answer: Carboxin (Vitavax)** (a systemic oxathiin fungicide that specifically controls smuts by inhibiting succinate dehydrogenase (KPSC AAO 2017)).

Q67. What are the key internal symptoms of red rot of sugarcane?

- **Correct Answer: Internal tissues of** split stalks show red discoloration with diagnostic white transverse bands, accompanied by a sour, acidic fermentation odor (KPSC AO 2017).

Q68. Name the quarantine nematode pest of potato in India and its restricted location.

- **Correct Answer: Golden cyst nematode (*Globodera rostochiensis* and *Globodera pallida*)**, which (restricted under domestic quarantine in the Nilgiri and Palni hills of Tamil Nadu (KPSC AO 2017)).

Q69. What are the modified balancing hindwings of Diptera called?

- **Correct Answer: They** (called halteres, which are small club-like structures that function as gyroscopic sensors for flight stability (KPSC AO 2017)).

Q70. Which type of antennae are elbowed and found in honeybees?

- **Correct Answer: Geniculate antennae**, which have a long basal segment (scape) and subsequent segments (pedicel and flagellum) bent at a sharp angle (KPSC AO 2017).

Q71. Which organism is used for the biological control of sugarcane pyrrilla?

- **Correct Answer: *Epiricania melanoleuca***, a lepidopteran ectoparasitoid whose larvae feed on the nymphs and adults of *Pyrilla perpusilla* (KPSC AO 2017).

Q72. Define Economic Injury Level (EIL) in IPM.

- **Correct Answer: EIL** (the lowest pest population density that will cause economic damage, where the cost of control equals the value of yield saved (KPSC AO 2017)).

Q73. Which hormone is responsible for maintaining larval characters in insects?

- **Correct Answer: Juvenile Hormone (JH)**, which (secreted by the endocrine glands corpora allata (KPSC AO 2017)).

Q74. Which insect vector transmits the Rice Tungro Virus?

- **Correct Answer: The green leafhopper**, *Nephotettix virescens*, transmits both components of the Rice Tungro Virus complex (KPSC AO 2017).

Q75. Name the microsporidian pathogen causing Pebrine disease in silkworms.

- **Correct Answer: *Nosema bombycis***, which causes fatal infection and (transmitted transovarially (vertically) (KPSC AAO 2017)).

Q76. What chemical group does the insecticide Imidacloprid belong to and what is its target site?

- **Correct Answer: Imidacloprid belongs to the Neonicotinoid group and acts as an agonist on postsynaptic Nicotinic Acetylcholine Receptors (nAChR) (KPSC AAO 2017).**

Q77. Which draft cattle breed of Karnataka was bred by Tippu Sultan for military transport?

- **Correct Answer: Amritmahal, which (developed under the patronage of the Kings of Mysore for speed and endurance (KPSC AAO 2017)).**

Q78. Which metabolic disease in dairy cows is caused by hypocalcemia and shows subnormal temperature?

- **Correct Answer: Milk fever (parturient paresis), caused by acute calcium loss into colostrum right after calving, causing hypothermia and recumbency (KPSC AO 2017).**

Q79. What is the standard temperature and time for HTST pasteurization?

- **Correct Answer: HTST pasteurization requires heating milk to 72°C and holding it for exactly 15 seconds before rapid cooling (KPSC AO 2017).**

Q80. Why is post-mortem examination prohibited for animals suspected of dying from Anthrax?

- **Correct Answer: To prevent exposure of vegetative Bacillus anthracis cells to oxygen, which induces sporulation, contaminating the soil and posing zoonotic risk (KPSC AO 2017).**

Q81. What is the largest and driest agro-climatic zone in Karnataka, and what is its headquarters?

- **Correct Answer: Northern Dry Zone (Zone 3) (the largest and driest, headquartered at Vijayapura; (KPSC AO 2017)).**

Q82. In which year was UAS Bangalore established under Mysore Act No. 22?

- **Correct Answer: UAS Bangalore (established in 1963 and inaugurated in 1965; (KPSC AO 2014)).**

Q83. Which unique Farmer ID system integrates land registry (Bhoomi) in Karnataka?

- **Correct Answer: The FRUITS portal (Farmer Registration and Unified Beneficiary Information System); (Latest 2020).**

Q84. What is the cash incentive for minor millets under Raitha Siri, and the maximum land limit?

- **Correct Answer: Rs. 10,000 per hectare, with a maximum limit of 2 hectares per farmer. (Latest 2019).**

Q85. What is the recommended polythene sheet thickness for lining farm ponds under Krishi Bhagya?

- **Correct Answer: 300 to 500 microns; (KPSC AAO 2017).**

Q86. What is the state incentive under Ksheera Dhare for supplying milk to cooperative societies?

- **Correct Answer: Rs. 5 per liter of milk. (KPSC AO 2017).**

Q87. Which project, assisted by the World Bank, is executed by the Karnataka Watershed Development Department?

- **Correct Answer: The REWARD project (Rejuvenating Watersheds for Agricultural Resilience through Innovative Development); (KPSC AO 2017).**

Q88. Which horticultural product from Uttara Kannada was awarded a GI tag in 2023?

- **Correct Answer: Kari Ishad Mango of Ankola; (Latest 2023).**

Q89. What is the formula for Cost C2, the comprehensive cost of cultivation?

- **Correct Answer: Cost C2 = Cost B2 (Cost B1 + rental value of owned land + rent for leased-in land) + imputed value of family labor; (KPSC AAO 2017).**

Q90. In which stage of the Law of Variable Proportions does the rational farm manager operate?

- **Correct Answer: Stage II (where average product (declining, and marginal product is positive but declining); (KPSC AO 2017)).**

Q91. What is the input optimization condition in the factor-product relationship?

- **Correct Answer: Marginal Value Product (MVP)** must equal Marginal Input Cost (MIC), i.e., $MVP = MIC$; (KPSC AO 2017).

Q92. What is the optimization condition for the least-cost combination of inputs in the factor-factor relationship?

- **Correct Answer: Marginal Rate of Technical Substitution of X for Y** must equal the input price ratio, i.e., $MRTS_{xy} = P_x / P_y$; (KPSC AAO 2015).

Q93. On which date was NABARD established and which committee recommended it?

- **Correct Answer: Established on July 12, 1982**, on the recommendation of the B; Sivaraman Committee.

Q94. What is the interest rate for prompt payers of short-term crop loans under the Kisan Credit Card (KCC) scheme?

- **Correct Answer: 4% per annum** (after 3% prompt repayment incentive on the base rate of 7%); (KPSC AAO 2017).

Q95. Which committee recommended the establishment of Regional Rural Banks (RRBs) in 1975?

- **Correct Answer: M. Narasimham Committee.** (KPSC AAO 2017).

Q96. Which agency recommends the Minimum Support Prices (MSP) for agricultural crops in India?

- **Correct Answer: Commission for Agricultural Costs and Prices (CACP);** (KPSC AO 2017).

Q97. What is the corpus of the AgriSURE fund launched by NABARD in 2024?

- **Correct Answer: Rs. 750 crore.** (Latest 2024).

Q98. What is the approved MSP for Paddy (Common) for the 2024-25 crop year?

- **Correct Answer: Rs. 2,300 per quintal.** (Latest 2024).

Q99. Which brand name is mandatory for all subsidized chemical fertilizers under the One Nation One Fertilizer scheme?

- **Correct Answer: BHARAT (e.g., Bharat Urea);** (Latest 2024).

Q100. What is the name of the central database portal developed for AgriStack implementations?

- **Correct Answer: Farmer Registry / Digital Agriculture Mission (AgriStack);** (Latest 2021).

--- Final Revision ---

Final Revision Section

Night Before Exam Revision Notes

Focus ONLY on these highly volatile, easy-to-forget numbers and facts:

[NOTE] Seed Tag Color Coding

- Breeder Seed: **Golden Yellow**
- Foundation Seed: **White**
- Certified Seed: **Azure Blue**
- Truthful Label Seed: **Opal Green**

[NOTE] Essential Nutrient Concentration & Forms

- Central atom in chlorophyll: **Magnesium (Mg)**
- Deficiency causing terminal bud death: **Calcium (Ca) and Boron (B)**
- Nitrogen source preferred by potato/rice: **Ammoniacal (NH₄⁺) form**
- Yellowing of tea leaves: **Sulfur (S) deficiency**

[NOTE] Major Milestones

- First Agri University: **Pantnagar, 1960**
- NABARD: **12th July 1982**
- Seeds Act: **1966 (enforced 1969)**
- Krishi Bhagya: **2014**

Last 100 Questions Before Exam

Revise these 100 quick check questions on the morning of the exam:

Q1. What is the soil moisture tension at the permanent wilting point and field capacity?

- **Correct Answer: Permanent wilting point** occurs at -15 bars (1.5 MPa) of soil moisture tension; Field capacity occurs at -0.33 bar (33 kPa) in medium-to-fine textured soils, and -0.1 bar (10 kPa) in sandy soils. [KPSC AO 2014]

Q2. List the nitrogen content and equivalent acidity of Ammonium Chloride.

- **Correct Answer: Ammonium chloride contains** 26% Nitrogen; It has an equivalent acidity of 128, meaning it requires 128 kg of calcium carbonate (CaCO₃) to neutralize the acidity produced in the soil by 100 kg of the fertilizer. [KPSC AAO 2015]

Q3. Define the critical weed-free period for finger millet (ragi) under transplanted conditions.

- **Correct Answer: Critical weed-free period for transplanted ragi** (25 to 35 days after transplanting (DAT)); Effective weed control during this window is essential to prevent

significant yield losses). [KPSC AAO 2017]

Q4. What is the chemical name and target of the herbicide Glyphosate?

- **Correct Answer: Glyphosate** (N-(phosphonomethyl) glycine; It is a non-selective, systemic, post-emergence herbicide that targets the EPSP synthase enzyme in the shikimic acid pathway, blocking aromatic amino acid synthesis). [KPSC AAO 2015]

Q5. Name the parents of the cotton hybrid variety 'DCH-32'.

- **Correct Answer: 'DCH-32'** (an interspecific cotton hybrid developed by crossing *Gossypium hirsutum* (DS-28) as the female parent and *Gossypium barbadense* (SB-289-E) as the male parent). [KPSC AAO 2017]

Q6. State the biuret limit in urea for soil and foliar applications.

- **Correct Answer: According to the Fertilizer Control Order (FCO)**, the biuret content in urea must not exceed 1.5% by weight for soil application and should be restricted to less than 1.0% for foliar spray applications. [KPSC AAO 2015]

Q7. What is the largest agro-climatic zone in Karnataka?

- **Correct Answer: Northern Dry Zone (Zone 3)** (the largest agro-climatic zone in Karnataka, covering districts like Bagalkot, Vijayapura, Gadag, Dharwad, Koppal, and parts of Belagavi). [KPSC AO 2014]

Q8. Which nutrient deficiency causes 'poppod' in groundnut, and how is it managed?

- **Correct Answer: Poppod (empty shells)** (calcium deficiency in the pod zone; It is managed by applying Gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) at 500 kg/ha at the pegging stage (40-45 DAS)). [KPSC AAO 2017]

Q9. What are the core seedling age and spacing guidelines in the SRI system of rice cultivation?

- **Correct Answer: In the System of Rice Intensification (SRI)**, seedlings should be transplanted at 10-12 days of age (2-leaf stage) at a spacing of 25 cm x 25 cm with a single seedling per hill. [KPSC AO 2019]

Q10. Explain the difference in potometer and lysimeter applications.

- **Correct Answer: Potometer** (used to measure transpiration from a plant shoot, whereas a lysimeter is used to measure total evapotranspiration (evaporation + transpiration) from a cropped field). [KPSC AAO 2017]

Q11. What are the tag colors for Breeder, Foundation, and Certified seeds?

- **Correct Answer: Breeder Seed** has a Golden Yellow tag; **Foundation Seed** has a White tag; and **Certified Seed** has an Azure Blue tag. [KPSC AO 2017]

Q12. Explain the biochemical basis of the Tetrazolium (TZ) test staining.

- **Correct Answer: Dehydrogenase enzymes** in living embryonic tissues reduce the colorless 2,3,5-triphenyl tetrazolium chloride solution to red triphenyl formazan; Dead tissues lack active enzymes and remain unstained. [KPSC AO 2014]

Q13. What are the isolation distance requirements for foundation seed of self-pollinated rice and hybrid maize?

- **Correct Answer: Foundation seed production** of self-pollinated rice requires a minimum isolation distance of 3 meters, whereas hybrid maize requires 400 meters. [KPSC AAO 2015 / 2017]

Q14. State the formula for the Real Value of Seed (RVS).

- **Correct Answer: Real Value of Seed (RVS)** (calculated as: $(\text{Purity \%} \times \text{Germination \%}) / 100$; It determines the utility value of a seed lot). [KPSC AO 2017]

Q15. Differentiate between epigeal and hypogeal seed germination.

- **Correct Answer: In epigeal germination**, the hypocotyl elongates and pushes the cotyledons above the soil surface (e.g., beans, castor); In hypogeal germination, the epicotyl

elongates while the cotyledons remain below the soil surface (e.g., chickpea, wheat). [KPSC AO 2019]

Q16. Name the physiological disorders in mango caused by Boron deficiency and convective heat.

- **Correct Answer: Boron deficiency (combined** with toxic gases from brick kilns) causes 'Black Tip' of mango; Convective soil heat during fruit development causes 'Spongy Tissue' in Alphonso mangoes. [KPSC AO 2014 / KPSC AAO 2017]

Q17. What are the commercial propagation methods for Mango and Sapota?

- **Correct Answer: Mango** (commercially propagated by Epicotyl (stone) grafting or veneer grafting; Sapota is propagated by Approach grafting or softwood grafting on Rayan / Khirni (Manilkara hexandra) rootstock). [KPSC AO 2014 / KPSC AAO 2015]

Q18. Which growth regulator is used to overcome alternate bearing in mango, and what is its mode of action?

- **Correct Answer: Paclobutrazol (Cultar)** (applied to the soil; It inhibits gibberellin biosynthesis, restricting vegetative growth and promoting early, regular flower bud differentiation). [KPSC AO 2017]

Q19. What causes Blossom End Rot in tomato and Whiptail in cauliflower?

- **Correct Answer: Blossom End Rot in tomato** (calcium deficiency; Whiptail in cauliflower is caused by molybdenum deficiency, which is common in acidic soils (pH < 5.5)). [KPSC AAO 2015 / KPSC AO 2017]

Q20. How does 1-Methylcyclopropene (1-MCP) extend the post-harvest shelf life of fruits?

- **Correct Answer: 1-MCP** (a gaseous compound that binds to ethylene receptors in plant cells, blocking ethylene action; This delays ripening, reduces respiration, and slows down softening and senescence). [KPSC AO 2017]

Q21. What are the pH ranges for Saline, Sodic, and Acid soils?

- **Correct Answer: Acid soils: pH < 6.5; Saline soils: pH < 8.5** (usually neutral to slightly alkaline).

Q22. Write down the electrical conductivity (EC) threshold for saline soils in dS/m at 25°C.

- **Correct Answer: Critical EC threshold for saline soils (> 4.0 dS/m** (KPSC AAO 2015)).

Q23. What is the C:N ratio of normal arable agricultural soils?

- **Correct Answer: The C:N ratio** of normal arable agricultural soils ranges from 10:1 to 12:1 (KPSC AAO 2015).

Q24. Name the primary clay mineral in red soils and state its structure type.

- **Correct Answer: Kaolinite** (the primary clay mineral in red soils, and it is a 1:1 type non-expanding silicate clay (KPSC AO 2014)).

Q25. What are the dominant soil orders of Karnataka's Red and Black soils respectively?

- **Correct Answer: Red soils belong** to Alfisols (and Inceptisols); Black soils belong to Vertisols (KPSC AAO 2017).

Q26. What is the standard particle density of mineral soils used for porosity calculations?

- **Correct Answer: Standard average particle density of mineral soils** (2.65 g/cm³ (KPSC AAO 2017)).

Q27. Which nutrient deficiency causes 'whip tail' in cauliflower?

- **Correct Answer: Molybdenum (Mo) deficiency** causes whip tail in cauliflower (KPSC AO 2017).

Q28. What is the biuret limit in urea for foliar application according to the FCO?

- **Correct Answer: Biuret limit in urea for foliar spray** (0.5% (soil application limit is 1.5%) (KPSC AO 2017)).

Q29. What is the equivalent basicity of calcium cyanamide?

- **Correct Answer: Equivalent basicity of calcium cyanamide** (63 (KPSC AAO 2015)).

Q30. Which indicator plant is used to detect Boron deficiency in fields?

- **Correct Answer: Sunflower** (the classic indicator plant used to detect Boron deficiency (KPSC AO 2017)).

Q31. What is the minimum viable cell count for Rhizobium in carrier-based biofertilizers at the time of expiry?

- **Correct Answer: Minimum viable count** (10^7 cells per gram of carrier (10^8 cells/g at manufacture) (KPSC AAO 2017)).

Q32. Which bacteria are responsible for the oxidation steps of Nitrification?

- **Correct Answer: Nitrosomonas oxidizes ammonium** (NH_4^+) to nitrite (NO_2^-), and Nitrobacter oxidizes nitrite (NO_2^-) to nitrate (NO_3^-) (KPSC AAO 2017).

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- **Correct Answer: The symbiotic association** between the water fern Azolla and the cyanobacterium Anabaena azollae (KPSC AO 2017).

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- **Correct Answer: Gluconic acid (and 2-ketogluconic acid)** (the major organic acid produced by PSB (KPSC AO 2017)).

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- **Correct Answer: AM fungi belong** to the phylum Glomeromycota (KPSC AO 2014).

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- **Correct Answer: A-line** (the male-sterile line; the B-line is the maintainer line (used to maintain the A-line); and the R-line is the restorer line (used to restore fertility in the F1 hybrid); (KPSC AAO 2017)).

Q48. How many backcrosses are typically required to recover over 99% of the recurrent parent's genome?

- **Correct Answer: Typically, 6 to 8 backcrosses (BC6 to BC8)** (required to recover approximately 99.2% to 99.6% of the recurrent parent's genetic background; (KPSC AAO 2017)).

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- **Correct Answer: Maternal inheritance (cytoplasmic inheritance)** (controlled by genes located in the organelle genomes, namely mitochondrial DNA (mtDNA) and chloroplast DNA (cpDNA), and is inherited through the female gamete; (KPSC AAO 2015)).

Q50. What is the difference between General Combining Ability (GCA) and Specific Combining Ability (SCA) in terms of gene action?

- **Correct Answer: GCA** (governed by additive genetic variance and additive $\times$ additive gene action, which is fixable; SCA is governed by non-additive gene action, including dominance and epistatic variances, which is non-fixable but exploitable in F1 hybrids).

Q51. What is the primary difference between somatic embryogenesis and organogenesis?

- **Correct Answer: Somatic embryogenesis regenerates** plants via bipolar structures containing both root and shoot apices, which develop directly or indirectly; Organogenesis produces unipolar shoots or roots that require separate media to develop the missing root or shoot pole.

Q52. What enzymes are used for plant protoplast isolation?

- **Correct Answer: Mixture of cellulase (to digest cellulose) and pectinase (to digest pectin)** (used to enzymatically remove the plant cell wall and isolate viable protoplasts; (KPSC AO 2017)).

Q53. What is the function of T4 DNA ligase in genetic engineering?

- **Correct Answer: T4 DNA ligase** joins double-stranded DNA fragments by catalyzing the formation of a phosphodiester bond between the 3'-OH and 5'-phosphate ends of the DNA strands; (KPSC AO 2014).

Q54. What plasmid is the classic cloning vector containing ampicillin and tetracycline resistance genes?

- **Correct Answer: pBR322** (the classic plasmid cloning vector containing two selectable marker genes: bla (ampicillin resistance) and tetR (tetracycline resistance); (KPSC AAO 2017)).

Q55. What is the role of the PAM sequence in CRISPR-Cas9 gene editing?

- **Correct Answer: Protospacer Adjacent Motif (PAM, typically 5'-NGG-3' for Cas9)** (a short DNA sequence next to the target site; It is required for Cas9 binding and cleavage; without it, Cas9 cannot cut the target DNA).

Q56. What is the thermostable DNA polymerase used in PCR, and what is its limitation?

- **Correct Answer: Taq DNA polymerase, isolated from *Thermus aquaticus*,** (the primary enzyme used in PCR; Its main limitation is that it lacks 3'-to-5' proofreading exonuclease activity, leading to a higher mutation error rate during amplification).

Q57. How does Sanger DNA sequencing terminate chain elongation?

- **Correct Answer: Sanger sequencing uses** dideoxynucleotide triphosphates (ddNTPs) as chain terminators because they lack the 3'-OH group needed for DNA polymerase to form the next phosphodiester bond; (KPSC AO 2017).

Q58. What is the encapsulating agent used to produce synthetic seeds?

- **Correct Answer: Sodium alginate (which polymerizes into calcium alginate beads upon exposure to calcium chloride)** (the standard matrix used to encapsulate somatic embryos or shoot buds; (KPSC AO 2014)).

Q59. Which apex regulatory body in India approves the commercial release of GM crops?

- **Correct Answer: Genetic Engineering Appraisal Committee (GEAC), which functions under the Ministry of Environment, Forest and Climate Change (MoEFCC),** (the statutory body in India; (KPSC AO 2014)).

Q60. What technology was used to develop the first commercial GM food crop, the Flavr Savr tomato?

- **Correct Answer: Antisense RNA technology** (used to downregulate the polygalacturonase (PG) enzyme, which degrades pectin in cell walls, thereby delaying fruit softening during ripening; (KPSC AAO 2015)).

Q61. What is the composition and ratio of Bordeaux mixture?

- **Correct Answer: Bordeaux mixture** (composed of Copper Sulphate, Quick Lime (Calcium Oxide), and Water; The standard concentration is 1% (ratio of 1 kg Copper Sulphate : 1 kg Quick Lime : 100 liters of Water) or the historic 5:5:50 formula (5 lbs Copper Sulphate : 5 lbs Lime : 50 gallons Water) (KPSC AO 2017)).

Q62. Which fungus causes the late blight of potato, responsible for the Irish Famine?

- **Correct Answer: Late blight of potato** (the oomycete pathogen *Phytophthora infestans* (KPSC AAO 2017)).

Q63. Which spore stage of *Puccinia graminis tritici* is the dikaryotic, repeating asexual stage on wheat?

- **Correct Answer: Urediniospore (Stage II)** (the repeating asexual dikaryotic spore stage on wheat stems (KPSC AO 2017)).

Q64. What is the causal agent and vector of little leaf of brinjal?

- **Correct Answer: Causal agent** (phytoplasma (a wall-less prokaryote) and the vector is the leafhopper *Hishimonus phycitis* (KPSC AO 2014)).

Q65. Explain the Luthra and Sattar solar heat treatment for loose smut of wheat.

- **Correct Answer: Seeds** (soaked in water for 4 hours in a summer morning to activate dormant internally seed-borne mycelium (*Ustilago nuda tritici*), then spread on concrete in the hot sun for 4 hours to kill the activated mycelium without affecting seed germination (KPSC AO 2017)).

Q66. Which systemic fungicide specifically controls smuts (Basidiomycetes)?

- **Correct Answer: Carboxin (Vitavax)** (a systemic oxathiin fungicide that specifically controls smuts by inhibiting succinate dehydrogenase (KPSC AAO 2017)).

Q67. What are the key internal symptoms of red rot of sugarcane?

- **Correct Answer: Internal tissues of** split stalks show red discoloration with diagnostic white transverse bands, accompanied by a sour, acidic fermentation odor (KPSC AO 2017).

Q68. Name the quarantine nematode pest of potato in India and its restricted location.

- **Correct Answer: Golden cyst nematode (*Globodera rostochiensis* and *Globodera pallida*), which** (restricted under domestic quarantine in the Nilgiri and Palni hills of Tamil Nadu (KPSC AO 2017)).

Q69. What are the modified balancing hindwings of Diptera called?

- **Correct Answer: They** (called halteres, which are small club-like structures that function as gyroscopic sensors for flight stability (KPSC AO 2017)).

Q70. Which type of antennae are elbowed and found in honeybees?

- **Correct Answer: Geniculate antennae, which** have a long basal segment (scape) and subsequent segments (pedicel and flagellum) bent at a sharp angle (KPSC AO 2017).

Q71. Which organism is used for the biological control of sugarcane pyrrilla?

- **Correct Answer: *Epiricania melanoleuca*, a** lepidopteran ectoparasitoid whose larvae feed on the nymphs and adults of *Pyrilla perpusilla* (KPSC AO 2017).

Q72. Define Economic Injury Level (EIL) in IPM.

- **Correct Answer: EIL** (the lowest pest population density that will cause economic damage, where the cost of control equals the value of yield saved (KPSC AO 2017)).

Q73. Which hormone is responsible for maintaining larval characters in insects?

- **Correct Answer: Juvenile Hormone (JH), which** (secreted by the endocrine glands corpora allata (KPSC AO 2017)).

Q74. Which insect vector transmits the Rice Tungro Virus?

- **Correct Answer: The green leafhopper, *Nephotettix virescens*,** transmits both components of the Rice Tungro Virus complex (KPSC AO 2017).

Q75. Name the microsporidian pathogen causing Pebrine disease in silkworms.

- **Correct Answer: *Nosema bombycis*, which causes fatal infection and** (transmitted transovarially (vertically) (KPSC AAO 2017)).

Q76. What chemical group does the insecticide Imidacloprid belong to and what is its target site?

- **Correct Answer: Imidacloprid belongs to** the Neonicotinoid group and acts as an agonist on postsynaptic Nicotinic Acetylcholine Receptors (nAChR) (KPSC AAO 2017).

Q77. Which draft cattle breed of Karnataka was bred by Tippu Sultan for military transport?

- **Correct Answer: Amritmahal, which** (developed under the patronage of the Kings of Mysore for speed and endurance (KPSC AAO 2017)).

Q78. Which metabolic disease in dairy cows is caused by hypocalcemia and shows subnormal temperature?

- **Correct Answer: Milk fever (parturient paresis),** caused by acute calcium loss into colostrum right after calving, causing hypothermia and recumbency (KPSC AO 2017).

Q79. What is the standard temperature and time for HTST pasteurization?

- **Correct Answer: HTST pasteurization requires** heating milk to 72°C and holding it for exactly 15 seconds before rapid cooling (KPSC AO 2017).

Q80. Why is post-mortem examination prohibited for animals suspected of dying from Anthrax?

- **Correct Answer: To prevent exposure** of vegetative *Bacillus anthracis* cells to oxygen, which induces sporulation, contaminating the soil and posing zoonotic risk (KPSC AO 2017).

Q81. What is the largest and driest agro-climatic zone in Karnataka, and what is its headquarters?

- **Correct Answer: Northern Dry Zone (Zone 3)** (the largest and driest, headquartered at Vijayapura; (KPSC AO 2017)).

Q82. In which year was UAS Bangalore established under Mysore Act No. 22?

- **Correct Answer: UAS Bangalore** (established in 1963 and inaugurated in 1965; (KPSC AO 2014)).

Q83. Which unique Farmer ID system integrates land registry (Bhoomi) in Karnataka?

- **Correct Answer: The FRUITS portal** (Farmer Registration and Unified Beneficiary Information System); (Latest 2020).

Q84. What is the cash incentive for minor millets under Raitha Siri, and the maximum land limit?

- **Correct Answer: Rs. 10,000 per hectare, with a maximum limit of 2 hectares per farmer. (Latest 2019).**

Q85. What is the recommended polythene sheet thickness for lining farm ponds under Krishi Bhagya?

- **Correct Answer: 300 to 500 microns;** (KPSC AAO 2017).

Q86. What is the state incentive under Ksheera Dhare for supplying milk to cooperative societies?

- **Correct Answer: Rs. 5 per liter of milk. (KPSC AO 2017).**

Q87. Which project, assisted by the World Bank, is executed by the Karnataka Watershed Development Department?

- **Correct Answer: The REWARD project** (Rejuvenating Watersheds for Agricultural Resilience through Innovative Development); (KPSC AO 2017).

Q88. Which horticultural product from Uttara Kannada was awarded a GI tag in 2023?

- **Correct Answer: Kari Ishad Mango** of Ankola; (Latest 2023).

Q89. What is the formula for Cost C2, the comprehensive cost of cultivation?

- **Correct Answer: Cost C2 = Cost B2 (Cost B1 + rental value of owned land + rent for leased-in land) + imputed value of family labor;** (KPSC AAO 2017).

Q90. In which stage of the Law of Variable Proportions does the rational farm manager operate?

- **Correct Answer: Stage II (where average product (declining, and marginal product is positive but declining);** (KPSC AO 2017)).

Q91. What is the input optimization condition in the factor-product relationship?

- **Correct Answer: Marginal Value Product (MVP) must equal Marginal Input Cost (MIC), i.e., MVP = MIC;** (KPSC AO 2017).

Q92. What is the optimization condition for the least-cost combination of inputs in the factor-factor relationship?

- **Correct Answer: Marginal Rate of Technical Substitution of X for Y must equal the input price ratio, i.e., $MRTS_{xy} = P_x / P_y$;** (KPSC AAO 2015).

Q93. On which date was NABARD established and which committee recommended it?

- **Correct Answer: Established on July 12, 1982, on the recommendation of the B; Sivaraman Committee.**

Q94. What is the interest rate for prompt payers of short-term crop loans under the Kisan Credit Card (KCC) scheme?

- **Correct Answer: 4% per annum (after 3% prompt repayment incentive on the base rate of 7%);** (KPSC AAO 2017).

Q95. Which committee recommended the establishment of Regional Rural Banks (RRBs) in 1975?

- **Correct Answer: M. Narasimham Committee. (KPSC AAO 2017).**

Q96. Which agency recommends the Minimum Support Prices (MSP) for agricultural crops in India?

- **Correct Answer: Commission for Agricultural Costs and Prices (CACP); (KPSC AO 2017).**

Q97. What is the corpus of the AgriSURE fund launched by NABARD in 2024?

- **Correct Answer: Rs. 750 crore. (Latest 2024).**

Q98. What is the approved MSP for Paddy (Common) for the 2024-25 crop year?

- **Correct Answer: Rs. 2,300 per quintal. (Latest 2024).**

Q99. Which brand name is mandatory for all subsidized chemical fertilizers under the One Nation One Fertilizer scheme?

- **Correct Answer: BHARAT (e.g., Bharat Urea); (Latest 2024).**

Q100. What is the name of the central database portal developed for AgriStack implementations?

- **Correct Answer: Farmer Registry / Digital Agriculture Mission (AgriStack); (Latest 2021).**

--- General Paper Booster ---

Paper 1: General Studies Revision Booster

This section compiles the most high-yield, frequently asked concepts for the KEA/KPSC Paper 1 (General Studies) exam, including Karnataka History, Geography, Polity, Science, and 10 expected MCQs.

Indian Constitution & State Administration

- **Preamble:** The words 'Socialist', 'Secular', and 'Integrity' were added to the Preamble by the **42nd Amendment Act of 1976**.
- **Fundamental Rights:** Articles 12-35 are in Part III. Article 21A guarantees the **Right to Education** (added by the 86th Amendment Act in 2002).
- **Directive Principles:** Article 40 mandates the organization of **Village Panchayats**, which forms the basis of rural self-governance.
- **Panchayat Raj Amendments:** The **73rd Amendment Act of 1992** added Part IX and the **11th Schedule** to the Constitution, giving constitutional status to Panchayats. Gram Sabha is the foundation of the Panchayat Raj system (Article 243A).

History of Karnataka & India

- **Halmidi Inscription:** The earliest written evidence of the Kannada language, dating back to **450 AD**, belonging to the Kadamba dynasty of Banavasi.
- **Kadambas:** The first ruling dynasty of Karnataka, founded by **Mayurasharma** with Banavasi as the capital.
- **Rashtrakutas:** Known for their patronage of art and literature. King **Amoghavarsha Nrupatunga I** authored **Kavirajamarga**, the earliest available work on Kannada poetics.
- **Vijayanagara Empire:** Founded in **1336 AD** on the banks of the Tungabhadra River by Harihara I and Bukka Raya I. Its greatest ruler, Krishnadevaraya, wrote **Amuktamalyada**.
- **Rajarshi Nalvadi Krishnaraja Wodeyar:** Ruler of Mysore (1902-1940) who implemented path-breaking social reforms, including free primary education, the first reservation policy, and establishing the Representative Assembly. Mahatma Gandhi called him a **Rajarshi** (royal saint).
- **Vidurashwatha:** Located in Chikkaballapur district, it is known as the '**Jallianwala Bagh of Karnataka**' because British police opened fire on peaceful freedom fighters in April 1938, killing around 35 people.

Geography of Karnataka & India

- **Jog Falls:** The highest untiered waterfall in India, formed by the **Sharavathi River** in Shivamogga district, Karnataka.
- **River Systems:** East-flowing rivers include Krishna (tributaries: Ghataprabha, Malaprabha, Bhima, Tungabhadra) and Cauvery (tributaries: Hemavati, Kabini, Harangi). West-flowing rivers include Sharavathi, Netravathi, and Kali.
- **National Parks:** Karnataka has **5 National Parks:** Bandipur (famous tiger reserve), Nagarhole (Kabini basin), Kudremukh (Western Ghats), Anshi (Kali Tiger Reserve), and Bannerghatta (Bengaluru).
- **Soils & Topography:** **Alfisols (Red Soils)** cover the largest geographical area in Karnataka, followed by Vertisols (Black Soils) in northern dry plains. The highest peak is **Mullayanagiri** (1,930 m) in Chikkamagaluru.

Rural Development & Panchayat Raj Schemes

- **Karnataka Gram Swaraj Act, 1993:** Establishes the 3-tier structure of local self-governance in the state: Gram Panchayat (village), Taluk Panchayat (intermediate), and Zilla Panchayat (district).
- **MGNREGA:** Guarantees **100 days of wage employment** per financial year to every rural household whose adult members volunteer for unskilled manual work.

- **Renaming of Mysore State:** Mysore State was renamed as **Karnataka** on **November 1, 1973**, during the tenure of Chief Minister D. Devaraj Urs.

General Science & Environment

- **Vitamins:** Vitamin A (Retinol) deficiency causes Night Blindness; Vitamin C (Ascorbic acid) deficiency causes Scurvy; Vitamin D deficiency causes Rickets.
- **Ecology:** The **Western Ghats** of Karnataka are recognized as one of the world's 8 'hottest hotspots' of biological diversity and a UNESCO World Heritage Site.
- **Greenhouse Effect:** Major greenhouse gases include Carbon dioxide (CO₂), Methane (CH₄), and Nitrous oxide (N₂O), which trap infra-red radiation in the atmosphere.

Expected General Studies MCQs

Q1. The earliest available lithic (stone) inscription in the Kannada language, the Halmidi Inscription, is associated with which dynasty?

- A. Kadambas of Banavasi
- B. Western Gangas
- C. Badami Chalukyas
- D. Rashtrakutas
- **Correct Answer: A**
- **Explanation:** The Halmidi inscription (dated to c. 450 CE) is the earliest known Kannada language stone inscription. It was discovered in the Halmidi village of Hassan district, Karnataka, and belongs to the reign of the Kadamba King Kakusthavarma.

Q2. Who among the following Kannada leaders was bestowed with the title 'Karnataka Kulapurohita' (High Priest of the Kannada family) for his spearheading role in the Karnataka Unification (Ekikarana) movement?

- A. Kengal Hanumanthaiah
- B. Alur Venkata Rao
- C. Gorur Ramaswamy Iyengar
- D. S. Nijalingappa
- **Correct Answer: B**
- **Explanation:** Alur Venkata Rao is known as 'Karnataka Kulapurohita'. He initiated the Karnataka Ekikarana movement, founded the Kannada Sahitya Parishat in 1915, and wrote the seminal historical work 'Karnataka Gatha Vaibhava' in 1917 to inspire Kannadigas.

Q3. The Vidurashwatha tragedy of April 25, 1938, which is popularly referred to as the 'Jallianwala Bagh of Karnataka', occurred in which of the following present-day districts?

- A. Chikkaballapur
- B. Tumakuru
- C. Kolar
- D. Mandya
- **Correct Answer: A**
- **Explanation:** On April 25, 1938, at Vidurashwatha in Gauribidanur taluk of Chikkaballapur district (formerly part of Kolar district), police opened fire on a crowd of freedom fighters gathered to hoist the national flag, killing around 32 people. This event is known as the 'Jallianwala Bagh of Karnataka'.

Q4. In 1918, the Maharaja of Mysore, Nalvadi Krishnaraja Wodeyar, appointed a committee under the chairmanship of

Chief Justice Leslie Miller. What was the primary objective of this committee?

- A. To recommend administrative reforms for the Representative Assembly
- B. To investigate the feasibility of the Krishnaraja Sagara Dam
- C. To recommend measures for the adequate representation of non-Brahmin communities in state services
- D. To reform primary and secondary education in the state
- **Correct Answer: C**
- **Explanation:** In August 1918, Nalvadi Krishnaraja Wodeyar appointed the Miller Committee (headed by Chief Justice Leslie Miller) to look into the representation of backward and non-Brahmin communities in public service. The committee submitted its report in 1919, recommending reservation in state services, which was accepted.

Q5. Under the administration of Nalvadi Krishnaraja Wodeyar, Mysore became the first state in Asia to generate hydroelectric power. On which river and at which site was this hydroelectric station established in 1902?

- A. Sharavathi River at Jog Falls
- B. Cauvery River at Shivanasamudra
- C. Krishna River at Almatti
- D. Kabini River at Heggadadevankote
- **Correct Answer: B**
- **Explanation:** In 1902, during the reign of Nalvadi Krishnaraja Wodeyar, the Shivanasamudra Hydroelectric Project was established on the Cauvery River. It was the first major hydroelectric power station in Asia, and the power generated was transmitted to the Kolar Gold Fields (KGF) first.

Q6. Who was the Kadamba ruler who established the independent Kadamba dynasty and made Banavasi his capital, as described in the Talagunda pillar inscription?

- A. Mayurasharma
- B. Kakusthavarma
- C. Ravivarma
- D. Bhagiratha
- **Correct Answer: A**
- **Explanation:** Mayurasharma (reigned c. 345-365 CE) was the founder of the Kadamba dynasty of Banavasi, the first native kingdom to rule over Kannada-speaking regions. The Talagunda pillar inscription details his life and how he defeated the Pallavas to establish his kingdom.

Q7. The historic publication 'Jaya Karnataka', which played a monumental role in mobilizing public support for the integration of Kannada-speaking regions, was founded and edited by whom?

- A. Huilgol Narayana Rao
- B. Alur Venkata Rao
- C. Deputy Channabasappa
- D. B. M. Srikanthiah
- **Correct Answer: B**
- **Explanation:** 'Jaya Karnataka' was a highly influential monthly journal started in 1922 by Alur Venkata Rao. It served as a powerful literary and political forum to advocate for the unification of Karnataka (Ekikarana) and to revive Kannada cultural pride.

Q8. Mullayanagiri, the highest peak in Karnataka, is located in which of the following hill ranges?

- A. Baba Budangiri Hills (Chandra Drona Range)
- B. Biligirirangana Hills
- C. Gopalaswamy Hills
- D. Kudremukh Range
- **Correct Answer: A**
- **Explanation:** Mullayanagiri is the highest peak in Karnataka (elevation of approximately 1,930 meters/6,330 feet). It is located in the Chandra Drona Hill Ranges (popularly known as the Baba Budangiri Range) of the Western Ghats in the Chikkamagaluru district.

Q9. Which of the following is the correct matching of the waterfalls in Karnataka with the rivers they are formed by?

- 1. Jog Falls - Sharavathi
- 2. Gokak Falls - Ghataprabha
- 3. Shivanasamudra Falls - Cauvery
- 4. Abbey Falls - Kaveri (tributary)
- A. 1 and 3 only
- B. 2 and 4 only
- C. 1, 2, and 3 only
- D. All of the above
- **Correct Answer: D**
- **Explanation:** Jog Falls is formed by the Sharavathi River in Shimoga; Gokak Falls is formed by the Ghataprabha River in Belagavi; Shivanasamudra Falls is formed by the Cauvery River in Chamaraajanagar/Mandya; Abbey Falls is located in Coorg on a tributary of the Cauvery.

Q10. The Almatti Dam, a major irrigation and power project in North Karnataka, is constructed across which river?

- A. Tungabhadra
- B. Krishna
- C. Ghataprabha
- D. Bhima
- **Correct Answer: B**
- **Explanation:** The Almatti Dam is the main reservoir of the Upper Krishna Project (UKP) and is built across the Krishna River in the Bagalkot district of Northern Karnataka.

Q11. Which type of soil covers the largest geographical area of Karnataka and is widely distributed in both the Southern and Northern Maidan regions?

- A. Black Soil
- B. Red Soil
- C. Laterite Soil
- D. Coastal Alluvial Soil
- **Correct Answer: B**
- **Explanation:** Red soils (specifically red loamy and red gravelly clay soils) cover the largest area in Karnataka, occupying over 50% of the state's territory. They are found extensively in South Karnataka and parts of North Karnataka. Black soils (Regur) are second and dominate North Karnataka.

Q12. Consider the following statements regarding the national parks in Karnataka:

- 1. Nagarahole National Park is also known as the Rajiv Gandhi National Park and is spread across Kodagu and Mysore districts.
- 2. Bandipur National Park is part of the Nilgiri Biosphere Reserve.
- Which of the statements given above is/are correct?
- A. 1 only
- B. 2 only
- C. Both 1 and 2
- D. Neither 1 nor 2
- **Correct Answer: C**
- **Explanation:** Both statements are correct. Nagarahole (Rajiv Gandhi) NP is located in Kodagu and Mysore districts. Bandipur NP, established under Project Tiger in 1973, is contiguous with Madumalai (Tamil Nadu), Wayanad (Kerala), and Nagarahole, forming part of the Nilgiri Biosphere Reserve.

Q13. Which river is famously known as the 'Dakshina Ganga' (Ganges of the South) and originates at Talakaveri in the Brahmagiri Range of the Western Ghats?

- A. Krishna
- B. Godavari
- C. Cauvery
- D. Pennar
- **Correct Answer: C**
- **Explanation:** The Cauvery River originates at Talakaveri in the Kodagu district's Brahmagiri Range. It is revered in South India as 'Dakshina Ganga' and is the lifeblood of southern Karnataka and Tamil Nadu.

Q14. Which tributary of the Krishna river is known for the Gokak Falls, often compared to Niagara Falls for its shape and beauty?

- A. Ghataprabha
- B. Malaprabha
- C. Bhima
- D. Tungabhadra
- **Correct Answer: A**
- **Explanation:** The Gokak Falls is formed by the Ghataprabha River in the Belagavi district of Karnataka. It drops from a height of about 52 meters (171 ft) over a horseshoe-shaped sandstone cliff, resembling a miniature Niagara Falls.

Q15. By which Constitutional Amendment Act was Article 21A, which makes free and compulsory education for children aged 6 to 14 years a Fundamental Right, inserted into the Indian Constitution?

- A. 44th Amendment Act, 1978
- B. 86th Amendment Act, 2002
- C. 91st Amendment Act, 2003
- D. 93rd Amendment Act, 2005
- **Correct Answer: B**

- **Explanation:** The 86th Constitutional Amendment Act, 2002 inserted Article 21A, making the right to education a Fundamental Right for children between the ages of 6 and 14.

Q16. The Karnataka Gram Swaraj and Panchayat Raj Act was originally enacted in 1993 to give effect to which Constitutional Amendment?

- A. 71st Constitutional Amendment Act
- B. 72nd Constitutional Amendment Act
- C. 73rd Constitutional Amendment Act
- D. 74th Constitutional Amendment Act
- **Correct Answer: C**

- **Explanation:** The Karnataka Panchayat Raj Act was enacted in 1993 (and renamed the Karnataka Gram Swaraj and Panchayat Raj Act in 2015) in conformity with the 73rd Constitutional Amendment Act of 1992, which constitutionalized rural local self-governments.

Q17. Which of the following constitutional articles deals with the constitution, powers, and functions of the 'Gram Sabha' under the Panchayati Raj system?

- A. Article 243
- B. Article 243A
- C. Article 243B
- D. Article 243C
- **Correct Answer: B**

- **Explanation:** Article 243A of the Constitution of India provides for the Gram Sabha. It states that a Gram Sabha may exercise such powers and perform such functions at the village level as the Legislature of a State may, by law, provide.

Q18. Under the Karnataka Gram Swaraj and Panchayat Raj Act, who is the administrative head of the Zilla Panchayat at the district level, appointed by the State Government?

- A. President of Zilla Panchayat
- B. Deputy Commissioner (DC)
- C. Chief Executive Officer (CEO)
- D. Chief Accounts Officer (CAO)
- **Correct Answer: C**

- **Explanation:** The Chief Executive Officer (CEO), an IAS officer appointed by the State Government, is the administrative head of the Zilla Panchayat. The President of the Zilla Panchayat is the political/elected head.

Q19. In the context of urban local bodies, the 74th Constitutional Amendment Act added which Part and Schedule to the Constitution of India?

- A. Part IX and 11th Schedule
- B. Part IX-A and 12th Schedule
- C. Part X and 11th Schedule
- D. Part IX-B and 12th Schedule
- **Correct Answer: B**

- **Explanation:** The 74th Amendment Act, 1992, added Part IX-A ('The Municipalities') and the 12th Schedule (which specifies 18 functional focus areas for urban local bodies) to the Indian Constitution. Part IX and the 11th Schedule were added by the 73rd

Amendment for Panchayats.

Q20. Which article of the Indian Constitution mandates the establishment of a State Finance Commission (SFC) every five years to review the financial position of Panchayats and recommend distribution of revenues?

- A. Article 243I
- B. Article 243K
- C. Article 243D
- D. Article 243G
- **Correct Answer: A**
- **Explanation:** Article 243I mandates the Governor of a State to constitute a State Finance Commission every five years to review the financial position of Panchayats and make recommendations. Article 243K deals with the State Election Commission.

Q21. The 'Ganga Kalyana' scheme implemented by the Government of Karnataka primarily aims to provide which of the following benefits to small and marginal farmers?

- A. Quality seeds and bio-fertilizers at subsidized rates
- B. Financial assistance for setting up dairy units
- C. Irrigation facilities through drilling borewells or open wells with pump sets
- D. Soil health cards and testing facilities
- **Correct Answer: C**
- **Explanation:** The Ganga Kalyana Scheme, implemented by various state development corporations (for SC/ST and minorities/OBCs), provides irrigation facilities to dry-land agricultural plots of small and marginal farmers by drilling borewells/digging open wells and installing pump sets.

Q22. The flagship agricultural scheme 'Krishi Bhagya' of the Karnataka Government is designed to promote sustainable agriculture in rain-fed areas. What is its central component?

- A. Establishment of polyhouses and solar pumps
- B. Creation of 'Krishi Hondas' (farm ponds) for rainwater harvesting
- C. Subsidizing tractor rentals for small-scale farmers
- D. Direct cash transfers to farmers during crop failure
- **Correct Answer: B**
- **Explanation:** The 'Krishi Bhagya' scheme focuses on rain-fed farming by promoting rainwater harvesting. Its key component is the construction of a 'Krishi Honda' (farm pond) along with micro-irrigation systems (drip/sprinkler) and diesel/solar pumpsets to provide protective irrigation to crops during critical dry spells.

Q23. The 'Raitha Vidya Nidhi' scheme launched by the Government of Karnataka in 2021 provides scholarships for higher education. Which registry/portal ID of the government is mandatory for farmers' children to claim benefits under this scheme?

- A. Kutumba ID
- B. FRUITS ID
- C. SSP (State Scholarship Portal) Student ID
- D. Kaushalya Kar ID
- **Correct Answer: B**

- **Explanation:** The Mukhya Mantri Raitha Vidya Nidhi scheme utilizes the FRUITS (Farmer Registration and Unified Beneficiary Information System) database. The student's parents must have a registered FRUITS ID to verify their status as farmers, which makes the child eligible for the DBT scholarship.

Q24. The 'Bhoochetna' program, launched in 2009 by the Government of Karnataka in technical collaboration with which international institute, aims to improve the productivity of dry-land crops through soil-nutrient management?

- A. IRRI (International Rice Research Institute)
- B. ICRISAT (International Crops Research Institute for the Semi-Arid Tropics)
- C. CIMMYT (International Maize and Wheat Improvement Center)
- D. FAO (Food and Agriculture Organization)
- **Correct Answer: B**
- **Explanation:** 'Bhoochetna' is a soil health management program launched in 2009 by the Department of Agriculture, Karnataka, in collaboration with ICRISAT (headquartered in Patancheru, Telangana), focusing on reviving soil fertility using micronutrient mapping and applications.

Q25. Under the 'Raitha Vidya Nidhi' scholarship scheme of the Karnataka Government, the target group of beneficiaries has been expanded to include children of:

- A. Agricultural labourers
- B. Weavers and fishermen
- C. Yellow board taxi and auto-rickshaw drivers
- D. All of the above
- **Correct Answer: D**
- **Explanation:** Initially launched for children of registered farmers, the 'Raitha Vidya Nidhi' scheme was subsequently expanded by the state government to include children of landless agricultural labourers, weavers, fishermen, auto-rickshaw/taxi drivers, and tailors to support their higher education.

--- 01 Agronomy ---

Chapter 1: Agronomy

[TIP] Top Repeated Questions

- **Question:** Which is the most critical growth stage for water in wheat?
- **Years Asked:** KPSC AO 2017, KPSC AAO 2017 (Frequency: 2 times)
- **Correct Answer:** Crown Root Initiation (CRI) stage (21 days after sowing)
- **Question:** What is the optimum depth of water to be maintained in a wet nursery of rice?
- **Years Asked:** KPSC AO 2014, KPSC AAO 2015 (Frequency: 2 times)
- **Correct Answer:** 5 cm
- **Question:** Nipping in chickpea is done to promote:
- **Years Asked:** KPSC AO 2017, KPSC AAO 2017 (Frequency: 2 times)
- **Correct Answer:** Lateral branching and more pod formation
- **Question:** The herbicide used for pre-emergence control of weeds in groundnut is:
- **Years Asked:** KPSC AO 2014, KPSC AO 2017 (Frequency: 2 times)
- **Correct Answer:** Pendimethalin

[NOTE] Frequently Asked Facts

- Water Use Efficiency (WUE) is highest in Finger Millet (Ragi) at 13.4 kg/ha-mm.
- Puddling in rice reduces percolation losses and destroys soil structure to create an impervious layer.
- Zero tillage is also known as no-till farming, where the soil is left undisturbed except for seed placement.
- Sorghum sickness is caused by the high C:N ratio of sorghum stubble, causing nitrogen immobilization.
- Relay cropping refers to sowing the next crop before the harvest of the preceding crop (e.g., potato in maize).

Important Crop Varieties

Crop	Varieties / Focus
Rice	IR-8 (Miracle Rice), Jaya (Miracle Rice of India), KRH-2 (First rice hybrid in Karnataka)
Ragi	GPU-28, GPU-48, Indaf-8, Indaf-9, MR-1
Sorghum	CSH-1 (First hybrid, 1964), M-35-1 (Maldandi - popular rabi variety)
Groundnut	TMV-2, GPBD-4 (Rust resistant), Kadiri-6, JL-24

Important Years & Milestones

Year	Event / Significance
1960	Establishment of India's first Agricultural University at Pantnagar (GBPUAT)
1966	High Yielding Varieties Programme (HYVP) launched, starting the Green Revolution
1973	Detailed soil survey of India initiated under AICRIP

Expected MCQs

Q1. The crop weed competition is most critical during:

- A. Sowing time
- B. First 1/3rd of crop duration
- C. Flowering stage
- D. Harvesting stage
- **Correct Answer: B**
- **Explanation:** The critical period of crop-weed competition is generally the first 30-45 days (first 1/3rd of the lifecycle).

Q2. Which crop is known as the 'Camel of Desert'?

- A. Maize
- B. Pearl millet
- C. Sorghum
- D. Finger millet
- **Correct Answer: C**
- **Explanation:** Sorghum is called the camel of desert due to its high drought tolerance and ability to go dormant during stress.

[QUICK] Last-Minute Revision Points

- Always check active weedicides: Pre-emergence (Pendimethalin, Alachlor), Post-emergence (2,4-D, Glyphosate).
- C3 Plants: Rice, Wheat, Soy, Potato. C4 Plants: Maize, Sorghum, Sugarcane, Pearl Millet. CAM: Pineapple, Agave.
- Harvest Index (HI) formula: $(\text{Economic Yield} / \text{Biological Yield}) * 100$.

--- 02 Soil Science ---

Chapter 2: Soil Science

[TIP] Top Repeated Questions

- **Question:** Which clay mineral shows the highest Cation Exchange Capacity (CEC)?
- **Years Asked:** KPSC AO 2017, KPSC AAO 2017 (Frequency: 2 times)
- **Correct Answer:** Vermiculite (followed by Montmorillonite)
- **Question:** The C:N ratio of average arable soil is:
- **Years Asked:** KPSC AO 2014, KPSC AAO 2015 (Frequency: 2 times)
- **Correct Answer:** 10:1 to 12:1
- **Question:** Yellowing of tea leaves is a deficiency symptom of:
- **Years Asked:** KPSC AO 2017, KPSC AO 2014 (Frequency: 2 times)
- **Correct Answer:** Sulfur
- **Question:** Which soil order is characterized by cracking clay soils?
- **Years Asked:** KPSC AO 2017, KPSC AAO 2017, KPSC AO 2014 (Frequency: 3 times)
- **Correct Answer:** Vertisols

[NOTE] Frequently Asked Facts

- The term 'soil' is derived from the Latin word 'Solum' which means floor or ground.
- Hans Jenny (1941) formulated the soil forming equation: $S = f(cl, o, r, p, t)$ where r is relief, p parent material.
- Montmorillonite is a 2:1 expanding type silicate clay mineral with high swelling and shrinkage properties.
- Bulk density of sandy soil (1.6 g/cm³) is higher than clay soil (1.1 g/cm³).
- Acid soils are reclaimed by applying lime (CaCO₃), while alkali/sodic soils are reclaimed by applying gypsum (CaSO₄·2H₂O).

Important Crop Varieties

Crop	Varieties / Focus
Gypsum Requirement	Calculated by Schoonover method to determine amount of Ca needed to replace Na.
Soil Textures	Sandy, Loamy, Clayey. Clay particles are < 0.002 mm in size.

Important Years & Milestones

Year	Event / Significance
1975	Introduction of Soil Taxonomy by USDA (comprehensive soil classification system)
2015	Launch of Soil Health Card Scheme in India (19th February)

Expected MCQs

Q1. Which element is a constituent of chlorophyll and is required for active photosynthesis?

- A. Nitrogen
- B. Iron
- C. Magnesium
- D. Manganese
- **Correct Answer: C**
- **Explanation:** Magnesium (Mg) is the central atom in the chlorophyll molecule.

Q2. The dominant soil order found in Karnataka is:

- A. Vertisols
- B. Alfisols
- C. Entisols
- D. Aridisols
- **Correct Answer: B**
- **Explanation:** Alfisols (Red soils) cover the largest area in Karnataka, followed by Vertisols (Black soils).

[QUICK] Last-Minute Revision Points

- CEC Order: Organic Matter (Humus) > Vermiculite > Montmorillonite > Illite > Kaolinite.
- Essential plant nutrients are 17. The 17th is Nickel (Ni), discovered by Dalton in 1987.
- Active acidity is due to H^+ ions in soil solution. Reserve/exchangeable acidity is due to Al^{3+} and H^+ on exchange complex.

--- 03 Genetics Plant Breeding ---

Chapter 3: Genetics & Plant Breeding

[TIP] Top Repeated Questions

- **Question:** Who is the Father of Genetics?
- **Years Asked:** KPSC AO 2014, KPSC AO 2017 (Frequency: 2 times)
- **Correct Answer:** Gregor Johann Mendel
- **Question:** Double cross hybrid technique in maize was first proposed by:
- **Years Asked:** KPSC AO 2017, KPSC AAO 2017, KPSC AO 2014 (Frequency: 3 times)
- **Correct Answer:** D. F. Jones (1918)
- **Question:** The term 'heterosis' was coined by:
- **Years Asked:** KPSC AO 2014, KPSC AAO 2015 (Frequency: 2 times)
- **Correct Answer:** G. H. Shull (1914)

[NOTE] Frequently Asked Facts

- Gregor Mendel presented his hybridization work in 1865, published in 1866; rediscovered in 1900 by Hugo de Vries, Carl Correns, and Erich von Tschermak.
- Pure Line Theory was proposed by W. Johannsen (1903) working on Princess variety of common bean (*Phaseolus vulgaris*).
- Cytoplasmic Male Sterility (CMS) is utilized for hybrid seed production in crops like onion and pearl millet.
- Emasculation is the removal of stamens/anthers from bisexual flowers before dehiscing without affecting the female reproductive organ.

Important Crop Varieties

Crop	Varieties / Focus
Maize Hybrids	Ganga-1, Ganga-101, Deccan (First double cross hybrids in India, 1961)
Wheat	Kalyansona, Sonalika (Semi-dwarf varieties developed from Mexican wheat)

Important Years & Milestones

Year	Event / Significance
1903	W. Johannsen coined the terms 'Gene', 'Genotype', and 'Phenotype'
1953	Watson and Crick proposed the Double Helix model of DNA

Expected MCQs

Q1. When F1 is crossed with either of its parents, it is known as:

- A. Test cross
- B. Back cross
- C. Reciprocal cross
- D. Top cross

• **Correct Answer: B**

• **Explanation:** Crossing F1 with any parent is a Back cross. Crossing F1 with homozygous recessive parent is a Test cross.

[QUICK] Last-Minute Revision Points

- Mitosis produces two genetically identical diploid cells. Meiosis produces four genetically diverse haploid cells.
- Law of Segregation is also known as the Law of Purity of Gametes.

--- 04 Plant Biotech ---

Chapter 4: Plant Biotechnology

[TIP] Top Repeated Questions

- **Question:** Which molecular marker is co-dominant and PCR-based?
- **Years Asked:** KPSC AO 2017, KPSC AAO 2017 (Frequency: 2 times)
- **Correct Answer:** Simple Sequence Repeats (SSR) or Microsatellites
- **Question:** Who is known as the Father of Plant Tissue Culture?
- **Years Asked:** KPSC AO 2014, KPSC AO 2017 (Frequency: 2 times)
- **Correct Answer:** Gottlieb Haberlandt
- **Question:** Which organism is known as 'Nature's Genetic Engineer'?
- **Years Asked:** KPSC AO 2014, KPSC AO 2017, KPSC AAO 2015 (Frequency: 3 times)
- **Correct Answer:** Agrobacterium tumefaciens

[NOTE] Frequently Asked Facts

- Restriction endonucleases are molecular scissors that cut DNA at specific palindromic sequences.
- Ti (Tumor-inducing) plasmid of Agrobacterium is widely used for plant transformation.
- Anther/Pollen culture is used to produce haploid plants, first demonstrated by Guha and Maheshwari (1964) in Datura.
- Biolistics or Gene Gun method uses gold or tungsten microparticles coated with DNA.

Important Crop Varieties

Crop	Varieties / Focus
Bt Cotton	Bollgard I (expressing Cry1Ac gene, approved in India in 2002), Bollgard II (Cry1Ac and Cry2Ab)
Golden Rice	Engineered with beta-carotene biosynthesis genes (phytoene synthase and carotene desaturase) to combat Vitamin A deficiency.

Important Years & Milestones

Year	Event / Significance
1983	First transgenic plant (tobacco resistant to antibiotics) produced
2002	Commercial approval of Bt Cotton in India

Expected MCQs

Q1. Which marker is dominant and non-locus specific?

- A. RFLP
- B. RAPD
- C. SSR
- D. SNP

- **Correct Answer: B**

- **Explanation:** Random Amplified Polymorphic DNA (RAPD) is a dominant marker utilizing short arbitrary primers.

[QUICK] Last-Minute Revision Points

- Totipotency is the cellular capacity to regenerate into a whole organism.
- PCR steps: Denaturation (94°C), Annealing (50-60°C), Extension (72°C using Taq polymerase).

--- 05 Agri Microbiology ---

Chapter 5: Agricultural Microbiology

[TIP] Top Repeated Questions

- **Question:** Which bacterium is responsible for symbiotic nitrogen fixation in legume root nodules?
- **Years Asked:** KPSC AO 2017, KPSC AAO 2017, KPSC AO 2014 (Frequency: 3 times)
- **Correct Answer:** Rhizobium species
- **Question:** Frankia is a nitrogen-fixing:
- **Years Asked:** KPSC AO 2014, KPSC AO 2017 (Frequency: 2 times)
- **Correct Answer:** Actinomycete (associated with non-leguminous plants like Casuarina)
- **Question:** Which biofertilizer is commonly used in low-land flooded rice cultivation?
- **Years Asked:** KPSC AO 2017, KPSC AAO 2017, KPSC AAO 2015 (Frequency: 3 times)
- **Correct Answer:** Azolla-Anabaena complex

[NOTE] Frequently Asked Facts

- Rhizobium forms symbiotic associations with legumes. Specificity: R. leguminosarum (Pea), R. meliloti (Alfalfa), R. japonicum (Soybean).
- Azotobacter is a free-living, aerobic nitrogen-fixing bacterium.
- Mycorrhizae (specifically VAM - Vesicular Arbuscular Mycorrhiza) help in mobilizing phosphorus from soil to plant roots.
- Phosphate Solubilizing Bacteria (PSB) include Pseudomonas striata and Bacillus polymyxa.

Important Crop Varieties

Crop	Varieties / Focus
Rhizobium cultures	Supplied in peat-based carriers; applied to seeds using jaggery solution as sticker.

Important Years & Milestones

Year	Event / Significance
1888	Beijerinck isolated Rhizobium under the name Bacillus radicicola

Expected MCQs

Q1. Which of the following is a non-symbiotic nitrogen fixing bacterium?

- A. Rhizobium
- B. Azotobacter

- C. Frankia
- D. Anabaena azollae
- **Correct Answer: B**
- **Explanation:** Azotobacter is free-living (non-symbiotic) and aerobic.

[QUICK] Last-Minute Revision Points

- Nitrosomonas converts ammonia to nitrite (NO_2^-). Nitrobacter converts nitrite to nitrate (NO_3^-).
- Denitrification is an anaerobic process carried out by Pseudomonas and Thiobacillus.

--- 06 Plant Pathology ---

Chapter 6: Plant Pathology

[TIP] Top Repeated Questions

- **Question:** Bordeaux mixture was discovered by:
- **Years Asked:** KPSC AO 2017, KPSC AAO 2017, KPSC AO 2014 (Frequency: 3 times)
- **Correct Answer:** P. M. A. Millardet (1885) for controlling Downy Mildew of Grapes
- **Question:** Late Blight of Potato is caused by:
- **Years Asked:** KPSC AO 2017, KPSC AAO 2017, KPSC AAO 2015 (Frequency: 3 times)
- **Correct Answer:** Phytophthora infestans (responsible for the Irish Famine of 1845)
- **Question:** The first plant virus discovered was:
- **Years Asked:** KPSC AO 2014, KPSC AO 2017 (Frequency: 2 times)
- **Correct Answer:** Tobacco Mosaic Virus (TMV) by Ivanowsky/Beijerinck

[NOTE] Frequently Asked Facts

- Bordeaux mixture consists of Copper Sulfate, Lime, and Water (5:5:50 ratio).
- Loose smut of wheat (*Ustilago nuda tritici*) is an internally seed-borne disease, controlled by hot water treatment (proposed by Jensen) or systemic fungicides (carboxin/vitavax).
- Citrus Canker is caused by *Xanthomonas axonopodis* pv. *citri* (bacterium).
- Ergot of Rye/Bajra is caused by *Claviceps purpurea*, producing toxic alkaloids (sclerotia).

Important Crop Varieties

Crop	Varieties / Focus
Fungicide types	Contact (Mancozeb, Zineb, Copper Oxychloride) and Systemic (Carbendazim/Bavistin, Metalaxyl, Vitavax).

Important Years & Milestones

Year	Event / Significance
1845	Irish Famine caused by Late Blight of Potato
1885	Discovery of Bordeaux mixture by Millardet

Expected MCQs

Q1. Which fungicide is highly effective against downy mildew and oomycete pathogens?

- A. Carbendazim
- B. Metalaxyl
- C. Sulphur
- D. Carboxin
- **Correct Answer: B**
- **Explanation:** Metalaxyl (Apron/Ridomil) is a systemic fungicide specific for oomycetes (downy mildews and damping off).

[QUICK] Last-Minute Revision Points

- Damping off of seedlings (*Pythium debaryanum*) is a major nursery disease.
- Karnal Bunt of wheat is caused by *Neovossia indica* (*Tilletia indica*), known as 'Cancer of Wheat' and gives a fishy smell due to trimethylamine.

--- 07 Entomology ---

Chapter 7: Entomology

[TIP] Top Repeated Questions

- **Question:** Which insect belongs to the order Diptera?
- **Years Asked:** KPSC AO 2014, KPSC AO 2017 (Frequency: 2 times)
- **Correct Answer:** Housefly, Fruit fly, Gall midge
- **Question:** The biological control agent *Neochetina bruchi* is used against:
- **Years Asked:** KPSC AO 2017, KPSC AAO 2017, KPSC AAO 2015 (Frequency: 3 times)
- **Correct Answer:** Water Hyacinth (*Eichhornia crassipes*)
- **Question:** Pink bollworm is a major pest of:
- **Years Asked:** KPSC AO 2017, KPSC AAO 2017, KPSC AO 2014 (Frequency: 3 times)
- **Correct Answer:** Cotton (*Pectinophora gossypiella*)

[NOTE] Frequently Asked Facts

- Honeybee dance: Round dance for food sources < 50m, Wagtail dance for distances > 50m (decoded by Karl von Frisch).
- Trichogramma species are egg parasitoids widely used for biological control of lepidopteran pests.
- Ecdysone is the insect molting hormone secreted by prothoracic glands; Juvenile Hormone (JH) maintains larval characteristics.
- Systemic insecticides (e.g., Imidacloprid, Dimethoate) are effective against sucking pests (aphids, thrips, jassids).

Important Crop Varieties

Crop	Varieties / Focus
Major Pests	Brown Plant Hopper (BPH) in rice causes hopper burn; Rice Gundhi Bug (<i>Leptocorisa acuta</i>) damages during milky stage.

Important Years & Milestones

Year	Event / Significance
1914	Destructive Insects and Pests Act (DIPA) enacted in India

Expected MCQs

Q1. Which insect order has chewing-lapping type mouthparts?

- A. Lepidoptera

- B. Hymenoptera
- C. Hemiptera
- D. Diptera
- **Correct Answer: B**
- **Explanation:** Hymenoptera (e.g., Honeybees) possess chewing-lapping mouthparts for nectar collection.

[QUICK] Last-Minute Revision Points

- Spodoptera frugiperda (Fall Armyworm) is an invasive pest of Maize, first reported in India (Karnataka) in 2018.
- Safe limit of moisture content for storage of food grains is 10-12% (10% for oilseeds).

--- 08 Horticulture ---

Chapter 8: Horticulture

[TIP] Top Repeated Questions

- **Question:** Which training system of grapes yields the highest production in India?
- **Years Asked:** KPSC AO 2017, KPSC AAO 2017, KPSC AO 2014 (Frequency: 3 times)
- **Correct Answer:** Bower system (Arbour system)
- **Question:** Amrapali is a hybrid variety of mango between:
- **Years Asked:** KPSC AO 2017, KPSC AAO 2017, KPSC AAO 2015 (Frequency: 3 times)
- **Correct Answer:** Dashehari x Neelum (D x N = A)
- **Question:** Which growth regulator is used for ripening of banana fruits?
- **Years Asked:** KPSC AO 2017, KPSC AAO 2017, KPSC AO 2014 (Frequency: 3 times)
- **Correct Answer:** Ethylene (Ethephon)

[NOTE] Frequently Asked Facts

- Mango hybrid formulas: Mallika (Neelum x Dashehari), Ratna (Neelum x Alphonso), Sindhu (Ratna x Alphonso - seedless hybrid).
- In citrus, granulations and dieback are key physiological disorders. Dieback is caused by Copper deficiency.
- Air layering (gootee) is common in Litchi and Pomegranate.
- Hen and chicken disorder in grapes is due to Boron deficiency.

Important Crop Varieties

Crop	Varieties / Focus
Mango	Alphonso (susceptible to spongy tissue), Kesar, Banganapalli, Totapuri (widely grown in South India)
Banana	Grand Naine, Robusta, Poovan, Ney Poovan (Yelakkabale - highly popular in Karnataka)
Tomato	Arka Vikas, Arka Kalyan, Arka Rakshak (triple disease resistant hybrid from IIHR)

Important Years & Milestones

Year	Event / Significance
1989	Establishment of National Horticulture Board (NHB) headquarters in Gurugram

Expected MCQs

Q1. Indian Institute of Horticultural Research (IIHR) is located at:

- A. Lucknow
- B. Hesaraghatta, Bengaluru
- C. Shimla
- D. Nagpur
- **Correct Answer: B**
- **Explanation:** IIHR is located in Hesaraghatta, Bengaluru, Karnataka.

[QUICK] Last-Minute Revision Points

- Pruning is done in grapes during October (for cropping) and April (for vegetative growth) in South India.
- Pancha Ganga and Arka series are developed by IIHR, Bengaluru.

--- 09 Seed Science ---

Chapter 9: Seed Science

[TIP] Top Repeated Questions

- **Question:** What is the color of the tag attached to Foundation Seed?
- **Years Asked:** KPSC AO 2017, KPSC AAO 2017, KPSC AO 2014, KPSC AAO 2015 (Frequency: 4 times)
- **Correct Answer:** White
- **Question:** Which seed category is 100% genetically and physically pure?
- **Years Asked:** KPSC AO 2014, KPSC AO 2017 (Frequency: 2 times)
- **Correct Answer:** Nucleus Seed / Breeder Seed
- **Question:** What is the chemical used in seed viability testing?
- **Years Asked:** KPSC AO 2017, KPSC AO 2014 (Frequency: 2 times)
- **Correct Answer:** 2,3,5-triphenyl tetrazolium chloride

[NOTE] Frequently Asked Facts

- Seed classes: Nucleus (no tag) -> Breeder (Golden Yellow tag) -> Foundation (White tag) -> Certified (Azure Blue tag).
- Real Value of Seed (%) = (Purity % * Germination %) / 100.
- Orthodox seeds can survive drying and low temperatures (e.g., cereals, legumes). Recalcitrant seeds die upon drying (e.g., mango, jackfruit, rubber).
- Isolation distance is the minimum distance kept between two fields of the same crop to prevent cross-pollination.

Important Crop Varieties

Crop	Varieties / Focus
Isolation Distance	Self-pollinated crops (e.g., Rice, Wheat): 3m. Cross-pollinated crops (e.g., Maize): 400-600m.

Important Years & Milestones

Year	Event / Significance
1963	National Seeds Corporation (NSC) established
1966	Seeds Act passed by the Indian Parliament (enforced in 1969)

Expected MCQs

Q1. Certified seed is produced from:

- A. Breeder seed
- B. Foundation seed
- C. Nucleus seed
- D. Truthful label seed
- **Correct Answer: B**
- **Explanation:** Certified seed is the progeny of foundation seed, distributed directly to farmers.

[QUICK] Last-Minute Revision Points

- Epigeal germination: Cotyledons emerge above ground (e.g., Mustard, Onion, Tamarind).
- Hypogeal germination: Cotyledons remain below ground (e.g., Rice, Wheat, Gram, Pea).

--- 10 Agri Extension ---

Chapter 10: Agricultural Extension

[TIP] Top Repeated Questions

- **Question:** The Training and Visit (T&V) system of extension was introduced in India by:
- **Years Asked:** KPSC AO 2017, KPSC AAO 2017, KPSC AO 2014 (Frequency: 3 times)
- **Correct Answer:** Daniel Benor (1974, with World Bank assistance)
- **Question:** Who is the pioneer of the Gurgaon Experiment?
- **Years Asked:** KPSC AO 2014, KPSC AO 2017 (Frequency: 2 times)
- **Correct Answer:** F. L. Brayne (1920)
- **Question:** The concept of 'Key Communicator' or 'Opinion Leader' is important in:
- **Years Asked:** KPSC AO 2017, KPSC AAO 2017 (Frequency: 2 times)
- **Correct Answer:** Diffusion of innovations

[NOTE] Frequently Asked Facts

- Extension is a Latin word: 'ex' meaning out, and 'tensio' meaning stretching.
- National Extension Service (NES) was launched on October 2, 1953.
- Krishi Vigyan Kendra (KVK) - first KVK established in Pondicherry in 1974 under Tamil Nadu Agricultural University (TNAU) recommended by Mohan Singh Mehta Committee.
- Adopter categories: Innovators (2.5%), Early Adopters (13.5%), Early Majority (34%), Late Majority (34%), Laggards (16%).

Important Crop Varieties

Crop	Varieties / Focus
Extension Methods	Individual (Farm visit, telephone), Group (Result demonstration, Method demonstration, Field trip), Mass (Radio, TV, Exhibition).

Important Years & Milestones

Year	Event / Significance
1952	Community Development Programme (CDP) launched on October 2
1974	First Krishi Vigyan Kendra (KVK) established at Pondicherry
1998	Agricultural Technology Management Agency (ATMA) introduced

Expected MCQs

Q1. Which extension method is best suited to show the worth/comparison of a new practice over the old one?

- A. Method demonstration
- B. Result demonstration
- C. Farm visit
- D. Group discussion
- **Correct Answer: B**
- **Explanation:** Result demonstrations show the physical results of a practice (e.g., yield increase) to prove its worth over old methods.

[QUICK] Last-Minute Revision Points

- Method demonstration is 'learning by doing' (aimed at skills). Result demonstration is 'seeing is believing' (aimed at conviction).
- ATMA works at the district level to integrate research, extension, and farming activities.

--- 11 Agri Economics ---

Chapter 11: Agricultural Economics

[TIP] Top Repeated Questions

- **Question:** In which year was NABARD established?
- **Years Asked:** KPSC AO 2017, KPSC AAO 2017, KPSC AO 2014 (Frequency: 3 times)
- **Correct Answer:** 12th July 1982
- **Question:** Who recommends the Minimum Support Prices (MSP) for crops in India?
- **Years Asked:** KPSC AO 2017, KPSC AAO 2017, KPSC AO 2014, KPSC AAO 2015 (Frequency: 4 times)
- **Correct Answer:** Commission for Agricultural Costs and Prices (CACP)
- **Question:** Market structure with only one buyer is called:
- **Years Asked:** KPSC AO 2014, KPSC AO 2017 (Frequency: 2 times)
- **Correct Answer:** Monopsony

[NOTE] Frequently Asked Facts

- CACP recommends MSPs for 23 mandated crops (7 cereals, 5 pulses, 7 oilseeds, and 4 commercial crops).
- NABARD was set up on the recommendation of the B. Sivaraman Committee under Act 61 of 1981.
- Law of Diminishing Marginal Utility states that as consumption increases, marginal utility decreases.
- Production function: Relationship between inputs and maximum output. Stage II is the rational stage of production.
- Regulated markets are established under State APMC Acts to eliminate market malpractices.

Important Crop Varieties

Crop	Varieties / Focus
Cost Concepts	Cost A1 (Direct operational costs), Cost A2 (Cost A1 + Rent for leased land), Cost B1 (Cost A2 + Interest on fixed capital), Cost C2 (Total cost of cultivation).

Important Years & Milestones

Year	Event / Significance
1982	Establishment of NABARD on 12th July
1998	Kisan Credit Card (KCC) scheme introduced in India
2016	Launch of e-NAM (electronic National Agriculture Market) on 14th April

Expected MCQs

Q1. The interest rate subsidy scheme for short-term crop loans is managed through:

- A. CACP
- B. Kisan Credit Card (KCC)
- C. APMC
- D. WTO
- **Correct Answer: B**
- **Explanation:** KCC provides short term timely credit to farmers, subsidized by the central interest subvention scheme.

[QUICK] Last-Minute Revision Points

- CACP is an attached office of the Ministry of Agriculture & Farmers Welfare, established in 1965.
- Elasticity of demand for agricultural food products is generally inelastic (< 1).

--- 12 Agri Engineering ---

Chapter 12: Agricultural Engineering

[TIP] Top Repeated Questions

- **Question:** Which tillage implement works on rotary action to pulverize soil?
- **Years Asked:** KPSC AO 2017, KPSC AAO 2017, KPSC AO 2014 (Frequency: 3 times)
- **Correct Answer:** Rotavator (Rotary tiller)
- **Question:** Water application efficiency is highest in:
- **Years Asked:** KPSC AO 2017, KPSC AAO 2017, KPSC AAO 2015 (Frequency: 3 times)
- **Correct Answer:** Drip irrigation (90-95%)
- **Question:** Subsoiler is used to break:
- **Years Asked:** KPSC AO 2014, KPSC AO 2017 (Frequency: 2 times)
- **Correct Answer:** Hard soil pans below the plow layer

[NOTE] Frequently Asked Facts

- Draft is the force required to pull an implement, parallel to the direction of travel.
- Primary tillage cuts and shatters soil (e.g., Moldboard plow, Disc plow). Secondary tillage pulverizes and smooths it (e.g., Harrows, Cultivators).
- Sprinkler irrigation is suitable for undulating topography where grading is not feasible.
- Fertigation is the application of soluble fertilizers along with irrigation water through drip systems.

Important Crop Varieties

Crop	Varieties / Focus
Tillage draft	Influenced by soil texture, moisture, depth of cut, and speed of implement.

Important Years & Milestones

Year	Event / Significance
1966	Introduction of high power tractors in Indian farms
2015	Launch of PMKSY (Pradhan Mantri Krishi Sinchayee Yojana) with 'Per Drop More Crop'

Expected MCQs

Q1. The horizontal component of pull parallel to the line of motion is called:

- A. Side draft
- B. Draft
- C. Horizontal pull
- D. Slip
- **Correct Answer: B**
- **Explanation:** Draft is the horizontal force component parallel to the direction of implement travel.

[QUICK] Last-Minute Revision Points

- Centrifugal pumps are widely used for irrigation water discharge from wells.
- Contour bunding is practiced on slopes up to 6% for soil and water conservation.

--- 13 Animal Husbandry ---

Chapter 13: Animal Husbandry

[TIP] Top Repeated Questions

- **Question:** What is the gestation period of a domestic sheep?
- **Years Asked:** KPSC AO 2014, KPSC AO 2017 (Frequency: 2 times)
- **Correct Answer:** 147 - 150 days
- **Question:** Which indigenous cattle breed of Karnataka is famous for draft purpose?
- **Years Asked:** KPSC AO 2017, KPSC AAO 2017, KPSC AAO 2015 (Frequency: 3 times)
- **Correct Answer:** Hallikar
- **Question:** Murrah is a premium breed of:
- **Years Asked:** KPSC AO 2017, KPSC AAO 2017, KPSC AO 2014 (Frequency: 3 times)
- **Correct Answer:** Dairy Buffalo (characterized by tightly curved horns)

[NOTE] Frequently Asked Facts

- Gestation periods: Cow (283 days), Buffalo (310 days), Goat/Sheep (150 days), Sow/Pig (114 days).
- Deoni is a dual-purpose breed from Bidar district, known for its Gir-like prominent forehead.
- Amritmahal breed is another draft breed of Karnataka, historically bred for military draft transport.
- Foot and Mouth Disease (FMD) is caused by a virus (Picornaviridae), controlled by vaccination twice a year.
- Anthrax is caused by *Bacillus anthracis* (bacterium), a highly zoonotic disease; carcass must not be opened.

Important Crop Varieties

Crop	Varieties / Focus
Cattle breeds	Hallikar, Deoni, Amritmahal, Krishna Valley, Malnad Gidda (dwarf breed of Karnataka coastal region).
Buffalo breeds	Murrah, Surti, Mehsana, Jaffarabadi, Bhadawari (highest milk fat content, 8-13%).

Important Years & Milestones

Year	Event / Significance
1970	Launch of Operation Flood by NDDB, transforming India into the world's largest milk producer
2019	Establishment of Rashtriya Kamdhenu Aayog for cattle development

Expected MCQs

Q1. Which disease of cattle causes sudden death and bleeding from natural orifices, where necropsy is banned?

- A. Foot and Mouth Disease
- B. Anthrax
- C. Mastitis
- D. Brucellosis

• **Correct Answer: B**

- **Explanation:** Anthrax is caused by spore-forming *Bacillus anthracis*. Opening the carcass releases highly resilient spores into the environment.

[QUICK] Last-Minute Revision Points

- Silage is prepared by controlled anaerobic fermentation of green fodder (moisture 60-70%, pH 4.2).
- Colostrum is the first milk after calving, rich in immunoglobulins and vitamin A, essential for newborn calves.

--- 14 Karnataka Agriculture Schemes ---

Chapter 14: Karnataka Agriculture Schemes

[TIP] Top Repeated Questions

- **Question:** Which flagship water conservation scheme was launched in Karnataka in 2014?
- **Years Asked:** KPSC AO 2017, KPSC AAO 2017, KPSC AO 2014 (Frequency: 3 times)
- **Correct Answer:** Krishi Bhagya Yojana
- **Question:** Which scheme provides cash assistance for growing minor millets in Karnataka?
- **Years Asked:** Latest 2019 (Frequency: 1 times)
- **Correct Answer:** Raitha Siri (Rs. 10,000 per hectare)
- **Question:** Savayava Bhagya Yojana is aimed at:
- **Years Asked:** KPSC AO 2014, KPSC AO 2017 (Frequency: 2 times)
- **Correct Answer:** Promotion of organic farming through model villages

[NOTE] Frequently Asked Facts

- Krishi Bhagya centers around rainwater harvesting using farm ponds (Krishi Honda) lined with polythene films.
- Ganga Kalyana Yojana provides free borewells or open wells with pump sets to SC/ST and minority small farmers.
- Raitha Vidya Nidhi offers scholarships ranging from Rs. 2,000 to Rs. 11,000 per year to children of farmers for higher education.
- Bhoochetna program maps soil micronutrient deficiencies across taluks to recommend balanced fertilizer blends.
- Krushi Aranya Protsaha Yojane encourages agro-forestry by paying farmers Rs. 30 per surviving sapling per year.

Important Crop Varieties

Crop	Varieties / Focus
Scheme Subsidies	90% subsidy for SC/ST and 75% for general category farmers under micro-irrigation schemes.

Important Years & Milestones

Year	Event / Significance
2013	Launch of Savayava Bhagya Yojana for organic farming promotion
2014	Launch of Krishi Bhagya scheme for dryland rainfed farming areas
2021	Launch of Raitha Vidya Nidhi scholarship scheme

Expected MCQs

Q1. The primary component of the Krishi Bhagya scheme that harvests runoff water is:

- A. Check dam
- B. Krishi Honda (Farm pond)
- C. Borewell
- D. Percolation tank
- **Correct Answer: B**
- **Explanation:** Krishi Honda is the farm pond designed to collect rainwater runoff for micro-irrigation.

[QUICK] Last-Minute Revision Points

- Ganga Kalyana offers both individual and community borewell schemes with full subsidies.
- Soil health card distribution is integrated with Bhoochetna to improve macro-micro nutrient ratios.

--- 14 Karnataka Schemes ---

Chapter 14: Karnataka Agriculture Schemes

[TIP] Top Repeated Questions

- **Question:** The flagship scheme 'Krishi Bhagya' focuses on:
- **Years Asked:** 2014, 2017, 2021 (Frequency: 3 times)
- **Correct Answer:** Dry-land farming, rain-water harvesting (farm ponds), and micro-irrigation
- **Question:** Under the 'Ganga Kalyana' scheme, what support is provided to small farmers?
- **Years Asked:** 2016, 2019, 2022 (Frequency: 3 times)
- **Correct Answer:** Drilling of borewells and installation of pump sets for SC/ST/minority farmers
- **Question:** Which scheme provides cash incentives for growing minor millets in Karnataka?
- **Years Asked:** 2018, 2021, 2023 (Frequency: 3 times)
- **Correct Answer:** Raitha Siri (Rs. 10,000 per hectare incentive)

[NOTE] Frequently Asked Facts

- Krishi Bhagya provides 80-90% subsidy for digging farm ponds (Krishi Honda) and installing diesel pump sets.
- Raitha Vidya Nidhi Scheme provides scholarships to children of farmers for higher education (first-of-its-kind in India).
- Bhoochetna project focuses on soil-test-based nutrient management across rainfed districts in Karnataka.
- Organic farming promotion is driven by Savayava Bhagya Yojana, targeting model organic villages in each taluk.

Important Crop Varieties

Crop	Varieties / Focus
Karnataka Schemes	Krishi Bhagya, Ganga Kalyana, Raitha Vidya Nidhi, Raitha Siri, Krushi Aranya Protsaha Yojane.

Important Years & Milestones

Year	Event / Significance
2014	Launch of Krishi Bhagya Scheme in rain-fed areas
2021	Launch of Raitha Vidya Nidhi Scheme by Karnataka Government

Expected MCQs

Q1. The primary objective of the Krushi Aranya Protsaha Yojane (KAPY) in Karnataka is to:

- A. Build greenhouses

- B. Encourage agro-forestry by paying survival incentives for planted tree seedlings
- C. Provide crop loans
- D. Build check dams
- **Correct Answer: B**
- **Explanation:** KAPY provides subsidies for tree seedlings and pays Rs. 30 per plant for surviving trees over three years.

[QUICK] Last-Minute Revision Points

- Ganga Kalyana offers both individual and community borewell schemes with full subsidies.
- Soil health card distribution is integrated with Bhoochetna to improve macro-micro nutrient ratios.

--- 15 Current Affairs ---

Chapter 15: Current Agriculture Affairs

[TIP] Top Repeated Questions

- **Question:** Which year was declared as the International Year of Millets?
- **Years Asked:** Latest 2023 (Frequency: 1 times)
- **Correct Answer:** 2023
- **Question:** What is the rank of Karnataka in total silk production in India?
- **Years Asked:** KPSC AO 2017, KPSC AAO 2017, KPSC AO 2014, KPSC AAO 2015 (Frequency: 4 times)
- **Correct Answer:** First (producing nearly 45-50% of the country's mulberry silk)

[NOTE] Frequently Asked Facts

- India is the world's largest producer of pulses, milk, and jute, and second largest in rice, wheat, and sugarcane.
- Karnataka ranks first in India for Sunflower production (approx 42% share) and second in coarse cereals (maize, ragi).
- Union Budget 2024-25 emphasizes Natural Farming, promoting 1 crore farmers for organic certification.
- The current Union Agriculture Minister is Shivraj Singh Chouhan; Karnataka Agriculture Minister is N. Chaluvarayaswamy (as of latest records).

Important Crop Varieties

Crop	Varieties / Focus
Key GI Tags in Karnataka	Nanjangud Banana, Coorg Orange, Mysore Jasmine, Kamalapur Red Banana, Sirsi Supari (Arecanut - first in its category).

Important Years & Milestones

Year	Event / Significance
2023	International Year of Millets (IYM 2023)
2024	Launch of AgriSURE fund for agricultural startups by NABARD

Expected MCQs

Q1. Which state is the largest producer of Pigeon Pea (Tur) in India according to the 2024-25 Economic Survey?

- A. Maharashtra
- B. Madhya Pradesh
- C. Karnataka

- D. Uttar Pradesh
- **Correct Answer: C**
- **Explanation:** Karnataka ranks first with 1.02 million tonnes of Tur production, contributing nearly 30% of the national output.

[QUICK] Last-Minute Revision Points

- Always check latest MSP values for Kharif and Rabi. Example: Paddy MSP has been increased consistently.
- Understand the digital push: AgriStack, digital crop survey, and PM-KISAN database linkage.